Chapter 12

Fundamentals of Electricity & Electronics

Introduction

This chapter addresses the fundamental concepts that are the building blocks for advanced electrical knowledge and practical troubleshooting. Some of the questions addressed are: How does energy travel through a copper wire and through space? What is electric current and electromotive force? What makes a landing light turn on or a hydraulic pump motor run? Each of these questions requires an understanding of many basic principles. By adding one basic idea on top of other basic ideas, it becomes possible to answer most of the interesting and practical questions about electricity or electronics.

Our understanding of electrical current must begin with the nature of matter. All matter is composed of molecules. All molecules are made up of atoms, which are themselves made up of electrons, protons, and neutrons.

General Composition of Matter

Matter

Matter can be defined as anything that has mass and volume and is the substance of which physical objects are composed. Essentially, it is anything that can be touched. Matter is what all things are made of; whatever occupies space, has mass, and is perceptible to the senses in some way. Weight is an indirect method of determining mass, but it is not the same. Weight is a measure of the pull of gravity acting on the mass of an object. The more mass an object has, the more it weighs under the earth's force of gravity. Mathematically, weight can be stated as follows:

$$\text{Weight} = \text{Mass} \times \text{Gravity}$$

Categories of matter are ordered by molecular activity. The four categories or states are: solids, liquids, gases, and plasma. For the purposes of the aircraft technician, only solids, liquids, and gases are considered.

Element

An element is a substance that cannot be reduced to a simpler form by chemical means. Iron, gold, silver, copper, and oxygen are examples of elements. Beyond this point of reduction, the element ceases to be what it is.

Compound

A compound is a chemical combination of two or more elements. Water is one of the most common compounds and is made up of two hydrogen atoms and one oxygen atom.

Molecule

The smallest particle of matter that can exist and still retain its identity, such as water (H_2O), is called a molecule. *[Figure 12-1]* Substances composed of only one type of atom are called elements. But most substances occur in nature as compounds, that is, combinations of two or more types of atoms. It would no longer retain the characteristics of water if it were compounded of one atom of hydrogen and two atoms of oxygen. If a drop of water is divided and then divided again and again until it cannot be divided any longer, it is still water.

Atom

The atom is considered to be the most basic building block of all matter. Atoms are composed of three subatomic particles: protons, neutrons, and electrons. These three particles determine the properties of the specific atoms. Elements are substances composed of the same atoms with specific properties. Oxygen is an example of this. The main property that defines each element is the number of neutrons, protons, and electrons. Hydrogen and helium are examples of elements. Both of these elements have neutrons, protons, and electrons but differ in the number of those items. This difference alone accounts for the variations in chemical and physical properties of these two different elements. There are over 100 known elements in the periodic table. *[Figure 12-2]* They are categorized according to their properties on that table. The kinetic theory of matter also states that the particles that make up the matter are always moving. Thermal expansion is considered in the kinetic theory and explains why matter contracts when it is cool and expands when it is hot, with the exception of water/ice.

Electrons, Protons, & Neutrons

At the center of the atom is the nucleus, which contains protons and neutrons. Protons are positively-charged particles, and neutrons are neutrally-charged particles. A neutron has approximately the same mass as the proton. The third particle of the atom is the electron that is a negatively-

charged particle with a very small mass compared to the proton. The proton's mass is approximately 1,837 times greater than the electron. Due to the proton and the neutron location in the central portion of the atom (nucleus) and the electron's position at the distant periphery of the atom, it is the electron that undergoes the change during chemical reactions. Since a proton weighs approximately 1,845 times as much as an electron, the number of protons and neutrons in its nucleus determines the overall weight of an atom. The weight of an electron is not considered in determining the weight of an atom. Indeed, the nature of electricity cannot be defined clearly because it is not certain whether the electron is a negative charge with no mass (weight) or a particle of matter with a negative charge.

Hydrogen represents the simplest form of an atom. *[Figure 12-3]* At the nucleus of the hydrogen atom is one proton and at the outer shell is one orbiting electron. At a more complex level is the oxygen atoms, which has eight electrons in two shells orbiting the nucleus with eight protons and eight neutrons. *[Figure 12-4]* When the total positive charge of the protons in the nucleus equals the total negative charge of the electrons in orbit around the nucleus, the atom is said to have a neutral charge.

Electron Shells & Energy Levels

Electrons require a certain amount of energy to stay in an orbit. This particular quantity is called the electron's energy level. By its motion alone, the electron possesses kinetic energy, while the electron's position in orbit determines its potential energy. The total energy of an electron is the main factor that determines the radius of the electron's orbit.
Electrons of an atom appear only at certain definite energy levels (shells). The spacing between energy levels is such that when the chemical properties of the various elements are cataloged, it is convenient to group several closely spaced permissible energy levels together into electron shells. The maximum number of electrons that can be contained in any shell or sub-shell is the same for all atoms and is defined as electron capacity = $2n^2$. In this equation, n represents the energy level in question. The first shell can only contain two electrons; the second shell can only contain eight electrons; the third, 18, and so on until we reach the seventh shell for the heaviest atoms, which have six energy levels. Because the innermost shell is the lowest energy level, the shell begins to fill up from the shell closest to the nucleus and fill outward as the atomic number of the element increases. However, an energy level does not need to be completely filled before electrons begin to fill the next level. The Periodic Table of Elements should be checked to determine an element's electron configuration.

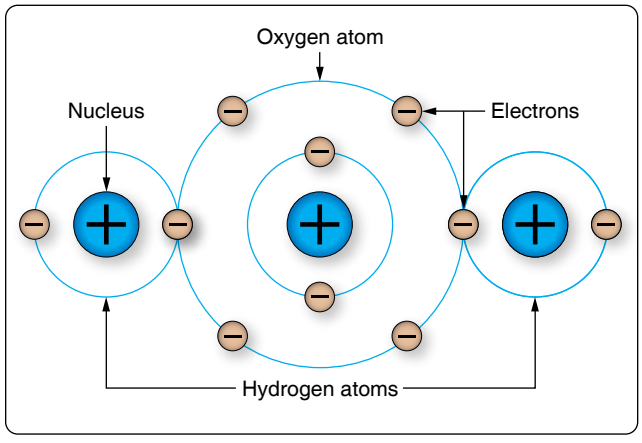


Figure 12-1. *A water molecule.*

Valence Electrons

Valence is the number of chemical bonds an atom can form. Valence electrons are electrons that can participate in chemical bonds with other atoms. The number of electrons in the outermost shell of the atom is the determining factor in its valence. Therefore, the electrons contained in this shell are called valence electrons.

Ions

Ionization is the process by which an atom loses or gains electrons. Dislodging an electron from an atom causes the atom to become positively charged. This net positively-charged atom is called a positive ion or a cation. An atom that has gained an extra number of electrons is negatively charged and is called a negative ion or an anion. When atoms are neutral, the positively-charged proton and the negatively-charged electron are equal.

Free Electrons

Valence electrons are found drifting midway between two nuclei. Some electrons are more tightly bound to the nucleus of their atom than others and are positioned in a shell or sphere closer to the nucleus, while others are more loosely bound and orbit at a greater distance from the nucleus. These outermost electrons are called "free" electrons because they can be easily dislodged from the positive attraction of the protons in the nucleus. Once freed from the atom, the electron can then travel from atom to atom, becoming the flow of electrons commonly called current in a practical electrical circuit.

Electron Movement

Conductors, Insulators, and Semiconductors

The valence of an atom determines its ability to gain or lose an electron, which ultimately determines the chemical and electrical properties of the atom. These properties can be categorized as being a conductor, semiconductor, or insulator, depending on the ability of the material to produce free electrons. When a material has a large number of free

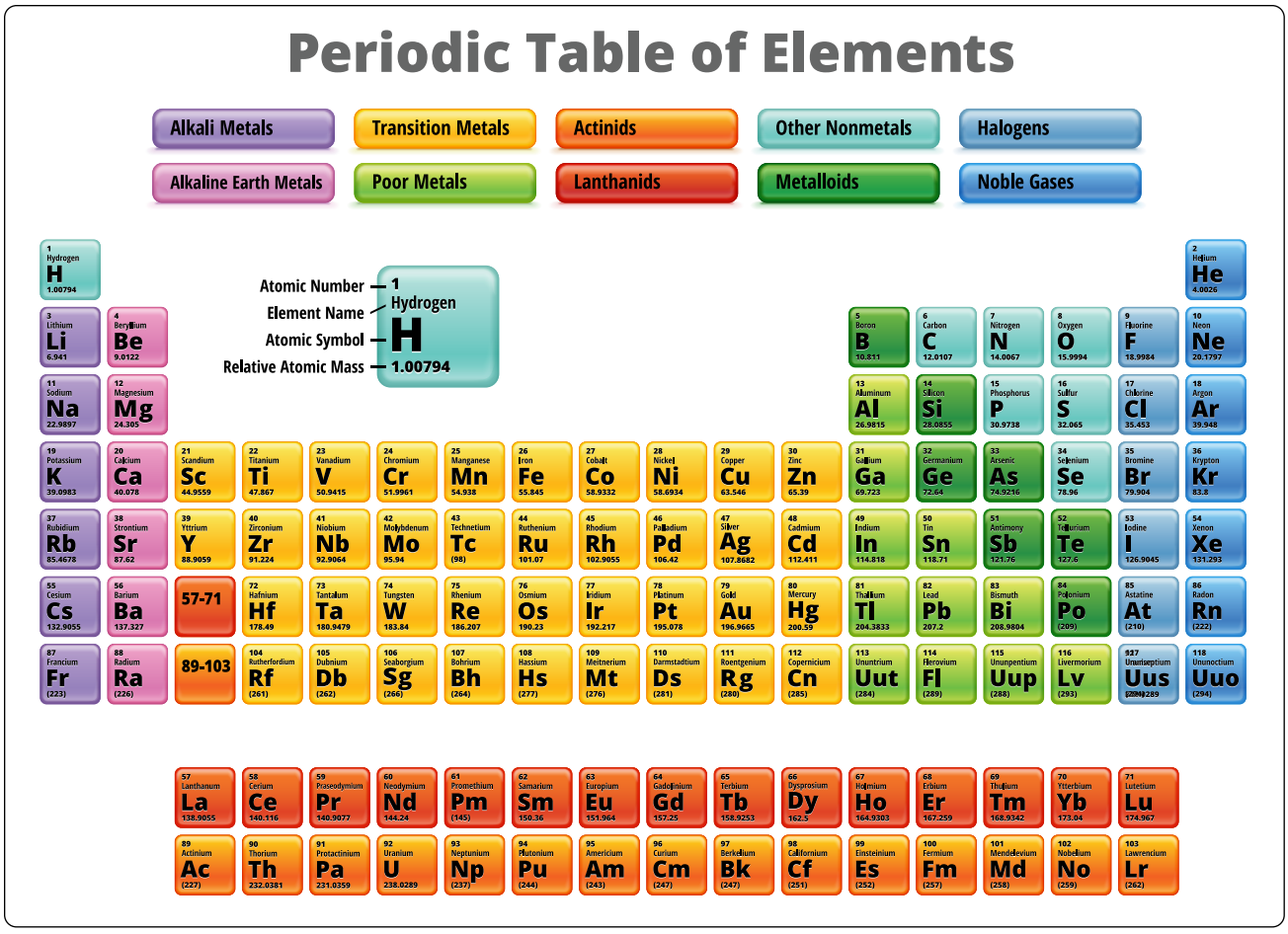


Figure 12-2. *Periodic table of elements.*

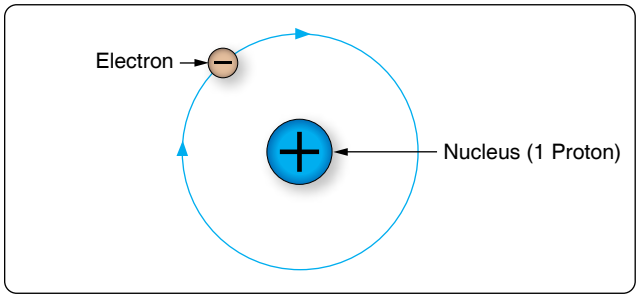


Figure 12-3. *Hydrogen atom.*

electrons available, a greater current can be conducted in the material.

Conductors

Elements such as gold, copper, and silver possess many free electrons and make good conductors. The atoms in these materials have a few loosely bound electrons in their outer orbits. Energy in the form of heat can cause these electrons in the outer orbit to break loose and drift throughout the material. Copper and silver have one electron in their outer orbits. At room temperature, a piece of silver wire has billions

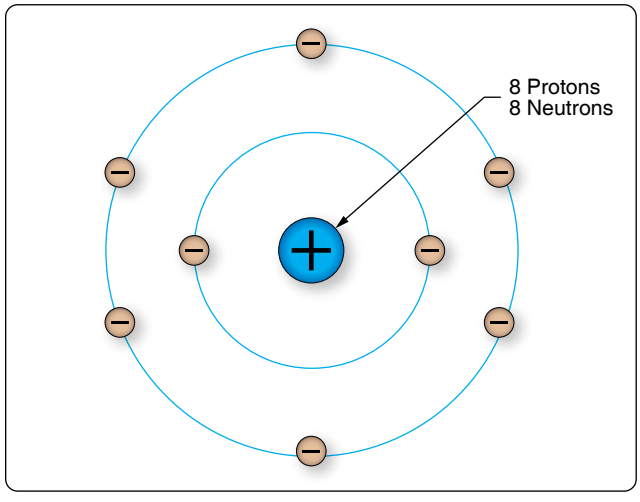


Figure 12-4. *Oxygen atom.*

of free electrons.

Insulators

Insulators are materials that do not conduct electrical current very well or not at all, such as glass, ceramic, and plastic.

Under normal conditions, atoms in these materials do not produce free electrons. The absence of free electrons means that electrical current cannot be conducted through the material. Only when the material is in an extremely strong electrical field will the outer electrons be dislodged. This action is called breakdown and usually causes physical damage to the insulator.

Semiconductors

The material properties of semiconductors fall in between conductors and insulators. In their pure state, they are not good at conducting or insulating. Semiconductors can operate like a conductor or insulator depending on what external load is placed on the material. Semiconductors are used to make transistors and integrated circuits. Silicon and germanium are the most widely used semiconductor materials. For a more detailed explanation on this topic, refer to page 12-98 in this chapter.

Metric Based Prefixes Used for Electrical Calculations

In any system of measurements, a single set of units is usually not sufficient for all the computations involved in electrical repair and maintenance. Small distances, for example, can usually be measured in inches, but larger distances are more meaningfully expressed in feet, yards, or miles. Since electrical values often vary from numbers that are a millionth part of a basic unit of measurement to very large values, it is often necessary to use a wide range of numbers to represent the values of units, such as volts, amperes, or ohms. A series of prefixes that appear with the name of the unit have been devised for the various multiples or submultiples of the basic units. There are 12 of these prefixes, which are also known as conversion factors. Four of the most commonly used prefixes in electrical work are:

Mega (M) means one million (1,000,000).

Kilo (k) means one thousand (1,000).

Milli (m) means one-thousandth (1⁄1,000).

Micro (µ) means one-millionth (1⁄1,000,000).

Kilo is one of the most extensively used conversion factors. It explains the use of prefixes with basic units of measurement. Kilo means 1,000, and when used with volts, is expressed as kilovolt, meaning 1,000 volts. The symbol for kilo is the letter "k." Thus, 1,000 volts is one kilovolt or 1 kV. Conversely, one volt would equal one-thousandth of a kV, or 1⁄1,000 kV. This could also be written 0.001 kV.

Similarly, the word "milli" means one-thousandth, and thus, 1 millivolt equals one-thousandth (1⁄1000) of a volt. *Figure 12-5* contains a complete list of the multiples used to express electrical quantities, together with the prefixes and symbols used to represent each number.

Static Electricity

Electricity is often described as being either static or dynamic. The difference between the two is based simply on whether the electrons are at rest (static) or in motion (dynamic). Static electricity is a buildup of an electrical charge on the surface of an object. It is considered "static" due to the fact that there is no current flowing as in alternate current (AC) or direct current (DC) electricity. Static electricity is usually caused when non-conductive materials, such as rubber, plastic, or glass, are rubbed together causing a transfer of electrons, which results in an imbalance of charges between the two materials. The fact that there is an imbalance of charges between the two materials means that the objects will exhibit an attractive or repulsive force.

Attractive and Repulsive Forces

One of the most fundamental laws of static electricity, as well as magnetics, deals with attraction and repulsion. Like charges repel each other and unlike charges attract each other. All electrons possess a negative charge and as such repel each other. Similarly, all protons possess a positive charge and as such repel each other. Electrons (negative) and protons (positive) are opposite in their charge and attract each other.

For example, if two pith balls are suspended, as shown in *Figure 12-6*, and each ball is touched with the charged glass rod, some of the charge from the rod is transferred to the balls. The balls now have similar charges and, consequently, repel each other as shown in part B of *Figure 12-6*. If a plastic rod is rubbed with fur, it becomes negatively charged and the fur is positively charged. By touching each ball with these differently charged sources, the balls obtain opposite charges and attract each other as shown in part C of *Figure 12-6*.

Although most objects become charged with static electricity by means of friction, a charged substance can also influence objects near it by contact. *[Figure 12-7]* If a positively-charged rod touches an uncharged metal bar, it draws electrons from the uncharged bar to the point of contact. Some electrons enter the rod, leaving the metal bar with a deficiency of electrons (positively charged) and making the rod less positive than it was or, perhaps, even neutralizing its charge completely.

A method of charging a metal bar by induction is demonstrated in *Figure 12-8*. A positively-charged rod is brought near, but does not touch, an uncharged metal bar. Electrons in the metal bar are attracted to the end of the bar nearest the positively-charged rod, leaving a deficiency of electrons at the opposite end of the bar. If this positively-charged end is touched by a neutral object, electrons will flow into the metal bar and

neutralize the charge. The metal bar is left with an overall excess of electrons.

Electrostatic Field

A field of force exists around a charged body. This field is an electrostatic field (sometimes called a dielectric field) and is represented by lines extending in all directions from the charged body and terminating where there is an equal and opposite charge.

To explain the action of an electrostatic field, lines are used to represent the direction and intensity of the electric field of force. As illustrated in *Figure 12-9*, the intensity of the field is indicated by the number of lines per unit area, and the direction is shown by arrowheads on the lines pointing in the direction in which a small test charge would move (or tend to move) if acted upon by the field of force.

Either a positive or negative test charge can be used, but it has been arbitrarily agreed that a small positive charge is always used in determining the direction of the field. Thus, the direction of the field around a positive charge is always away from the charge because a positive test charge would be repelled. *[Figure 12-9]* On the other hand, the direction of the lines about a negative charge is toward the charge, since a positive test charge is attracted toward it.

Figure 12-10 illustrates the field around bodies having like charges. Positive charges are shown, but regardless of the type of charge, the lines of force would repel each other if the charges were alike. The lines terminate on material objects and always extend from a positive charge to a negative charge. These are imaginary lines used to show the direction a real force takes.

It is important to know how a charge is distributed on an object. *Figure 12-11* shows a small metal disk on which a concentrated negative charge has been placed. By using an electrostatic detector, it can be shown that the charge is spread evenly over the entire surface of the disk. Since the metal disk provides uniform resistance everywhere on its surface, the mutual repulsion of electrons results in an even distribution over the entire surface.

Number	Prefix	Symbol
1,000,000,000,000	tera	t
1,000,000,000	giga	g
1,000,000	mega	M
1,000	kilo	k
100	hecto	h
10	deka	dk
0.1	deci	d
0.01	centi	c
0.001	milli	m
0.000001	micro	μ

Figure 12-5. *Prefixes and symbols for multiples of basic quantities.*

Another example, shown in *Figure 12-12*, is the charge on a hollow sphere. Although the sphere is made of conducting material, the charge is evenly distributed over the outside surface. The inner surface is completely neutral. This phenomenon is used to safeguard operating personnel of the large Van de Graaff static generators used for atom smashing. The safest area for the operators is inside the large sphere, where millions of volts are being generated.

The distribution of the charge on an irregularly-shaped object differs from that on a regularly-shaped object. *Figure 12-13* shows that the charge on such objects is not evenly distributed. The greatest charge is at the points, or areas of sharpest curvature, of the objects.

Electrostatic Discharge (ESD) Considerations

One of the most frequent causes of damage to a solid-state component or integrated circuits is the electrostatic discharge (ESD) from the human body when one of these devices is handled. Careless handling of line replaceable units (LRUs), circuit cards, and discrete components can cause unnecessarily time consuming and expensive repairs. This damage can occur if a technician touches the mating pins for a card or box. Other sources for ESD can be the top of a toolbox that is covered with a carpet. Damage can be avoided by discharging the static electricity from your body by touching the chassis of the removed box, by wearing a grounding wrist strap, and exercising good professional handling of the components in the aircraft. This can include placing protective caps over open connectors and not placing an ESD-sensitive component in an environment that causes damage. Parts that are ESD sensitive are typically shipped in bags specially designed to protect components from electrostatic damage.

Other precautions that should be taken with working with electronic components are:

1. Always connect a ground between test equipment and circuit before attempting to inject or monitor a signal.
2. Ensure test voltages do not exceed maximum allowable voltage for the circuit components and transistors.
3. Ohmmeter ranges that require a current of more than one milliampere in the test circuit should not be used for testing transistors.

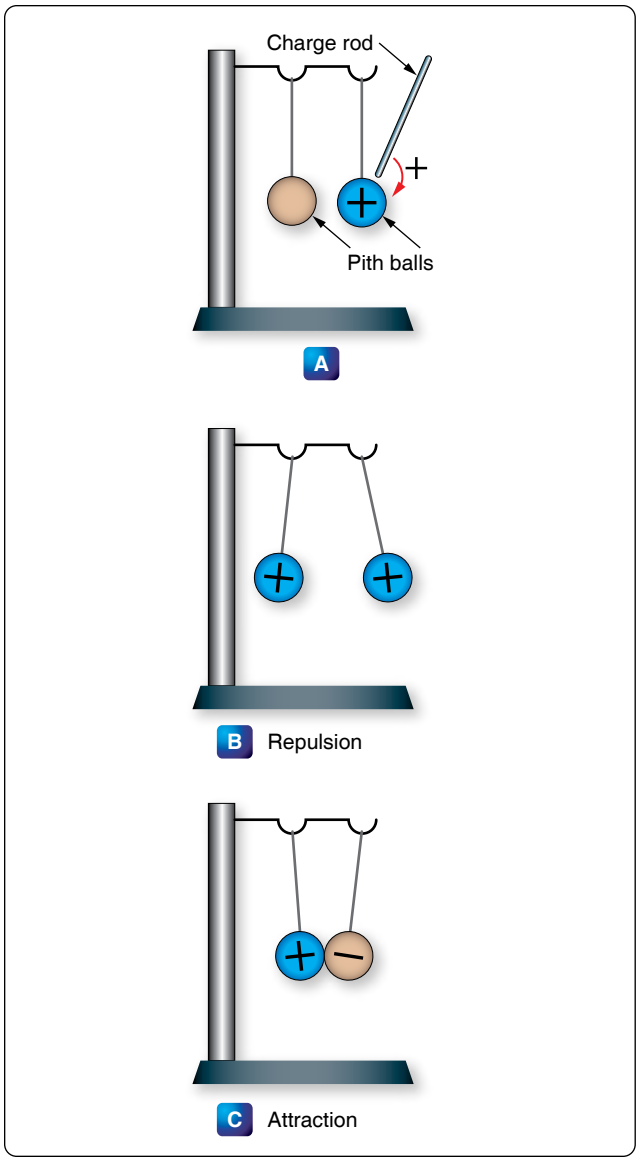


Figure 12-6. *Reaction of like and unlike charges.*

4. The heat applied to a diode or transistor, when soldering is required, should be kept to a minimum by using low-wattage soldering irons and heat sinks.
5. Do not pry components of a circuit board.
6. Power must be removed from a circuit before replacing a component.
7. When using test probes on equipment and the space between the test points is very close, keep the exposed portion of the leads as small as possible to prevent shorting.

Magnetism

Magnetism is defined as the property of an object to attract certain metallic substances. In general, these substances are ferrous materials; that is, materials composed of iron or iron

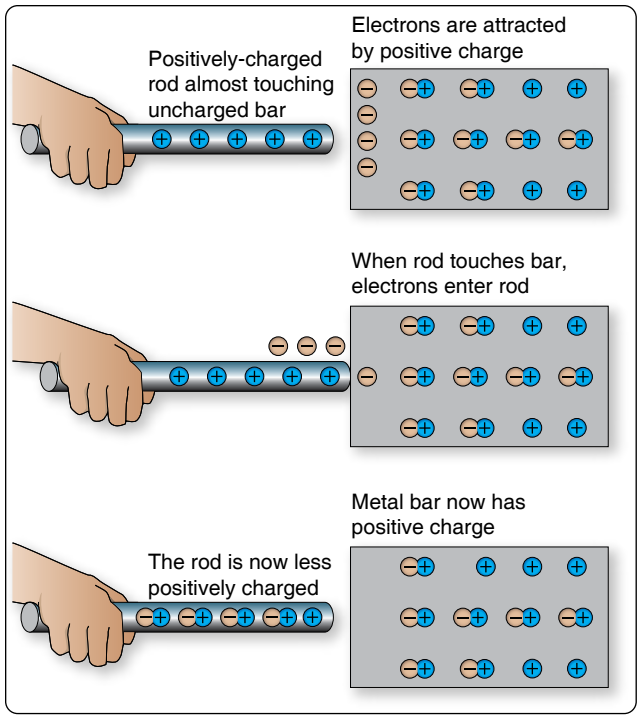


Figure 12-7. *Charging by contact.*

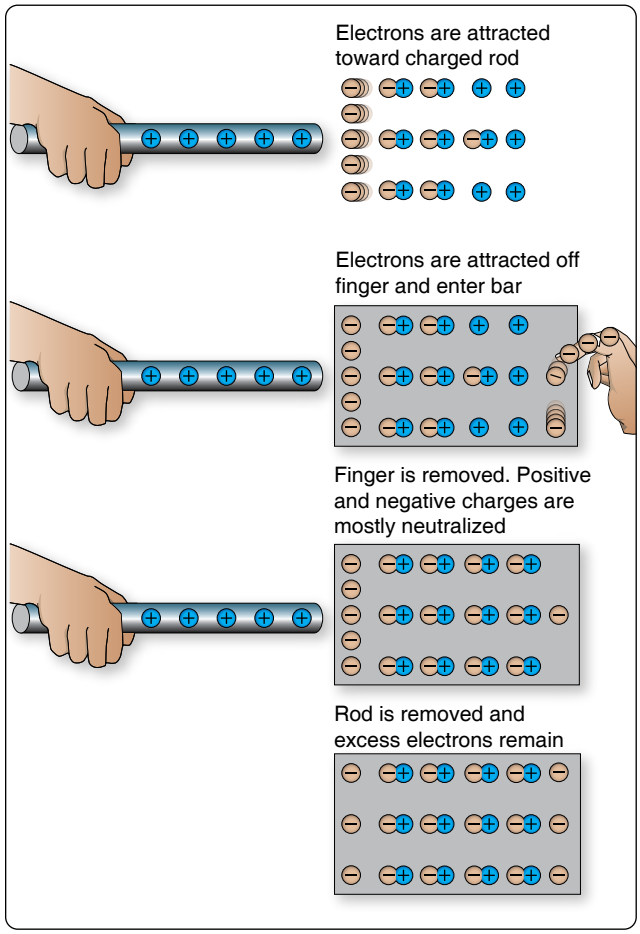


Figure 12-8. *Charging a bar by induction.*

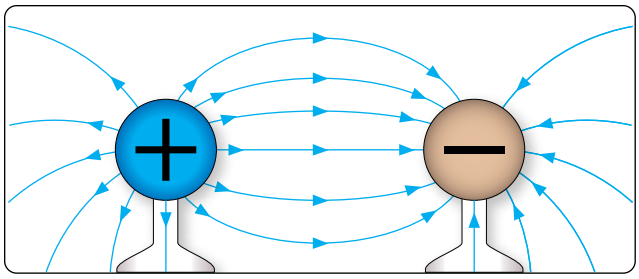


Figure 12-9. *Direction of electric field around positive and negative charges.*

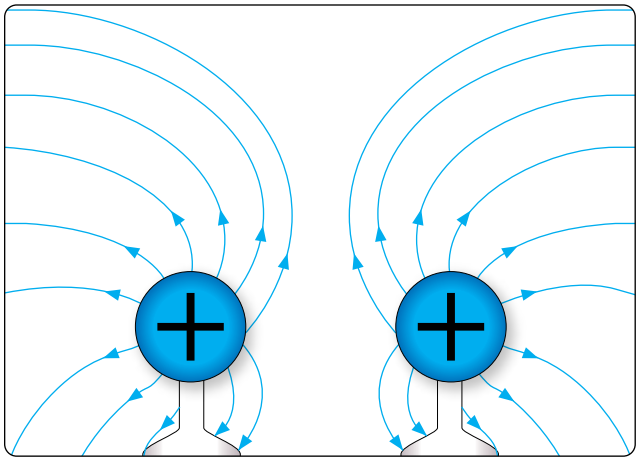


Figure 12-10. *Field around two positively-charged bodies.*

alloys, such as soft iron, steel, and alnico. These materials, sometimes called magnetic materials, include at least three nonferrous materials: nickel, cobalt, and gadolinium, which are magnetic to a limited degree. All other substances are considered nonmagnetic. A few of these non-magnetic substances can be classified as diamagnetic since they are repelled by both poles of a magnet.

Magnetism is an invisible force, the ultimate nature of which has not been fully determined. It can best be described by the effects it produces. Examination of a simple bar magnet similar to that illustrated in *Figure 12-14* discloses some basic characteristics of all magnets. If the magnet is suspended to

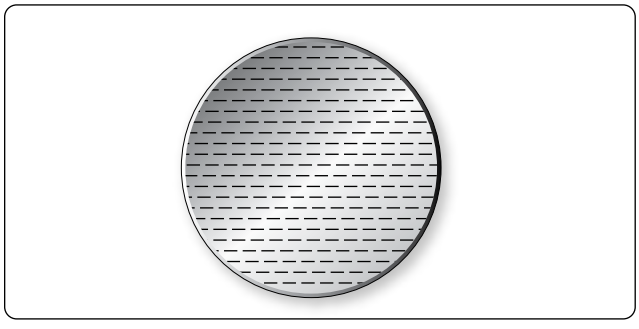


Figure 12-11. *Even distribution of charge on metal disk.*

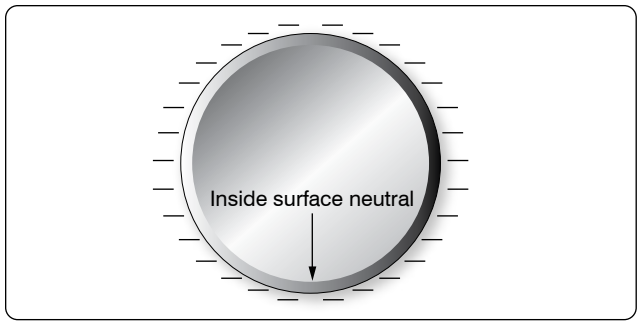


Figure 12-12. *Charge on a hollow sphere.*

swing freely, it aligns itself with the earth's magnetic poles. One end is labeled "N," meaning the North seeking end or pole of the magnet. If the "N" end of a compass or magnet is referred to as North seeking rather than North, there is no conflict in referring to the pole it seeks, which is the North magnetic pole. The opposite end of the magnet, marked "S" is the South seeking end and points to the South magnetic pole. Since the earth is a giant magnet, its poles attract the ends of the magnet. These poles are not located at the geographic poles.

The somewhat mysterious and completely invisible force of a magnet depends on a magnetic field that surrounds the magnet. *[Figure 12-15]* This field always exists between the poles of a magnet and arranges itself to conform to the shape of any magnet.

The theory that explains the action of a magnet holds that each molecule making up the iron bar is itself a tiny magnet, with both North and South poles as illustrated in *Figure 12-16A*. These molecular magnets each possess a magnetic field, but in an unmagnetized state, the molecules are arranged at random throughout the iron bar. If a magnetizing force, such as stroking with a lodestone, is applied to the unmagnetized

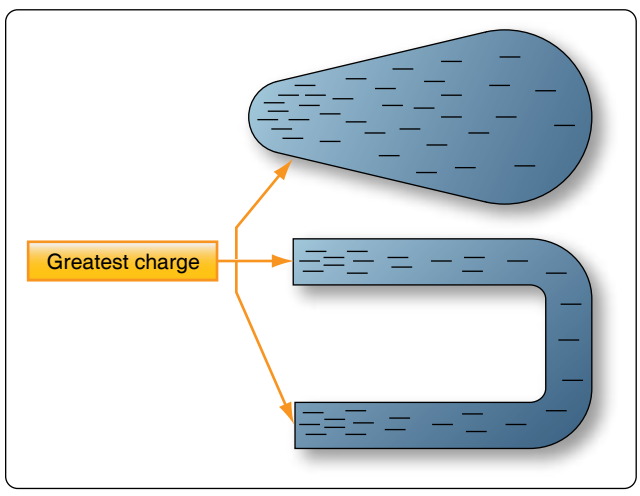


Figure 12-13. *Charge on irregularly-shaped objects.*

bar, the molecular magnets rearrange themselves in line with the magnetic field of the lodestone, with all North ends of the magnets pointing in one direction and all South ends in the opposite direction. *[Figure 12-16B]* In such a configuration, the magnetic fields of the magnets combine to produce the total field of the magnetized bar.

When handling a magnet, avoid applying direct heat, hammering, or dropping it. Heating or sudden shock creates misalignment of the molecules, causing the strength of a magnet to decrease. When a magnet is to be stored, devices known as "keeper bars" are installed to provide an easy path for flux lines from one pole to the other. This promotes the retention of the molecules in their North-South alignment.

The presence of the magnetic force or field around a magnet can best be demonstrated by the experiment illustrated in *Figure 12-17*. A sheet of transparent material, such as glass or Lucite™, is placed over a bar magnet and iron filings are sprinkled slowly on this transparent shield. If the glass or Lucite™ is tapped lightly, the iron filings arrange themselves in a definite pattern around the bar, forming a series of lines from the North to South end of the bar to indicate the pattern of the magnetic field.

As shown, the field of a magnet is made up of many individual forces that appear as lines in the iron filing demonstration. Although they are not "lines" in the ordinary sense, this word is used to describe the individual nature of the separate forces making up the entire magnetic field. These lines of force are also referred to as magnetic flux.

They are separate and individual forces, since one line will never cross another; indeed, they actually repel one another. They remain parallel to one another and resemble stretched rubber bands, since they are held in place around the bar by the internal magnetizing force of the magnet.

The demonstration with iron filings further shows that the magnetic field of a magnet is concentrated at the ends of the magnet. These areas of concentrated flux are called the North and South poles of the magnet. There is a limit to the number of lines of force that can be crowded into a magnet of a given size. When a magnetizing force is applied to a piece of magnetic material, a point is reached where no more lines of force can be induced or introduced. The material is then said to be saturated.

The characteristics of the magnetic flux can be demonstrated by tracing the flux patterns of two bar magnets with like poles together. *[Figure 12-18]* The two like poles repel one another because the lines of force will not cross each other. As the arrows on the individual lines indicate, the lines turn aside as

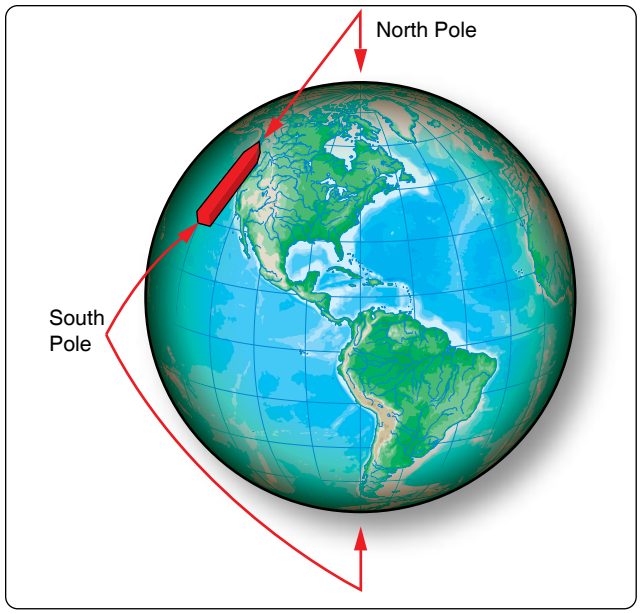


Figure 12-14. *One end of magnetized strip points to the magnetic North pole.*

the two like poles are brought near each other and travel in a path parallel to each other. Lines moving in this manner repel each other, causing the magnets as a whole to repel each other.

By reversing the position of one of the magnets, the attraction of unlike poles can be demonstrated. *[Figure 12-19]* As the unlike poles are brought near each other, the lines of force rearrange their paths and most of the flux leaving the North pole of one magnet enters the South pole of the other. The tendency of lines of force to repel each other is indicated by the bulging of the flux in the air gap between the two magnets.

To further demonstrate that lines of force do not cross one another, a bar magnet and a horseshoe magnet can be positioned to display a magnetic field similar to that of

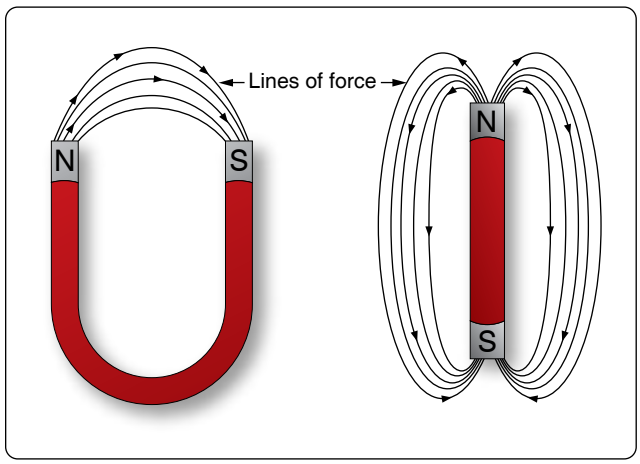


Figure 12-15. *Magnetic field around magnets.*

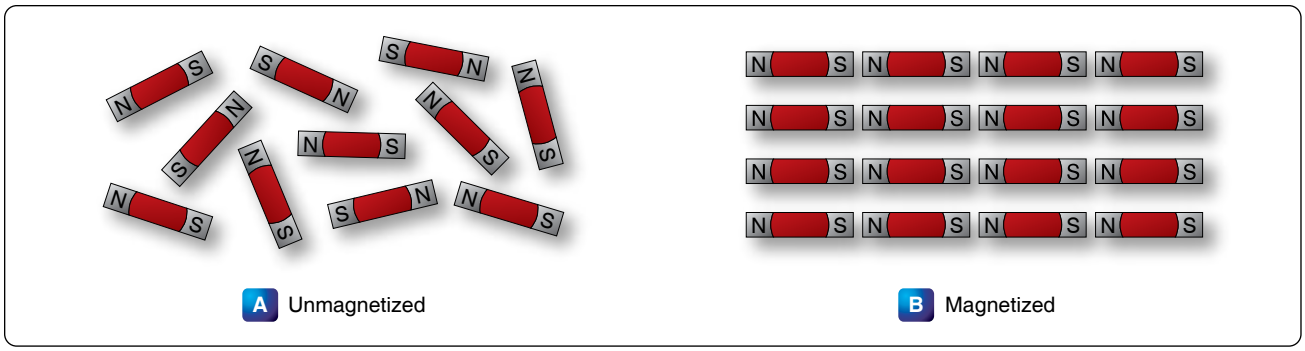


Figure 12-16. *Arrangement of molecules in a piece of magnetic material.*

Figure 12-20. The magnetic fields of the two magnets do not combine, but are rearranged into a distorted flux pattern.

The two bar magnets may be held in the hands and the North poles brought near each other to demonstrate the force of repulsion between like poles. In a similar manner, the two South poles can demonstrate this force. The force of attraction between unlike poles can be felt by bringing a South and a North end together. *[Figure 12-21]*

Figure 12-22 illustrates another characteristic of magnets. If the bar magnet is cut or broken into pieces, each piece immediately becomes a magnet itself, with a North and South pole. This feature supports the theory that each molecule is a magnet, since each successive division of the magnet produces more magnets.

Since the magnetic lines of force form a continuous loop, they form a magnetic circuit. It is impossible to say where in the magnet they originate or start. Arbitrarily, it is assumed that all lines of force leave the North pole of any magnet and enter at the South pole.

There is no known insulator for magnetic flux, or lines of force, since they pass through all materials. However, they do pass through some materials more easily than others. Thus, it is possible to shield items, such as instruments, from the effects of the flux by surrounding them with a material that offers an easier path for the lines of force. *Figure 12-23* shows an instrument surrounded by a path of soft iron, which offers very little opposition to magnetic flux. The lines of force take the easier path, the path of greater permeability, and are guided away from the instrument.

Materials such as soft iron and other ferrous metals are said to have a high permeability, the measure of the ease with which magnetic flux can penetrate a material. The permeability scale is based on a perfect vacuum with a rating of one. Air and other nonmagnetic materials are so close to this that they are also considered to have a rating of one. The nonferrous metals with a permeability greater than one, such as nickel and cobalt, are called paramagnetic. The term ferromagnetic is applied to iron and its alloys, which have by far the greatest permeability. Any substance, such as bismuth, having a permeability of less than one, is considered diamagnetic.

Reluctance, the measure of opposition to the lines of force through a material, can be compared to the resistance of an electrical circuit. The reluctance of soft iron, for instance, is much lower than that of air. *Figure 12-24* demonstrates that a piece of soft iron placed near the field of a magnet can distort the lines of force, which follow the path of lowest reluctance through the soft iron.

The magnetic circuit can be compared in many respects to an electrical circuit. The magnetomotive force, causing lines of force in the magnetic circuit, can be compared to the electromotive force (emf) or electrical pressure of an electrical circuit. The magnetomotive force is measured in gilberts, symbolized by the capital letter "F." The symbol for the intensity of the lines of force, or flux, is the Greek letter phi, and the unit of field intensity is the gauss. An individual line of force, called a maxwell, in an area of one square centimeter produces a field intensity of one gauss. Using reluctance rather than permeability, the law for magnetic circuits can be stated: a magnetomotive force of one gilbert causes one maxwell, or line of force, to be set up in a material when the reluctance of the material is one.

Types of Magnets

Magnets are either natural or artificial. Since naturally occurring magnets or lodestones have no practical use, all magnets considered in this chapter are artificial or manmade. Artificial magnets can be further classified as permanent magnets, which retain their magnetism long after the magnetizing force has been removed, and temporary magnets, which quickly lose most of their magnetism when the external magnetizing force is removed.

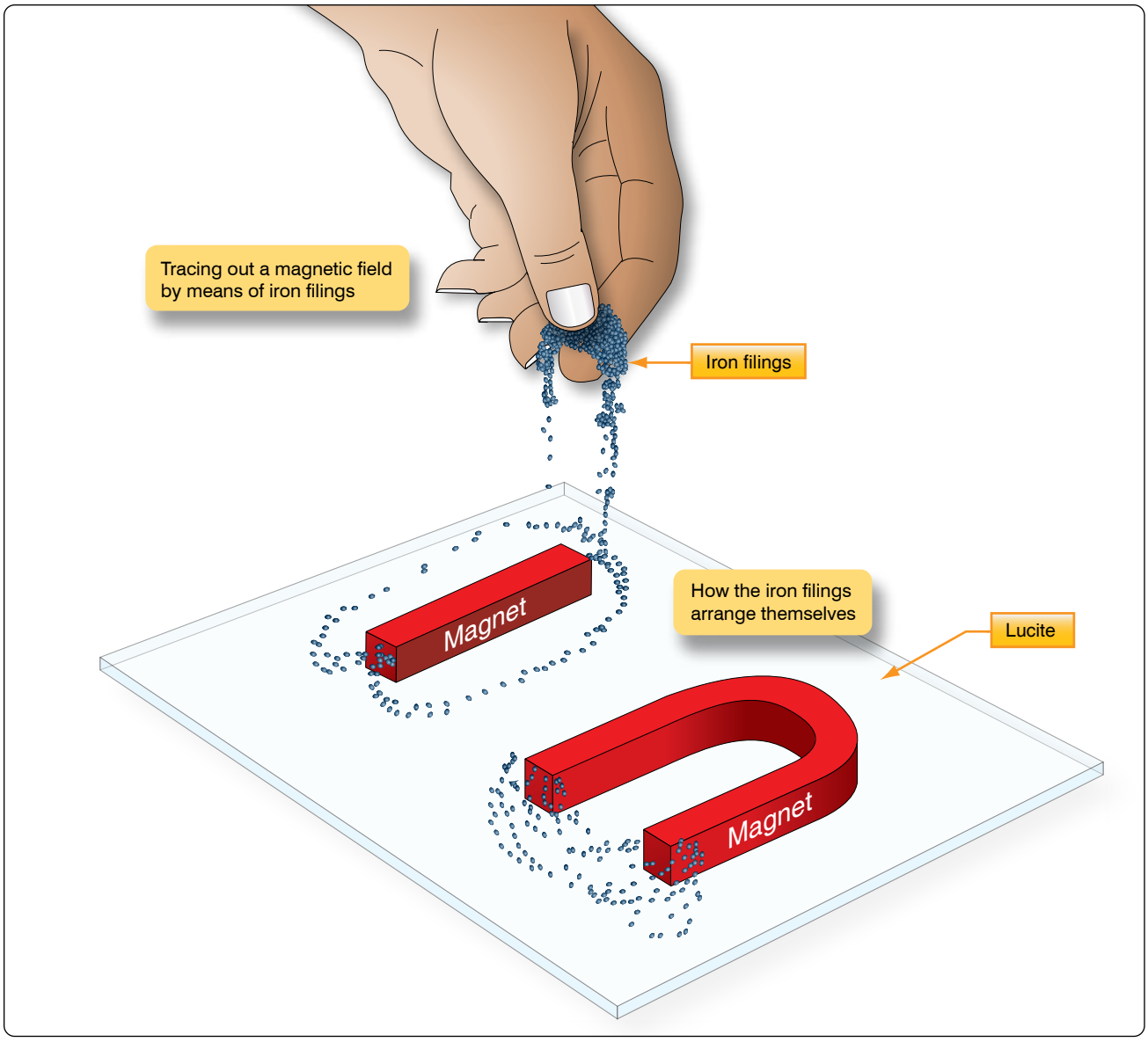


Figure 12-17. *Tracing out a magnetic field with iron filings.*

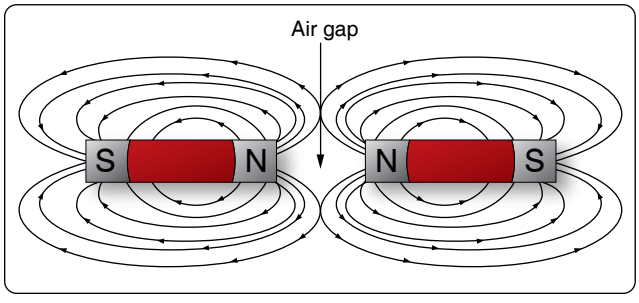


Figure 12-18. *Like poles repel.*

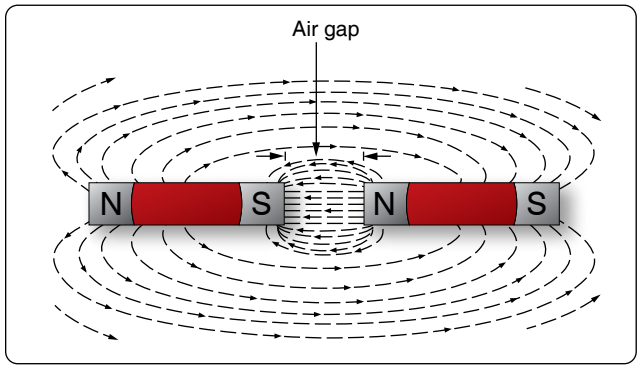


Figure 12-19. *Unlike poles attract.*

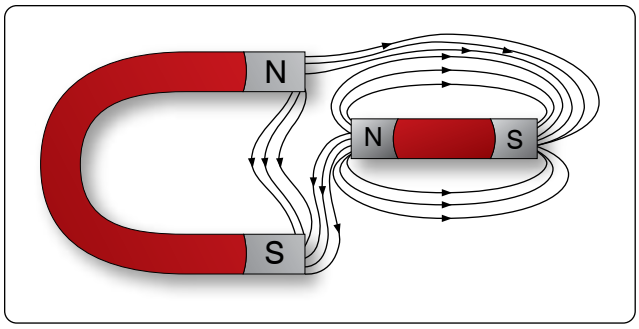


Figure 12-20. *Bypassing flux lines.*

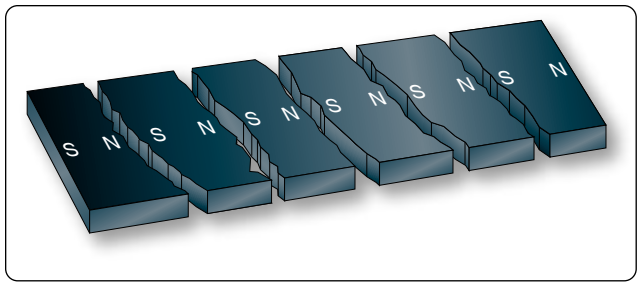


Figure 12-22. *Magnetic poles in a broken magnet.*

Modern permanent magnets are made of special alloys that have been found through research to create increasingly better magnets. The most common categories of magnet materials are made out of aluminum-nickel-cobalt (alnicos), strontium-iron (ferrites, also known as ceramics), neodymium-iron-boron (neo magnets), and samarium-cobalt. Alnico, an alloy of iron, aluminum, nickel and cobalt, is considered one of the very best. Others with excellent magnetic qualities are alloys such as Remalloy™ and Permendur™.

The ability of a magnet to hold its magnetism varies greatly with the type of metal and is known as retentivity. Magnets made of soft iron are very easily magnetized but quickly lose most of their magnetism when the external magnetizing force is removed. The small amount of magnetism remaining, called residual magnetism, is of great importance in such electrical applications as generator operation.

Horseshoe magnets are commonly manufactured in two forms. *[Figure 12-25]* The most common type is made from a long bar curved into a horseshoe shape, while a variation of this type consists of two bars connected by a third bar, or yoke.

Magnets can be made in many different shapes, such as balls, cylinders, or disks. One special type of magnet is the ring magnet, or Gramme ring, often used in instruments. This is a closed loop magnet, similar to the type used in transformer cores, and is the only type that has no poles.

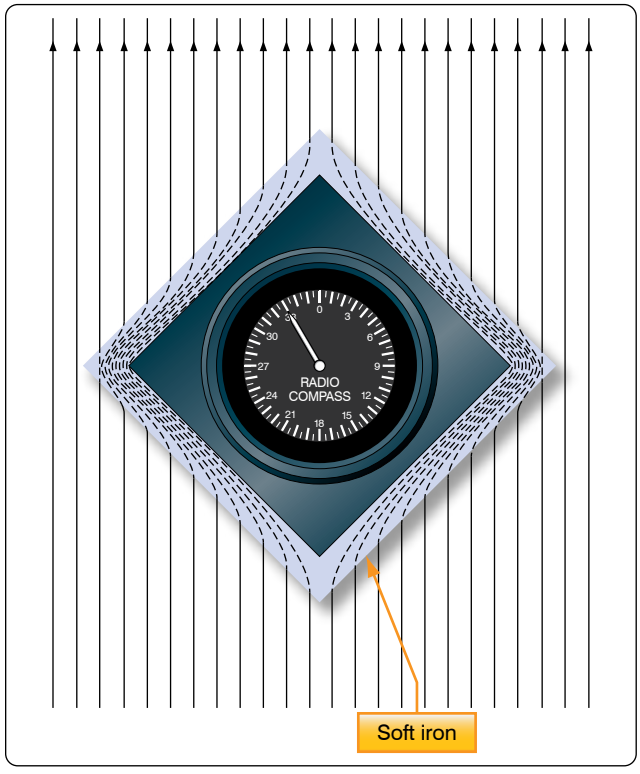


Figure 12-23. *Magnetic shield.*

Sometimes special applications require that the field of force lie through the thickness rather than the length of a piece of metal. Such magnets are called flat magnets and are used as pole pieces in generators and motors.

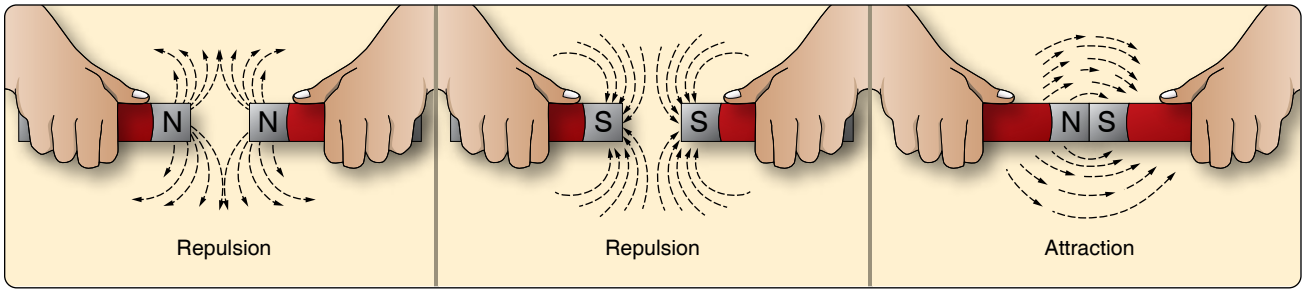


Figure 12-21. *Repulsion and attraction of magnet poles.*

Electromagnetism

In 1820, the Danish physicist, Hans Christian Oersted, discovered that the needle of a compass brought near a current carrying conductor would be deflected. When the current flow stopped, the compass needle returned to its original position. This important discovery demonstrated a relationship between electricity and magnetism that led to the electromagnet and to many of the inventions on which modern industry is based.

Oersted discovered that the magnetic field had no connection with the conductor in which the electrons were flowing, because the conductor was made of nonmagnetic copper. The electrons moving through the wire created the magnetic field around the conductor. Since a magnetic field accompanies a charged particle, the greater the current flow, the greater the magnetic field. *Figure 12-26 i*llustrates the magnetic field around a current carrying wire. A series of concentric circles around the conductor represent the field, which if all the lines were shown would appear more as a continuous cylinder of such circles around the conductor.

As long as current flows in the conductor, the lines of force remain around it. *[Figure 12-27]* If a small current flows through the conductor, there will be a line of force extending out to circle A. If the current flow is increased, the line of force increases in size to circle B, and a further increase in current expands it to circle C. As the original line (circle) of force expands from circle A to B, a new line of force appears at circle A. As the current flow increases, the number of circles of force increases, expanding the outer circles farther from the surface of the current carrying conductor.

If the current flow is a steady nonvarying direct current, the magnetic field remains stationary. When the current stops, the magnetic field collapses and the magnetism around the conductor disappears.

A compass needle is used to demonstrate the direction of the magnetic field around a current carrying conductor. *Figure 12-28A* shows a compass needle positioned at right angles to, and approximately one inch from, a current carrying conductor. If no current were flowing, the North seeking end of the compass needle would point toward the earth's magnetic pole. When current flows, the needle lines itself up at right angles to a radius drawn from the conductor. Since the compass needle is a small magnet, with lines of force extending from South to North inside the metal, it turns until the direction of these lines agrees with the direction of the lines of force around the conductor. As the compass needle is moved around the conductor, it maintains itself in a position at right angles to the conductor, indicating that the magnetic field around a current carrying conductor is circular. As shown in *Figure 12-28B*, when the direction of current flow through the conductor is reversed, the compass needle points in the opposite direction, indicating the magnetic field has reversed its direction.

A method used to determine the direction of the lines of force when the direction of the current flow is known is shown in *Figure 12-29*. If the conductor is grasped in the left hand, with the thumb pointing in the direction of current flow, the fingers will be wrapped around the conductor in the same direction as the lines of the magnetic field. This is called the left-hand rule.

Although it has been stated that the lines of force have direction, this should not be construed to mean that the lines have motion in a circular direction around the conductor. Although the lines of force tend to act in a clockwise or counterclockwise direction, they are not revolving around the conductor.

Since current flows from negative to positive, many illustrations indicate current direction with a dot symbol on the end of the conductor when the electrons are flowing toward and a plus sign when the current is flowing away from the observer. *[Figure 12-30]*

When a wire is bent into a loop and an electric current flows through it, the left-hand rule remains valid. *[Figure 12-31]* If the wire is coiled into two loops, many of the lines of force become large enough to include both loops. Lines of force

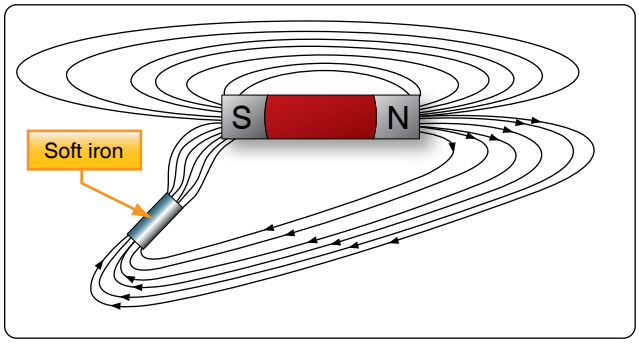


Figure 12-24. *Effect of a magnetic substance in a magnetic field.*

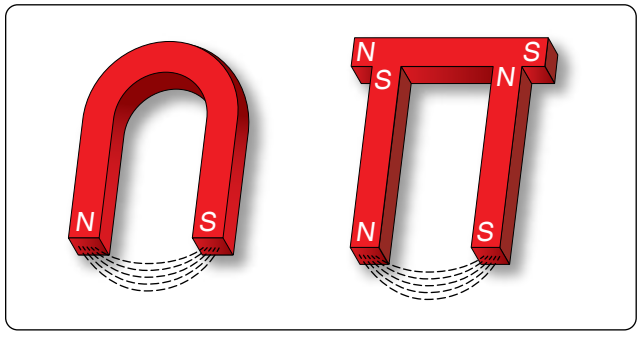


Figure 12-25. *Two forms of horseshoe magnets.*

go through the loops in the same direction, circle around the outside of the two coils, and come in at the opposite end. *[Figure 12-32]*

When a wire contains many such loops, it is called a coil. The lines of force form a pattern through all the loops causing a high concentration of flux lines through the center of the coil. *[Figure 12-33]*

In a coil made from loops of a conductor, many of the lines of force are dissipated between the loops of the coil. By placing a soft iron bar inside the coil, the lines of force are concentrated in the center of the coil, since soft iron has a greater permeability than air. *[Figure 12-34]* This combination of an iron core in a coil of wire loops, or turns, is called an electromagnet, since the poles (ends) of the coil possess the characteristics of a bar magnet.

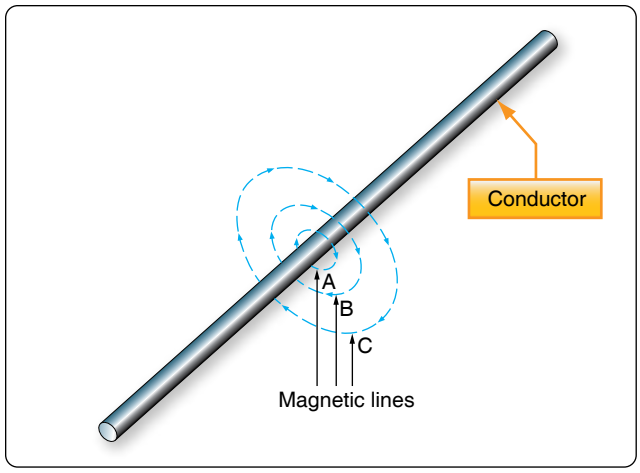


Figure 12-27. *Expansion of magnetic field as current increases.*

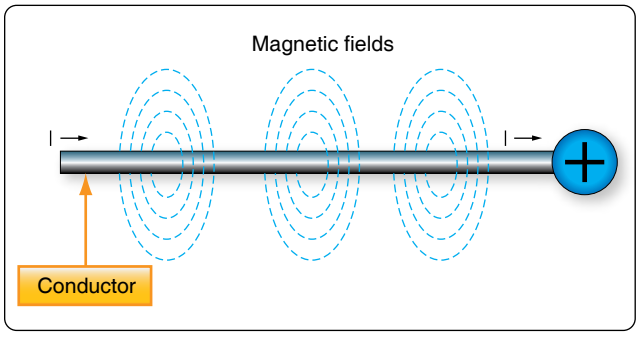


Figure 12-26. *Magnetic field formed around a conductor in which current is flowing.*

The addition of the soft iron core does two things for the current carrying coil. First, the magnetic flux is increased. Second, the flux lines are more highly concentrated.

When direct current flows through the coil, the core becomes magnetized with the same polarity (location of North and South poles) as the coil would have without the core. If the current is reversed, the polarity is also reversed.

The polarity of the electromagnet is determined by the left-hand rule in the same manner as the polarity of the coil without the core was determined. If the coil is grasped in the left hand in such a manner that the fingers curve

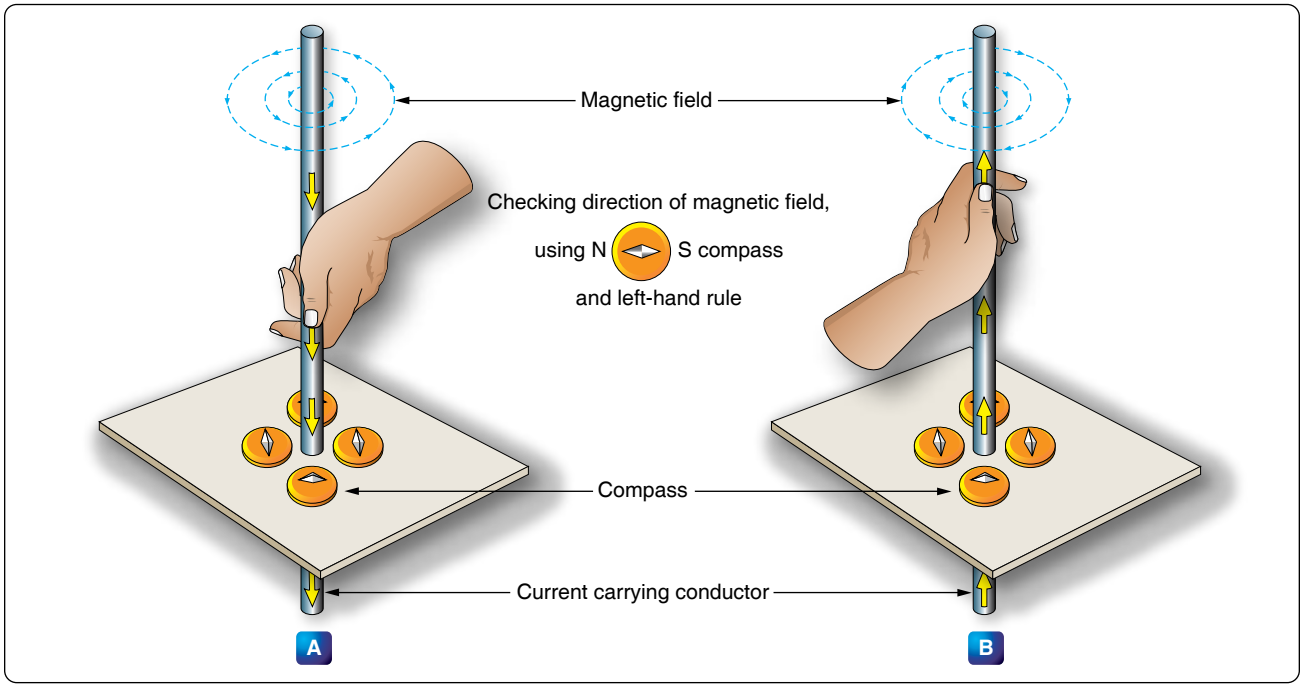


Figure 12-28. *Magnetic field around a current-carrying conductor.*

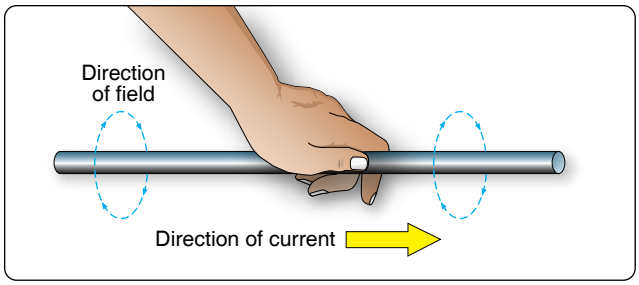


Figure 12-29. *Left-hand rule.*

around the coil in the direction of electron flow (minus to plus), the thumb points in the direction of the North pole. *[Figure 12-35]*

The strength of the magnetic field of the electromagnet can be increased by either increasing the flow of current or the number of loops in the wire. Doubling the current flow approximately doubles the strength of the field. In a similar manner, doubling the number of loops approximately doubles magnetic field strength. Finally, the type of metal in the core is a factor in the field strength of the electromagnet.

A soft iron bar is attracted to either pole of a permanent magnet and, likewise, is attracted by a current carrying coil. The lines of force extend through the soft iron, magnetizing it by induction and pulling the iron bar toward the coil. If the bar is free to move, it is drawn into the coil to a position near the center where the field is strongest. *[Figure 12-36]*

Electromagnets are used in electrical instruments, motors, generators, relays, and other devices. Some electromagnetic devices operate on the principle that an iron core held away from the center of a coil is rapidly pulled into a center position when the coil is energized. This principle is used in the solenoid, also called solenoid switch or relay, in which the iron core is spring-loaded off center and moves to complete a circuit when the coil is energized.

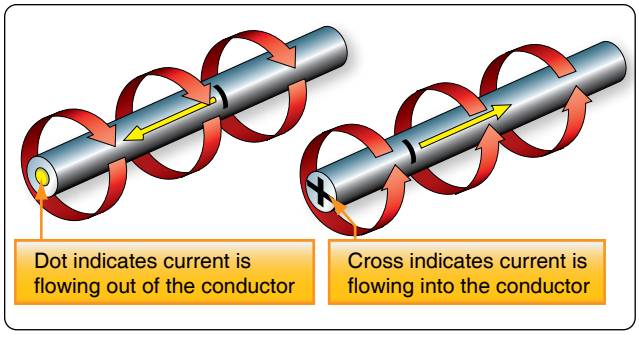


Figure 12-30. *Direction of current flow in a conductor.*

Conventional Flow & Electron Flow

Today's technician will find that there are two competing schools of thought and analytical practices regarding the flow of electricity. The two are called the conventional current theory and the electron theory.

Conventional Flow

Of the two, the conventional current theory was the first to be developed and, through many years of use, this method has become ingrained in electrical publications. The theory was initially advanced by Benjamin Franklin who reasoned that current flowed out of a positive source into a negative source or an area that lacked an abundance of charge. The notation assigned to the electric charges was positive (+) for the abundance of charge and negative (−) for a lack of charge. It then seemed natural to visualize the flow of current as being from the positive (+) to the negative (−).

Electron Flow

Later discoveries were made that proved just the opposite is true. Electron flow is what actually happens when an abundance of electrons flow out of the negative (−) source to an area that lacks electrons or the positive (+) source. Both conventional flow and electron flow are used in industry. Many publications in current use employ both electron flow and conventional flow methods. From the practical standpoint of the technician troubleshooting a system, it makes little to no difference which way current is flowing as long as it is used consistently in the analysis. The Federal Aviation Administration (FAA) officially defines current flow using electron theory (negative to positive).

Electromotive Force (Voltage)

Unlike current, which is easy to visualize as a flow, voltage is a variable that is determined between two points. Often, we refer to voltage as a value across two points. It is the electromotive force (emf) or the push or pressure felt in a

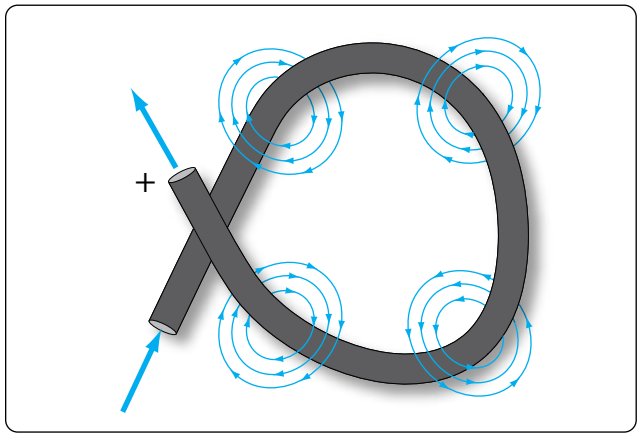


Figure 12-31. *Magnetic field around a looped conductor.*

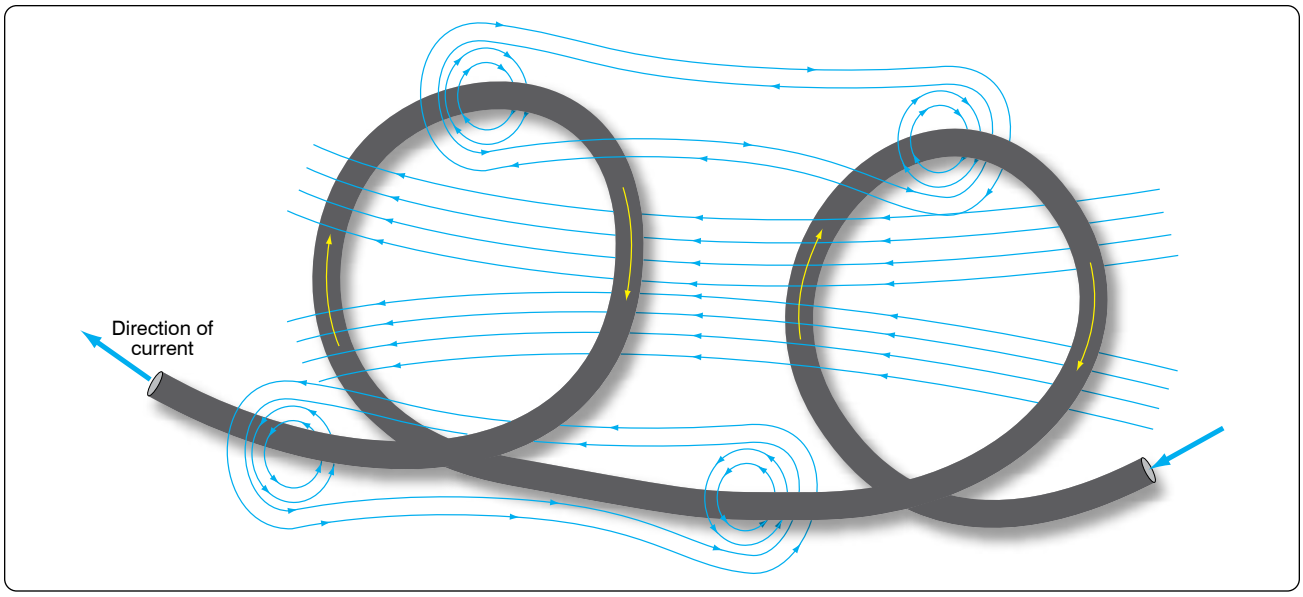


Figure 12-32. *Magnetic field around a conductor with two loops.*

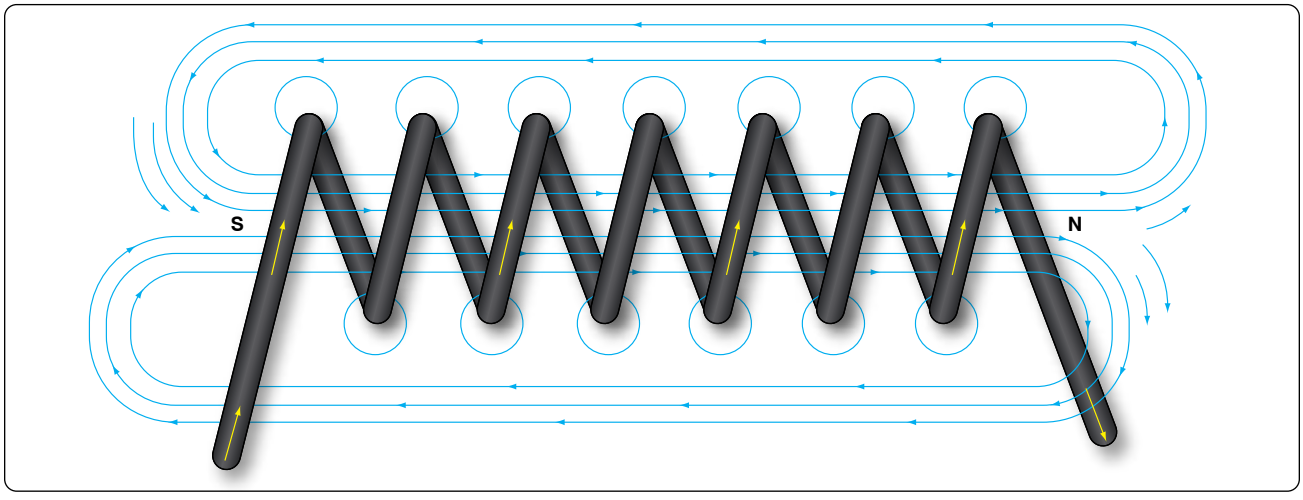


Figure 12-33. *Magnetic field of a coil.*

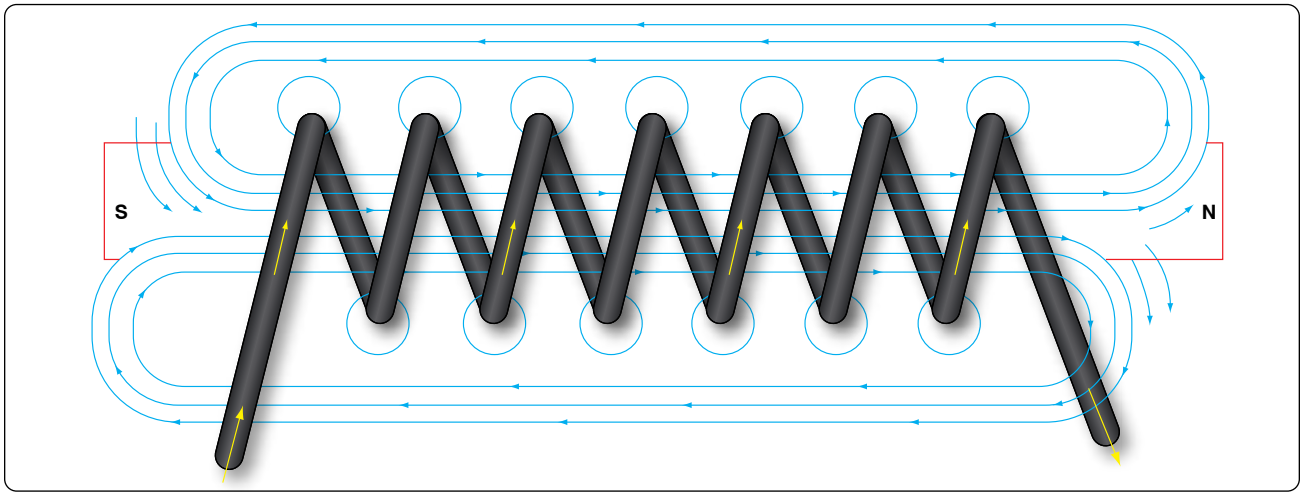


Figure 12-34. *Electromagnet.*

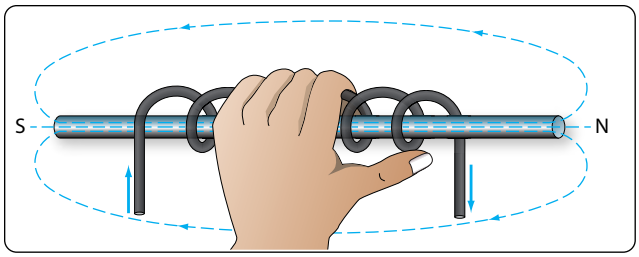


Figure 12-35. *Left-hand rule applied to a coil.*

conductor that ultimately moves the electrons in a flow. The symbol for emf is the capital letter "E."

Across the terminals of the typical aircraft battery, voltage can be measured as the potential difference of 12 volts or 24 volts. That is to say that between the two terminal posts of the battery, there is an emf of 12 or 24 volts available to push current through a circuit. Relatively free electrons in the negative terminal move toward the excessive number of positive charges in the positive terminal. Recall from the discussion on static electricity that like charges repel each other but opposite charges attract each other. The net result is a flow or current through a conductor. There cannot be a flow in a conductor unless there is an applied voltage from a battery, generator, or ground power unit. The potential difference, or the voltage across any two points in an electrical system, can be determined by:
Where

$$E = \frac{\mathcal{E}}{Q}$$

E = Potential difference in volts
$\mathcal{E}$ = Energy expanded or absorbed in joules (J)
Q = Charge measured in coulombs

Figure 12-37 illustrates the flow of electrons of electric current. Two interconnected water tanks demonstrate that

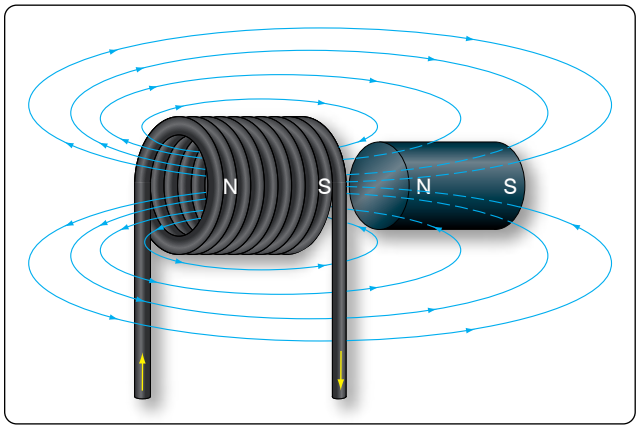


Figure 12-36. *Solenoid with iron core.*

when a difference of pressure exists between the two tanks, water flows until the two tanks are equalized. The illustration shows the level of water in tank A to be at a higher level, reading 10 psi (higher potential energy) than the water level in tank B, reading 2 psi (lower potential energy). Between the two tanks, there is 8-psi potential difference. If the valve in the interconnecting line between the tanks is opened, water flows from tank A into tank B until the level of water (potential energy) of both tanks is equalized. It is important to note that it was not the pressure in tank A that caused the water to flow; rather, it was the difference in pressure between tank A and tank B that caused the flow.

This comparison illustrates the principle that electrons move, when a path is available, from a point of excess electrons (higher potential energy) to a point deficient in electrons (lower potential energy). The force that causes this movement is the potential difference in electrical energy between the two points. This force is called the electrical pressure or the potential difference or the electromotive force (electron moving force).

Current

Electrons in motion make up an electric current. This electric current is usually referred to as "current" or "current flow," no matter how many electrons are moving. Current is a measurement of a rate at which a charge flows through some region of space or a conductor. The moving charges are the free electrons found in conductors, such as copper, silver, aluminum, and gold. The term "free electron" describes a condition in some atoms where the outer electrons are loosely bound to their parent atom. These loosely bound electrons can be easily motivated to move in a given direction when an external source, such as a battery, is applied to the circuit. These electrons are attracted to the positive terminal of the battery, while the negative terminal is the source of the electrons. The greater amount of charge moving through the conductor in a given amount of time translates into a current.

$$\text{Current} = \frac{\text{Charge}}{\text{Time}}$$

or

$$I = \frac{Q}{t}$$

Where:
I = current in amperes (A)
Q = charge in coulombs (C)
T = time

The System International (SI) unit for current is the ampere (A), where

$$1A = 1\,\frac{C}{s}$$

One ampere (A) of current is equivalent to 1 coulomb (C) of charge passing through a conductor in 1 second. One coulomb of charge equals 6.28 billion electrons. The symbol used to indicate current in formulas or on schematics is the capital letter "I."

When current flow is one direction, it is called direct current (DC). Later in the handbook, the form of current that periodically oscillates back and forth within the circuit is discussed. The present discussion is only concerned with the use of DC.

The velocity of the charge is actually an average velocity and is called drift velocity. To understand the idea of drift velocity, think of a conductor in which the charge carriers are free electrons. These electrons are always in a state of random motion similar to that of gas molecules. When a voltage is applied across the conductor, an emf creates an electric field within the conductor and a current is established. The electrons do not move in a straight direction but undergo repeated collisions with other nearby atoms. These collisions usually knock other free electrons from their atoms, and these electrons move on toward the positive end of the conductor with an average velocity called the drift velocity, which is relatively a slow speed. To understand the nearly instantaneous speed of the effect of the current, it is helpful to visualize a long tube filled with steel balls as shown in *Figure 12-38*. It can be seen that a ball introduced in one end of the tube, which represents the conductor, will immediately cause a ball to be emitted at the opposite end of the tube. Thus, electric current can be viewed as instantaneous, even though it is the result of a relatively slow drift of electrons.

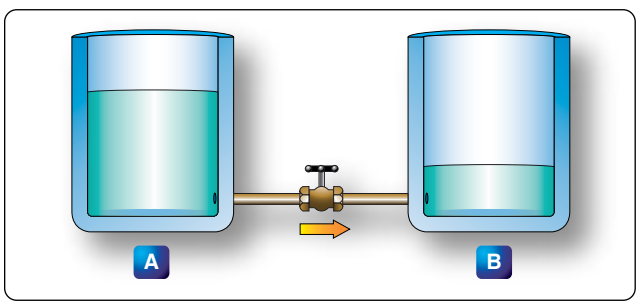


Figure 12-37. *Difference of pressure.*

Ohm's Law (Resistance)

The two fundamental properties of current and voltage are related by a third property known as resistance. In any electrical circuit, when voltage is applied to it, a current results. The resistance of the conductor determines the amount of current that flows under the given voltage. In most cases, the greater the circuit resistance, the less the current. If the resistance is reduced, then the current increases. This relation is linear in nature and is known as Ohm's Law.

By having a linearly proportional characteristic, it is meant that if one unit in the relationship increases or decreases by a certain percentage, the other variables in the relationship increase or decrease by the same percentage. An example would be if the voltage across a resistor is doubled, then the current through the resistor doubles. It should be added that this relationship is true only if the resistance in the circuit remains constant. If the resistance changes, current also changes. A graph of this relationship is shown in *Figure 12-39*, which uses a constant resistance of 20Ω. The relationship between voltage and current in this example shows voltage plotted horizontally along the X axis in values from 0 to 120 volts, and the corresponding values of current are plotted vertically in values from 0 to 6.0 amperes along the Y axis. A straight line drawn through all the points where the voltage and current lines meet represents the equation $I = E/20$ and is called a linear relationship.

If $E = 10$ V

Then $\frac{10\text{ V}}{20\ \Omega} = 0.5$ A

If $E = 60$ V

Then $\frac{60\text{ V}}{20\ \Omega} = 3$ A

If $E = 120$ V

Then $\frac{120\text{V}}{20\ \Omega} = 6$ A

Ohm's Law may be expressed as an equation, as follows:

Equation 1

$$I = \frac{E}{R}$$

I = current in amperes (A)
E = voltage (V)
R = resistance (Ω)

Where I is current in amperes, E is the potential difference measured in volts, and R is the resistance measured in ohms.

If any two of these circuit quantities are known, the third may be found by simple algebraic transposition. With this equation, we can calculate current in a circuit if the voltage and resistance are known. This same formula can be used to calculate voltage. By multiplying both sides of the equation 1 by R, we get an equivalent form of Ohm's Law, which is:

Equation 2
$E = I (R)$

Finally, if we divide equation 2 by I, we solve for resistance, This relationship is true only if the resistance in the circuit remains constant. If the resistance changes, current also changes. A graph of this relationship is shown in *Figure 12-39*, which uses a constant resistance of 20Ω. The relationship between voltage and current in this example shows voltage plotted horizontally along the X axis in values from 0 to 120 volts. The corresponding values of current are plotted vertically in values from 0 to 6.0 amperes along the Y axis. A straight line drawn through all the points where the voltage and current lines meet represents the equation $I = E/20$ and is called a linear relationship.

Equation 3

$$R = \frac{E}{I}$$

All three formulas presented in this section are equivalent to each other and are simply different ways of expressing Ohm's Law.

The various equations, which may be derived by transposing the basic law, can be easily obtained by using the triangles in *Figure 12-40*.

The triangles containing E, R, and I are divided into two parts, with E above the line and I × R below it. To determine an unknown circuit quantity when the other two are known, cover the unknown quantity with a thumb. The location of the remaining uncovered letters in the triangle indicate the mathematical operation to be performed. For example, to find I, refer to *Figure 12-40A*, and cover I with the thumb. The uncovered letters indicate that E is to be divided by R, or $I = E/R$. To find R, refer to *Figure 12-40B*, and cover R with the thumb. The result indicates that E is to be divided by I, or $R = E/I$. To find E, refer to *Figure 12-40C*, and cover E with the thumb. The result indicates I is to be multiplied by R, or $E = I \times R$. This chart is useful when learning to use Ohm's Law. It should be used to supplement the beginner's knowledge of the algebraic method.

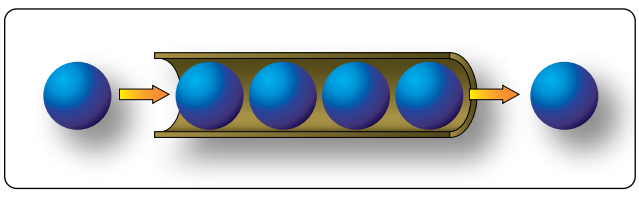


Figure 12-38. *Electron movement.*

Resistance of a Conductor

While wire of any size or resistance value may be used, the word "conductor" usually refers to materials that offer low resistance to current flow, and the word "insulator" describes materials that offer high resistance to current. There is no distinct dividing line between conductors and insulators; under the proper conditions, all types of material conduct some current. Materials offering a resistance to current flow midway between the best conductors and the poorest conductors (insulators) are sometimes referred to as "semiconductors," and find their greatest application in the field of transistors.

The best conductors are materials, chiefly metals, which possess a large number of free electrons; conversely, insulators are materials having few free electrons. The best conductors are silver, copper, gold, and aluminum; but some nonmetals, such as carbon and water, can be used as conductors. Materials such as rubber, glass, ceramics, and plastics are such poor conductors that they are usually used as insulators. The current flow in some of these materials is so low that it is usually considered zero. The unit used to measure resistance is called the ohm. The symbol for the ohm is the Greek letter omega (Ω). In mathematical formulas, the capital letter "R" refers to resistance. The resistance of a conductor and the voltage applied to it determine the number of amperes of current flowing through the conductor. Thus, 1 ohm of resistance limits the current flow to 1 ampere in a conductor to which a voltage of 1 volt is applied.

Factors Affecting Resistance

1. The resistance of a metallic conductor is dependent on the type of conductor material. It has been pointed out that certain metals are commonly used as conductors because of the large number of free electrons in their outer orbits. Copper is usually considered the best available conductor material, since a copper wire of a particular diameter offers a lower resistance to current flow than an aluminum wire of the same diameter. However, aluminum is much lighter than copper, and for this reason, as well as cost considerations, aluminum is often used when the weight factor is important.
2. The resistance of a metallic conductor is directly proportional to its length. The longer the length of a given size of wire, the greater the resistance. *Figure 12-41* shows two wire conductors of different lengths. If 1 volt of electrical pressure is applied across

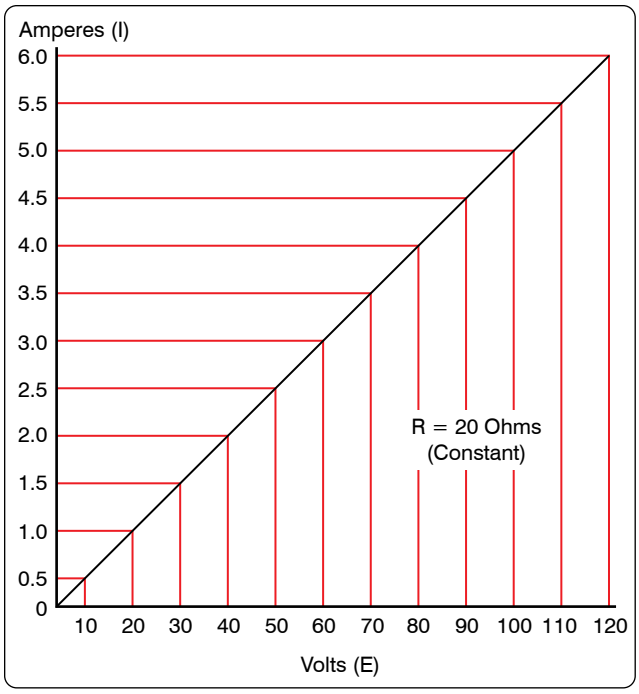


Figure 12-39. *Voltage vs. current in a constant-resistance circuit.*

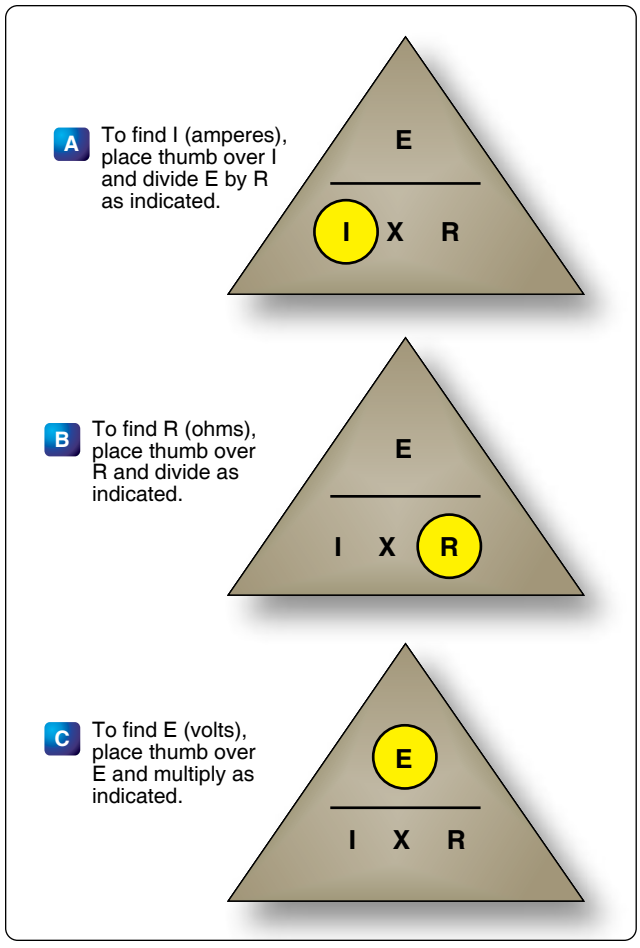


Figure 12-40. *Ohm's law chart.*

the two ends of the conductor that is 1 foot in length, and the resistance to the movement of free electrons is assumed to be 1 ohm, the current flow is limited to 1 ampere. If the same size conductor is doubled in length, the same electrons set in motion by the 1 volt applied now find twice the resistance; consequently, the current flow is reduced by one-half.

3. The resistance of a metallic conductor is inversely proportional to the cross-sectional area. This area may be triangular or even square, but is usually circular. If the cross-sectional area of a conductor is doubled, the resistance to current flow is reduced in half. This is true because of the increased area in which an electron can move without collision or capture by an atom. Thus, the resistance varies inversely with the cross-sectional area of a conductor.

4. The fourth major factor influencing the resistance of a conductor is temperature. Although some substances, such as carbon, show a decrease in resistance as the ambient (surrounding) temperature increases, most materials used as conductors increase in resistance as temperature increases. The resistance of a few alloys, such as constantan and Manganin™, change very little as the temperature changes. The amount of increase in the resistance of a 1 ohm sample of a conductor, per degree rise in temperature above 0° Centigrade (C), the assumed standard, is called the temperature coefficient of resistance. For each metal, this is a different value. For example, for copper the value is approximately 0.00427 ohm. Thus, a copper wire having a resistance of 50 ohms at a temperature of 0 °C has an increase in resistance of 50 × 0.00427, or 0.214 ohm, for each degree rise in temperature above 0 °C. The temperature coefficient of resistance must be considered where there is an appreciable change in temperature of a conductor during operation. Charts listing the temperature coefficient of resistance for different materials are available. *Figure 12-42* shows a table for "resistivity" of some common electric conductors.

The resistance of a material is determined by four properties: material, length, area, and temperature. The first three properties are related by the following equation at T = 20 °C (room temperature):

$$R = \frac{(\rho \times 1)}{A}$$

Where
R = resistance in ohms
ρ = resistivity of the material in circular mil-ohms per foot

l = length of the sample in feet
A = area in circular mils

Resistance and Relation to Wire Sizing

Circular Conductors (Wires/Cables)

Because it is known that the resistance of a conductor is directly proportional to its length, and if we are given the resistance of the unit length of wire, we can readily calculate the resistance of any length of wire of that particular material having the same diameter. Also, because it is known that the resistance of a conductor is inversely proportional to its cross-sectional area, and if we are given the resistance of a length of wire with unit cross-sectional area, we can calculate the resistance of a similar length of wire of the same material with any cross-sectional area. Therefore, if we know the resistance of a given conductor, we can calculate the resistance for any conductor of the same material at the same temperature. From the relationship:

$$R = \frac{(\rho \times 1)}{A}$$

It can also be written:

$$\frac{R_1}{R_2} = \frac{1_1}{1_2} = \frac{A_1}{A_2}$$

If we have a conductor that is 1 meter (m) long with a cross-sectional area of 1 (millimeter) mm^2 and has a resistance of 0.017 ohm, what is the resistance of 50 m of wire from the same material but with a cross-sectional area of 0.25 mm^2?

$$\frac{R_1}{R_2} = \frac{1_1}{1_2} = \frac{A_1}{A_2}$$

$$R^2 = 0.017\ \Omega \times \frac{50\ m}{1\ m} \times \frac{1\ mm^2}{0.25\ mm^2} = 3.4\ \Omega$$

While the SI units are commonly used in the analysis of electric circuits, electrical conductors in North America are still being manufactured using the foot as the unit length and the mil (one thousandth of an inch) as the unit of diameter. Before using the equation $R = (\rho \times l)/A$ to calculate the resistance of a conductor of a given American wire gauge (AWG) size, the cross-sectional area in square meters must be determined using the conversion factor 1 mil = 0.0254 mm. The most convenient unit of wire length is the foot. Using these standards, the unit of size is the mil-foot. Thus, a wire has unit size if it has a diameter of 1 mil and length of 1 foot.

In the case of using copper conductors, we are spared the task of tedious calculations by using a table as shown in *Figure 12-43*. Note that cross-sectional dimensions listed on the table are such that each decrease of one gauge number equals a 25 percent increase in the cross-sectional area. Because of this, a decrease of three gauge numbers represents an increase in cross-sectional area of approximately 2:1. Likewise, change of ten wire gauge numbers represents a 10:1 change in cross-sectional area—also, by doubling the cross-sectional area of the conductor, the resistance is cut in half. A decrease of three wire gauge numbers cuts the resistance of the conductor of a given length in half.

Rectangular Conductors (Bus Bars)

To compute the cross-sectional area of a conductor in square mils, the length in mils of one side is squared. In the case of a rectangular conductor, the length of one side is multiplied by the length of the other. For example, a common rectangular bus bar (large, special conductor) is ⅜ inch thick and 4 inches wide. The ⅜-inch thickness may be expressed as 0.375 inch. Since 1,000 mils equal 1 inch, the width in inches can be converted to 4,000 mils. The cross-sectional area of the rectangular conductor is found by converting 0.375 to mils (375 mils × 4,000 mils = 1,500,000 square mils).

Power and Energy

Power in an Electrical Circuit

This section covers power in the DC circuit and energy consumption. Whether referring to mechanical or electrical systems, power is defined as the rate of energy consumption

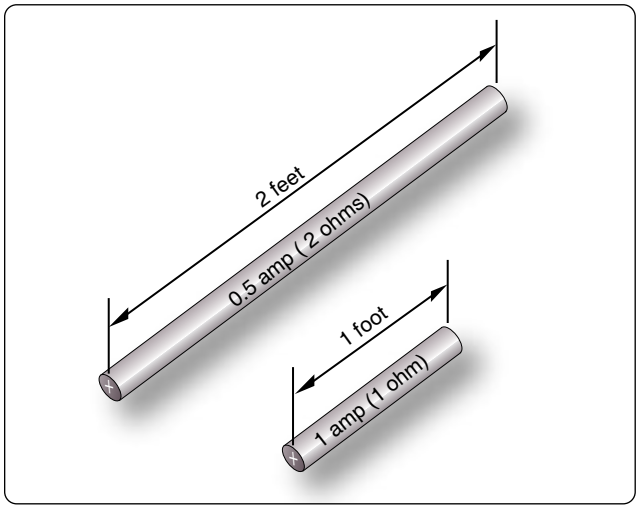


Figure 12-41. *Resistance varies with length of conductor.*

Conductor Material	Resistivity (ohm meters @ 20 °C)
Silver	1.64×10^{-8}
Copper	1.72×10^{-8}
Aluminum	2.83×10^{-8}
Tungsten	5.50×10^{-8}
Nickel	7.80×10^{-8}
Iron	12.0×10^{-8}
Constantan	49.0×10^{-8}
Nichrome II	110×10^{-8}

Figure 12-42. *Resistivity table.*

or conversion within that system—that is, the amount of energy used or converted in a given amount of time.

From the scientific discipline of physics, the fundamental expression for power is:

$$P = \frac{\mathcal{E}}{t}$$

Where
P = power measured in watts (W)
$\mathcal{E}$ = energy ($\mathcal{E}$ is a script E) measured in joules (J)
and
t = time measured in seconds (s)

The unit measurement for power is the watt (W), which refers to a rate of energy conversion of 1 joule (J)/second. Therefore, the number of joules consumed in 1 second is equal to the number of watts. A simple example is given below.

Suppose 300 joules of energy is consumed in 10 seconds. What would be the power in watts?

$$\text{General formula} \quad P = \frac{\text{energy}}{\text{time}}$$

$$P = \frac{300\ \text{J}}{10\ \text{s}}$$

$$P = 30\ \text{W}$$

The watt is named for James Watt, the inventor of the steam engine. Watt devised an experiment to measure the power of a horse in order to find a means of measuring the mechanical power of his steam engine. One horsepower is required to move 33,000 pounds 1 foot in 1 minute. Since power is the rate of doing work, it is equivalent to the work divided by time. Stated as a formula, this is:

$$\text{Power} = \frac{33{,}000\ \text{ft-lb}}{60\ \text{s}}$$

$$P = 550\ \text{ft-lb/s}$$

Electrical power can be rated in a similar manner. For example, an electric motor rated as a 1 horsepower motor requires 746 watts of electrical energy.

Power Formulas Used in the Study of Electricity

When current flows through a resistive circuit, energy is dissipated in the form of heat. Recall that voltage can be expressed in the terms of energy and charge as given in the expression:

$$E = \frac{\mathcal{E}}{Q}$$

Where

AWG Number	Diameter in mils	Ohms per 1,000 ft.
0000	460.0	0.04901
000	409.6	0.06180
00	364.8	0.07793
0	324.9	0.09827
1	289.3	0.1239
2	257.6	0.1563
3	229.4	0.1970
4	204.3	0.2485
5	181.9	0.3133
6	162.0	0.3951
8	128.5	0.6282
10	101.9	0.9989
12	80.81	1.588
14	64.08	2.525
16	50.82	4.016
18	40.30	6.385
20	31.96	10.15
22	25.35	16.14
24	20.10	25.67
26	15.94	40.81
28	12.64	64.9
30	10.03	103.2

Figure 12-43. *Conversion table when using copper conductors.*

E = potential difference in volts
$\mathcal{E}$ = energy expanded or absorbed in joules (J)
Q = charge measured in coulombs

Current I, can also be expressed in terms of charge and time as given by the expression:

$$\text{Current} = \frac{\text{Charge}}{\text{Time}}$$

or

$$I = \frac{Q}{T}$$

Where:
I = current in amperes (A)
Q = charge in coulombs (C)
T = time

When voltage % and current % are multiplied, the charge Q is divided out leaving the basic expression from physics:

$$E \times I = \frac{\mathcal{E}}{Q} \times \frac{\mathcal{E}}{t} = \frac{\mathcal{E}}{t} = \text{power}$$

For a simple DC electrical system, power dissipation

can then be given by the equation:

General Power Formula $P = I (E)$

Where
P = Power
I = Current
E = Volts

If a circuit has a known voltage of 24 volts and a current of 2 amps, then the power in the circuit is:

$P = I (E)$
$P = 2A (24V)$
$P = 48W$

Now recall Ohm's Laws that state $E = I(R)$. If we now substitute IR for E in the general formula, we get a formula that uses only current I and resistance R to determine the power in a circuit.

$P = I (IR)$

Second Form of Power Equation

$P = I^2R$

If a circuit has a known current of 2 amps and a resistance of 100 Ω, then the power in the circuit is:

$P = I^2R$
$P = (2A)^2\ 100\ \Omega$
$P = 400\ W$

Using Ohm's Law, which can be stated as $I = E/R$, we can again make a substitution such that power can be determined by knowing only the voltage (E) and resistance (R) of the circuit.

$$P = \left(\frac{E}{R}\right)\left(E\right)$$

Third Form of Power Equation

$$P = \frac{E^2}{R}$$

If a circuit has a known voltage of 24 volts and a resistance of 20 Ω, then the power in the circuit is:

$$P = \frac{E^2}{R}$$

$$P = \frac{(24\ V)^2}{20\ \Omega}$$

$$P = 28.8\ W$$

Power in a Series & Parallel Circuit

The total power dissipated in both a series and parallel circuit is equal to the sum of the power dissipated in each resistor in the circuit. Power is simply additive and can be stated as:

$$P_T = P_1 + P_2 + P_3 + \ldots\ldots P_N$$

Figure 12-44 provides a summary of all the possible transpositions of the Ohm's Law formula and the power formula.

Energy in an Electrical Circuit

Energy is defined as the ability to do work. Because power is the rate of energy usage, power used over a span of time is actually energy consumption. If power and time are multiplied together, we get energy.

The joule is defined as a unit of energy. There is another unit of measure which is perhaps more familiar. Because power is expressed in watts and time in seconds, a unit of energy can be called a watt-second (Ws) or more recognizable from the electric bill, a kilowatt-hour (kWh). Refer to Chapter 5, Physics, for further discussion on energy.

Sources of Electricity

Electrical energy can be produced in a number of methods. The four most common are pressure, chemical, thermal, and light.

Pressure Source

This form of electrical generation is commonly known as piezoelectric (piezo or piez taken from Greek: to press; pressure; to squeeze) is a result of the application of mechanical pressure on a dielectric or non-conducting crystal. The most common piezoelectric materials used today are crystalline quartz and Rochelle salt. However, Rochelle salt is being superseded by other materials, such as barium titanate.

The application of a mechanical stress produces an electric polarization, which is proportional to this stress. This polarization establishes a voltage across the crystal. If a circuit is connected across the crystal, a flow of current can be observed when the crystal is loaded (pressure is applied). An opposite condition can occur, where an application of a voltage between certain faces of the crystal can produce a mechanical distortion. This effect is commonly referred to as the piezoelectric effect.

Piezoelectric materials are used extensively in transducers for converting a mechanical strain into an electrical signal. Such devices include microphones, phonograph pickups, and vibration-sensing elements. The opposite effect, in which a mechanical output is derived from an electrical signal input, is also widely used in headphones and loudspeakers.

Chemical Source

Chemical energy can be converted into electricity; the most common form of this is the battery. A primary battery produces electricity using two different metals in a chemical solution like alkaline electrolyte, where a chemical reaction between the metals and the chemicals frees more electrons in one metal than in the other. One terminal of the battery is attached to one of the metals, such as zinc; the other terminal is attached to the other metal, such as manganese oxide. The end that frees more electrons develops a positive charge and the other end develops a negative charge. If a wire is attached from one end of the battery to the other, electrons flow through the wire to balance the electrical charge.

Thermal Sources

The most common source of thermal electricity found in the aviation industry comes from thermocouples. Thermocouples are widely used as temperature sensors. They are cheap and interchangeable, have standard connectors, and can measure a wide range of temperatures. Thermocouples are pairs of dissimilar metal wires joined at least at one end, which generate a voltage between the two wires that is proportional to the temperature at the junction. This is called the Seebeck effect, in honor of Thomas Seebeck who first noticed the phenomena in 1821. It was also noticed that different metal combinations have a different voltage difference. Thermocouples are used to measure cylinder head temperatures and Turbine Inlet Temperature (TIT).

Light Sources

A solar cell or a photovoltaic cell is a device that converts light energy into electricity. Fundamentally, the device contains certain chemical elements that, when exposed to light energy, release electrons.

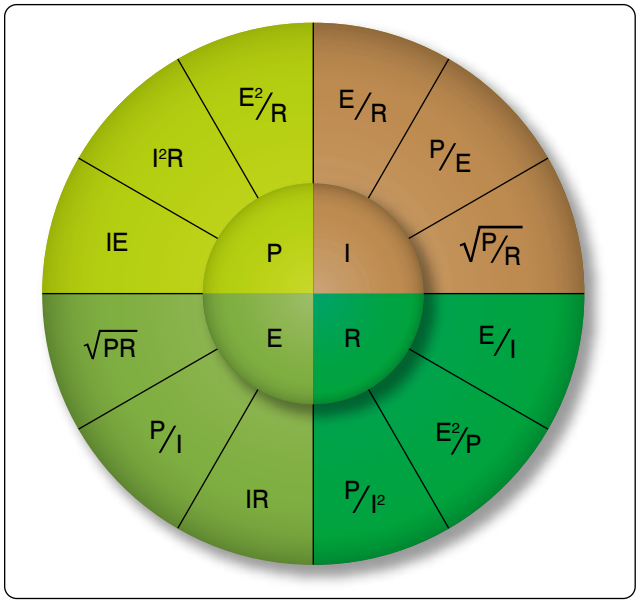


Figure 12-44. *Ohm's Law formula.*

Photons in sunlight are taken in by the solar panel or cell, where they are absorbed by semiconducting materials, such as silicon. Electrons in the cell are broken loose from their atoms, allowing them to flow through the material to produce electricity. The complementary positive charges that are also created are called holes (absence of electron) and flow in the direction opposite of the electrons in a silicon solar panel.

Solar cells have many applications and have historically been used in earth orbiting satellites or space probes, handheld calculators, and wrist watches.

Schematic Representation of Electrical Components

The schematic is the most common place where the technician finds electronic symbols. The schematic is a diagram that depicts the interconnection and logic of an electronic or electrical circuit. Many symbols are employed for use in the schematic drawings, blueprints, and illustrations. This section briefly outlines some of the more common symbols and explains how to interpret them.

Conductors

The schematic depiction of a conductor is simple enough. This is generally shown as a solid line. However, the line types may vary depending on who drew the schematics and what exactly the line represents. While the solid line is used to depict the wire or conductor, schematics used for aircraft modifications can also use other line types, such as a dashed to represent "existing" wires prior to modification and solid lines for "new" wires.

There are two methods employed to show wire crossovers and wire connections. *Figure 12-45* shows the two methods of drawing wires that cross: version A and version B. *Figure 12-46* shows the two methods for drawing wire that connect version A and version B. If version A in *Figure 12-45* is used to depict crossovers, then version A for wire connections in *Figure 12-46* is used. The same can then be said about the use of version B methods. The technician encounters both in common use.

Figure 12-47 shows a few examples of the more common wire types that the technician encounters in schematics. They are the single wire, single shielded, shielded twisted pair or double, and the shielded triple. This is not an exhaustive list of wire types but a fair representation of how they are depicted. *Figure 12-47* also shows the wires having a wire number. These are shown for the sake of illustration and vary from one

installation agency to another. For further understanding of a wire numbering system, consult the appropriate wiring guide published by the agency that drew the prints. Regardless of the specifics of the wire numbering system, organizations that exercise professional wiring practices have the installed wire marked in some manner. This is an aid to the technician who has to troubleshoot or modify the system at a later date.

Types of Resistors

Fixed Resistor

Fixed resistors have built into the design a means of opposing current. *[Figure 12-48]* The general use of a resistor in a circuit is to limit the amount of current flow. There are a number of methods used in construction and sizing of a resistor that control properties, such as resistance value, the precision of the resistance value, and the ability to dissipate heat. While in some applications the purpose of the resistive element is used to generate heat, such as in propeller anti-ice boots, heat typically is the unwanted loss of energy.

Carbon Composition

The carbon composed resistor is constructed from a mixture of finely grouped carbon/graphite, an insulation material for filler, and a substance for binding the material together. The amount of graphite in relation to the insulation material determines the ohmic or resistive value of the resistor. This mixture is compressed into a rod, which is fitted with axial leads or "pigtails." The finished product is sealed in an insulating coating for isolation and physical protection.

There are other types of fixed resistors in common use. Included in this group are:

- Carbon film
- Metal-oxide
- Metal film
- Metal glaze

The construction of a film resistor is accomplished by depositing a resistive material evenly on a ceramic rod. This resistive material can be graphite for the carbon film resistor, nickel chromium for the metal film resistor, metal and glass for the metal glaze resistor, and metal and an insulating oxide for the metal-oxide resistor.

Resistor Ratings

Color Code

It is very difficult to manufacture a resistor to an exact standard of ohmic values. Fortunately, most circuit requirements are not extremely critical. For many uses, the actual resistance in ohms can be 20 percent higher or lower than the value marked on the resistor without causing difficulty. The percentage variation between the marked value and the actual value of a resistor is known as the "tolerance" of a resistor. A resistor coded for a 5 percent tolerance is not more than 5 percent higher or lower than the value indicated by the color code.

The resistor color code is made up of a group of colors, numbers, and tolerance values. Each color is represented by a number, and in most cases, by a tolerance value. *[Figure 12-49]*

When the color code is used with the end-to-center band marking system, the resistor is normally marked with bands of color at one end of the resistor. The body or base color of the resistor has nothing to do with the color code, and in no way indicates a resistance value. To prevent confusion, this body is never the same color as any of the bands indicating resistance value.

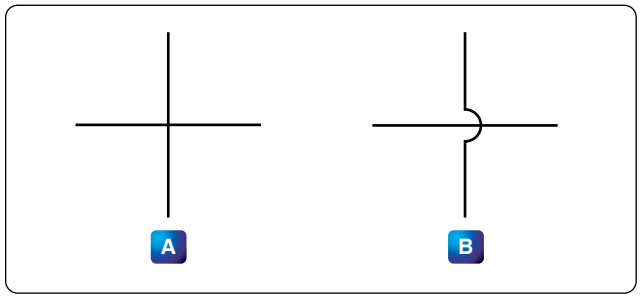


Figure 12-45. *Unconnected crossover wires.*

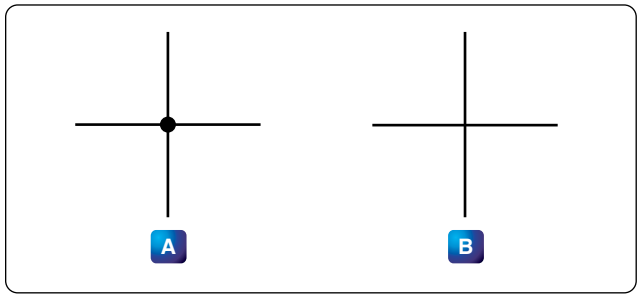


Figure 12-46. *Connected wires.*

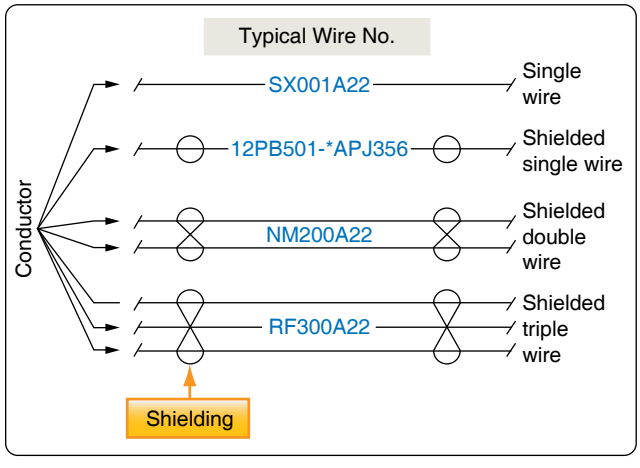


Figure 12-47. *Common wire types.*

Color Band Decoding

When the end-to-center band marking system is used, either three or four bands mark the resistor.

1. The first color band (nearest the end of the resistor) indicates the first digit in the numerical resistance value. This band is never gold or silver in color.
2. The second color band always indicates the second digit of ohmic value. It is never gold or silver in color. *[Figure 12-50]*
3. The third color band indicates the number of zeros to be added to the two digits derived from the first and second bands, except in the following two cases:
   (a) If the third band is gold in color, the first two digits must be multiplied by 10 percent.
   (b) If the third band is silver in color, the first two digits must be multiplied by 1 percent.
4. If there is a fourth color band, it is used as a multiplier for percentage of tolerance, as indicated in the color code chart in *Figure 12-49*. If there is no fourth band, the tolerance is understood to be 20 percent.

Figure 12-50 illustrates the rules for reading the resistance value of a resistor marked with the end-to-center band system. This resistor is marked with three bands of color, which must be read from the end toward the center.

There is no fourth color band; therefore, the tolerance is understood to be 20 percent. 20 percent of 250,000 Ω, equals 50,000 Ω.

Since the 20 percent tolerance is plus or minus,

Maximum resistance
= 250,000 Ω + 50,000 Ω
= 300,000 Ω

Minimum resistance
= 250,000 Ω − 50,000 Ω
= 200,000 Ω

The following paragraphs provide a few extra examples of resistor color band decoding. *Figure 12-51* contains a resistor with another set of colors. This resistor code should be read as follows:

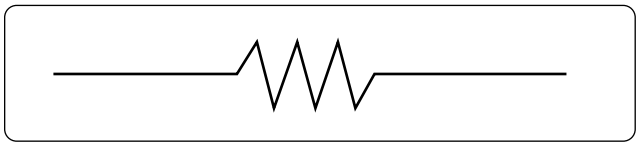


Figure 12-48. *Fixed resistor schematic.*

The resistance of this resistor is 86,000 ±10 percent ohms. The maximum resistance is 94,600 ohms, and the minimum resistance is 77,400 ohms.

Another example is the resistance of the resistor in *Figure 12-52* is 960 ±5 percent ohms. The maximum resistance is 1,008 ohms, and the minimum resistance is 912 ohms.
Sometimes circuit considerations dictate that the tolerance must be smaller than 20 percent. *Figure 12-53* shows an example of a resistor with a 2 percent tolerance. The resistance value of this resistor is 2,500 ±2 percent ohms. The maximum resistance is 2,550 ohms, and the minimum resistance is 2,450 ohms.

Figure 12-54 contains an example of a resistor with a black third color band. The color code value of black is zero, and the third band indicates the number of zeros to be added to the first two digits.

In this case, a zero number of zeros must be added to the first two digits; therefore, no zeros are added. Thus, the resistance value is 10 ±1 percent ohms. The maximum resistance is 10.1 ohms, and the minimum resistance is 9.9 ohms. There are two exceptions to the rule stating the third color band indicates the number of zeros. The first of these exceptions is illustrated in *Figure 12-55*. When the third band is gold in color, it indicates that the first two digits must be multiplied by 10 percent. The value of this resistor in this case is:

10 × 0.10 ±2% = 1 = 0.02 ohms

When the third band is silver, as is the case in *Figure 12-56*, the first two digits must be multiplied by 1 percent. The value

Resistor color code		
Color	**Number**	**Tolerance**
Black	0	—
Brown	1	1%
Red	2	2%
Orange	3	3%
Yellow	4	4%
Green	5	5%
Blue	6	6%
Violet	7	7%
Gray	8	8%
White	9	9%
Gold	—	5%
Silver	—	10%
No color	—	20%

Figure 12-49. *Resistor color code.*

of the resistor is 0.45 ±10 percent ohms.

Wire-Wound

Wire-wound resistors typically control large amounts of current and have high power ratings. Resistors of this type are constructed by winding a resistance wire around an insulating rod usually made of porcelain. The windings are coated with an insulation material for physical protection and heat conduction. Both ends of the windings are connected to terminals, which are used to connect the resistor to a circuit. *[Figure 12-57]*

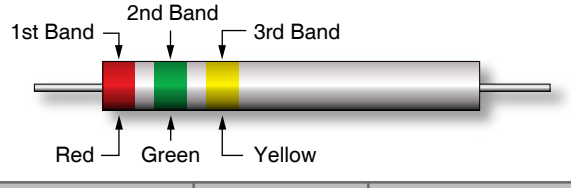
A wire-wound resistor with tap is a special type of fixed resistor that can be adjusted. These adjustments can be made by moving a slide bar tap or by moving the tap to a preset incremental position. While the tap may be adjustable, the adjustments are usually set at the time of installation to a specific value and then operated in service as a fixed resistor. Another type of wire-wound resistor is constructed of Manganin wire, used when high precision is needed.

Variable Resistors

Variable resistors are constructed so that the resistive value can be changed easily. This adjustment can be manual or automatic, and the adjustments can be made while the system that it is connected to is in operation. There are two basic types of manual adjustors: the rheostat and the potentiometer.

Rheostat

A rheostat is a variable resistor used to vary the amount of current flowing in a circuit. *[Figure 12-58] Figure 12-59* shows a rheostat connected in series with an ordinary resistance in a series circuit. As the slider arm moves from point A to B, the amount of rheostat resistance (AB) is increased. Since the rheostat resistance and the fixed resistance are in series, the total resistance in the circuit also increases, and the current in the circuit decreases. On the other hand, if the slider arm is moved toward point A, the total resistance decreases and the current in the circuit increases.



Color	Numerical Value	Significance
1st band — red	2	1st digit
2nd band — green	5	2nd digit
3rd band — yellow	4	No. of zeroes to add

Figure 12-50. *End-to-center band marking.*

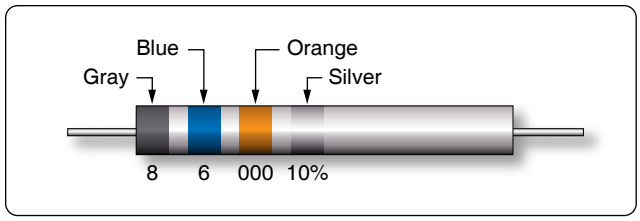


Figure 12-51. *Resistor color code example.*

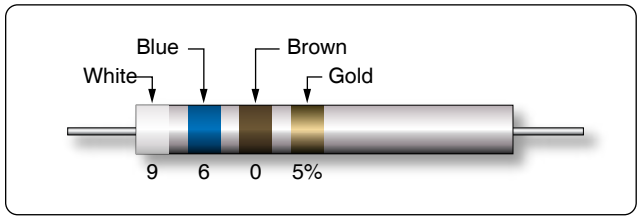


Figure 12-52. *Resistor color code example.*

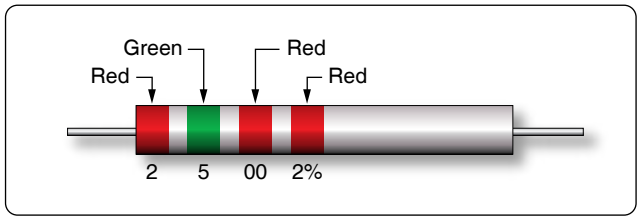


Figure 12-53. *Resistor with two percent tolerance.*

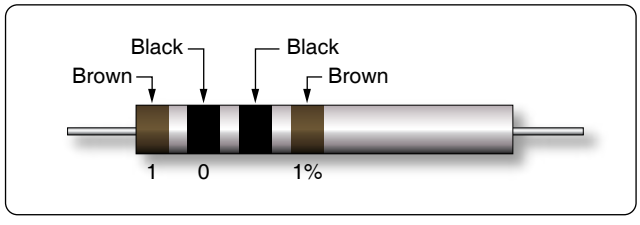


Figure 12-54. *Resistor with black third color band.*

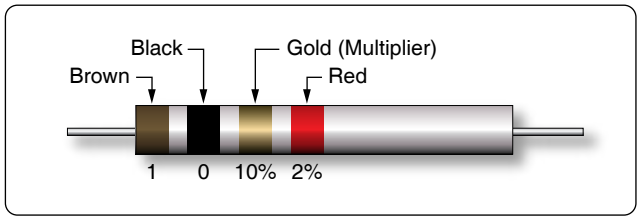


Figure 12-55. *Resistor with gold third band.*

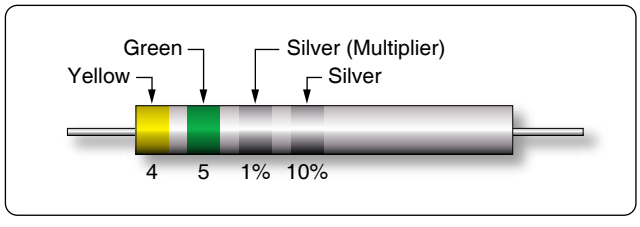


Figure 12-56. *Resistor with a silver third band.*

Potentiometer

The potentiometer is considered a three-terminal device. *[Figure 12-60]* As illustrated, terminals 1 and 2 have the entire value of the potentiometer resistance between them. Terminal 3 is the wiper or moving contact. Through this wiper, the resistance between terminals 1 and 3 or terminals 2 and 3 can be varied. While the rheostat is used to vary the current in a circuit, the potentiometer is used to vary the voltage in a circuit. A typical use for this component can be found in the volume controls on an audio panel and input devices for flight data recorders, among many other applications.

In *Figure 12-61A*, a potentiometer is used to obtain a variable voltage from a fixed voltage source to apply to an electrical load. The voltage applied to the load is the voltage between points 2 and 3. When the slider arm is moved to point 1, the entire voltage is applied to the electrical device (load); when the arm is moved to point 3, the voltage applied to the load is zero. The potentiometer makes possible the application of any voltage between zero and full voltage to the load.

The current flowing through the circuit of *Figure 12-61* leaves the negative terminal electron flow of the battery and divides one part flowing through the lower portion of the potentiometer (points 3 to 2) and the other part through the load. Both parts combine at point 2 and flow through the upper portion of the potentiometer (points 2 to 1) back to the positive terminal of the battery. In View B of *Figure 12-61*, a potentiometer and its schematic symbol are shown.

In choosing a potentiometer resistance, the amount of current drawn by the load should be considered, as well as the current flow through the potentiometer at all settings of the slider arm. The energy of the current through the potentiometer is dissipated in the form of heat.

It is important to keep this wasted current as small as possible by making the resistance of the potentiometer as large as practicable. In most cases, the resistance of the potentiometer can be several times the resistance of the load. *Figure 12-62* shows how a potentiometer can be wired to function as a rheostat.

Linear Potentiometers

In a linear potentiometer, the resistance between both terminal and the wiper varies linearly with the position of the wiper. To illustrate, one quarter of a turn on the potentiometer results in one quarter of the total resistance. The same relationship exists when one-half or three-quarters of potentiometer movement. *[Figure 12-63]*

Tapered Potentiometers

Resistance varies in a nonlinear manner in the case of the tapered potentiometer. *[Figure 12-64]* Keep in mind that one-half of full potentiometer travel does not necessarily correspond to one-half the total resistance of the potentiometer.

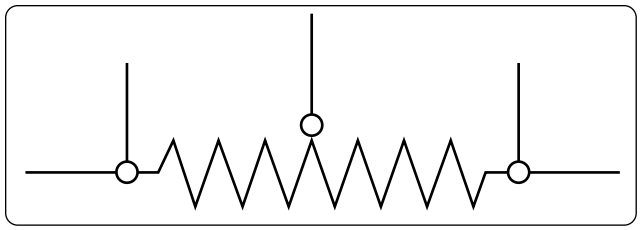


Figure 12-57. *Wire-wound resistors.*

Thermistors

The thermistor is a type of a variable resistor that is temperature sensitive. *[Figure 12-65]* This component has a negative temperature coefficient, which means that as the sensed temperature increases, the resistance of the thermistor decreases.

Photoconductive Cells

The photoconductive cell is similar to the thermistor. Like the thermistor, it has a negative temperature coefficient. Unlike the thermistor, the resistance is controlled by light intensity. This kind of component can be found in radio control heads where the intensity of the ambient light is sensed through the photoconductive cell resulting in the backlighting of the control heads to adjust to the flight deck lighting conditions. *[Figure 12-66]*

Circuit Protection Devices

Perhaps the most serious trouble in a circuit is a direct short. The term "direct short" describes a situation in which some point in the circuit, where full system voltage is present, comes in direct contact with the ground or return side of the circuit. This establishes a path for current flow that contains no resistance other than that present in the wires carrying the current, and these wires have very little resistance.

Most wires used in aircraft electrical circuits are small gauge, and their current carrying capacity is quite limited. The size of the wires used in any given circuit is determined

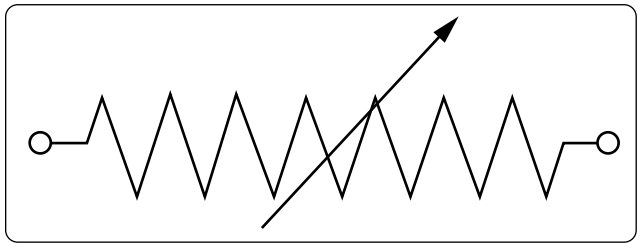


Figure 12-58. *Rheostat schematic symbol.*

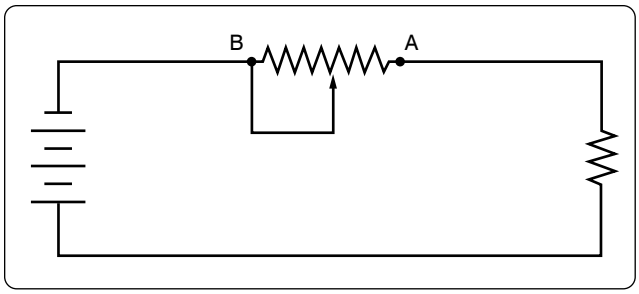


Figure 12-59. *Rheostat connected in series.*

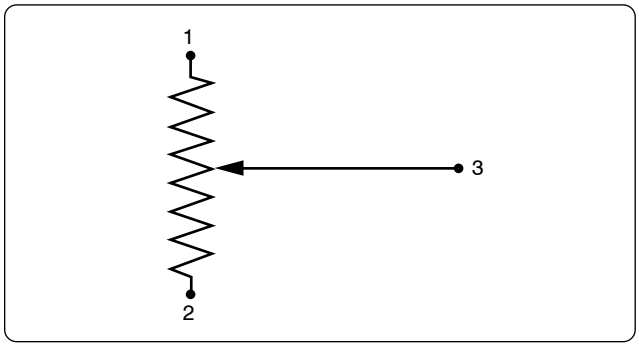


Figure 12-60. *Potentiometer schematic symbol.*

by the amount of current the wires are expected to carry under normal operating conditions. Any current flow in excess of normal, such as the case of a direct short, would cause a rapid generation of heat. If the excessive current flow caused by the short is left unchecked, the heat in the wire could cause a portion of the wire to melt and at the very least, open the circuit.

To protect aircraft electrical systems from damage and failure caused by excessive current, several kinds of protective devices are installed in the systems. Fuses, circuit breakers, thermal protectors, and arc fault circuit breakers are used for this purpose.

Circuit protective devices, as the name implies, all have a common purpose—to protect the units and the wires in the circuit. Some are designed primarily to protect the wiring and to open the circuit in such a way as to stop the current flow when the current becomes greater than the wires can safely carry. Other devices are designed to protect a unit in the circuit by stopping the current flow to it when the unit becomes excessively warm.

Fuse

Fuses are used to protect the circuit from over current conditions. *[Figure 12-67A]* The fuse is installed in the circuit so that all the current in the circuit passes through it. In most fuses, the strip of metal is made of an alloy of tin and bismuth, which melts and opens the circuit when the current exceeds

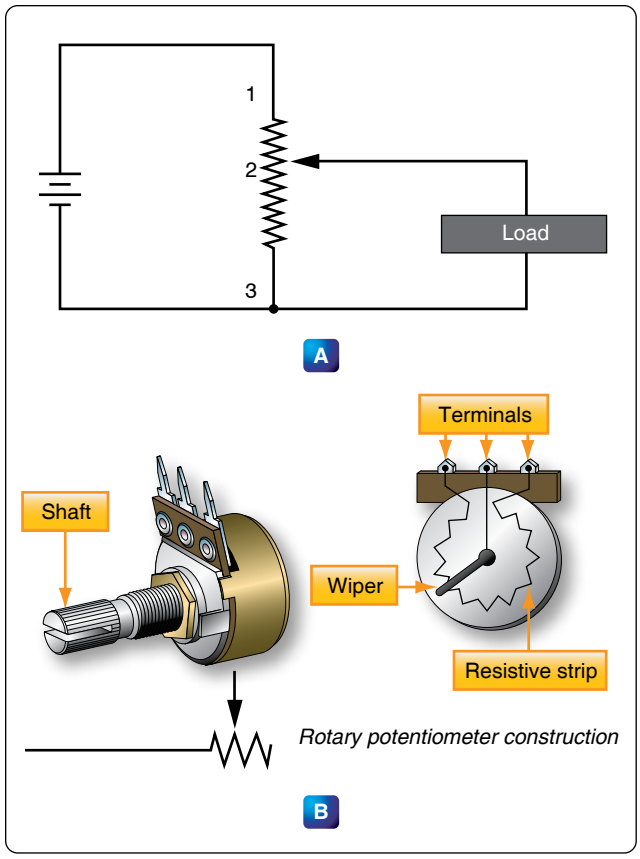


Figure 12-61. *Potentiometer and schematic symbol.*

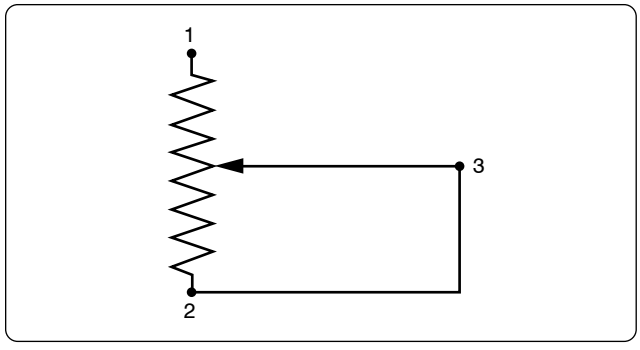


Figure 12-62. *Potentiometer wired to function as rheostat.*

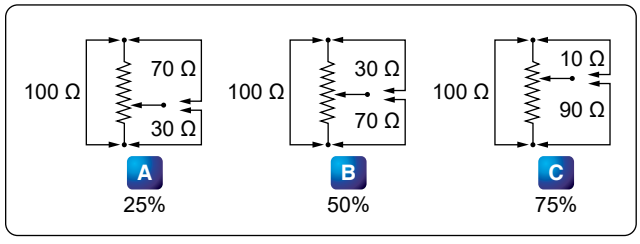


Figure 12-63. *Linear potentiometer schematic.*

the rated capacity of the fuse. For example, if a 5-amp fuse is placed into a circuit, the fuse allows currents up to 5 amps to pass. Because the fuse is intended to protect the circuit, it is quite important that its capacity match the needs of the circuit in which it is used.

When replacing a fuse, consult the applicable manufacturer's instructions to be sure a fuse of the correct type and capacity is installed. Fuses are installed in two types of fuse holders in aircraft. "Plug-in holders" or in-line holders are used for small and low capacity fuses. "Clip" type holders are used for heavy high capacity fuses and current limiters.

Current Limiter

The current limiter is very much like the fuse. However, the current limiter link is usually made of copper and will stand a considerable overload for a short period of time. Like the fuse, it opens up in an over current condition in heavy current circuits such as 30 amp or greater. These are used primarily to sectionalize an aircraft circuit or bus. Once the limiter is opened, it must be replaced. The schematic symbol for the current limiter shows two triangles pointing to each other with a line on both sides of the triangles. [*Figure 12-67B*].

Circuit Breaker

The circuit breaker is commonly used in place of a fuse and is designed to break the circuit and stop the current flow when the current exceeds a predetermined value. Unlike the fuse, the circuit breaker can be reset; whereas the fuse or current limiter must be replaced. *[Figure 12-68]*

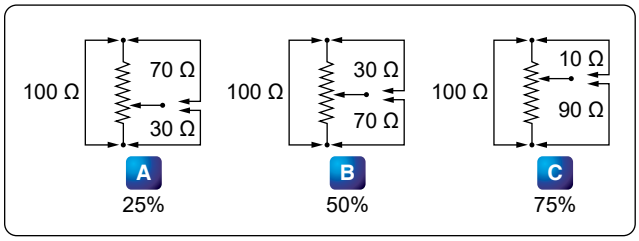


Figure 12-64. *Tapered potentiometer.*

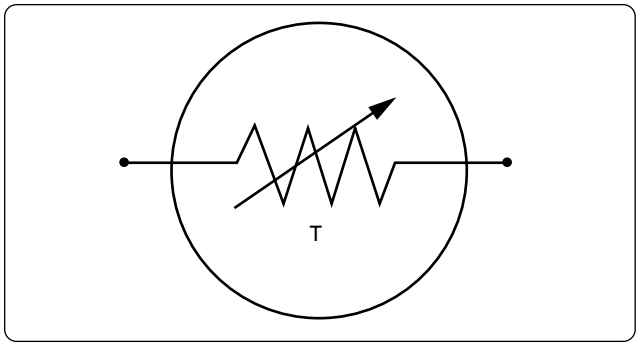


Figure 12-65. *Schematic symbol for thermistor.*

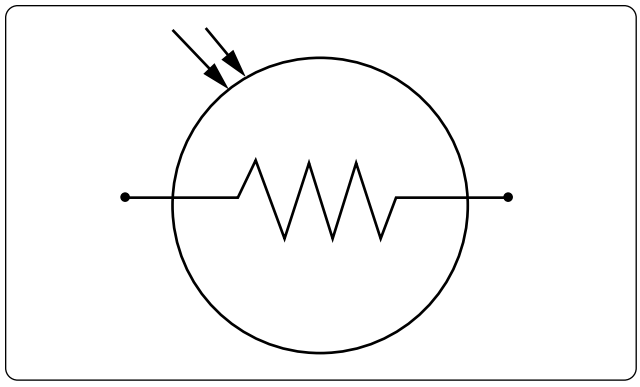


Figure 12-66. *Photoconductive cell schematic symbol component.*

There are several types of circuit breakers in general use in aircraft systems. One is a magnetic type. When excessive current flows in the circuit, it makes an electromagnet strong enough to move a small armature, which trips the breaker. Another type is the thermal overload switch or breaker. This consists of a bimetallic strip which, when it becomes overheated from excessive current, bends away from a catch on the switch lever and permits the switch to trip open.

Most circuit breakers must be reset by hand. If the overload condition still exists, the circuit breaker trips again to prevent damage to the circuit. At this point, it is usually not advisable to continue resetting the circuit breaker, but to initiate troubleshooting to determine the cause. Repeated resetting of a circuit breaker can lead to circuit or component damage or worse, the possibility of a fire or explosion. Automatic reset type circuit breakers are not allowed in aircraft. Circuit breakers are commonly grouped on a circuit breaker panel that is accessible to the flight crew. *Figure 12-69* shows a circuit breaker and a circuit breaker panel.

Arc Fault Circuit Breaker

In recent years, the arc fault circuit breaker has begun to provide an additional layer of protection beyond that of the thermal protection already provided by conventional circuit breakers. The arc fault circuit breaker monitors the circuit for an electrical arcing signature, which can indicate possible wiring faults and unsafe conditions. These conditions can lead to fires or loss of power to critical systems. The arc fault circuit breaker is only beginning to make an appearance in the aircraft industry and is not widely used like the thermal type of circuit breaker.

Thermal Protectors

A thermal protector, or switch, is used to protect a motor. It is designed to open the circuit automatically whenever the temperature of the motor becomes excessively high. It has two positions—open and closed. The most common use for a thermal switch is to keep a motor from overheating. If a

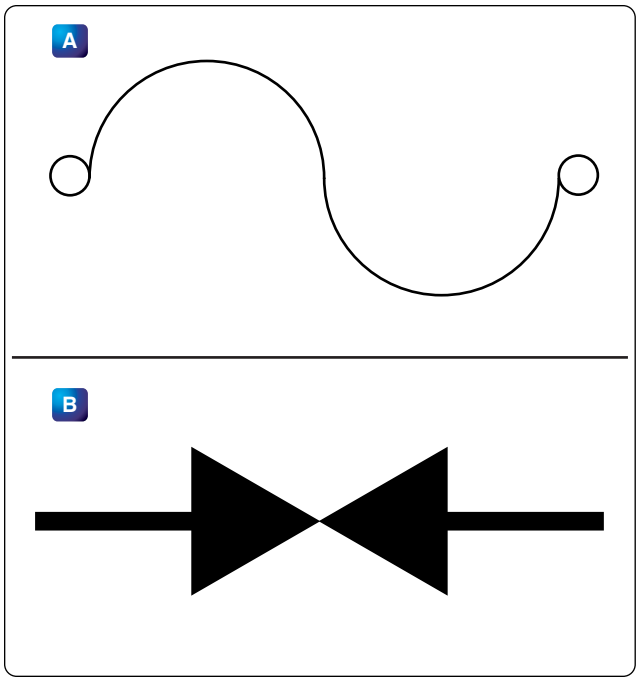


Figure 12-67. *Schematic symbol for fuse (A) and current limiter (B).*

malfunction in the motor causes it to overheat, the thermal switch breaks the circuit intermittently.

The thermal switch contains a bimetallic disk, or strip, that bends and breaks the circuit when it is heated. This occurs because one of the metals expands more than the other when they are subjected to the same temperature. When the strip or disk cools, the metals contract and the strip returns to its original position and closes the circuit.

Control Devices

Components in the electrical circuits are typically not all intended to operate continuously or automatically. Most of them are meant to operate at certain times, under certain conditions, to perform very definite functions. There must be some means of controlling their operation. Either a switch, or a relay, or both may be included in the circuit for this purpose.

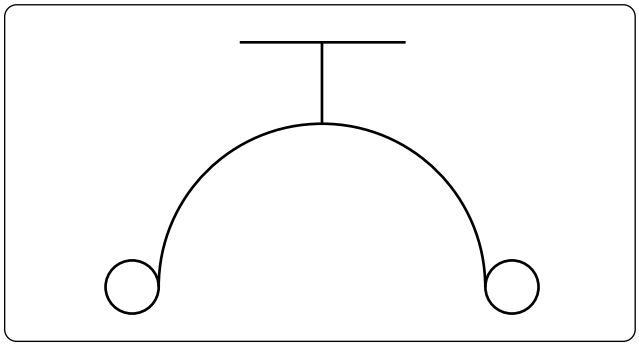


Figure 12-68. *Schematic symbol for circuit breaker.*

Switches

Switches control the current flow in most aircraft electrical circuits. A switch is used to start, to stop, or to change the direction of the current flow in the circuit. The switch in each circuit must be able to carry the normal current of the circuit and must be insulated heavily enough for the voltage of the circuit. *Figure 12-70* shows various switches used in aircraft electrical systems.

An understanding of some basic definitions of the switch is necessary before any of the switch types are discussed. The number of poles, throws, and positions they have designates toggle switches, as well as some other type of switches.

Pole—the switch's movable blade or contactor. The number of poles is equal to the number of circuits, or paths for current flow, that can be completed through the switch at any one time.

Throw—indicates the number of circuits, or paths for current, that it is possible to complete through the switch with each pole or contactor.

Positions—indicates the number of places at which the operating device (toggle, plunger, and so forth) comes to rest and at the same time open or close one or more circuits.

Toggle Switch

Single-Pole, Single-Throw (SPST)

The single-pole, single-throw (SPST) switch allows a connection between two contacts. One of two conditions exist. Either the circuit is open in one position or closed in the other position. *[Figure 12-71]*

Single-Pole, Double-Throw (SPDT)

The single-pole, double-throw (SPDT) switch is shown in *Figure 12-72*. With this switch, contact between one contact can be made between one contact and the other.

Double-Pole, Single-Throw (DPST)

The double-pole, single-throw (DPST) switch connection can be made between one set of contacts and either of two other sets of contacts. *[Figure 12-73]*

Double-Pole, Double-Throw (DPDT)

The schematic symbol for the double-pole, double-throw (DPDT) switch is shown in *Figure 12-74*. This type of switch makes a connection from one set of contacts to either of two other sets of contacts.

A toggle switch that is spring-loaded to the OFF position

Figure 12-69. *Circuit breaker assembly for aircraft electrical system.*

and must be held in the ON position to complete the circuit is a momentary contact two-position switch. One that comes to rest at either of two positions, opening the circuit in one position and closing it in another, is a two-position switch. A toggle switch that comes to rest at any one of three positions is a three-position switch.

A switch that stays open, except when it is held in the closed position, is a normally open switch (usually identified as NO). One that stays closed, except when it is held in the open position is a normally closed switch (NC). Both kinds are spring loaded to their normal position and return to that position as soon as they are released.

Locking toggles require the operator to pull out on the switch toggle before moving it in to another position. Once in the new position, the switch toggle is release back into a lock, which then prevents the switch from inadvertently being moved.

Microswitches

A microswitch opens or closes a circuit with a very small movement of the tripping device (1/16 inch or less). This is what gives the switch its name, since micro means small.

Microswitches are usually pushbutton switches. They are used primarily as limit switches to provide automatic control of landing gears, actuator motors, and the like. *Figure 12-75* shows a normally closed microswitch in cross-section and illustrates how these switches operate. When the operating plunger is pressed in, the spring and the movable contact are pushed, opening the contacts and the circuit. *Figure 12-76* shows a pushbutton microswitch.

Rotary Selector Switches

A rotary selector switch takes the place of several switches. When the knob of the switch is rotated, the switch opens one circuit and closes another. Ignition switches and voltmeter selector switches are typical examples of this kind of switch. *[Figure 12-77]*

Pushbutton Switches

Pushbutton switches have one stationary contact and one

Figure 12-70. *Various types of switches used in modern aircraft.*

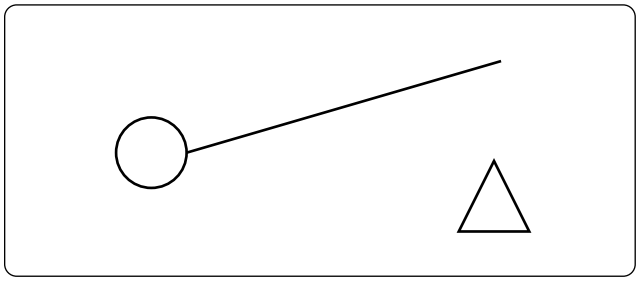


Figure 12-71. *Single-pole, single-throw switch schematic symbol.*

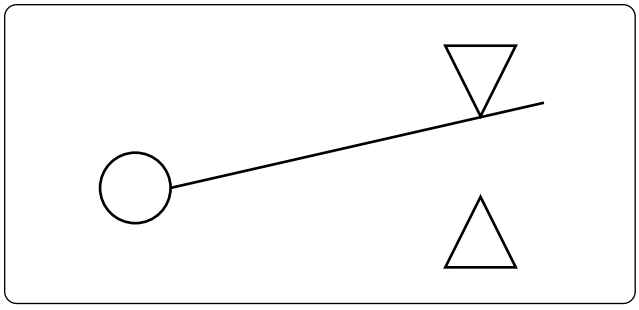


Figure 12-72. *Single-pole, double-throw switch schematic symbol.*

movable contact. The movable contact is attached to the pushbutton. The pushbutton is either an insulator itself or is insulated from the contact. This switch is spring loaded and designed for momentary contact.

Lighted Pushbutton Switches

Another more common switch found in today's aircraft is the lighted pushbutton switch. This type of switch takes the form of a ⅝-inch to 1-inch cube with incandescent or LED lights to indicate the function of the switch. Switch designs come in a number of configurations; the two most common are the alternate action and momentary action and usually have a two-pole or four-pole switch body. Other less common switch actions are the alternate and momentary holding coil configurations. The less known holding or latching coil switch bodies are designed to have a magnetic coil inside the switch body that is energized through two contacts in the base of the switch. When the coil is energized and the switch is pressed, the switch contacts remain latched until power is removed from the coil. This type of design allows for some degree of remote control over the switch body. *[Figure 12-78]*

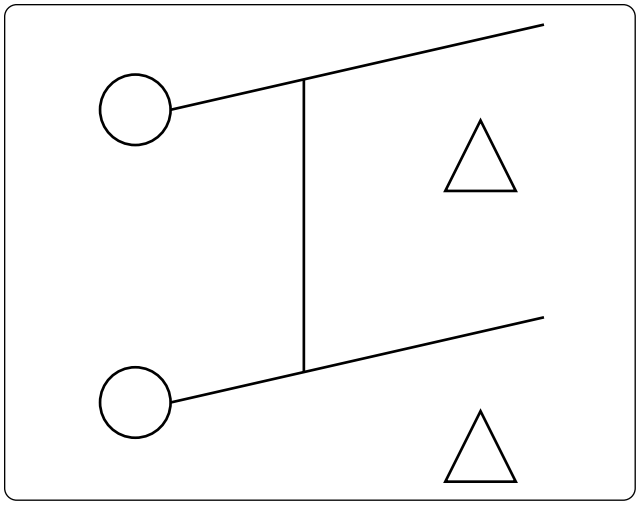


Figure 12-73. *Double-pole, single-throw switch schematic symbol.*

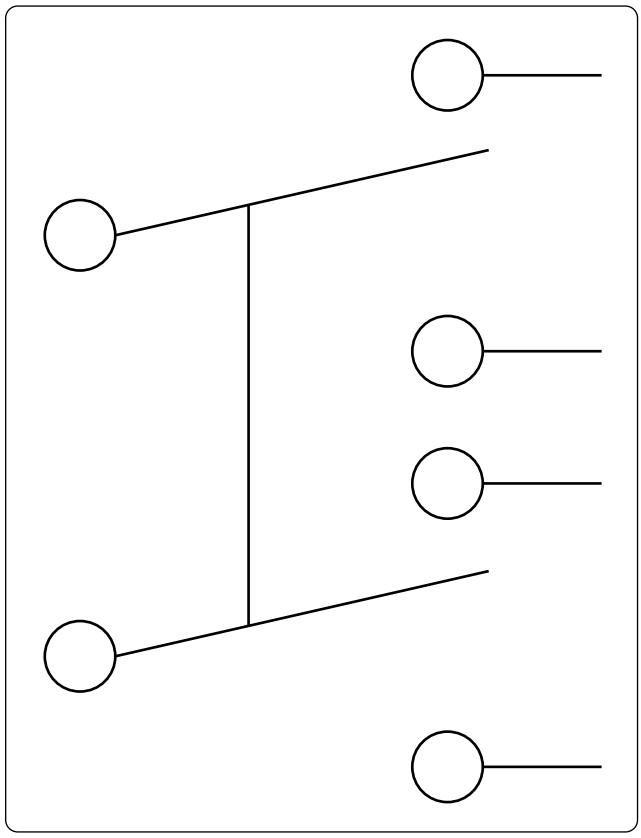


Figure 12-74. *Double-pole, double-throw switch schematic symbol.*

The display optics of the lighted pushbutton switch provide the crew with a clear message that is visible under a wide range of lighting conditions with very high luminance and wide viewing angles. While some displays are simply a transparent screen that is backlit by an incandescent light, the higher quality and more reliable switches are available in sunlight readable displays and night vision (NVIS) versions. Due to the sunlight environment of the flight deck, displays utilizing standard lighting techniques "washout" when viewed in direct sunlight. Sunlight readable displays are designed to minimize this effect.

Lighted pushbutton switches can also be used in applications where a switch is not required and the optics are only for indications. This type of an indicator is commonly called an annunciator.

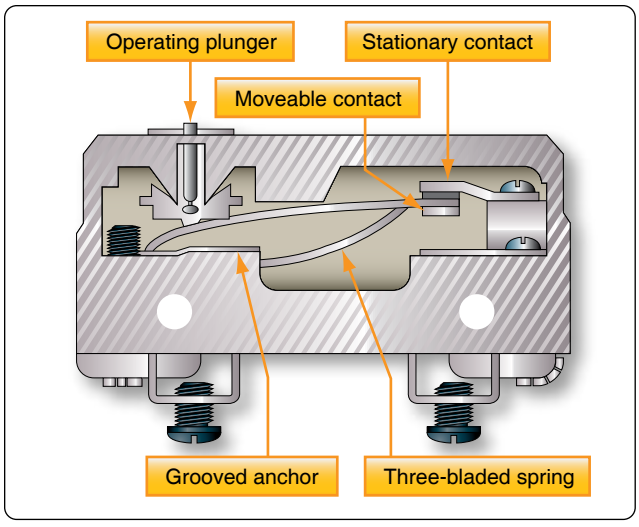


Figure 12-75. *Cross-section of a microswitch.*

Dual In-Line Parallel (DIP) Switches

The acronym "DIP" switch is defined as Dual In-Line Parallel switch in reference to the physical layout. DIP switches are commonly found in card cages, and line replaceable units (LRU) and are used in most cases to adjust gains, control configurations, and so forth. Each one of the switches is generally an SPST slide or rocker switch. The technician may find this switch in packages ranging in size from DIP2 through DIP32. Some of the more common sizes are DIP4 and DIP8.

Switch Guards

Switch guards are covers that protect a switch from unintended operation. Prior to the operation of the switch, the guard is usually lifted. Switch guards are commonly found on systems such as fire suppression and override logics for various systems. *Figure 12-79A* shows a traditional switch with a guard while *Figure 12-79B* shows a pushbutton switch with a guard. The guard needs to be moved before the switch can be pushed.

Relays

A relay is simply an electromechanical switch where a small amount of current can control a large amount of current. *[Figure 12-80]* When a voltage is applied to the coil of the relay, the electromagnet is energized due to the current. When energized, an electromagnetic field pulls the common (C) or arm of the relay down. When the arm or common is pulled down, the circuit between the arm and the normally closed (NC) contacts is opened and the circuit between the arm and the normally open (NO) contacts are closed. When the energizing voltage is removed, the spring returns the arm contacts back to the normally closed (NC) contacts. The relay usually has two connections for the coil. The (+) side is designated as X1 and the ground-side of the coil is designated as X2.

Figure 12-76. *Pushbutton microswitch.*

Figure 12-77. *Rotary switches.*

Series DC Circuits

The series circuit is the most basic electrical circuit and provides a good introduction to basic circuit analysis. The series circuit represents the first building block for all of the circuits to be studied and analyzed. *Figure 12-81* shows this simple circuit with nothing more than a voltage source or battery, a conductor, and a resistor. This is classified as a series circuit because the components are connected end-to-end, so that the same current flows through each component equally. There is only one path for the current to take and the battery and resistor are in series with each other. Next is to make a few additions to the simple circuit in *Figure 12-81*.

Figure 12-82 shows an additional resistor and a little more

Figure 12-78. *Lighted pushbutton switches.*

detail regarding the values. With these values, we can now begin to learn more about the nature of the circuit. In this configuration, there is a 12-volt DC source in series with two resistors, $R_1 = 10\ \Omega$ and $R_2 = 30\ \Omega$. For resistors in a series configuration, the total resistance of the circuit is equal to the sum of the individual resistors. The basic formula is:

$$R_T = R_1 + R_2 + R_3 + \ldots\ldots\ldots R_N$$

For *Figure 12-82*, this will be:

$$R_T = 10\ \Omega + 30\ \Omega$$
$$R_T = 40\ \Omega$$

Now that the total resistance of the circuit is known, the current for the circuit can be determined. In a series circuit, the current cannot be different at different points within the circuit. The current through a series circuit is always the same through each element and at any point. Therefore, the current in the simple circuit can now be determined using Ohm's Law:

Formula, $E = I\ (R)$

Solve for current, $I = \frac{E}{R}$

The variables, $E = 12$ V and $R_T = 40\ \Omega$

Substitute variables, $I = \frac{12\text{ V}}{40\ \Omega}$

Current in circuits, $I = 0.3$ A

Ohm's Law describes a relationship between the variables of voltage, current, and resistance that is linear and easy to illustrate with a few extra calculations. First is the act of changing the total resistance of the circuit while the other two remain constant. In this example, the R_T of the circuit in *Figure 12-82* is doubled.

The effects on the total current in the circuit are:

Figure 12-79. *(A) Red guarded switch. (B) Pushbutton switch with a guard.*

Formula, $E = I\ (R)$

Solve for current, $I = \frac{E}{R}$

The variables, $E = 12$ V and $R_T = 80\ \Omega$

Substitute variables, $I = \frac{12\text{ V}}{80\ \Omega}$

Current in circuits, $I = 0.15$ A

It can be seen quantitatively and intuitively that when the resistance of the circuit is doubled, the current is reduced by half the original value.
Next, reduce the R_T of the circuit in *Figure 12-82* to half of its original value. The effects on the total current are:

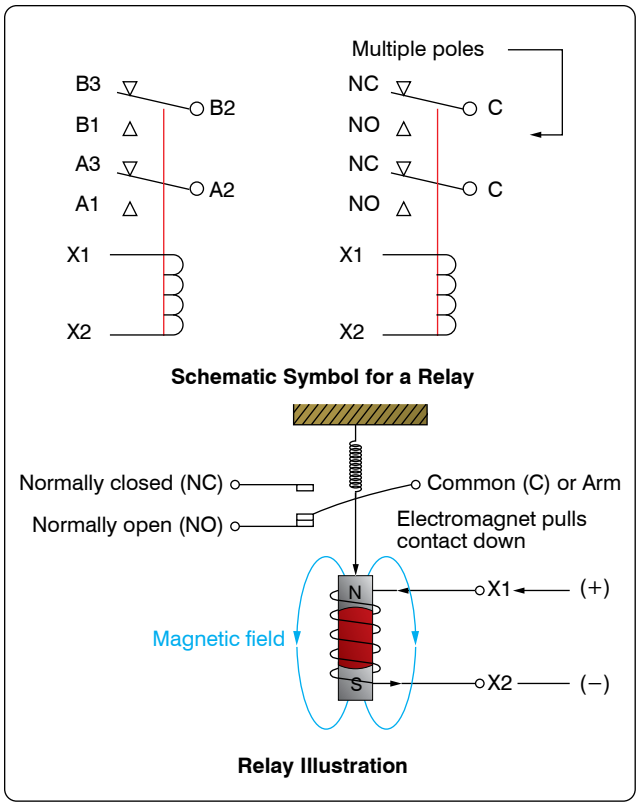


Figure 12-80. *Basic relay.*

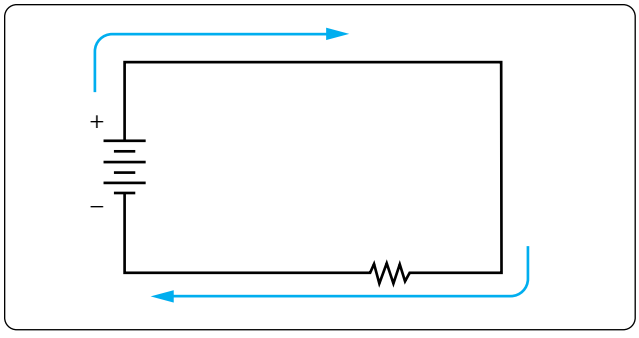


Figure 12-81. *Simple DC circuit.*

Formula, $E = I\,(R)$

Solve for current, $I = \frac{E}{R}$

The variables, $E = 12$ V and $R_T = 20\ \Omega$

Substitute variables, $I = \frac{12\text{ V}}{20\ \Omega}$

Current in circuits, $I = 0.6$ A

Voltage Drops & Further Application of Ohm's Law

The example circuit in *Figure 12-83* is used to illustrate the idea of voltage drop. It is important to differentiate between voltage and voltage drop when discussing series circuits. Voltage drop refers to the loss in electrical pressure or emf caused by forcing electrons through a resistor. Because there are two resistors in the example, there are separate voltage drops. Each drop is associated with each individual resistor. The amount of electrical pressure required to force a given number of electrons through a resistance is proportional to the size of the resistor.

In *Figure 12-83*, the values used to illustrate the idea of voltage drop are:

Current, $I = 1$ mA
$R_1 = 1\ \text{k}\Omega$
$R_2 = 3\ \text{k}\Omega$
$R_3 = 5\ \text{k}\Omega$

The voltage drop across each resistor is calculated using Ohm's Law. The drop for each resistor is the product of each resistance and the total current in the circuit. Keep in mind that the same current flows through series resistor.

Formula: $E = I\,(R)$

Voltage across R_1: $E_1 = I_T\,(R_1)$
$E_1 = 1\text{ mA}\,(1\ \text{k}\Omega) = 1$ volt

Voltage across R_2: $E_2 = I_T\,(R_2)$
$E_2 = 1\text{ mA}\,(3\ \text{k}\Omega) = 3$ volt

Voltage across R_3: $E_3 = I_T\,(R_3)$
$E_3 = 1\text{ mA}\,(5\ \text{k}\Omega) = 5$ volt

The source voltage can now be determined, which can then be used to confirm the calculations for each voltage drop. Using Ohm's Law:

Formula: $E = I\,(R)$

Source voltage = current × the total resistance
$E_S = I\,(R_T)$

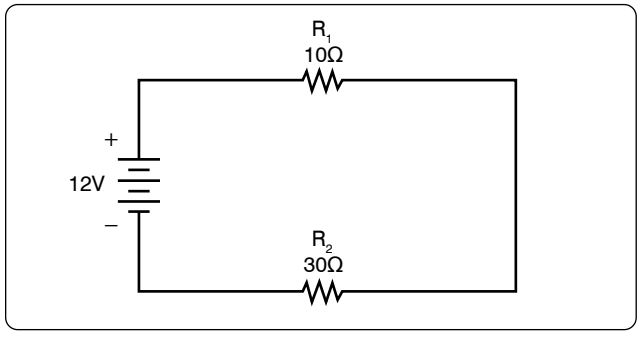


Figure 12-82. *Simple DC circuit with additional resistor.*

$R_T = 1\ k\Omega + 3\ k\Omega + 5\ k\Omega$
$R_T = 9\ k\Omega$

Now: $E_S = I\ (R_T)$

Substitute $E_S = 1\ mA\ (9\ k\Omega)$
$E_S = 9$ volts

Simple checks to confirm the calculation and to illustrate the concept of the voltage drop add up the individual values of the voltage drops and compare them to the results of the above calculation.

1 volt + 3 volts + 5 volts = 9 volts

Voltage Sources in Series

A voltage source is an energy source that provides a constant voltage to a load. Two or more of these sources in series equals the algebraic sum of all the sources connected in series. The significance of pointing out the algebraic sum is to indicate that the polarity of the sources must be considered when adding up the sources. The polarity is indicated by a plus or minus sign depending on the source's position in the circuit.

In *Figure 12-84,* all of the sources are in the same direction in terms of their polarity. All of the voltages have the same sign when added up. In the case of *Figure 12-84*, three cells of a value of 1.5 volts are in series with the polarity in the same direction. The addition is simple enough:

$E_T = 1.5v + 1.5v + 1.5v = +4.5$ volts

However, in *Figure 12-85*, one of the three sources has been turned around, and the polarity opposes the other two sources. Again the addition is simple:

$E_T = +1.5v - 1.5v + 1.5v = +1.5$ volts

Kirchhoff's Voltage Law

A law of basic importance to the analysis of an electrical circuit is Kirchhoff's Voltage Law. This law simply states that the algebraic sum of all voltages around a closed path or loop is zero. Another way of saying it: the sum of all the voltage drops equals the total source voltage. A simplified formula showing this law is shown below:

With three resistors in the circuit:
$E_S - E_1 - E_2 - E_3 \ldots - E_N = 0$ volts

Notice that the sign of the source is opposite that of the individual voltage drops. Therefore, the algebraic sum equals zero. Written another way:

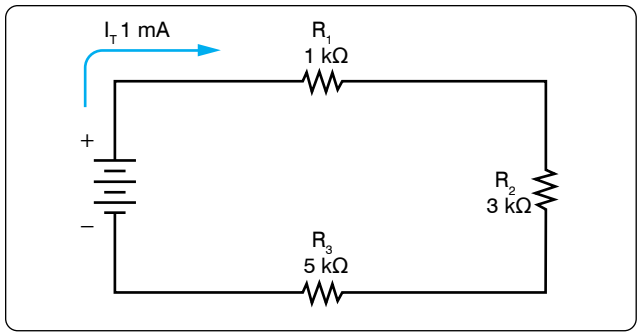


Figure 12-83. *Example of three resistors in series.*

$E_S = E_1 + E_2 + E_3 \ldots + E_N$

The source voltage equals the sum of the voltage drops. The polarity of the voltage drop is determined by the direction of the current flow. When going around the circuit, notice that the polarity of the resistor is opposite that of the source voltage. The positive on the resistor is facing the positive on the source, and the negative on the resistor is facing the negative on the source.

Figure 12-86 illustrates the very basic idea of Kirchhoff's Voltage Law. There are two resistors in this example. One has a drop of 14 volts and the other has a drop of 10 volts. The source voltage must equal the sum of the voltage drops around the circuit. By inspection, it is easy to determine the source voltage as 24 volts.

Figure 12-87 shows a series circuit with three voltage drops and one voltage source rated at 24 volts. Two of the voltage drops are known. However, the third is not known. Using Kirchhoff's Voltage Law, the third voltage drop can be determined.

With three resistors in the circuit:
$E_S - E_1 - E_2 - E_3 = 0$ volts

Substitute the known values:
$24v - 12v - 10v - E_3 = 0$

Collect known values: $2v - E_3 = 0$

Solve for the unknown: $E_3 = 2$ volts

Determine the value of E_4 in *Figure 12-88*. For this example, I = 200mA.

First, the voltage drop across each of the individual resistors must be determined.

$E_1 = I\ (R_1)$
$E_1 = (200\ mA)\ (10\ \Omega)$

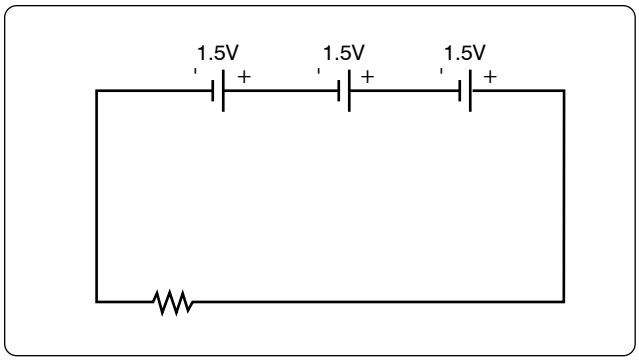


Figure 12-84. *Voltage sources in series add algebraically.*

Voltage drop across R_1 $E_1 = 2$ volts

$E_2 = I (R_2)$
$E_2 = (200 \text{ mA}) (50 \Omega)$
Voltage drop across R_2 $E_2 = 10$ volts

$E_3 = I (R_3)$
$E_3 = (200 \text{ mA}) (100 \Omega)$
Voltage drop across R_3 $E_3 = 20$ volts

Kirchhoff's Voltage Law is now employed to determine the voltage drop across E_4.

With four resistors in the circuit
$E_S - E_1 - E_2 - E_3 - E_4 = 0$ volts

Substituting values:
$100v - 2v - 10v - 20v - E_4 = 0$

Combine: $68v - E_4 = 0$

Solve for unknown: $E_4 = 68v$

Using Ohm's Law and substituting in E_4, the value for R_4 can now be determined.

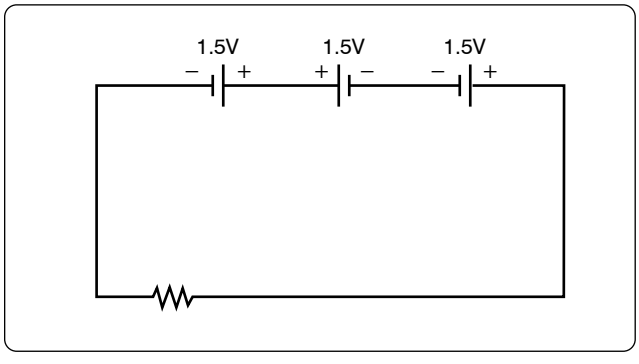


Figure 12-85. *Voltage sources add algebraically; one source reversed.*

Ohm's Law: $R = \frac{E}{I}$

Specific application: $R_4 = \frac{E_4}{I}$

Substitute values: $R_4 = \frac{68 \text{ V}}{200 \text{ mA}}$

Value for R_4: $R_4 = 340 \Omega$

Voltage Dividers

Voltage dividers are devices that make it possible to obtain more than one voltage from a single power source. A voltage divider usually consists of a resistor, or resistors connected in series, with fixed or movable contacts and two fixed terminal contacts. As current flows through the resistor, different voltages can be obtained between the contacts.

Series circuits are used for voltage dividers. The voltage divider rule allows the technician to calculate the voltage across one or a combination of series resistors without having to first calculate the current in the circuit. *[Figure 12-89]* Because the current flows through each resistor, the voltage drops are proportional to the ohmic values of the constituent resistors. To understand how a voltage divider works, examine *Figure 12-90* carefully and observe the following:

Each load draws a given amount of current: I_1, I_2, I_3. In

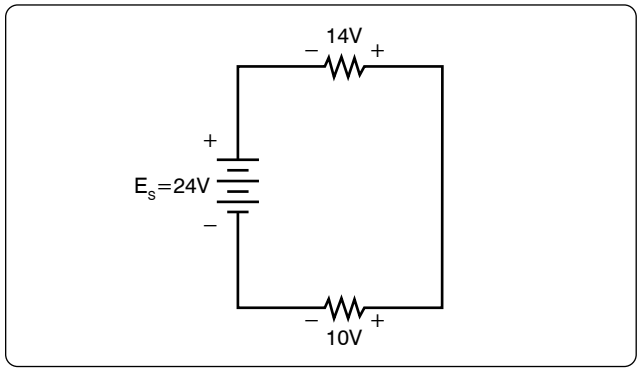


Figure 12-86. *Kirchhoff's Voltage Law.*

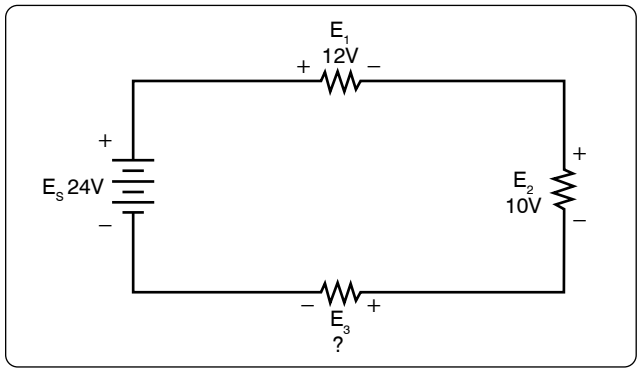


Figure 12-87. *Determine the unknown voltage drop.*

addition to the load currents, some bleeder current (I_B) flows. The current (I_T) is drawn from the power source and is equal to the sum of all currents.

The voltage at each point is measured with respect to a common point. Note that the common point is the point at which the total current (I_T) divides into separate currents (I_1, I_2, I_3). Each part of the voltage divider has a different current flowing in it. The current distribution is as follows:

Through R_1 — bleeder current (I_B)
Through R_2 — $I_B + I_1$
Through R_3 — $I_B + I_1$, + I_2

The voltage across each resistor of the voltage divider is:

90 volts across R_1
60 volts across R_2
50 volts across R_3

The voltage divider circuit discussed up to this point has had one side of the power supply (battery) at ground potential. In *Figure 12-91,* the common reference point (ground symbol) has been moved to a different point on the voltage divider. The voltage drop across R_1 is 20 volts; however, since tap A is connected to a point in the circuit that is at the same potential as the negative side of the battery, the voltage between tap A and the reference point is a negative (−) 20 volts. Since resistors R_2 and R_3 are connected to the positive side of the battery, the voltages between the reference point and tap B or C are positive.

The following rules provide a simple method of determining negative and positive voltages: (1) If current enters a resistance flowing away from the reference point, the voltage drop across that resistance is positive in respect to the reference point; (2) if current flows out of a resistance toward the reference point, the voltage drop across that resistance is negative in respect to the reference point. It is the location of the reference point that determines whether a voltage is negative or positive.

Tracing the current flow provides a means for determining the voltage polarity. *Figure 12-92* shows the same circuit with the polarities of the voltage drops and the direction of current flow indicated.

The current flows from the negative side of the battery to R_1. Tap A is at the same potential as the negative terminal of the battery since the slight voltage drop caused by the resistance of the conductor is disregarded; however, 20 volts of the source voltage are required to force the current through R_1 and this 20-volt drop has the polarity indicated. Stated another way, there are only 80 volts of electrical pressure left in the circuit on the ground side of R_1.

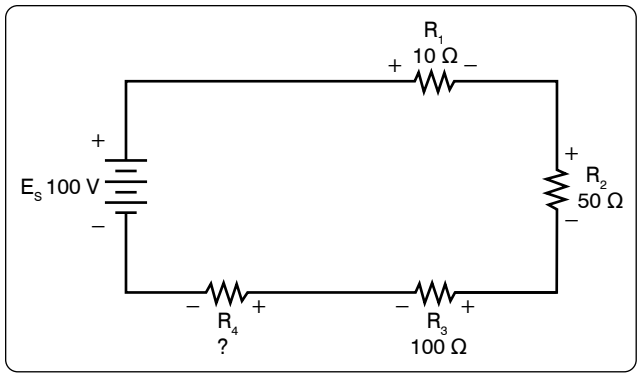


Figure 12-88. *Determine the unknown voltage drop.*

When the current reaches tap B, 30 more volts have been used to move the electrons through R_2, and in a similar manner the remaining 50 volts are used for R_3. But the voltages across R_2 and R_3 are positive voltages, since they are above ground potential.

Figure 12-93 shows the voltage divider used previously. The voltage drops across the resistances are the same; however, the reference point (ground) has been changed. The voltage between ground and tap A is now a negative 100 volts, or the applied voltage.

The voltage between ground and tap B is a negative 80 volts, and the voltage between ground and tap C is a negative 50 volts.

Determining the Voltage Divider Formula

Figure 12-94 shows the example network of four resistors and a voltage source. With a few simple calculations, a formula for determining the voltage divisions in a series circuit can be determined.

The voltage drop across any particular resistor shall be called E_X, where the subscript x is the value of a particular resistor (1, 2, 3, or 4). Using Ohm's Law, the voltage drop across any

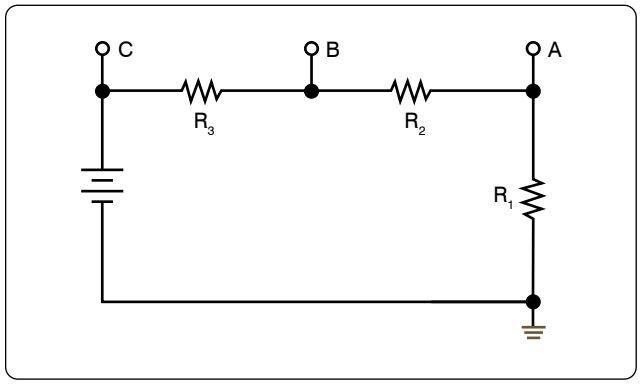


Figure 12-89. *A voltage divider circuit.*

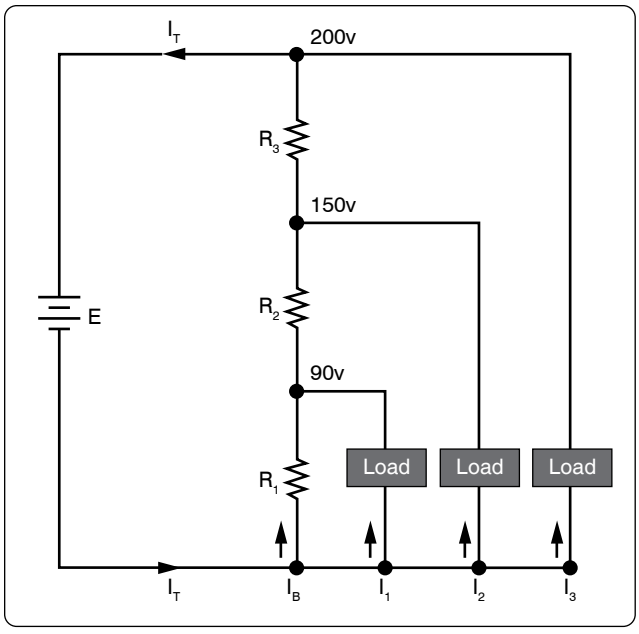


Figure 12-90. *A typical voltage divider.*

resistor can be determined.

Ohm's Law: $E_X = I\,(R_X)$

As seen earlier in the handbook, the current is equal to the source voltage divided by the total resistance of the series circuit.

Current: $I = \dfrac{E_S}{R_T}$

The current equation can now be substituted into the equation for Ohm's Law.

Substitute: $E_X = \left(\dfrac{E_S}{R_T}\right)\left(R_X\right)$

Algebraic rearrange: $E_X = \left(\dfrac{R_X}{R_T}\right)\left(E_S\right)$

This equation is the general voltage divider formula. The explanation of this formula is that the voltage drop across any resistor or combination of resistors in a series circuit is equal to the ratio of the resistance value to the total resistance, divided by the value of the source voltage. *Figure 12-95* illustrates this with a network of three resistors and one voltage source.

$$E_X = \left(\frac{R_X}{R_T}\right) E_S$$

$R_T = 100\ \Omega + 300\ \Omega + 600\ \Omega = 1{,}000\ \Omega$

$E_S = 10$ V

Voltage drop over 100 Ω resistor is:

$$E_X = \left(\frac{100\ \Omega}{1{,}000\ \Omega}\right) 100\text{ V}$$

$E_{100\Omega} = 10$ V

Voltage drop over 300 Ω resistor is:

$$E_X = \left(\frac{300\ \Omega}{1{,}000\ \Omega}\right) 100\text{ V}$$

$E_{100\,\Omega} = 30$ V

Voltage drop over 600 Ω resistor is:

$$E_X = \left(\frac{600\ \Omega}{1{,}000\ \Omega}\right) 100\text{ V}$$

$E_{100\Omega} = 60$ V

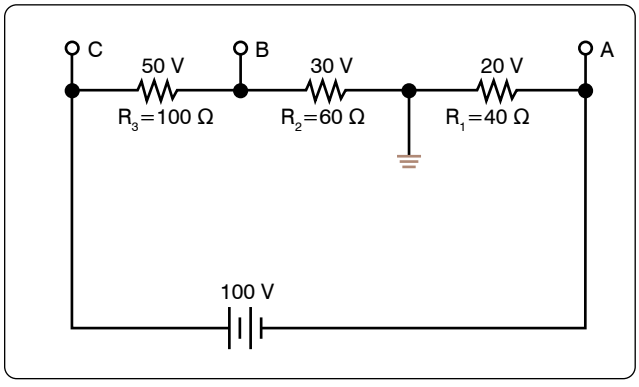


Figure 12-91. *Positive and negative voltage on a voltage divider.*

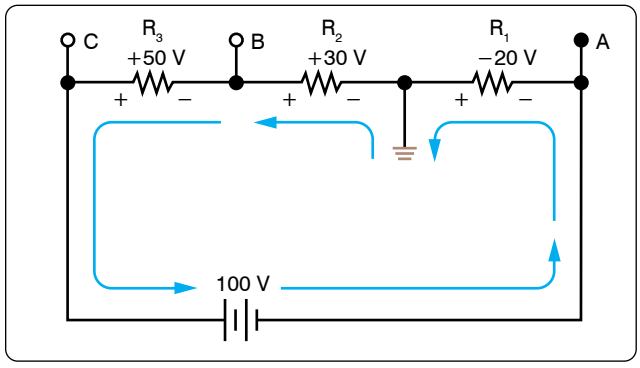


Figure 12-92. *Current flow through a voltage divider.*

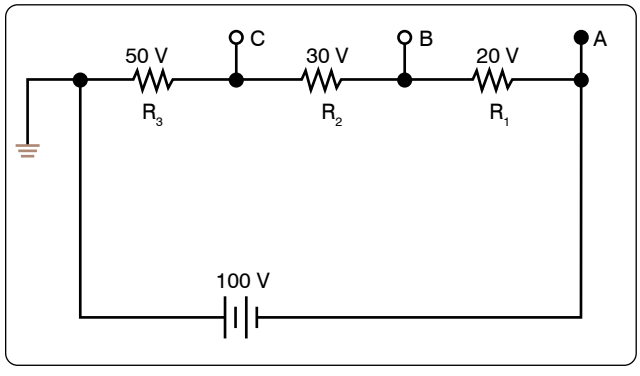


Figure 12-93. *Voltage divider with changed ground.*

Checking work
$E_T = 10\ V + 30\ V + 60\ V = 100\ V$

Parallel DC Circuits

A circuit in which two of more electrical resistances or loads are connected across the same voltage source is called a parallel circuit. The primary difference between the series circuit and the parallel circuit is that more than one path is provided for the current in the parallel circuit. Each of these parallel paths is called a branch. The minimum requirements for a parallel circuit are the following:

- A power source
- Conductors
- A resistance or load for each current path
- Two or more paths for current flow

Figure 12-96 depicts the most basic parallel circuit. Current flowing out of the source divides at point A in the diagram and goes through R_1 and R_2. As more branches are added to the circuit, more paths for the source current are provided.

Voltage Drops

The first point to understand is that the voltage across any branch is equal to the voltage across all of the other branches.

Total Parallel Resistance

The parallel circuit consists of two or more resistors connected in such a way as to allow current flow to pass through all of the resistors at once. This eliminates the need for current to pass one resistor before passing through the next. When resistors are connected in parallel, the total resistance of the circuit decreases. The total resistance of a parallel combination is always less than the value of the smallest resistor in the circuit. In the series circuit, the current has to pass through the resistors one at a time. This gave a resistance to the current equal the sum of all the resistors. In the parallel circuit, the current has several resistors that it can pass through, actually reducing the total resistance of the circuit in relation to any one resistor value.

The amount of current passing through each resistor varies according to its individual resistance. The total current of the circuit is the sum of the current in all branches. It can be determined by inspection that the total current is greater than that of any given branch. Using Ohm's Law to calculate the total resistance based on the applied voltage and the total current, it can be determined that the total resistance is less than any branch.

An example of this is if there was a circuit with a 100 Ω resistor and a 5 Ω resistor; while the exact value must be calculated, it still can be said that the combined resistance between the two is less than the 5 Ω.

Resistors in Parallel

The formula for the total parallel resistance is as follows:

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \ldots \frac{1}{R_N}$$

If the reciprocal of both sides is taken, then the general formula for the total parallel resistance is:

$$R_T = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \ldots \frac{1}{R_N}}$$

Two Resistors in Parallel

Typically, it is more convenient to consider only two resistors at a time because this setup occurs in common practice. Any number of resistors in a circuit can be broken down into pairs. Therefore, the most common method is to use the formula

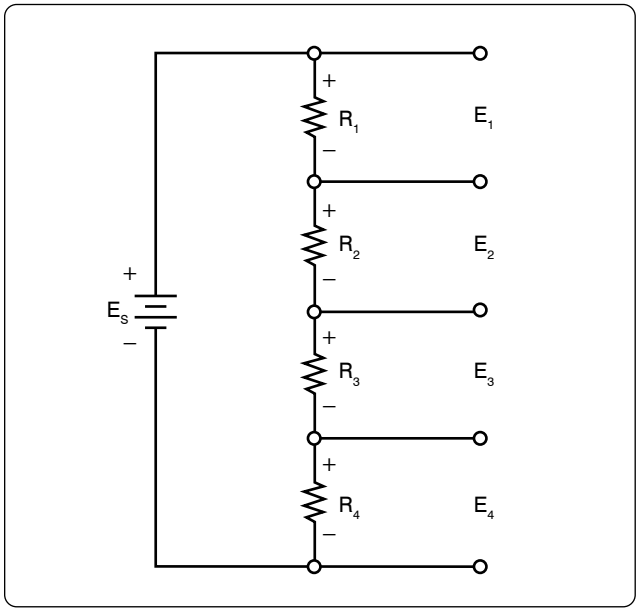


Figure 12-94. *Four resistor voltage divider.*

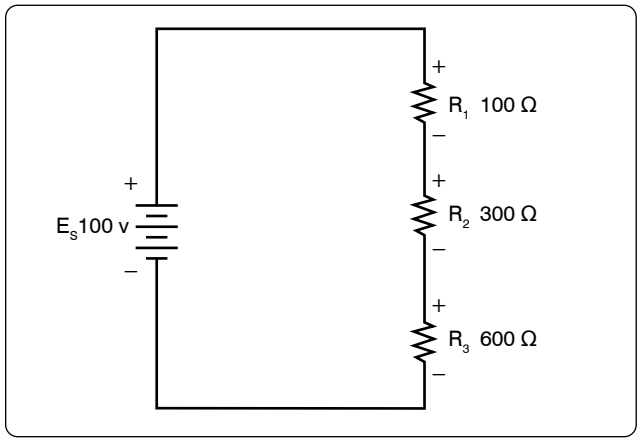


Figure 12-95. *Network of three resistors and one voltage source.*

for two resistors in parallel.

$$R_T = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2}}$$

Combining the terms in the denominator and rewriting:

$$R_T = \frac{R_1 R_2}{R_1 + R_2}$$

Put in words, this states that the total resistance for two resistors in parallel is equal to the product of both resistors divided by the sum of the two resistors. In the formula below, calculate the total resistance.

General formula $R_T = \frac{R_1 R_2}{R_1 + R_2}$

Known values $R_1 = 500\ \Omega$

$R_2 = 400\ \Omega$

$R_T = \frac{500\ \Omega\ 400\ \Omega}{500\ \Omega + 400\ \Omega}$

$R_T = \frac{200{,}000\ \Omega}{900\ \Omega}$

$R_T = 222.22\ \Omega$

Current Source

A current source is an energy source that provides a constant value of current to a load even when the load changes in resistive value. The general rule to remember is that the total current produced by current sources in parallel is equal to the algebraic sum of the individual sources.

Kirchhoff's Current Law

Kirchhoff's Current Law can be stated as: the sum of the currents into a junction or node is equal to the sum of the currents flowing out of that same junction or node. A junction can be defined as a point in the circuit where two or more circuit paths come together. In the case of the parallel circuit, it is the point in the circuit where the individual branches join.

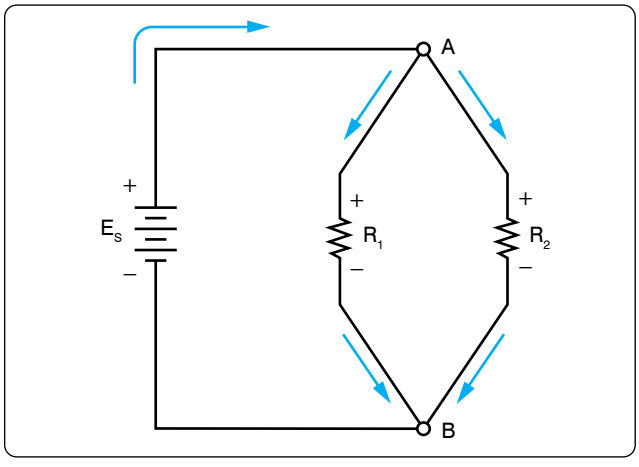


Figure 12-96. *Basic parallel circuit.*

General formula $I_T = I_1 + I_2 + I_3$

Refer to *Figure 12-97* for an example. Point A and point B represent two junctions or nodes in the circuit with three resistive branches in between. The voltage source provides a total current I_T into node A. At this point, the current must divide, flowing out of node A into each of the branches according to the resistive value of each branch. Kirchhoff's Current Law states that the current going in must equal that going out. Following the current through the three branches and back into node B, the total current I_T entering node B and leaving node B is the same as that which entered node A. The current then continues back to the voltage source.

Figure 12-98 shows that the individual branch currents are:

$I_1 = 5$ mA
$I_2 = 12$ mA

The total current flow into the node A equals the sum of the branch currents, which is: $I_T = I_1 + I_2$

Substitute $I_T = 5\text{ mA} + 12\text{ mA}$
$I_T = 17$ mA

The total current entering node B is also the same.

Figure 12-99 illustrates how to determine an unknown current in one branch. Note that the total current into a junction of the three branches is known. Two of the branch currents are known. By rearranging the general formula, the current in branch two can be determined.

General formula $I_T = I_1 + I_2 + I_3$
Substitute $75\text{ mA} = 30\text{ mA} + I_2 + 20\text{ mA}$
Solve I_2 $I_2 = 75\text{ mA} - 30\text{ mA} - 20\text{ mA}$
$I_2 = 25$ mA

Current Dividers

It can now be easily seen that the parallel circuit is a current divider. As shown in *Figure 12-96*, there is a current through each of the two resistors. Because the same voltage is applied across both resistors in parallel, the branch currents are inversely proportional to the ohmic values of the resistors. Branches with higher resistance have less current than those with lower resistance. For example, if the resistive value of R_2 is twice as high as that of R_1, the current in R_2 is half of that of R_1. All of this can be determined with Ohm's Law.

By Ohm's Law, the current through any one of the branches can be written as:

$$I_X = E_S/R_X$$

The voltage source appears across each of the parallel resistors and R_X represents any one the resistors. The source voltage is equal to the total current times the total parallel resistance.

$$E_S = I_T R_T$$

Substituting $I_T R_T$ for E_S

$$I_X = \frac{I_T R_T}{R_X}$$

Rearranging

$$I_X = \left(\frac{R_T}{R_X}\right) I_T$$

$$I_2 = \left(\frac{R_2}{R_T}\right) I_T$$

And

$$I_1 = \left(\frac{R_1}{R_T}\right) I_T$$

This formula is the general current divider formula. The current through any branch equals the total parallel resistance divided by the individual branch resistance, multiplied by the total current.

Series-Parallel DC Circuits

Most of the circuits that the technician encounters will not be a simple series or parallel circuit. Circuits are usually a combination of both, known as series-parallel circuits, which are groups consisting of resistors in parallel and in series. An example of this type of circuit can be seen in *Figure 12-100*. While the series-parallel circuit can initially appear to be complex, the same rules that have been used for the series and parallel circuits can be applied to these circuits.

The voltage source provides a current out to resistor R_1, then to the group of resistors R_2 and R_3 and then to the next resistor R_4 before returning to the voltage source. The first step in the simplification process is to isolate the group R_2 and R_3 and recognize that they are a parallel network that can be reduced to an equivalent resistor. Using the formula for parallel resistance,

$$R_{23} = \frac{R_2 R_3}{R_2 + R_3}$$

R_2 and R_3 can be reduced to R_{23}. *Figure 12-101* now shows an equivalent circuit with three series connected resistors. The total resistance of the circuit can now be simply determined by adding up the values of resistors R_1, R_{23}, and R_4.

Determining the Total Resistance

A more quantitative example for determining total resistance and the current in each branch in a combination circuit is shown in the following example. *[Figure 12-102]*

The first step is to determine the current at junction A, leading into the parallel branch. To determine the I_T, the total

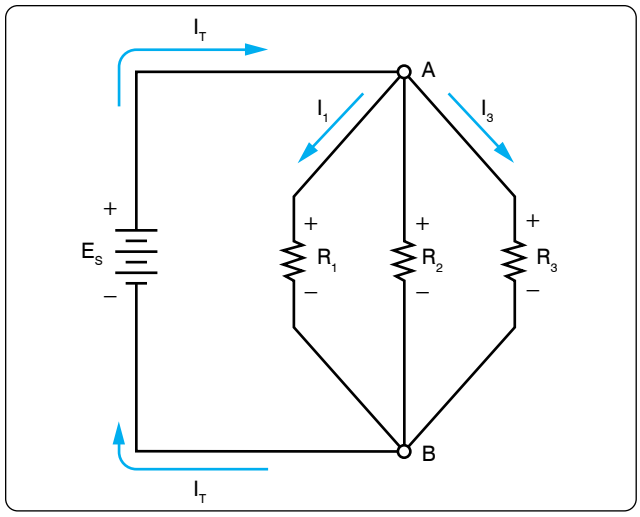


Figure 12-97. *Kirchhoff's Current Law.*

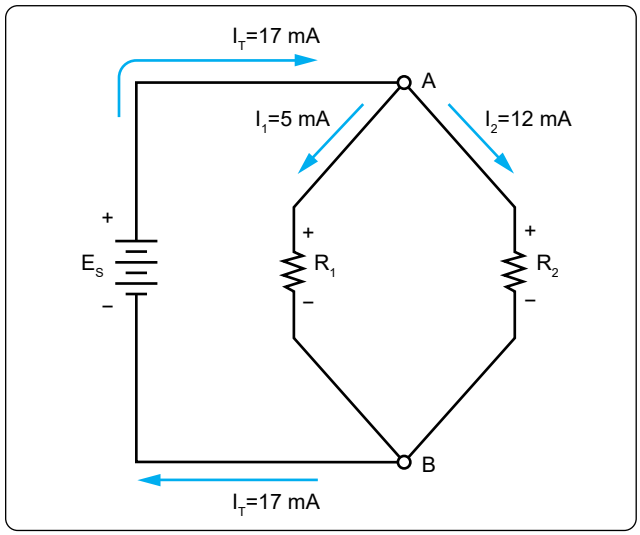


Figure 12-98. *Individual branch currents.*

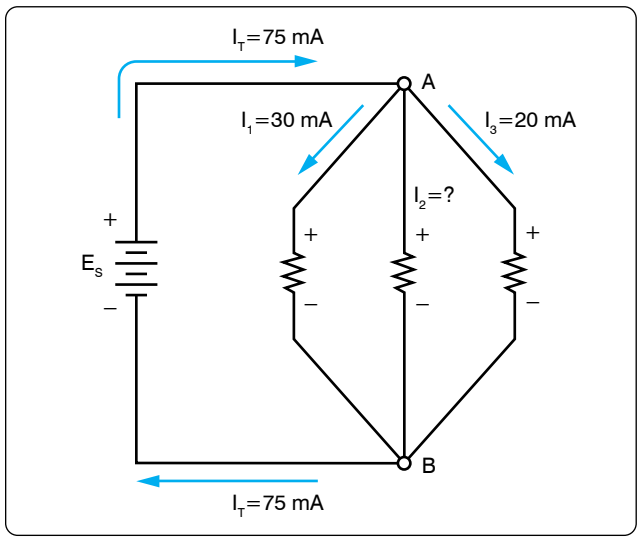


Figure 12-99. *Determining an unknown circuit in branch 2.*

resistance R_T of the entire circuit must be known. The total resistance of the circuit is given as:

$$R_T = R_1 + R_{23}$$

Where $$R_{23} = \left(\frac{R_2 R_3}{R_2 + R_3}\right)$$ Parallel network

Find R_{EQ} $$R_{23} = \frac{2k\ \Omega\ 3k\ \Omega}{2k\ \Omega + 3k\ \Omega}$$

Solve for R_{EQ} $$R_{23} = \frac{6{,}000k\ \Omega}{5k\ \Omega}$$
$$R_{23} = 1.2k\ \Omega$$

Solve for R_T $$R_T = 1k\ \Omega + 1.2k\ \Omega$$
$$R_T = 2.2k\ \Omega$$

With the total resistance R_T now determined, the total I_T can be determined. Using Ohm's Law:

$$I_T = \frac{E_S}{R_T}$$

Substitute values $$I_T = \frac{24\ V}{2.2k\ \Omega}$$

$$I_T = 10.9\ mA$$

The current through the parallel branches of R_2 and R_3 can be determined using the current divider rule discussed earlier in this handbook.

Recall Parallel Branch Resistance:

$$R_{23} = \frac{6{,}000\ \Omega}{5{,}000\ \Omega}$$
$$R_{23} = 1.2k\ \Omega$$

Substitute values for I_2:

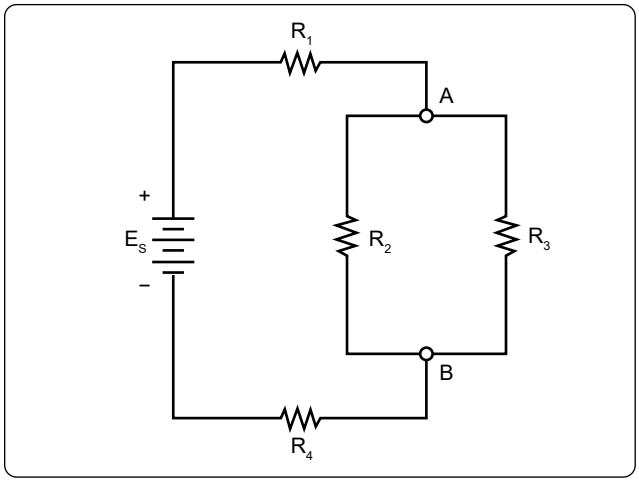


Figure 12-100. *Series-parallel circuits.*

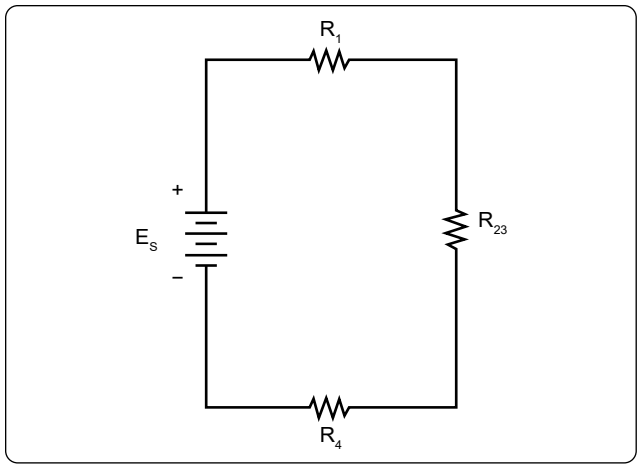


Figure 12-101. *Equivalent circuit with three series connected resistors.*

$$I_2 = \frac{1{,}200\ \Omega}{2{,}000\ \Omega} \times (10.9\ mA)$$
$$I_2 = 6.54\ mA$$

Now using Kirchhoff's Current Law, the current in the branch with R_3 can be determined.

$I_2 + I_3 = I_T$
Recall that $I_T = 10.9$ mA
$I_T - I_2 = I_3$
Also recall that $I_2 = 6.54$ mA
Subtract I_2 from I_T to get I_3
$I_3 = 4.36$ mA

Alternating Current (AC) & Voltage

Alternating current (AC) has largely replaced direct current (DC) in commercial power systems for a number of reasons. It can be transmitted over long distances more readily and more economically than DC, since AC voltages can be

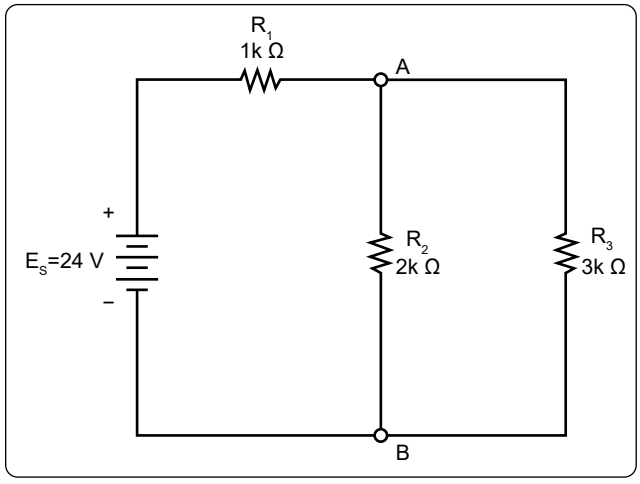


Figure 12-102. *Determining total resistance.*

increased or decreased by means of transformers.

Because more and more units are being operated electrically in airplanes, the power requirements are such that a number of advantages can be realized by using AC. Space and weight can be saved since AC devices, especially motors, are smaller and simpler than DC devices. In most AC motors, no brushes are required, and commutation trouble at high altitude is eliminated. Circuit breakers operate satisfactorily under load at high altitudes in an AC system, whereas arcing is so excessive on DC systems that circuit breakers must be replaced frequently. Finally, most airplanes using a 24-volt DC system have special equipment that requires a certain amount of 400-cycle AC current.

AC and DC Compared

Many of the principles, characteristics, and effects of AC are similar to those of DC. Similarly, there are a number of differences. DC flows constantly in only one direction with a constant polarity. It changes magnitude only when the circuit is opened or closed, as shown in the DC waveform in *Figure 12-103*. AC changes direction at regular intervals, increases in value at a definite rate from zero to a maximum positive strength, and decreases back to zero; then it flows in the opposite direction, similarly increasing to a maximum negative value, and again decreasing to zero. DC and AC waveforms are compared in *Figure 12-103*.

Since AC constantly changes direction and intensity, the following two effects (to be discussed later) take place in AC circuits that do not occur in DC circuits:

1. Inductive reactance
2. Capacitive reactance

Generator Principles

After the discovery that an electric current flowing through a conductor creates a magnetic field around the conductor, there was considerable scientific speculation about whether a magnetic field could create a current flow in a conductor. In 1831, Faraday discovered that this could be accomplished.

To show how an electric current can be created by a magnetic field, a demonstration similar to *Figure 12-104* can be used. Several turns of a conductor are wrapped around a cylindrical form, and the ends of the conductor are connected together to form a complete circuit, which includes a galvanometer. If a simple bar magnet is plunged into the cylinder, the galvanometer can be observed to deflect in one direction from its zero (center) position. *[Figure 12-104A]*

When the magnet is at rest inside the cylinder, the galvanometer shows a reading of zero, indicating that no current is flowing. *[Figure 12-104B]*

In *Figure 12-104C*, the galvanometer indicates a current flow in the opposite direction when the magnet is pulled from the cylinder.

The same results may be obtained by holding the magnet stationary and moving the cylinder over the magnet, indicating that a current flows when there is relative motion between the wire coil and the magnetic field. These results obey a law first stated by the German scientist, Heinrich Lenz. Lenz's Law states that the induced current caused by the relative motion of a conductor and a magnetic field always flows in such a direction that its magnetic field opposes the motion.

When a conductor is moved through a magnetic field, an emf is induced in the conductor. *[Figure 12-105]* The direction (polarity) of the induced emf is determined by the magnetic lines of force and the direction the conductor is moved through the magnetic field. The generator left-hand rule (not to be confused with the left-hand rules used with a coil) can be used to determine the direction of the induced emf. *[Figure 12-106]* The left-hand rule is summed up as follows:

The first finger of the left hand is pointed in the direction of the magnetic lines of force (North to South), the thumb is pointed in the direction of movement of the conductor through the magnetic field, and the second finger points in the direction of the induced emf.

When a loop conductor is rotated in a magnetic field, a voltage is induced in each side of the loop. *[Figure 12-107]* The two sides cut the magnetic field in opposite directions, and although the current flow is continuous, it moves in opposite directions with respect to the two sides of the loop. If sides A and B and the loop are rotated half a turn and the sides of the conductor have exchanged positions, the induced emf in each wire reverses its direction, since the wire formerly cutting the lines of force in an upward direction is now moving downward.

The value of an induced emf depends on three factors:

1. Number of wires moving through the magnetic field
2. Strength of the magnetic field
3. Speed of rotation

Generators of Alternating Current

Generators used to produce an alternating current are called AC generators or alternators.

The simple generator constitutes one method of generating an alternating voltage. *[Figure 12-108]* It consists of

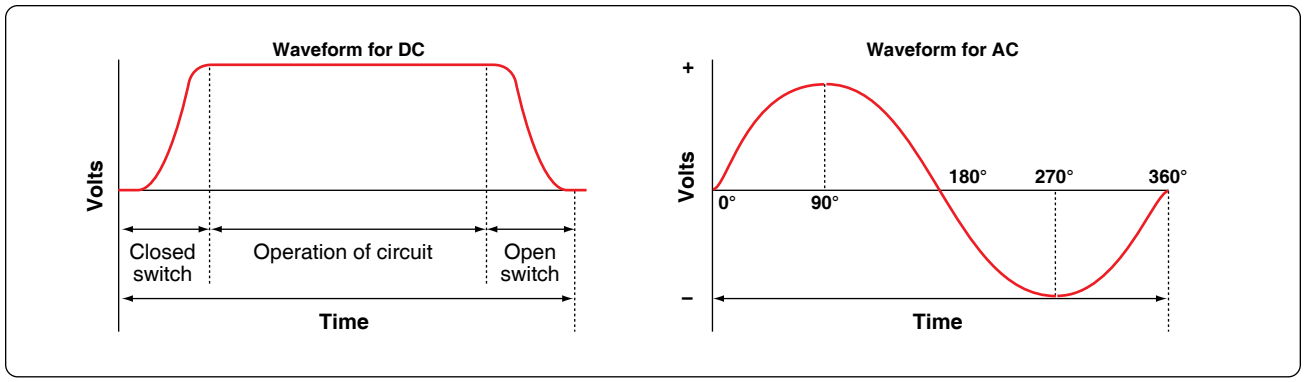


Figure 12-103. *DC and AC voltage curves.*

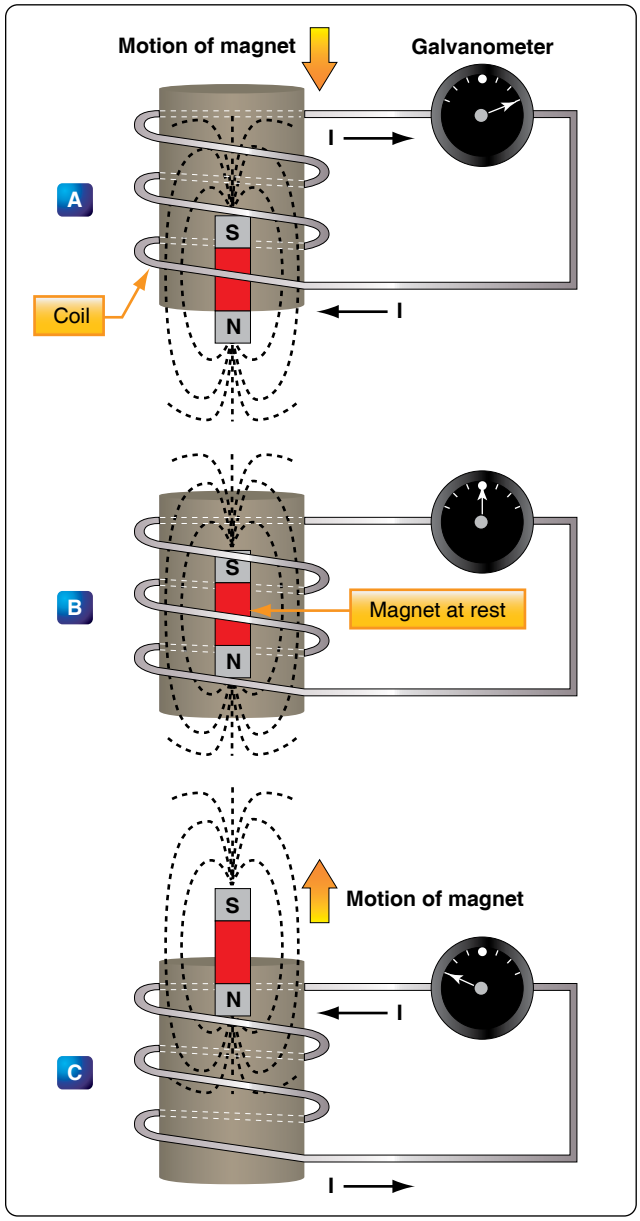


Figure 12-104. *Inducing a current flow.*

a rotating loop, marked A and B, placed between two magnetic poles, N and S. The ends of the loop are connected to two metal slip rings (collector rings), C_1 and C_2. Current is taken from the collector rings by brushes. If the loop is considered as separate wires A and B, and the left-hand rule for generators is applied, then it can be observed that as wire A moves up across the field, a voltage is induced which causes the current to flow inward. As wire B moves down across the field, a voltage is induced which causes the current to flow outward. When the wires are formed into a loop, the voltages induced in the two sides of the loop are combined. Therefore, for explanatory purposes, the action of either conductor, A or B, while rotating in the magnetic field is similar to the action of the loop.

Figure 12-109 illustrates the generation of AC with a simple loop conductor rotating in a magnetic field. As it is rotated in a counterclockwise direction, varying values of voltages are induced in it.

Position 1

The conductor A moves parallel to the lines of force. Since it cuts no lines of force, the induced voltage is zero. As the conductor advances from position 1 to position 2, the voltage induced gradually increases.

Position 2

The conductor is now moving perpendicular to the flux and cuts a maximum number of lines of force; therefore, a maximum voltage is induced. As the conductor moves beyond position 2, it cuts a decreasing amount of flux at each instant, and the induced voltage decreases.

Position 3

At this point, the conductor has made one-half of a revolution and again moves parallel to the lines of force, and no voltage is induced in the conductor. As the A conductor passes position 3, the direction of induced voltage now reverses since the A conductor is moving downward, cutting flux

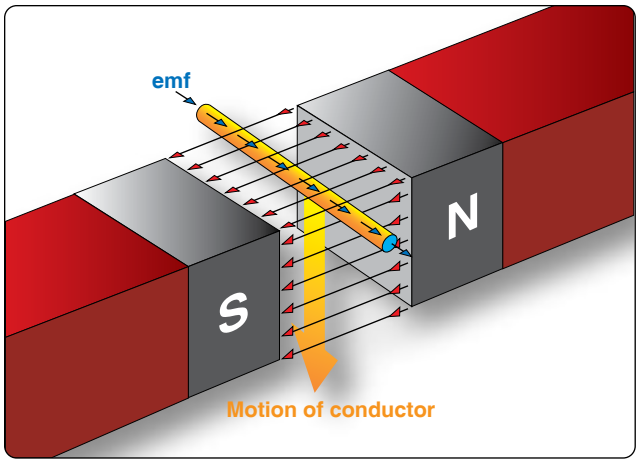


Figure 12-105. *Inducing an emf in a conductor.*

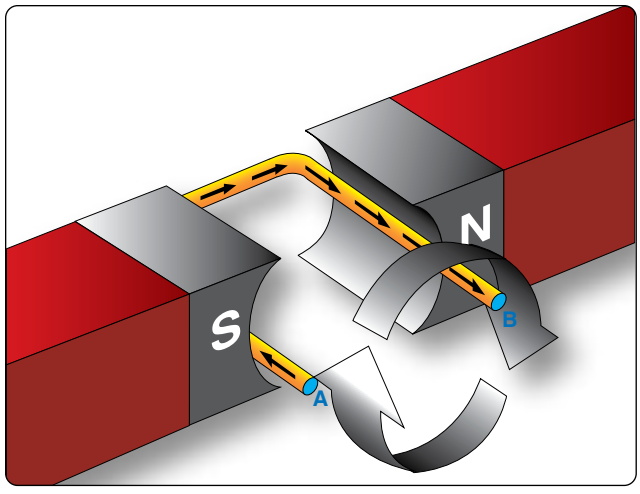


Figure 12-107. *Voltage induced in a loop.*

in the opposite direction. As the A conductor moves across the South pole, the induced voltage gradually increases in a negative direction, until it reaches position 4.

Position 4

Like position 2, the conductor is again moving perpendicular to the flux and generates a maximum negative voltage. From position 4 to 5, the induced voltage gradually decreases until the voltage is zero, and the conductor and wave are ready to start another cycle.

Position 5

The curve shown at position 5 is called a sine wave. It represents the polarity and the magnitude of the instantaneous values of the voltages generated. The horizontal base line is divided into degrees, or time, and the vertical distance above or below the base line represents the value of voltage at each particular point in the rotation of the loop.

Cycle and Frequency

Cycle Defined

A cycle is a repetition of a pattern. Whenever a voltage or current passes through a series of changes, returns to the starting point, and then again starts the same series of changes, the series is called a cycle. The cycle is represented by the symbol of a wavy line in a circle ⏦. In the cycle of voltage shown in *Figure 12-110*, the voltage increases from zero to a maximum positive value, decreases to zero; then increases to a maximum negative value, and again decreases to zero. At this point, it is ready to go through the same series of changes. There are two alternations in a complete cycle: the positive alternation and the negative. Each is half a cycle.

Frequency Defined

The frequency is the number of cycles of AC per second (1

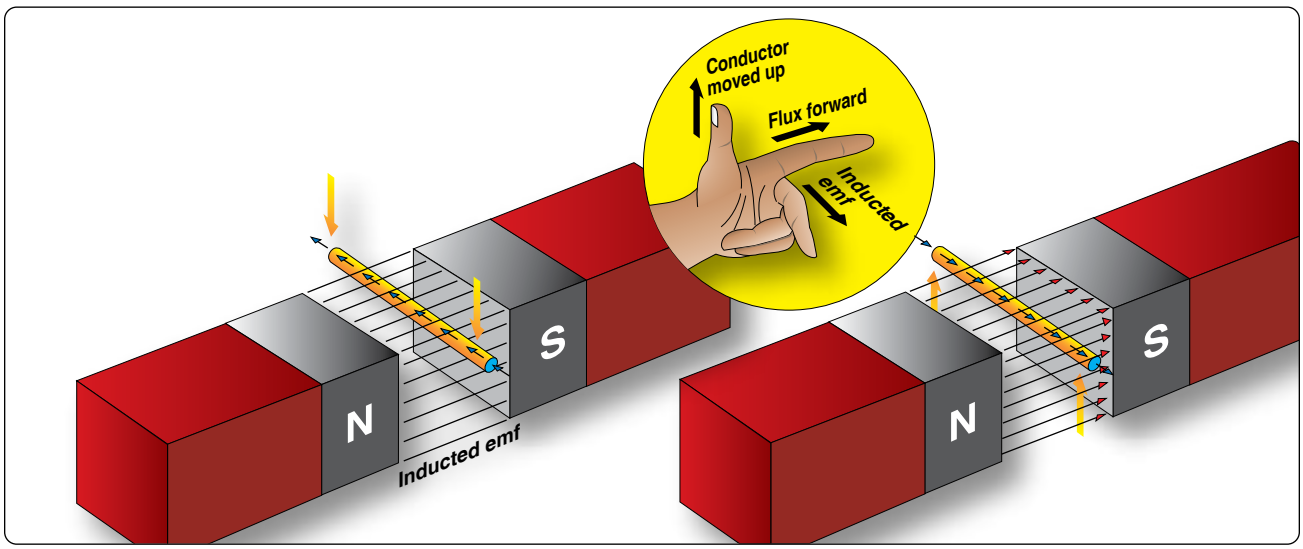


Figure 12-106. *An application of the generator left-hand rule.*

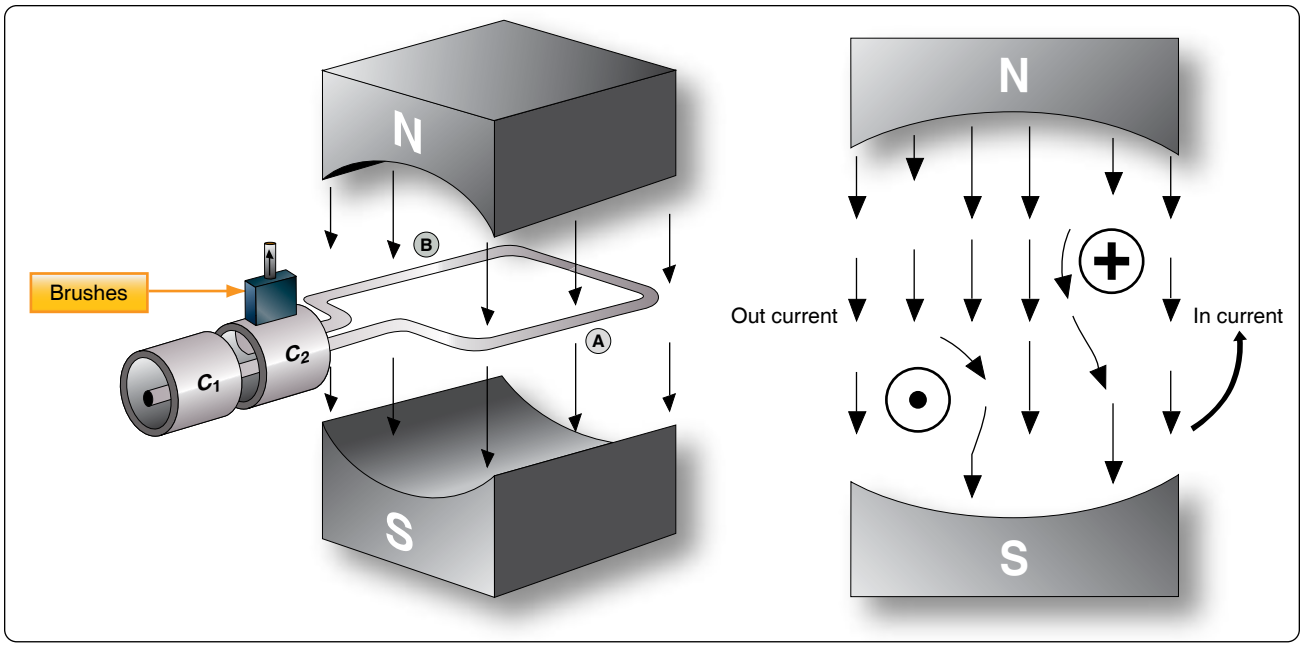


Figure 12-108. *Simple generator.*

second). The standard unit of frequency measurement is the hertz (Hz). *[Figure 12-111]* In a generator, the voltage and current pass through a complete cycle of values each time a coil or conductor passes under a North and South pole of the magnet. The number of cycles for each revolution of the coil or conductor is equal to the number of pairs of poles. The frequency, then, is equal to the number of cycles in one revolution multiplied by the number of revolutions per second (rps).

Expressed in equation form:

$$F = \frac{\text{Number of poles}}{2} \times \frac{\text{rpm}}{60}$$

where P/2 is the number of pairs of poles, and rpm/60 the number of revolutions per second. If in a 2-pole generator, the conductor is turning at 3,600 rpm, the revolutions per second are:

$$\text{rps} = \frac{3{,}600}{60} = 60 \text{ revolutions per second}$$

Since there are 2 poles, P/2 is 1, and the frequency is 60 cycles per second (cps). In a 4-pole generator with an armature speed of 1,800 rpm, substitute in the equation:

$$F = \frac{P}{2} \times \frac{\text{rpm}}{60} \text{ as follows}$$

$$F = \frac{4}{2} \times \frac{1{,}800}{60}$$

$$F = 2 \times 30$$

$$F = 60 \text{ cps}$$

Period Defined

The time required for a sine wave to complete one full cycle is called a period. *[Figure 12-110]* The period of a sine wave is inversely proportional to the frequency: the higher the frequency, the shorter the period. The mathematical relationship between frequency and period is given as:

$$\text{Period is } t = \frac{1}{f}$$

$$\text{Frequency is } f = \frac{1}{t}$$

Wavelength Defined

The distance that a waveform travels during a period is commonly referred to as a wavelength and is indicated by the Greek letter lambda (λ). The measurement of wavelength is taken from one point on the waveform to a corresponding point on the next waveform. *[Figure 12-110]*

Phase Relationships

In addition to frequency and cycle characteristics, alternating voltage and current also have a relationship called "phase." In a circuit that is fed (supplied) by one alternator, there must be a certain phase relationship between voltage and current if the circuit is to function efficiently. In a system fed by two or more alternators, not only must there be a certain phase relationship between voltage and current of one alternator, but there must be a phase relationship between the individual voltages and the individual currents. Also, two separate circuits can be compared by comparing the phase

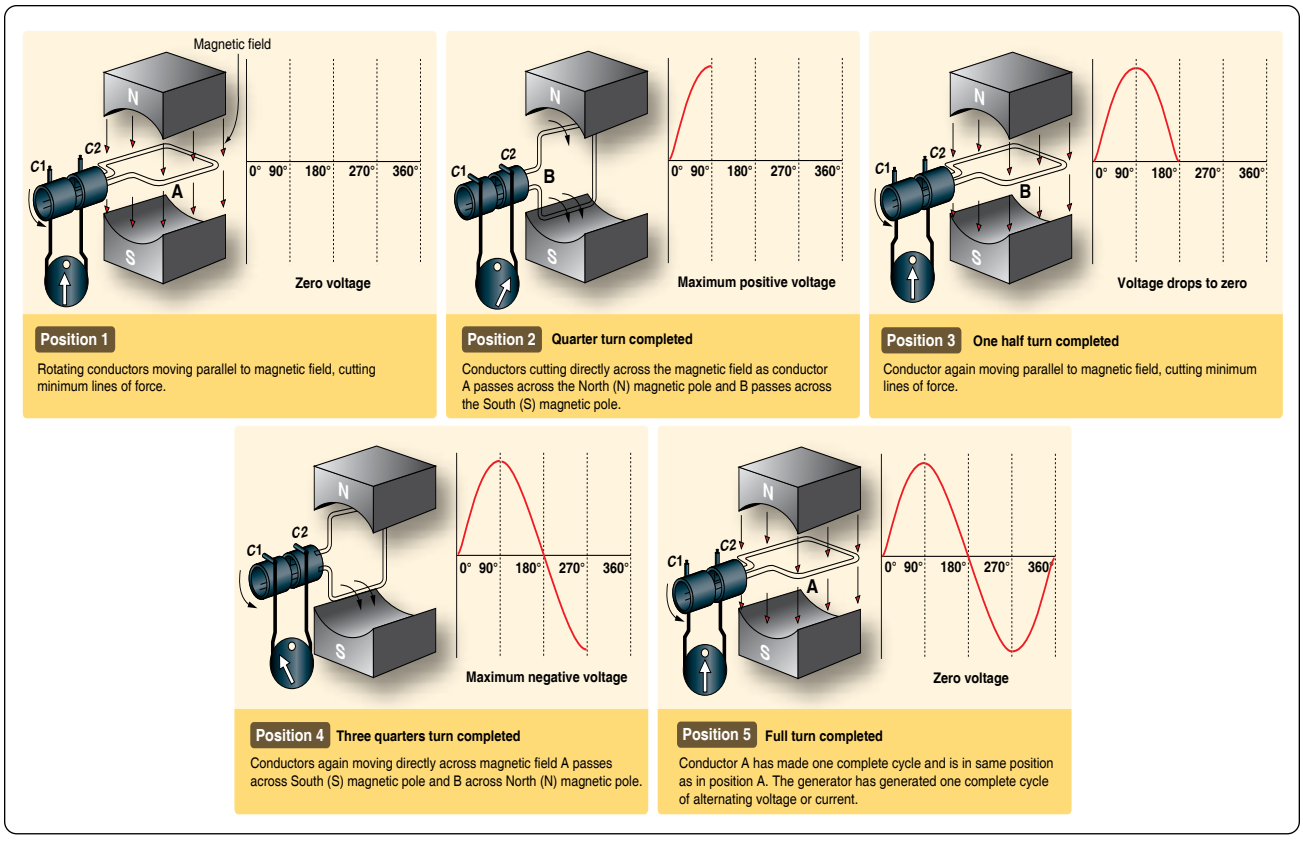


Figure 12-109. *Generation of a sine wave.*

characteristics of one to the phase characteristics of the other.

In Phase Condition

Figure 12-112A shows a voltage signal and a current signal superimposed on the same time axis. Notice that when the voltage increases in the positive alternation that the current also increases. When the voltage reaches its peak value, so does the current. Both waveforms then reverse and decrease back to a zero magnitude, then proceed in the same manner in the negative direction as they did in the positive direction. When two waves, such as these in *Figure 12-112A*, are exactly in step with each other, they are said to be in phase. To be in phase, the two waveforms must go through their maximum and minimum points at the same time and in the same direction.

Out of Phase Condition

When two waveforms go through their maximum and minimum points at different times, a phase difference exists between the two. In this case, the two wave-forms are said to be out of phase with each other. The terms lead and lag are often used to describe the phase difference between waveforms. The waveform that reaches its maximum or minimum value first is said to lead the other waveform. *Figure 12-112B* shows this relationship. Voltage source one starts to rise at the 0° position and voltage source two starts to rise at the 90° position. Because voltage source one begins its rise earlier in time (90°) in relation to the second voltage source, it is said to be leading the second source. On the other hand, the second source is said to be lagging the first source. When a waveform is said to be leading or lagging, the difference in degrees is usually stated. If the two waveforms differ by 360°, they are said to be in phase with each other. If there is a 180° difference between the two signals, then they are still out of phase even though they are both reaching their minimum and maximum values at the same time. *[Figure 12-112C]*

A practical note of caution: When encountering an aircraft that has two or more AC busses in use, it is possible that they may be split and not synchronized to be in phase with each other. When two signals that are not locked in phase are mixed, much damage can occur to aircraft systems or avionics.

Values of Alternating Current

There are three values of AC: instantaneous, peak, and effective root mean square (RMS).

Instantaneous Value

An instantaneous value of voltage or current is the induced voltage or current flowing at any instant during a cycle. The sine wave represents a series of these values. The instantaneous

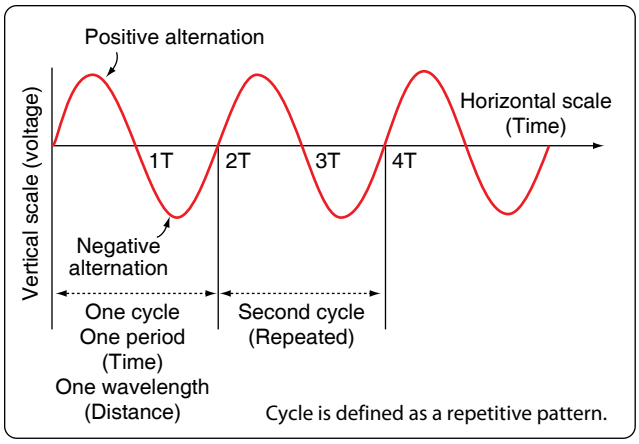


Figure 12-110. *Cycle of voltage.*

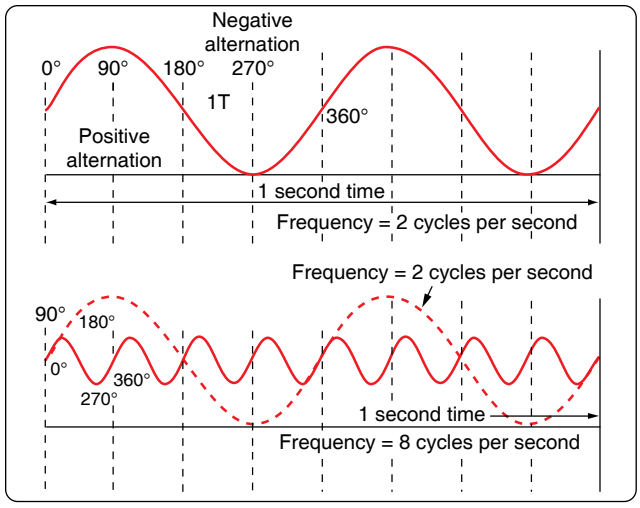


Figure 12-111. *Frequency in cycles per second.*

value of the voltage varies from zero at 0° to maximum at 90°, back to zero at 180°, to maximum in the opposite direction at 270°, and to zero again at 360°. Any point on the sine wave is considered the instantaneous value of voltage.

Peak Value

The peak value is the largest instantaneous value. The largest single positive value occurs when the sine wave of voltage is at 90°, and the largest single negative value occurs when it is at 270°. Maximum value is 1.41 times the effective value. These are called peak values.

Effective Value

The effective value is also known as the RMS value or root mean square, which refers to the mathematical process by which the value is derived. Most AC voltmeters display the effective or RMS value when used. The effective value is less than the maximum value, being equal to .707 times the maximum value.

The effective value of a sine wave is actually a measure of the heating effect of the sine wave. *Figure 12-113* illustrates what happens when a resistor is connected across an AC voltage source. In *Figure 12-113A*, a certain amount of heat is generated by the power in the resistor. *Figure 12-113B* shows the same resistor now inserted into a DC voltage source. The value of the DC voltage source can now be adjusted so that the resistor dissipates the same amount of heat as it did when it was in the AC circuit. The RMS or effective value of a sine wave is equal to the DC voltage that produces the same amount of heat as the sinusoidal voltage.

The peak value of a sine wave can be converted to the corresponding RMS value using the following relationship.

$$Vrms = (\sqrt{0.5}) \times Vp$$
$$Vrms = 0.707 \times Vp$$

This can be applied to either voltage or current.

Algebraically rearranging the formula and solving for Vp can also determine the peak voltage. The resulting formula is:

$$Vp = 1.414 \times Vrms$$

Thus, the 110 volt value given for AC supplied to homes is only 0.707 of the maximum voltage of this supply. The maximum voltage is approximately 155 volts (110 × 1.41 = 155 volts maximum).

In the study of AC, any values given for current or voltage are assumed to be effective values unless otherwise specified. In practice, only the effective values of voltage and current are used. Similarly, AC voltmeters and ammeters measure the effective value.

Opposition to Current Flow of AC

There are three factors that can create an opposition to the flow of electrons (current) in an AC circuit. Resistance, similar to resistance of DC circuits, is measured in ohms and has a direct influence on AC regardless of frequency. Inductive reactance and capacitive reactance, on the other hand, oppose current flow only in AC circuits, not in DC circuits. Since AC constantly changes direction and intensity, inductors and capacitors may also create an opposition to current flow in AC circuits. It should also be noted that inductive reactance and capacitive reactance may create a phase shift between the voltage and current in an AC circuit. Whenever analyzing an AC circuit, it is very important to consider the resistance, inductive reactance, and the capacitive reactance. All three have an effect on the current of that circuit.

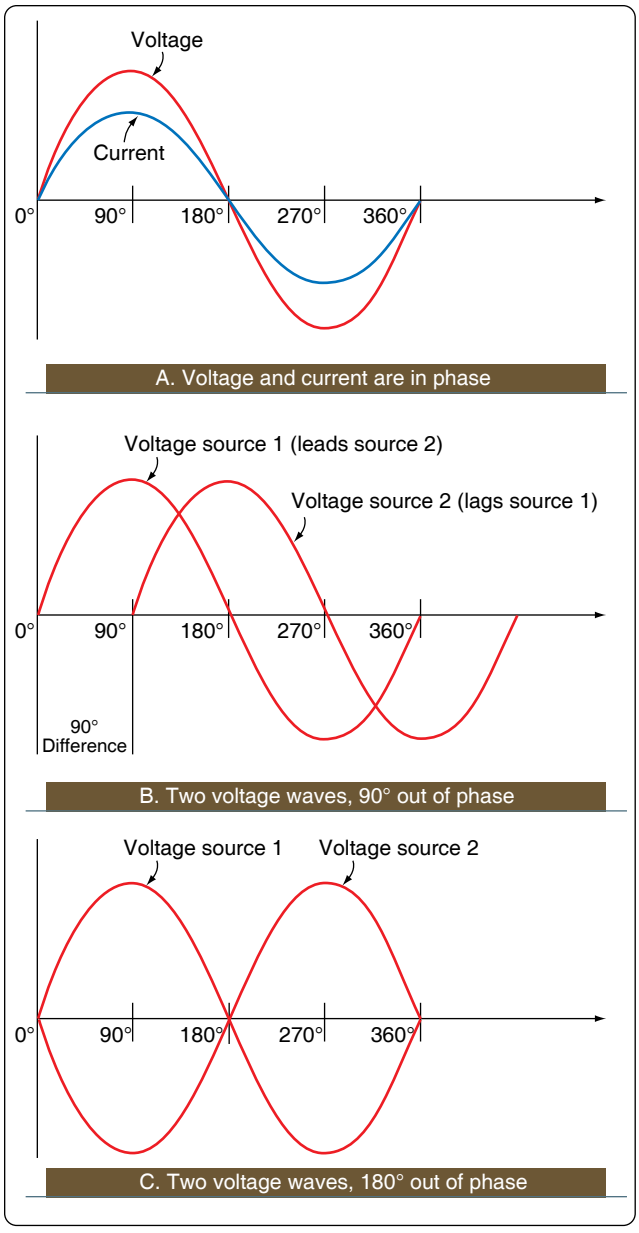


Figure 12-112. *In phase and out of phase conditions.*

Capacitance

Another important property in AC circuits, besides resistance and inductance, is capacitance. While inductance is represented in a circuit by a coil, capacitance is represented by a capacitor. In its most basic form, the capacitor is constructed of two parallel plates separated by a nonconductor called a dielectric. In an electrical circuit, a capacitor serves as a reservoir or storehouse for electricity.

Capacitors in Direct Current

When a capacitor is connected across a source of DC, such as a storage battery in the circuit shown in *Figure 12-114A*, and the switch is then closed, the plate marked B becomes positively charged, and the A plate negatively charged. Current flows in the external circuit during the time the electrons are moving from B to A. The current flow in the circuit is at a maximum the instant the switch is closed, but continually decreases thereafter until it reaches zero. The current becomes zero as soon as the difference in voltage of A and B becomes the same as the voltage of the battery. If the switch is opened as shown in *Figure 12-114B*, the plates remain charged. Once the capacitor is shorted, it discharges quickly as shown *Figure 12-114C*.

It should be clear that during the time the capacitor is being charged or discharged, there is current in the circuit, even though the circuit is broken by the gap between the capacitor plates. Current is present only during the time of charge and discharge, and this period of time is usually short.

The Resistor/Capacitor (RC) Time Constant

The time required for a capacitor to attain a full charge is proportional to the capacitance and the resistance of the circuit. The resistance of the circuit introduces the element of time into the charging and discharging of a capacitor.

When a capacitor charges or discharges through a resistance, a certain amount of time is required for a full charge or discharge. The voltage across the capacitor does not change instantaneously. The rate of charging or discharging is determined by the time constant of the circuit. The time constant of a series resistor/capacitor (RC) circuit is a time interval that equals the product of the resistance in ohms and the capacitance in farad and is symbolized by the Greek letter tau (τ).

$$\tau = RC$$

The time in the formula is the time required to charge to 63 percent of the voltage of the source. The time required to bring the charge to about 99 percent of the source voltage is approximately 5 τ. *[Figure 12-115]*

The measure of a capacitor's ability to store charge is its capacitance. The symbol used for capacitance is the letter C.

As can be seen from *Figure 12-115,* there can be no continuous movement of DC through a capacitor. A good capacitor blocks DC and passes the effects of pulsing DC or AC.

Units of Capacitance

Electrical charge, which is symbolized by the letter Q, is measured in units of coulombs. The coulomb is given by the letter C, as with capacitance. Unfortunately, this can be confusing. One coulomb of charge is defined as a charge having 6.28×10^{18} electrons. The basic unit of capacitance is the farad and is given by the letter f. By definition, one

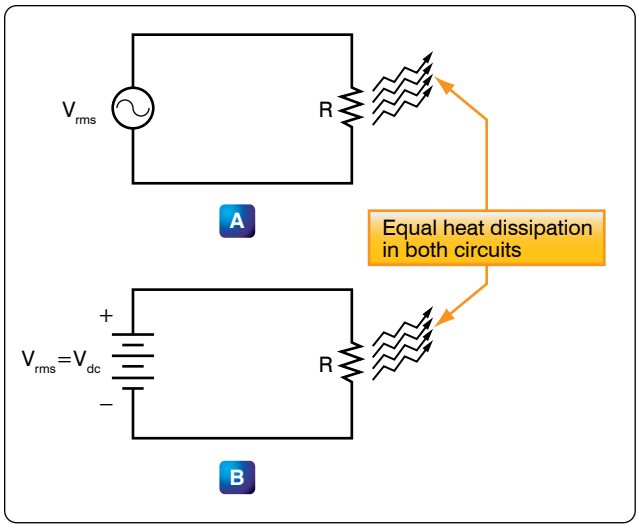


Figure 12-113. *Sine wave effective value.*

farad is one coulomb of charge stored with one volt across the plates of the capacitor. The general formula for capacitance in terms of charge and voltage is:

$$\text{Where} \quad C = \frac{Q}{E}$$

C = capacitance measured in farads
E = applied voltage measured in volts
Q = charge measured in coulombs

In practical terms, one farad is a large amount of capacitance. Typically, in electronics, much smaller units are used. The two more common smaller units are the microfarad (µF), which is 10^{-6} farad, and the picofarad (pF), which is 10^{-12} farad.

Voltage Rating of a Capacitor

Capacitors have their limits as to how much voltage can be applied across the plates. The aircraft technician must be aware of the voltage rating, which specifies the maximum DC voltage that can be applied without the risk of damage to the device. This voltage rating is typically called the breakdown voltage, the working voltage, or simply the voltage rating. If the voltage applied across the plates is too great, the dielectric breaks down and arcing occurs between the plates. The capacitor is then short circuited, and the possible flow of DC through it can cause damage to other parts of the equipment.

A capacitor that can be safely charged to 500 volts DC cannot be safely subjected to AC or pulsating DC whose effective values are 500 volts. An alternating voltage of 500 volts (RMS) has a peak voltage of 707 volts, and a capacitor to which it is applied should have a working voltage of at least 750 volts. The capacitor should be selected so that its working voltage is at least 50 percent greater than the highest voltage to be applied.

The voltage rating of the capacitor is a factor in determining the actual capacitance, because capacitance decreases as the thickness of the dielectric increases. A high-voltage capacitor that has a thick dielectric must have a larger plate area in order to have the same capacitance as a similar low voltage capacitor having a thin dielectric.

Factors Affecting Capacitance

1. The capacitance of parallel plates is directly proportional to their area. A larger plate area produces a larger capacitance and a smaller area produces less capacitance. If we double the area of the plates, there is room for twice as much charge. The charge that a capacitor can hold at a given potential difference is doubled, and since C = Q/E, the capacitance is doubled.
2. The capacitance of parallel plates is inversely proportional to their spacing.
3. The dielectric material affects the capacitance of parallel plates. The dielectric constant of a vacuum is defined as 1, and that of air is very close to 1. These values are used as a reference, and all other materials have values specified in relation to air (vacuum).

The strength of some commonly used dielectric materials is listed in *Figure 12-116*. The voltage rating also depends on frequency because the losses, and the resultant heating effect, increase as the frequency increases.

Types of Capacitors

Capacitors come in all shapes and sizes and are usually marked with their value in farads. They may also be divided into two groups: fixed and variable. The fixed capacitors, which have approximately constant capacitance, may then be further divided according to the type of dielectric used. Some varieties are: paper, oil, mica, electrolytic and ceramic capacitors. *Figure 12-117* shows the schematic symbols for a fixed and variable capacitor.

Fixed Capacitors

Mica Capacitors

The fixed mica capacitor is made of metal foil plates that are separated by sheets of mica, which form the dielectric. The whole assembly is covered in molded plastic, which keeps out moisture. Mica is an excellent dielectric and withstands higher voltages than paper without allowing arcing between the plates. Common values of mica capacitors range from approximately 50 microfarads to about 0.02 microfarads. *[Figure 12-118]*

Ceramic

The ceramic capacitor is constructed with materials, such as titanium acid barium for a dielectric. Internally these

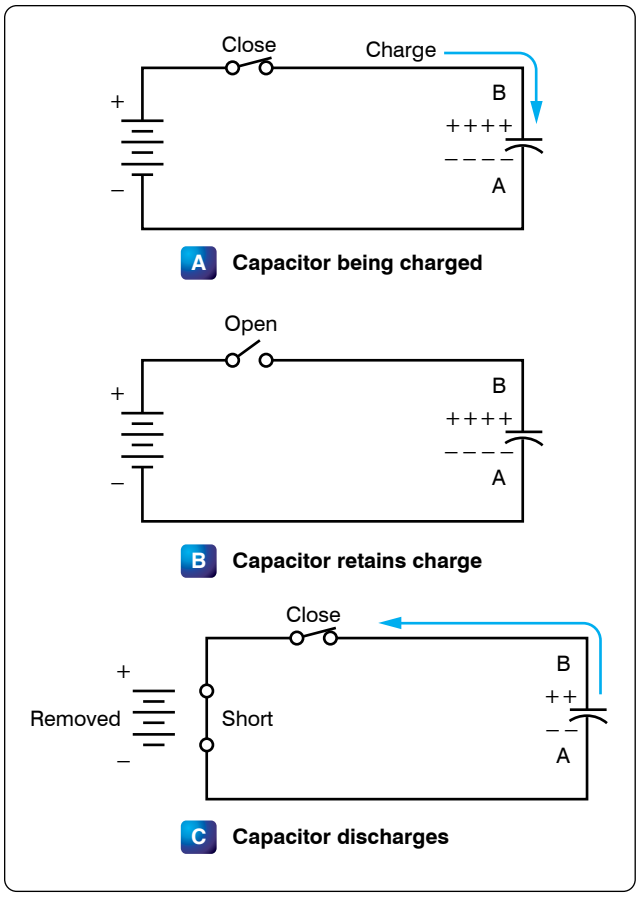


Figure 12-114. *Capacitors in direct current.*

capacitors are not constructed as a coil, so they are well suited for use in high-frequency applications. They are shaped like a disk, available in very small capacitance values, and very small sizes. This type is fairly small, inexpensive, and reliable. Both the ceramic and the electrolytic are the most widely available and used capacitor.

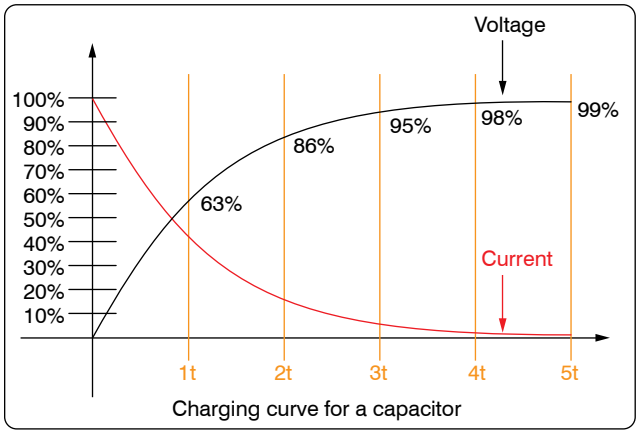


Figure 12-115. *Capacitance discharge curve.*

Electrolytic

Two kinds of electrolytic capacitors are in use: wet electrolytic and dry electrolytic. The wet electrolytic capacitor is designed of two metal plates separated by an electrolyte with an electrolyte dielectric, which is basically conductive salt in solvent. For capacitances greater than a few microfarads, the plate areas of paper or mica capacitors must become very large; thus, electrolytic capacitors are usually used instead. These units provide large capacitance in small physical sizes. Their values range from 1 to about 1,500 microfarads. Unlike the other types, electrolytic capacitors are generally polarized, with the positive lead marked with a "+" and the negative lead marked with a "–" and should only be subjected to direct voltage or pulsating direct voltage only.

The electrolyte in contact with the negative terminal, either in paste or liquid form, comprises the negative electrode. The dielectric is an exceedingly thin film of oxide deposited on the positive electrode of the capacitor. The positive electrode, which is an aluminum sheet, is folded to achieve maximum area. The capacitor is subjected to a forming process during manufacture in which current is passed through it. The flow of current results in the deposit of the thin coating of oxide on the aluminum plate.

The close spacing of the negative and positive electrodes gives rise to the comparatively high-capacitance value, but allows greater possibility of voltage breakdown and leakage of electrons from one electrode to the other.

The electrolyte of the dry electrolytic unit is a paste contained in a separator made of an absorbent material, such as gauze or paper. The separator not only holds the electrolyte in place but also prevents it from short circuiting the plates. Dry electrolytic capacitors are made in both cylindrical and rectangular block form and may be contained either within cardboard or metal covers. Since the electrolyte cannot spill, the dry capacitor may be mounted in any convenient position. *[Figure 12-119]*

Tantalum

Similar to the electrolytic, these capacitors are constructed with a material called tantalum, which is used for the electrodes. They are superior to electrolytic capacitors, having better temperature and frequency characteristics. When tantalum powder is baked in order to solidify it, a crack forms inside. This crack is used to store an electrical charge. Like electrolytic capacitors, the tantalum capacitors are also polarized and are indicated with the "+" and "–" symbols.

Polyester Film

In this capacitor, a thin polyester film is used as a dielectric. These components are inexpensive, temperature stable, and widely used. Tolerance is approximately 5–10 percent. It

can be quite large depending on capacity or rated voltage.

Oil Capacitors

In radio and radar transmitters, voltages high enough to cause arcing, or breakdown, of paper dielectrics are often used. Consequently, in these applications capacitors that use oil or oil impregnated paper for the dielectric material are preferred. Capacitors of this type are considerably more expensive than ordinary paper capacitors, and their use is generally restricted to radio and radar transmitting equipment. *[Figure 12-120]*

Variable Capacitors

Variable capacitors are mostly used in radio tuning circuits, and they are sometimes called "tuning capacitors." They have very small capacitance values, typically between 100 pF and 500 pF.

Trimmers

The trimmer is actually an adjustable or variable capacitor, which uses ceramic or plastic as a dielectric. Most of them are color coded to easily recognize their tunable size. The ceramic type has the value printed on them. Colors are: yellow (5 pF), blue (7 pF), white (10 pF), green (30 pF), and brown (60 pF).

Varactors

A voltage-variable capacitor or varactor is also known as a variable capacitance diode or a varicap. This device utilizes the variation of the barrier width in a reversed-biased diode. Because the barrier width of a diode acts as a non-conductor, a diode forms a capacitor when reversed biased. Essentially, the N-type material becomes one plate and the junctions are the dielectric. If the reversed-bias voltage is increased, then the barrier width widens, effectively separating the two capacitor plates and reducing the capacitance.

Capacitors in Series

When capacitors are placed in series, the effective plate separation is increased and the total capacitance is less than that of the smallest capacitor. Additionally, the series combination is capable of withstanding a higher total potential difference than any of the individual capacitors. *Figure 12-121* is a simple series circuit. The bottom plate of C_1 and the top plate of C_2 is charged by electrostatic induction. The capacitors charge as current is established through the circuit. Since this is a series circuit, the current must be the same at all points. Since the current is the rate of flow of charge, the amount of charge (Q) stored by each capacitor is equal to the total charge.

$$Q_T = Q_1 + Q_2 + Q_3$$

According to Kirchhoff's Voltage Law, the sum of the voltages across the charged capacitors must equal the total voltage, E_T. This is expressed as:

$$E_T = E_1 + E_2 + E_3$$

Equation E = Q/C can now be substituted into the voltage equation where we now get:

$$\frac{Q_T}{C_T} = \frac{Q_1}{C_1} + \frac{Q_2}{C_2} + \frac{Q_3}{C_3}$$

Since the charge on all capacitors is equal, the Q terms can be factored out, leaving us with the equation:

$$\frac{1}{C_T} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$$

Dielectric	K	Dielectric Strength (volts per .001 inch)
Air	1.0	80
Paper		
(1) Paraffined	2.2	1,200
(2) Beeswaxed	3.1	1,800
Glass	4.2	200
Castor Oil	4.7	380
Bakelite	6.0	500
Mica	6.0	2,000
Fiber	6.5	50

Figure 12-116. *Strength of some dielectric materials.*

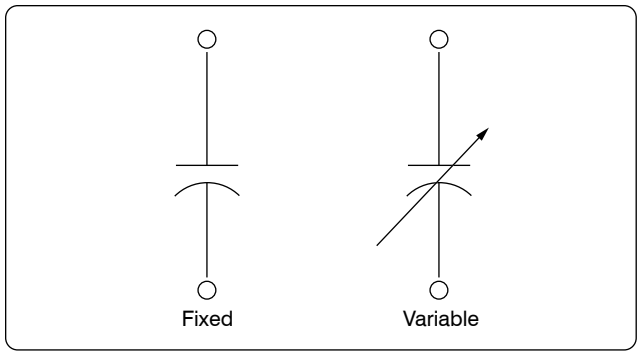


Figure 12-117. *Schematic symbols for a fixed and variable capacitor.*

Figure 12-118. *Fixed capacitors.*

Consider the following example:

If $C_1 = 10\ \mu F$, $C_2 = 5\ \mu F$ and $C_3 = 8\ \mu F$

Then $$\frac{1}{C_T} = \frac{1}{10\ \mu F} + \frac{1}{5\ \mu F} + \frac{1}{8\ \mu F}$$

$$C_T = \frac{1}{0.425\ \mu F} = 2.35\ \mu F$$

Capacitors in Parallel

When capacitors are connected in parallel, the effective plate area increases, and the total capacitance is the sum of the individual capacitances. *Figure 12-122* shows a simplified parallel circuit. The total charging current from the source divides at the junction of the parallel branches. There is a separate charging current through each branch so that a different charge can be stored by each capacitor. Using Kirchhoff's Current Law, the sum of all of the charging currents is then equal to the total current. The sum of the charges (Q) on the capacitors is equal to the total charge. The voltages (E) across all of the parallel branches are equal. With all of this in mind, a general equation for capacitors in parallel can be determined as:

$$Q_T = Q_1 + Q_2 + Q_3$$

Because $Q = CE$: $C_T E_T = C_1 E_1 + C_2 E_2 + C_3 E_3$

Voltages can be factored out because:

$$E_T = E_1 + E_2 + E_3$$

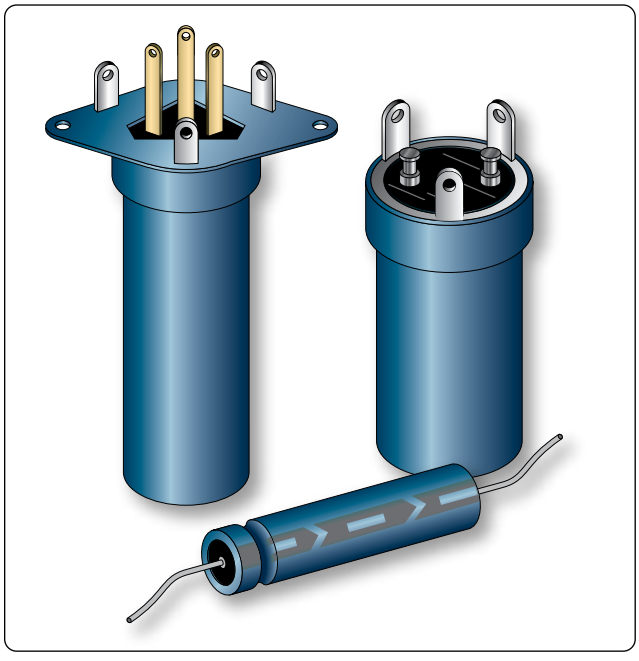


Figure 12-119. *Electrolytic capacitors.*

Leaving us with the equation for capacitors in parallel:

$$C_T = C_1 + C_2 + C_3$$

Consider the following example:

If $C_1 = 330\ \mu F$, $C_2 = 220\ \mu F$

Then $C_T = 330\ \mu F + 220\ \mu F = 550\ \mu F$

Capacitors in Alternating Current

If a source of AC is substituted for the battery, the capacitor acts quite differently than it does with DC. When AC is applied in the circuit, the charge on the plates constantly changes. *[Figure 12-123]* This means that electricity must flow first from Y clockwise around to X, then from X counterclockwise around to Y, then from Y clockwise around to X, and so on. Although no current flows through the insulator between the plates of the capacitor, it constantly flows in the remainder of the circuit between X and Y. In a circuit where there is only capacitance, current leads the applied voltage as contrasted with a circuit in which there is inductance, where the current lags the voltage.

Capacitive Reactance Xc

The effectiveness of a capacitor in allowing an AC flow to pass depends upon the capacitance of the circuit and the applied frequency. To what degree a capacitor allows an AC flow to pass depends largely upon the capacitive value of the capacitor given in farads (f). The greater the capacitance of the capacitor, the greater the number of electrons, measured in Coulombs, necessary to bring the capacitor to a fully charged state. Once the capacitor approaches or actually reaches a fully charged condition, the polarity of the capacitor opposes the polarity of the applied voltage, essentially acting then as an open circuit. To further illustrate this characteristic and how it manifests itself in an AC circuit, consider the following. If a capacitor has a large capacitive value, meaning that it requires a relatively large number of electrons to bring it to a fully charged state,

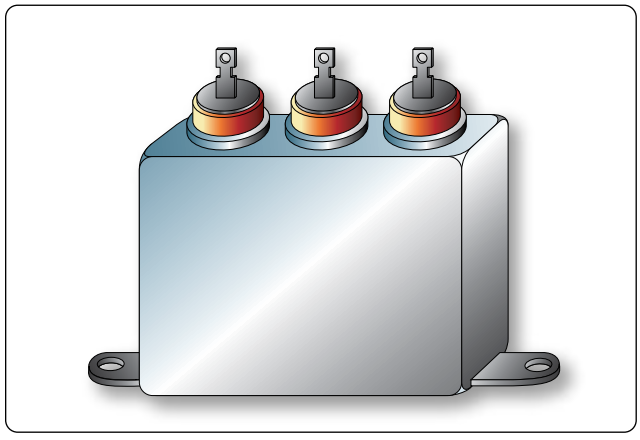


Figure 12-120. *Oil capacitor.*

then a rather high-frequency current can alternate through the capacitor without the capacitor ever reaching a full charge. In this case, if the frequency is high enough and the capacitance large enough that there is never enough time for the capacitor to ever reach a full charge, it is possible that the capacitor may offer very little or no resistance to the current. However, the smaller the capacitance, the fewer electrons are required to bring it up to a full charge and it is more likely that the capacitor will build up enough of an opposing charge that it can present a great deal of resistance to the current if not to the point of behaving like an open circuit. In between these two extreme conditions lies a continuum of possibilities of current opposition depending on the combination of applied frequency and the selected capacitance. Current in an AC circuit can be controlled by changing the circuit capacitance in a similar manner that resistance can control the current. The actual AC reactance Xc, which just like resistance, is measured in ohms (Ω). Capacitive reactance Xc is determined by the following:

$$Xc = \frac{1}{2\pi fC}$$

Where Xc = capacitive reactance
f = frequency in cps
C = capacity in farads
$2\pi = 6.28$

Sample Problem:

A series circuit is assumed in which the impressed voltage is 110 volts at 60 cps, and the capacitance of a condenser is 80 Mf. Find the capacitive reactance and the current flow.

Solution:

To find capacitive reactance, the equation $Xc = 1/(2\pi fC)$ is used. First, the capacitance, 80 Mf, is changed to farads by dividing 80 by 1,000,000, since 1 million microfarads is equal to 1 farad. This quotient equals 0.000080 farad. This is substituted in the equation and:

$$Xc = \frac{1}{6.28 \times 60 \times 0.000080}$$

Xc = 33.2 ohms reactance

Once the reactance has been determined, Ohm's Law can then be used in the same manner as it is used in DC circuits to determine the current.

$$\text{Current} = \frac{\text{Voltage}}{\text{Capacitive reactance}}, \text{ or}$$

$$I = \frac{E}{Xc}$$

Find the current flow:

$$I = \frac{E}{Xc}$$

$$I = \frac{110}{33.2}$$

I = 3.31 amperes

Capacitive Reactances in Series and in Parallel

When capacitors are connected in series, the total reactance is equal to the sum of the individual reactances. Thus,

$$Xct = (Xc)_1 + (Xc)_2$$

The total reactance of capacitors connected in parallel is found in the same way total resistance is computed in a parallel circuit:

$$(Xc)t = \frac{1}{\frac{1}{(Xc)_1} + \frac{1}{(Xc)_2} + \frac{1}{(Xc)_3}}$$

Phase of Current and Voltage in Reactive Circuits

Unlike a purely resistive circuit, the capacitive and inductive reactance has a significant effect on the phase relationship between the applied AC voltage and the corresponding

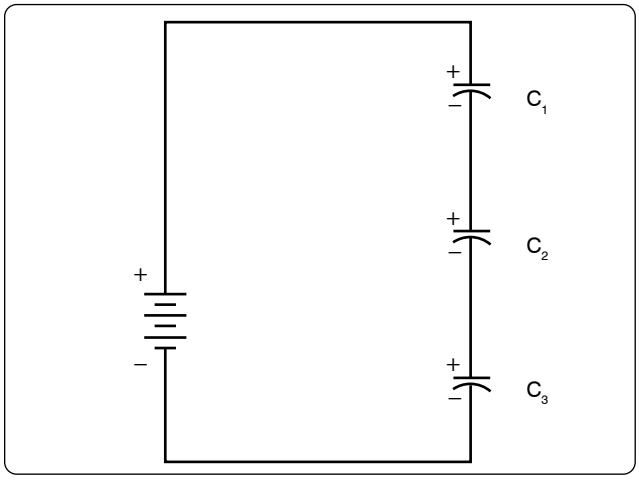


Figure 12-121. *Simple series circuit.*

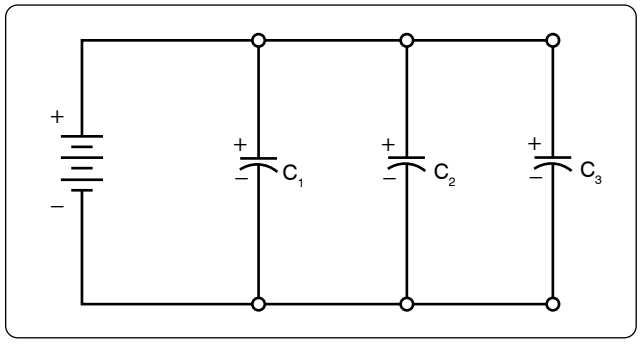


Figure 12-122. *Simplified parallel circuit.*

current in the circuit.

In review, when current and voltage pass through zero and reach maximum value at the same time, the current and voltage are said to be in phase. *[Figure 12-124A]* If the current and voltage pass through zero and reach the maximum values at different times, the current and voltage are said to be out of phase. In a circuit containing only inductance, the current reaches a maximum value later than the voltage, lagging the voltage by 90°, or one-fourth cycle. *[Figure 12-124B]*

In a circuit containing only capacitance, the current reaches its maximum value ahead of the voltage and the current leads the voltage by 90°, or one-fourth cycle. *[Figure 12-124C]* The amount the current lags or leads the voltage in a circuit depends on the relative amounts of resistance, inductance, and capacitance in the circuit.

Inductance

Characteristics of Inductance

Michael Faraday discovered that by moving a magnet through a coil of wire, a voltage was induced across the coil. If a complete circuit was provided, then a current was also induced. The amount of induced voltage is directly proportional to the rate of change of the magnetic field with respect to the coil. The simplest of experiments can prove that when a bar magnet is moved through a coil of wire, a voltage is induced and can be measured on a voltmeter. This is commonly known as Faraday's Law or the Law of Electromagnetic Induction, which states that the induced emf or electromagnetic force in a closed loop of wire is proportional to the rate of change of the magnetic flux through a coil of wire.

Conversely, current flowing through a coil of wire produces a magnetic field. When this wire is formed into a coil, it then becomes a basic inductor. The magnetic lines of force around each loop or turn in the coil effectively add to the lines of force around the adjoining loops. This forms a strong magnetic field within and around the coil. *Figure 12-125A* shows a coil of wire strengthening a magnetic field. The magnetic lines of force around adjacent loops are deflected into an outer path when the loops are brought close together. This happens because the magnetic lines of force between adjacent loops are in opposition with each other. The total magnetic field for the two loops is shown in *Figure 12-125B*. As more loops are added close together, the strength of the magnetic field increases. *Figure 12-125C* illustrates the combined effects of many loops of a coil. The result is a strong electromagnet.

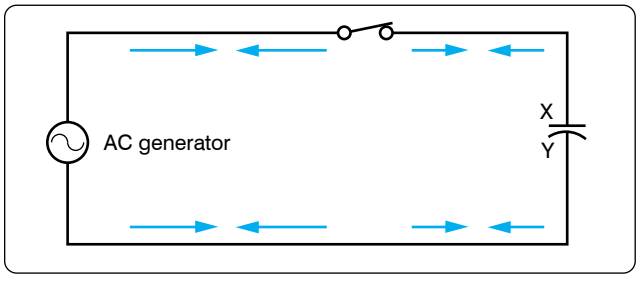


Figure 12-123. *Capacitor in an AC circuit.*

The primary aspect of the operation of a coil is its property to oppose any change in current through it. This property is called inductance. When current flows through any conductor, a magnetic field starts to expand from the center of the wire. As the lines of magnetic force grow outward through the conductor, they induce an emf in the conductor itself. The induced voltage is always in the direction opposite to the direction of the current flow. The effects of this countering emf are to oppose the immediate establishment of the maximum current. This effect is only a temporary condition. Once the current reaches a steady value in the conductor, the lines of magnetic force no longer expand and the countering emf is no longer present.

At the starting instant, the countering emf nearly equals the applied voltage, resulting in a small current flow. However, as the lines of force move outward, the number of lines cutting the conductor per second becomes progressively smaller, resulting in a diminished counter emf. Eventually, the counter emf drops to zero and the only voltage in the circuit is the applied voltage and the current is at its maximum value.

The RL Time Constant

Because the inductors basic action is to oppose a change in its current, it then follows that the current cannot change instantaneously in the inductor. A certain time is required for the current to make a change from one value to another. The rate at which the current changes is determined by a time constant represented by the Greek letter τ. The time constant for the RL circuit is:

$$\tau = \frac{L}{R}$$

Where τ = seconds
L = inductance (H)
R = resistance (Ω)

In a series RL circuit, the current increases to 63 percent of its full value in 1 time constant after the circuit is closed. This buildup is similar to the buildup of voltage in a capacitor when charging an RC circuit. Both follow an exponential curve and reach 99 percent value after the 5th time constant. *[Figure 12-126]*

Physical Parameters

Some of the physical factors that affect inductance are:

1. The number of turns: Doubling the number of turns in a coil produces a field twice as strong if the same current is used. As a general rule, the inductance varies as the square of the number of turns.
2. The cross-sectional area of the coil: The inductance of a coil increases directly as the cross-sectional area of the core increases. Doubling the radius of a coil increases the inductance by a factor of four.
3. The length of a coil: Doubling the length of a coil, while keeping the same number of turns, halves the value of inductance.
4. The core material around which the coil is formed: Coils are wound on either magnetic or nonmagnetic materials. Some nonmagnetic materials include air, copper, plastic, and glass. Magnetic materials include nickel, iron, steel, or cobalt, which have a permeability that provides a better path for the magnetic lines of force and permit a stronger magnetic field.

Self-Inductance

The characteristic of self-inductance was summarized by German physicist Heinrich Lenz in 1833, and gives the direction of the induced emf resulting from electromagnetic induction. This is commonly known as Lenz's Law, which states: The emf induced in an electric circuit always acts in such a direction that the current it drives around a closed circuit produces a magnetic field, which opposes the change in magnetic flux.

Self-inductance is the generation of a voltage in an electric circuit by a changing current in the same circuit. Even a straight piece of wire has some degree of inductance because current in a conductor produces a magnetic field. When the current in a conductor changes direction, there is a corresponding change in the polarity of the magnetic field around the conductor. Therefore, a changing current produces a changing magnetic field around the wire. To further intensify the magnetic field, the wire can be rolled into a coil, which is called an inductor. The changing magnetic field around the inductor induces a voltage across the coil. This induced emf is called self-inductance and tends to oppose any change in current within the circuit. This property is usually called inductance and symbolized with the letter L.

Types of Inductors

Inductors used in radios can range from a straight wire at UHF to large chokes and transformers used for filtering the ripple from the output of power supplies and in audio amplifiers. *Figure 12-127* shows the schematic symbols for common inductors. Values of inductors range from nano-henries to tens of henries.

Inductors are classified by the type of core and the method of winding them. The number of turns in the inductor winding and the core material determine the capacity of the inductor. Cores made of dielectric material like ceramics, wood, and paper provide small amounts of stored energy while cores made of ferrite substances have a much higher degree of stored energy. The core material is usually the most important aspect of the inductors construction. The conductors typically used in the construction of an inductor offer little resistance to the flow of current. However, with the introduction of a core, resistance is introduced in the circuit and the current now builds up in the windings until the resistance of the core is overcome. This buildup is stored as magnetic energy in the core. Depending on the core resistance, the buildup soon reaches a point of magnetic saturation, and it can be released when necessary. The most common core materials are: air, solid ferrite, powdered ferrite, steel, toroid, and ferrite toroid.

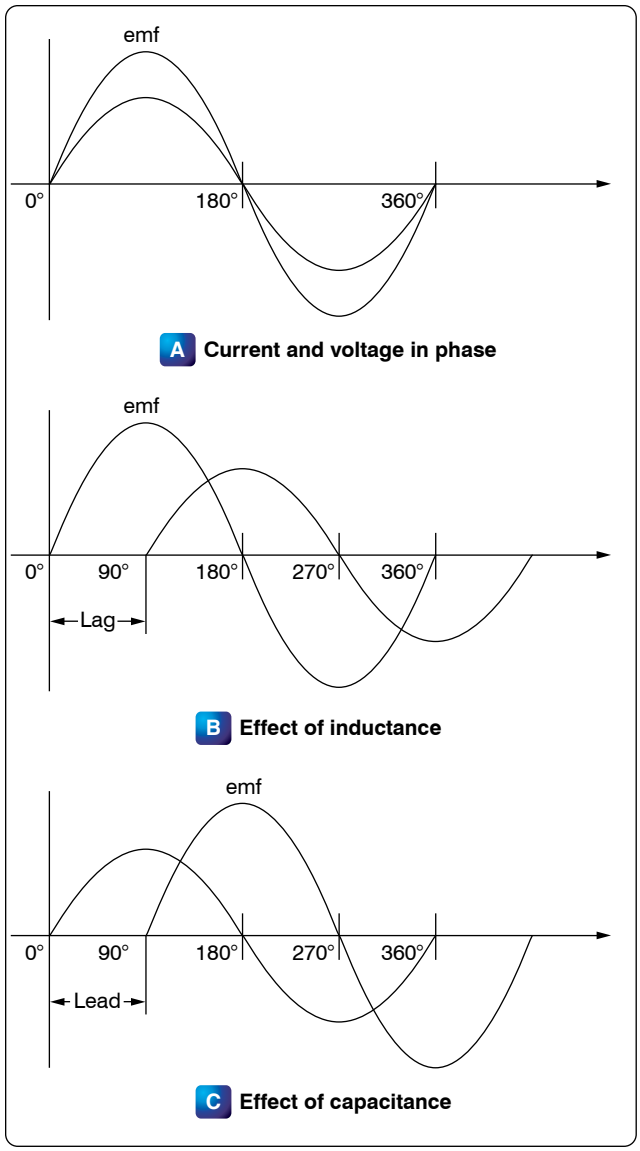


Figure 12-124. *Phase of current and voltage.*

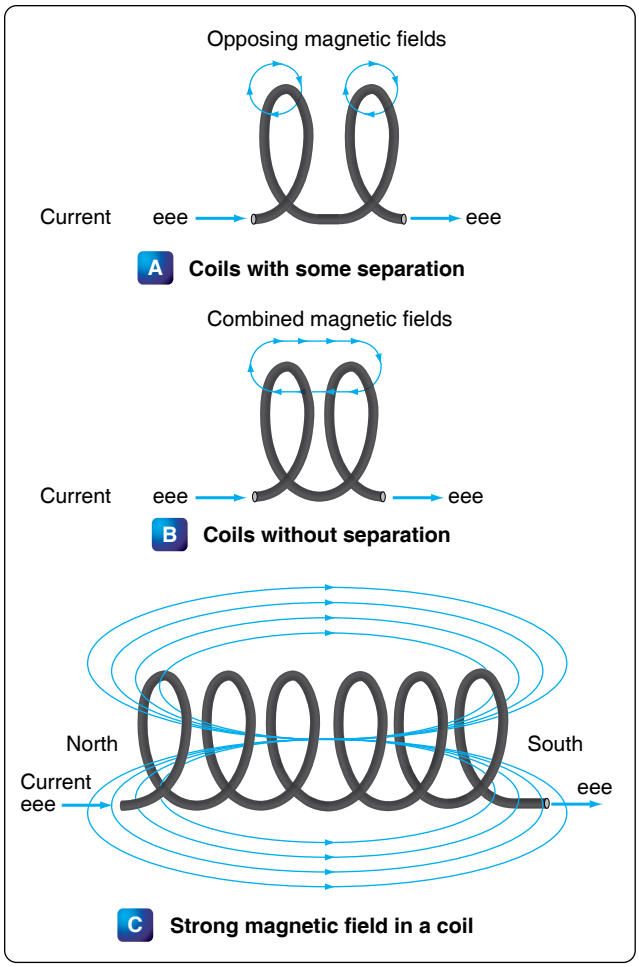


Figure 12-125. *Many loops of a coil.*

Units of Inductance

The henry is the basic unit of inductance and is symbolized with the letter H. An electric circuit has an inductance of one henry when current changing at the rate of one ampere per second induces a voltage of one volt into the circuit. In many practical applications, millihenries (mH) and microhenries (μH) are more common units. The typical symbol for an inductor is shown in *Figure 12-127*.

Inductors in Series

If we connect two inductors in series, the same current flows through both inductors and, therefore, both are subject to the same rate of change of current. *[Figure 12-128]* When inductors are connected in series, the total inductance L_T, is the sum of the individual inductors. The general equation for n number of inductors in series is:

$$L_T = L_1 + L_2 + L_3 + \ldots L_N$$

Inductors in Parallel

When two inductors are connected in parallel, each must have the same potential difference between the terminals. *[Figure 12-129]* When inductors are connected in parallel, the total inductance is less than the smallest inductance. The general equation for n number of inductors in parallel is:

$$L_T = \frac{1}{\frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{L_3} + \ldots \frac{1}{L_N}}$$

A simple example would be:

$$L_1 = 10 \text{ mH}, L_2 = 5 \text{ mH}, L_3 = 2 \text{ mH}$$

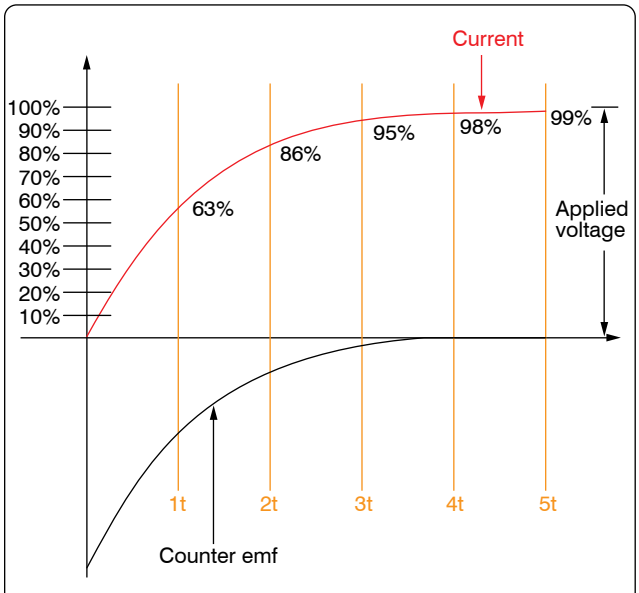
$$L_T = \frac{1}{\frac{1}{10\,\text{mH}} + \frac{1}{5\text{ mH}} + \frac{1}{2\text{ mH}}}$$

$$L_T = \frac{1}{0.8\text{ mH}}$$

$$L_T = 1.25 \text{ mH}$$

Inductive Reactance

Alternating current is in a constant state of change; the effects of the magnetic fields are a continuously inducted voltage opposition to the current in the circuit. This opposition is called inductive reactance, symbolized by X_L, and is measured in ohms just as resistance is measured. Inductance is the property of a circuit to oppose any change in current and is measured in henries. Inductive reactance is a measure of how much the countering emf in the circuit opposes current



Current, counter emf, and applied voltage in an inductive circuit.

Figure 12-126. *Inductor curve.*

variations.

The inductive reactance of a component is directly proportional to the inductance of the component and the applied frequency to the circuit. By increasing either the inductance or applied frequency, the inductive reactance likewise increases and presents more opposition to current in the circuit. This relationship is given as:

$$X_L = 2\pi fL$$

Where: X_L = inductive reactance in ohms
f = frequency in cycles per second
π = 3.1416
L = inductance

In *Figure 12-130*, an AC series circuit is shown in which the inductance is 0.146 henry and the voltage is 110 volts at a frequency of 60 cps. Inductive reactance is determined by the following method.

$$X_L = 2\pi \times f \times L$$
$$X_L = 6.28 \times 60 \times 0.146$$
$$X_L = 55 \text{ ohm}$$

In any circuit where there is only resistance, the expression for the relationship of voltage and current is given by Ohm's Law: I = E/R. Similarly, when there is inductance in an AC circuit, the relationship between voltage and current can be expressed as:

$$\text{Current} = \frac{\text{Voltage}}{\text{Reactance}} \text{ or } I = \frac{E}{X_L}$$

Where:
X_L = inductive reactance of the circuit in ohms

$$I = \frac{E}{X_L}$$
$$I = \frac{110}{55}$$
$$I = 2 \text{ amperes}$$

In AC series circuits, inductive reactances are added like resistances in series in a DC circuit. *[Figure 12-131]* Thus, the total reactance in the illustrated circuit equals the sum of the individual reactances.

The total reactance of inductors connected in parallel is found the same way as the total resistance in a parallel circuit. *[Figure 12-132]* Thus, the total reactance of inductances connected in parallel, as shown, is expressed as:

$$(X_L)T = \frac{1}{\frac{1}{(X_L)_1} + \frac{1}{(X_L)_2} + \frac{1}{(X_L)_3}}$$

AC Circuits

Ohm's Law for AC Circuits

The rules and equations for DC circuits apply to AC circuits only when the circuits contain resistance alone, as in the case of lamps and heating elements. In order to use effective values of voltage and current in AC circuits, the effect of inductance and capacitance with resistance must be considered.

The combined effects of resistance, inductive reactance, and capacitive reactance make up the total opposition to current flow in an AC circuit. This total opposition is called impedance and is represented by the letter Z. The unit for the measurement of impedance is the ohm.

Series AC Circuits

If an AC circuit consists of resistance only, the value of the impedance is the same as the resistance, and Ohm's Law for an AC circuit, I = E/Z, is exactly the same as for a DC circuit. In *Figure 12-133,* a series circuit containing a lamp with 11 ohms resistance connected across a source is illustrated. To find how much current flows if 110 volts DC is applied and how much current flows if 110 volts AC are applied, the following examples are solved:

$$I = \frac{E}{R} \qquad I = \frac{E}{Z} \text{ (where Z = R)}$$
$$I = \frac{110 \text{ V}}{11 \text{ W}} \qquad I = \frac{110 \text{ V}}{11 \text{ W}}$$
$$I = 10 \text{ amperes DC} \qquad I = 10 \text{ amperes AC}$$

When AC circuits contain resistance and either inductance or capacitance, the impedance, Z, is not the same as the

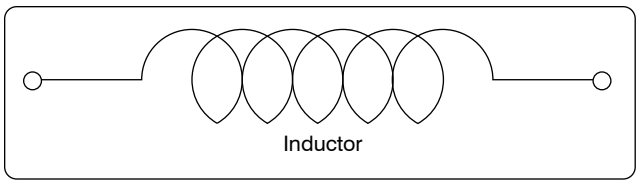


Figure 12-127. *Typical symbol for an inductor.*

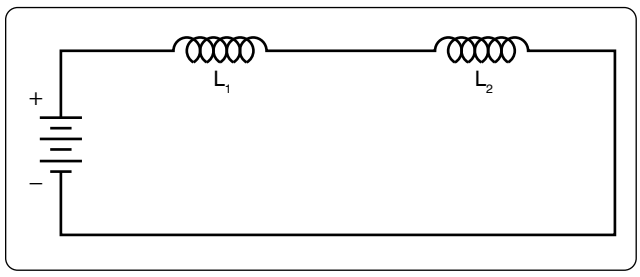


Figure 12-128. *Two inductors in series.*

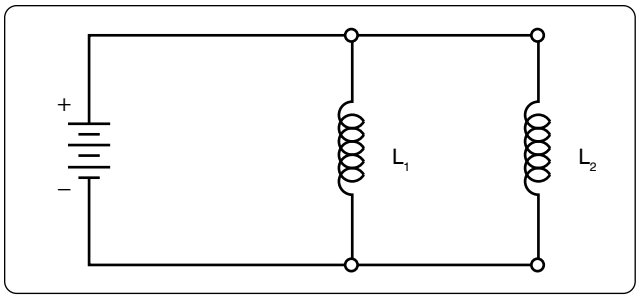


Figure 12-129. *Two inductors in parallel.*

resistance, R. The impedance of a circuit is the circuit's total opposition to the flow of current. In an AC circuit, this opposition consists of resistance and reactance, either inductive or capacitive or elements of both.

Resistance and reactance cannot be added directly, but they can be considered as two forces acting at right angles to each other. Thus, the relation between resistance, reactance, and impedance may be illustrated by a right triangle. *[Figure 12-134]*

Since these quantities may be related to the sides of a right triangle, the formula for finding the impedance, or total opposition to current flow in an AC circuit, can be found by using the law of right triangles. This theorem, called the Pythagorean theorem, applies to any right triangle. It states that the square of the hypotenuse is equal to the sum of the squares of the other two sides. Thus, the value of any side of a right triangle can be found if the other two sides are known. If an AC circuit contains resistance and inductance, as shown in *Figure 12-135*, the relation between the sides can be stated as:

$$Z^2 = R^2 + X_L{}^2$$

The square root of both sides of the equation gives

$$Z = \sqrt{R^2 + X_L{}^2}$$

This formula can be used to determine the impedance when the values of inductive reactance and resistance are known. It can be modified to solve for impedance in circuits containing capacitive reactance and resistance by substituting X_C in the formula in place of X_L. In circuits containing resistance with both inductive and capacitive reactance, the reactances can be combined, but because their effects in the circuit are exactly opposite, they are combined by subtraction:

$X = X_L - X_C$ or $X = X_C - X_L$ (the smaller number is always subtracted from the larger)

In *Figure 12-135*, a series circuit consisting of resistance and inductance connected in series is connected to a source of 110 volts at 60 cps. The resistive element is a lamp with 6 ohms resistance, and the inductive element is a coil with an inductance of 0.021 henry. What is the value of the impedance and the current through the lamp and the coil?

Solution:

First, the inductive reactance of the coil is computed:

$X_L = 2\pi \times f \times L$
$X_L = 6.28 \times 60 \times 0.021$
$X_L = 8$ ohms inductive reactance

Next, the total impedance is computed:

$Z = \sqrt{R^2 + X_L{}^2}$
$Z = \sqrt{6^2 + 8^2}$
$Z = \sqrt{36 + 64}$
$Z = \sqrt{100}$
$Z = 10$ ohms impedance

Then the current flow,

$$I = \frac{E}{Z}$$

$$I = \frac{110}{10}$$

$I = 11$ amperes current

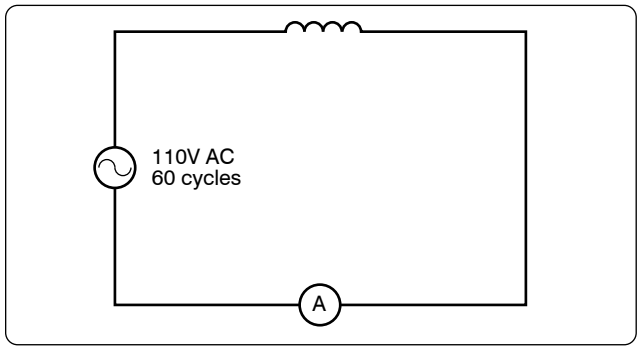


Figure 12-130. *AC circuit containing inductance.*

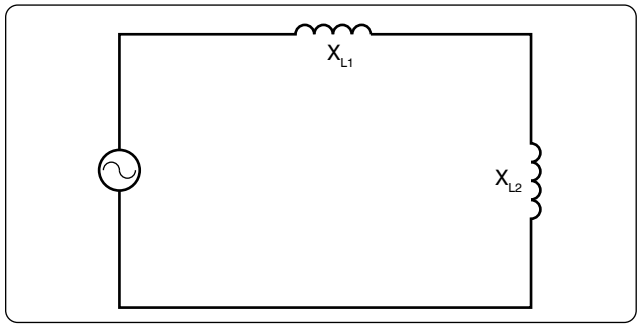


Figure 12-131. *Inductances in series.*

The voltage drop across the resistance (E^R) is:

$$E^R = I \times R$$
$$E^R = 11 \times 6 = 66 \text{ volts}$$

The voltage drop across the inductance (EX_L) is:

$$EX_L = I \times X_L$$
$$EX_L = 11 \times 8 = 88 \text{ volts}$$

The sum of the two voltages is greater than the impressed voltage. This results from the fact that the two voltages are out of phase and, as such, represent the maximum voltage. If the voltage in the circuit is measured by a voltmeter, it is approximately 110 volts, the impressed voltage. This can be proved by the equation:

$$E = \sqrt{(E_R)^2 + (EX_L)^2}$$
$$E = \sqrt{66^2 + 88^2}$$
$$E = \sqrt{4{,}356 + 7{,}744}$$
$$E = \sqrt{12{,}100}$$
$$E = 110 \text{ volts}$$

In *Figure 12-136*, a series circuit is illustrated in which a capacitor of 200 µf is connected in series with a 10 ohm lamp. What is the value of the impedance, the current flow, and the voltage drop across the lamp?

Solution:

First, the capacitance is changed from microfarads to farads. Since 1 million microfarads equal 1 farad, then:

$$200 \text{ µf} = \frac{200}{1{,}000{,}000} = 0.000200 \text{ farads}$$
$$X_C = \frac{1}{2\pi fC}$$
$$X_C = \frac{1}{6.28 \times 60 \times 0.000200 \text{ farads}}$$
$$X_C = \frac{1}{0.07536}$$

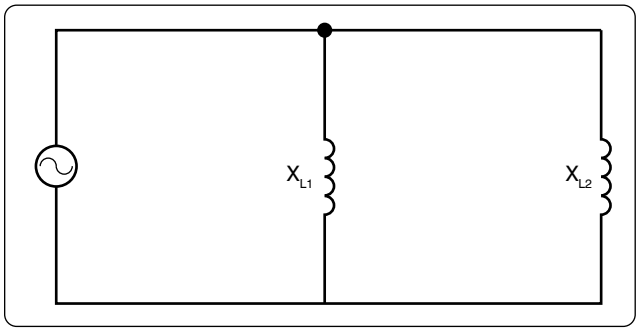


Figure 12-132. *Inductances in parallel.*

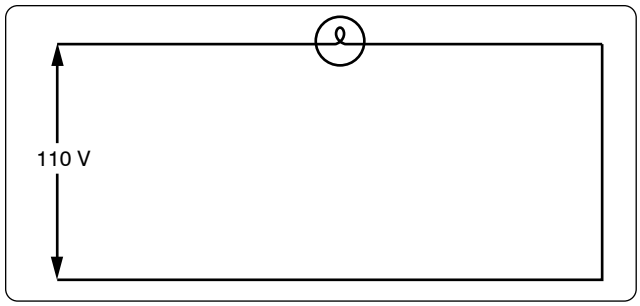


Figure 12-133. *Applying DC and AC to a circuit.*

$X_C = 13$ ohms capacitive reactance

To find the impedance,

$$Z = \sqrt{R^2 + X_C^2}$$
$$Z = \sqrt{10^2 + 13^2}$$
$$Z = \sqrt{100 + 169}$$
$$Z = \sqrt{269}$$
$$Z = 16.4 \text{ ohms capacitive reactance}$$

To find the current,

$$I = \frac{E}{Z}$$
$$I = \frac{110}{16.4}$$
$$I = 6.7 \text{ amperes}$$

The voltage drop across the lamp (E^R) is:

$$E^R = 6.7 \times 10$$
$$E^R = 67 \text{ volts}$$

The voltage drop across the capacitor (EX_C) is

$$EX_C = I \times X_C$$
$$EX_C = 6.7 \times 13$$
$$EX_C = 86.1 \text{ volts}$$

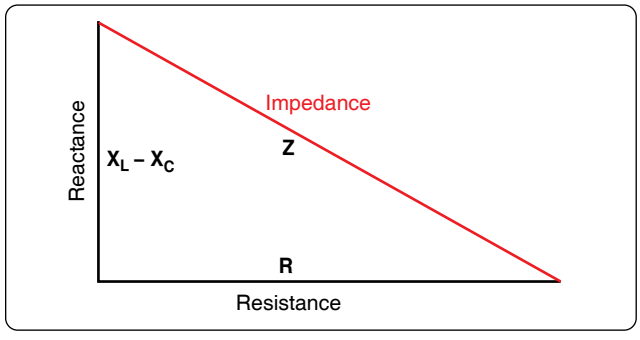


Figure 12-134. *Impedance triangle.*

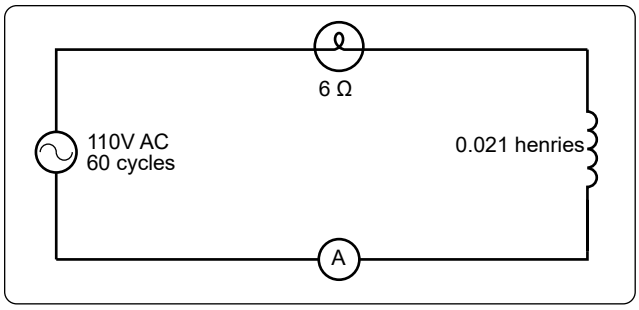


Figure 12-135. *A circuit containing resistance and inductance.*

The sum of these two voltages does not equal the applied voltage, since the current leads the voltage. To find the applied voltage, use the following formula:

$$E_T = \sqrt{(E_R)^2 + (EX_C)^2}$$
$$E_T = \sqrt{67^2 + 86.1^2}$$
$$E_T = \sqrt{4{,}489 + 7{,}413}$$
$$E_T = \sqrt{11{,}902}$$
$$E_T = 110 \text{ volts}$$

When the circuit contains resistance, inductance, and capacitance, the following equation is used to find the impedance:

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

Example: What is the impedance of a series circuit, consisting of a capacitor with a reactance of 7 ohms, an inductor with a reactance of 10 ohms, and a resistor with a resistance of 4 ohms? *[Figure 12-137]*

Solution:

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$
$$Z = \sqrt{4^2 + (10 - 7)^2}$$
$$Z = \sqrt{4^2 + 3^2}$$
$$Z = \sqrt{25}$$
$$Z = 5 \text{ ohms}$$

Assuming that the reactance of the capacitor is 10 ohms and the reactance of the inductor is 7 ohms, then X_C is greater than X_L. Thus,

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$
$$Z = \sqrt{4^2 + (7 - 10)^2}$$
$$Z = \sqrt{4^2 + (-3)^2}$$
$$Z = \sqrt{16 + 9}$$
$$Z = \sqrt{25}$$
$$Z = 5 \text{ ohms}$$

Parallel AC Circuits

The methods used in solving parallel AC circuit problems are basically the same as those used for series AC circuits. Out of phase voltages and currents can be added by using the law of right triangles. However, in solving circuit problems, the currents through the branches are added since the voltage drops across the various branches are the same and are equal to the applied voltage. In *Figure 12-138*, a parallel AC circuit containing an inductance and a resistance is shown schematically. The current flowing through the inductance, I_L, is 0.0584 ampere, and the current flowing through the resistance is 0.11 ampere. What is the total current in the circuit?

Solution:

$$I_T = \sqrt{I_L^2 + I_R^2}$$
$$= \sqrt{(0.0584)^2 + (0.11)^2}$$
$$= \sqrt{0.0155}$$
$$= 0.1245 \text{ ampere}$$

Since inductive reactance causes voltage to lead the current, the total current, which contains a component of inductive current, lags the applied voltage. If the current and voltages are plotted, the angle between the two, called the phase angle, illustrates the amount the current lags the voltage.

In *Figure 12-139*, a 112-volt generator is connected to a load consisting of a 2 µf capacitance and a 10,000-ohm resistance in parallel. What is the value of the impedance and total current flow?

Solution:

First, find the capacitive reactance of the circuit:

$$X_C = \frac{1}{2\pi fC}$$

Changing 2 µf to farads and entering the values into the formula given:

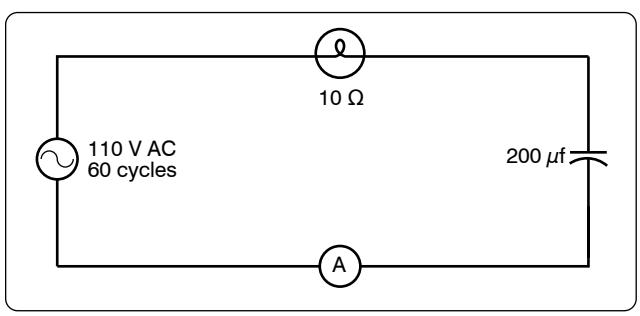


Figure 12-136. *A circuit containing resistance and capacitance.*

$$= \frac{1}{2 \times 3.14 \times 60 \times 0.000002}$$

$$= \frac{1}{0.00075360} \text{ or } \frac{10{,}000}{7.536}$$

$= 1{,}327\ X_C$ capacitive reactance

To find the impedance, the impedance formula used in a series AC circuit must be modified to fit the parallel circuit:

$$Z = \frac{RX_C}{\sqrt{R^2 + X_C^2}}$$

$$= \frac{10{,}000 \times 1{,}327}{\sqrt{(10{,}000)^2 + (1{,}327)^2}}$$

$= 0.1315$ W (approximately)

To find the current through the capacitance:

$$I_C = \frac{E}{X_C}$$

$$I_C = \frac{110}{1{,}327}$$

$= 0.0829$ ampere

To find the current flowing through the resistance:

$$I_R = \frac{E}{R}$$

$$= \frac{110}{10{,}000}$$

$= 0.011$ ampere

To find the total current in the circuit:

$$I_T^2 = \sqrt{I_R^2 + I_C^2}$$

$$I_T = \sqrt{I_L^2 + I_R^2}$$

$= 0.0836$ ampere (approximately)

Resonance

It has been shown that both inductive reactance:

$$(X_L = 2\pi fL)$$

and capacitive reactance:

$$X_C = \frac{1}{2\pi fC}$$

are functions of an AC frequency. Decreasing the frequency decreases the ohmic value of the inductive reactance, but a decrease in frequency increases the capacitive reactance. At some particular frequency, known as the resonant frequency, the reactive effects of a capacitor and an inductor is equal.

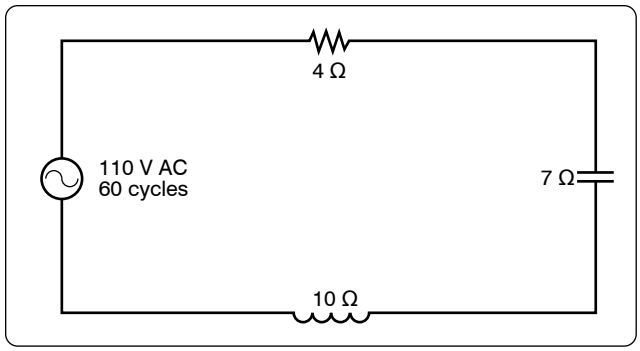


Figure 12-137. *A circuit containing resistance, inductance, and capacitance.*

Since these effects are the opposite of one another, they will cancel, leaving only the ohmic value of the resistance to oppose current flow in a circuit. If the value of resistance is small or consists only of the resistance in the conductors, the value of current flow can become very high.

In a circuit where the inductor and capacitor are in series, and the frequency is the resonant frequency, or frequency of resonance, the circuit is said to be "in resonance" and is referred to as a series resonant circuit. The symbol for resonant frequency is Fn.

If, at the frequency of resonance, the inductive reactance is equal to the capacitive reactance, then:

$$X_L = X_C, \text{ or}$$

$$2\pi fL = \frac{1}{2\pi fC}$$

Dividing both sides by 2 fL,

$$Fn2 = \frac{1}{(2\pi)\ 2LC}$$

Extracting the square root of both sides gives:

$$Fn = \frac{1}{2\pi\ \sqrt{LC}}$$

Where Fn is the resonant frequency in cps, C is the capacitance in farads, and L is the inductance in henries. With this formula, the frequency at which a capacitor and inductor is resonant can be determined.

To find the inductive reactance of a circuit use:

$$X_L = 2\pi fL$$

The impedance formula used in a series AC circuit must be modified to fit a parallel circuit.

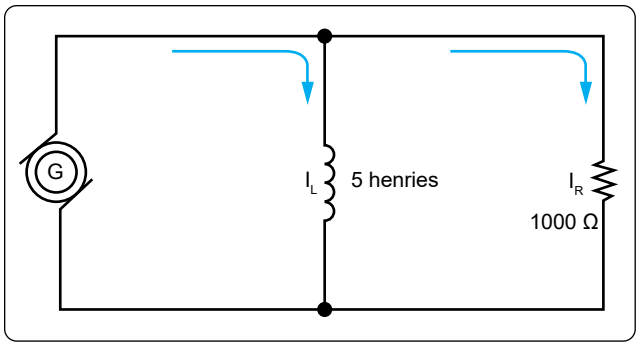


Figure 12-138. *AC parallel circuit containing inductance and resistance.*

$$Z = \frac{X_L}{\sqrt{R^2 = X_L{}^2}}$$

To find the parallel networks of inductance and capacitive reactors, use:

$$X = \frac{X_L + X_C}{\sqrt{X_L + X_C}}$$

To find the parallel networks with resistance capacitive and inductance, use:

$$Z = \frac{R\, X_L\, X_C}{\sqrt{X_L{}^2\, X_C{}^2 + (RX_L - RX_C)^2}}$$

Since at the resonant frequency X_L cancels X_C, the current can become very large, depending on the amount of resistance. In such cases, the voltage drop across the inductor or capacitor is often higher than the applied voltage.

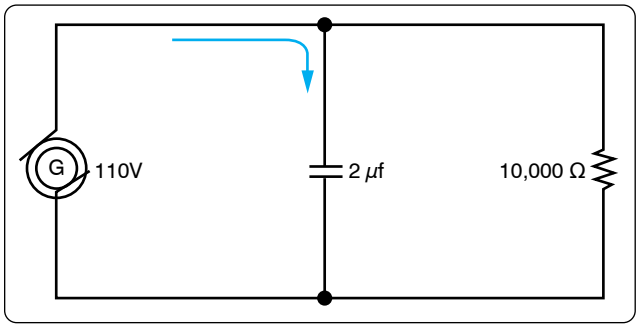


Figure 12-139. *A parallel AC circuit containing capacitance and resistance.*

In a parallel resonant circuit, the reactances are equal and equal currents flow through the coil and the capacitor. *[Figure 12-140]*

Since the inductive reactance causes the current through the coil to lag the voltage by 90°, and the capacitive reactance causes the current through the capacitor to lead the voltage by 90°, the two currents are 180° out of phase. The canceling effect of such currents would mean that no current would flow from the generator and the parallel combination of the inductor and the capacitor would appear as infinite impedance. In practice, no such circuit is possible, since some value of resistance is always present, and the parallel circuit, sometimes called a tank circuit, acts as very high impedance. It is also called an antiresonant circuit, since its effect in a circuit is opposite to that of a series resonant circuit, in which the impedance is very low.

Power in AC Circuits

In a DC circuit, power is obtained by the equation, P = EI, (watts equal volts × amperes). Thus, if 1 ampere of current flows in a circuit at a pressure of 200 volts, the power is 200 watts. The product of the volts and the amperes is the true power in the circuit.

True Power Defined

True power of any AC circuit is commonly referred to as the working power of the circuit. True power is the power consumed by the resistance portion of the circuit and is measured in watts. True power is symbolized by the letter P and is indicated by any wattmeter in the circuit. True power is calculated by the formula:

$$P = I^2 \times Z$$

Apparent Power Defined

Apparent power in an AC circuit is sometimes referred to as the reactive power of a circuit. Apparent power is the power consumed by the entire circuit, including both the resistance and the reactance. Apparent power is symbolized by the letter S and is measured in volt-amps (VA). Apparent power is a product of the effective voltage multiplied by the effective current. Apparent power is calculated by the formula:

$$S = I^2 \times Z$$

Only when the AC circuit is made up of pure resistance is the apparent power equal to the true power. *[Figure 12-141]* When there is capacitance or inductance in the circuit, the current and voltage are not exactly in phase, and the true power is less than the apparent power. The true power is obtained by a wattmeter reading. The ratio of the true power to the apparent power is called the power factor and is usually expressed in percent. In equation form, the relationship is:

$$\text{Power Factor (PF)} = \frac{100 \times \text{watts (True Power)}}{\text{volts} \times \text{amperes (Apparent Power)}}$$

Example: A 220-volt AC motor takes 50 amperes from the line, but a wattmeter in the line shows that only 9,350 watts

are taken by the motor. What are the apparent power and the power factor?

Solution:

Apparent power = Volts × Amperes

Apparent power = 220 × 50 = 11,000 watts or volt-amperes.

$$(PF) = \frac{\text{Watts (True Power)} \times 100}{\text{VA (Apparent Power)}}$$

$$(PF) = \frac{9{,}350 \times 100}{11{,}000}$$

(PF) = 85, or 85%

Transformers

A transformer changes electrical energy of a given voltage into electrical energy at a different voltage level. It consists of two coils that are not electrically connected, but are arranged so that the magnetic field surrounding one coil cuts through the other coil. When an alternating voltage is applied to (across) one coil, the varying magnetic field set up around that coil creates an alternating voltage in the other coil by mutual induction. A transformer can also be used with pulsating DC, but a pure DC voltage cannot be used, since only a varying voltage creates the varying magnetic field that is the basis of the mutual induction process.

A transformer consists of three basic parts. *[Figure 12-142]* These are an iron core, which provides a circuit of low reluctance for magnetic lines of force; a primary winding, which receives the electrical energy from the source of applied voltage; and a secondary winding, which receives electrical energy by induction from the primary coil.

The primary and secondary of this closed core transformer are wound on a closed core to obtain maximum inductive effect between the two coils.

There are two classes of transformers: voltage transformers, used for stepping up or stepping down voltages; and current transformers used in instrument circuits. In voltage transformers, the primary coils are connected in parallel across the supply voltage. *[Figure 12-143A]* The primary windings of current transformers are connected in series in the primary circuit. *[Figure 12-143B]* Of the two types, the voltage transformer is the more common.

There are many types of voltage transformers. Most of these are either step-up or step-down transformers. The factor that determines whether a transformer is a step-up or step-down type is the "turns" ratio. The turns ratio is the ratio of the number of turns in the primary winding to the number of turns in the secondary winding. For example, the turns ratio of the step-down transformer is 5 to 1, since there are five times as many turns in the primary as in the secondary. *[Figure 12-144A]* The step-up transformer has a 1 to 4 turns ratio. *[Figure 12-144B]*

The ratio of the transformer input voltage to the output voltage is the same as the turns ratio if the transformer is 100 percent efficient. Thus, when 10 volts are applied to the primary of the transformer, two volts are induced in the secondary. *[Figure 12-144A]* If 10 volts are applied to the primary of the transformer, the output voltage across the terminals of the secondary is 40 volts. *[Figure 12-144B]*

No transformer can be constructed that is 100 percent efficient, although iron core transformers can approach this figure. This is because all the magnetic lines of force set up in the primary do not cut across the turns of the secondary coil. A certain amount of the magnetic flux, called leakage flux, leaks out of the magnetic circuit. The measure of how well the flux of the primary is coupled into the secondary is called the "coefficient of coupling." For example, if it is assumed that the primary of a transformer develops 10,000 lines of force and only 9,000 cut across the secondary, the coefficient of coupling would be 0.9. Stated another way, the transformer would be 90 percent efficient.

When an AC voltage is connected across the primary terminals of a transformer, an AC flows and self induces a voltage in the primary coil that is opposite and nearly equal to the applied voltage. The difference between these two voltages allows just enough current in the primary to magnetize its core. This is called the exciting, or magnetizing, current. The magnetic field caused by this exciting current cuts across the secondary coil and induces a voltage by mutual induction.

If a load is connected across the secondary coil, the load current flowing through the secondary coil produces a magnetic field that tends to neutralize the magnetic field produced by the primary current. This reduces the self-induced (opposition) voltage in the primary coil and allows more primary current

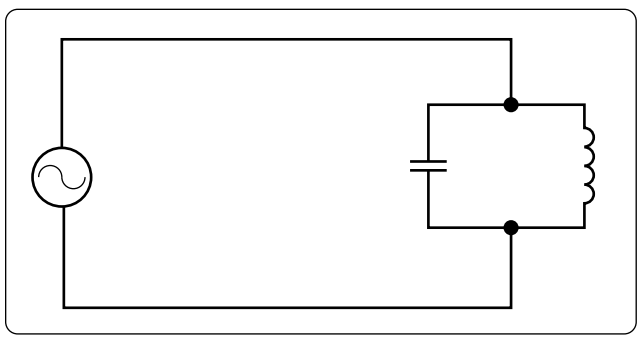


Figure 12-140. *A parallel resonant circuit.*

to flow. The primary current increases as the secondary load current increases, and decreases as the secondary load current decreases. When the secondary load is removed, the primary current is again reduced to the small exciting current sufficient only to magnetize the iron core of the transformer.

If a transformer steps up the voltage, it steps down the current by the same ratio. This should be evident if the power formula is considered, for the power (I × E) of the output (secondary) electrical energy is the same as the input (primary) power minus that energy loss in the transforming process. Thus, if 10 volts and 4 amps (40 watts of power) are used in the primary to produce a magnetic field, there is 40 watts of power developed in the secondary (disregarding any loss). If the transformer has a step-up ratio of 4 to 1, the voltage across the secondary is 40 volts and the current is 1 amp. The voltage is 4 times greater and the current is one-fourth the primary circuit value, but the power (I × E value) is the same.

When the turns ratio and the input voltage are known, the output voltage can be determined as follows:

$$\frac{E_2}{E_1} = \frac{N_2}{N_1}$$

Where E is the voltage of the primary, E_2 is the output voltage of the secondary, and N_1 and N_2 are the number of turns of the primary and secondary, respectively.

Transposing the equation to find the output voltage gives:

$$E_2 = \frac{E_1 N_2}{N_1}$$

The most commonly used types of voltage transformers are:

1. Power transformers are used to step up or step down voltages and current in many types of power supplies. They range in size from the small power transformer *[Figure 12-145]* used in a radio receiver to the large transformers used to step down high power line voltage to the 110–120 volt level used in homes.

   Figure 12-146 shows the schematic symbol for an iron core transformer. In this case, the secondary is made up of three separate windings. Each winding supplies a different circuit with a specific voltage, which saves the weight, space, and expense of three separate transformers. Each secondary has a midpoint connection called a "center tap," which provides a selection of half the voltage across the whole winding. The leads from the various windings are color coded by the manufacturer. *[Figure 12-146]* This is a standard color code, but other codes or numbers may be used.
2. Audio transformers resemble power transformers. They have only one secondary and are designed to operate over the range of audio frequencies (20 to 20,000 cps).
3. RF transformers are designed to operate in equipment that functions in the radio range of frequencies. The symbol for the RF transformer is the same as for an RF choke coil. It has an air core as shown in *Figure 12-147*.
4. Autotransformers are normally used in power circuits; however, they may be designed for other uses. Two different symbols for autotransformers used in power or audio circuits are shown in *Figure 12-148*. If used in an RF communication or navigation circuit *[Figure 12-148B]*, it is the same, except there is no symbol for an iron core. The autotransformer uses part of a winding as a primary; and, depending on whether it is step up or step down, it uses all or part of the same winding as the secondary. For example, the autotransformer shown in *Figure 12-148A* could use the following possible choices for primary and secondary terminals.

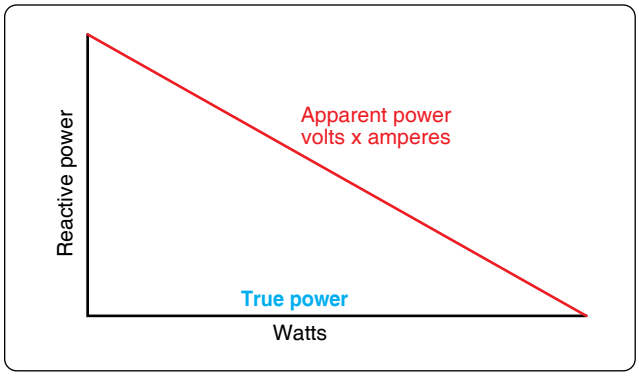


Figure 12-141. *Power relations in AC circuit.*

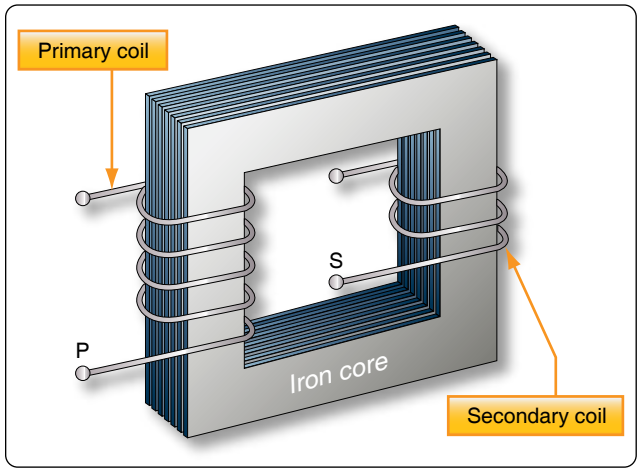


Figure 12-142. *An iron-core transformer.*

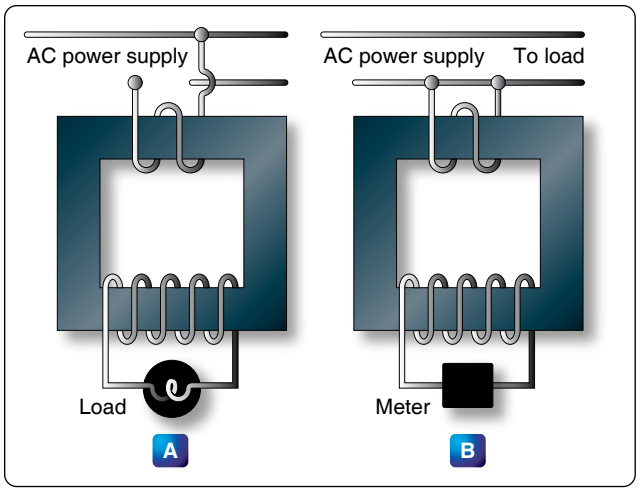


Figure 12-143. *Voltage and current transformers.*

Current Transformers

Current transformers are used in AC power supply systems to sense generator line current and to provide a current, proportional to the line current, for circuit protection and control devices.

The current transformer is a ring-type transformer using a current carrying power lead as a primary (either the power lead or the ground lead of the AC generator). The current in the primary induces a current in the secondary by magnetic induction. The sides of all current transformers are marked "H1" and "H2" on the unit base. The transformers must be installed with the "H1" side toward the generator in the circuit in order to have proper polarity. The secondary of the transformer should never be left open while the system is being operated; to do so could cause dangerously high voltages and could overheat the transformer. Therefore, the transformer output connections should always be connected with a jumper when the transformer is not being used but is left in the system.

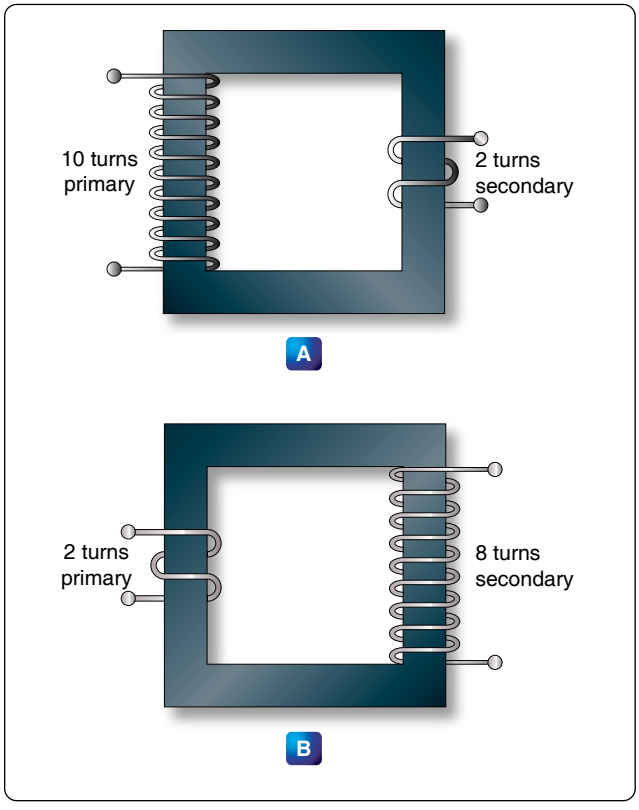


Figure 12-144. *A step-down and a step-up transformer.*

Transformer Losses

In addition to the power loss caused by imperfect coupling, transformers are subject to "copper" and "iron" losses. The resistance of the conductor comprising the turns of the coil causes copper loss. The iron losses are of two types: hysteresis loss and eddy current loss. Hysteresis loss is the electrical energy required to magnetize the transformer core, first in one direction and then in the other, in step with the applied alternating voltage. Eddy current loss is caused by electric currents (eddy currents) induced in the transformer core by the varying magnetic fields. To reduce eddy current losses, cores are made of laminations coated with an insulation, which reduces the circulation of induced currents.

Power in Transformers

Since a transformer does not add any electricity to the circuit but merely changes or transforms the electricity that already exists in the circuit from one voltage to another, the total amount of energy in a circuit must remain the same. If it were possible to construct a perfect transformer, there would be no loss of power in it; power would be transferred undiminished from one voltage to another.

Since power is the product of volts times amperes, an increase in voltage by the transformer must result in a decrease in current and vice versa. There cannot be more power in the secondary side of a transformer than there is in the primary. The product of amperes times volts remains the same.

The transmission of power over long distances is accomplished by using transformers. At the power source, the voltage is stepped up in order to reduce the line loss during transmission. At the point of utilization, the voltage is stepped down, since it is not feasible to use high voltage to operate motors, lights, or other electrical appliances.

DC Measuring Instruments

Understanding the functional design and operation of electrical measuring instruments is very important, since they are used in repairing, maintaining, and troubleshooting electrical circuits. The best and most expensive measuring instrument is of no use unless the technician knows what is

being measured and what each reading indicates. The purpose of the meter is to measure quantities existing in a circuit. For this reason, when a meter is connected to a circuit, it must not change the characteristics of that circuit.

Meters are either self-excited or externally excited. Those that are self-excited operate from a power source within the meter. Externally-excited meters get their power source from the circuit that they are connected to. The most common analog meters in use today are the voltmeter, ammeter, and ohmmeter. All of which operate on the principles of electromagnetism. The fundamental principle behind the operation of the meter is the interaction between magnetic fields created by a current gathered from the circuit in some manner. This interaction is between the magnetic fields of a permanent magnet and the coils of a rotating magnet. The greater the current through the coils of the rotating magnet, the stronger the magnetic field produced. A stronger field produces greater rotation of the coil. While some meters can be used for both DC and AC circuit measurement, only those used as DC instruments are discussed in this section. The meters used for AC, or for both AC and DC, are discussed in the study of AC theory and circuitry.

D'Arsonval Meter Movement

This basic DC type of meter movement—first employed by the French scientist, d'Arsonval, in making electrical measurement—is a current measuring device, which is used in the ammeter, voltmeter, and ohmmeter. The pointer is deflected in proportion to the amount of current through the coil. Basically, both the ammeter and the voltmeter are current measuring instruments, the principal difference being the method in which they are connected in a circuit. While an ohmmeter is also basically a current measuring instrument, it differs from the ammeter and voltmeter in that it provides its own source (self-excited) of power and contains other auxiliary circuits.

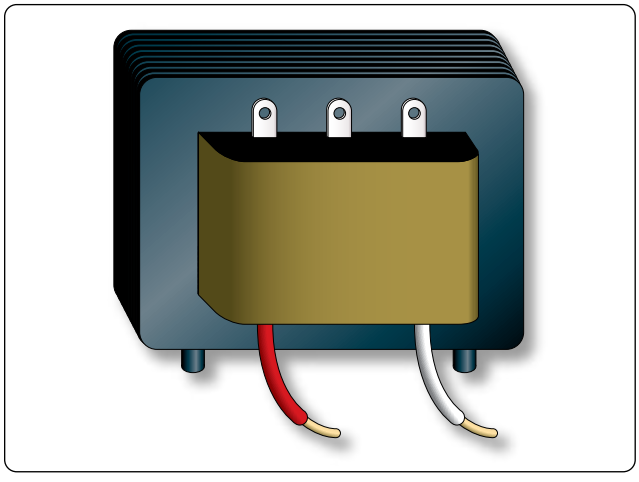


Figure 12-145. *Power supply transformer.*

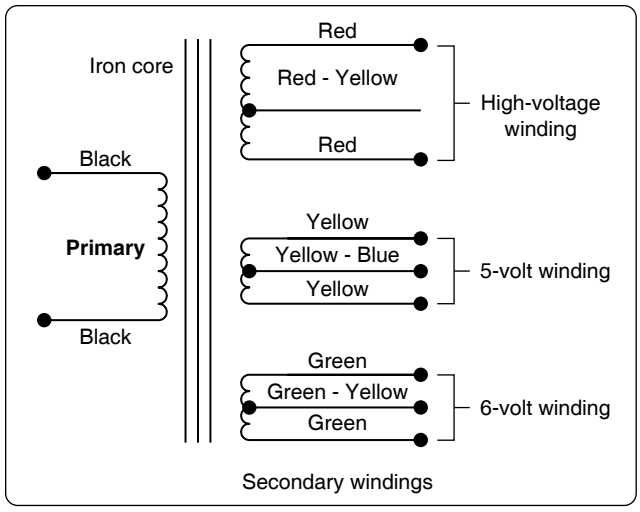


Figure 12-146. *Schematic symbol for an iron-core power transformer.*

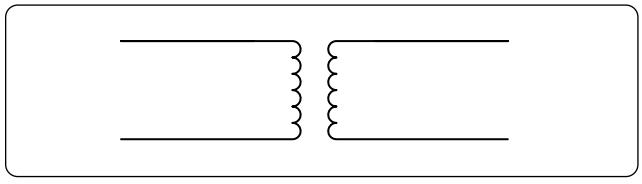


Figure 12-147. *An air-core transformer.*

Current Sensitivity and Resistance

The current sensitivity of a meter movement is the amount of current required to drive the meter movement to a full-scale deflection. A simple example would be a meter movement that has 1 mA sensitivity. What this indicates is that meter movement requires 1 mA of current to move the needle to a full-scale indication. Likewise, a half-scale deflection requires only 0.5 mA of current. Additionally, what is called

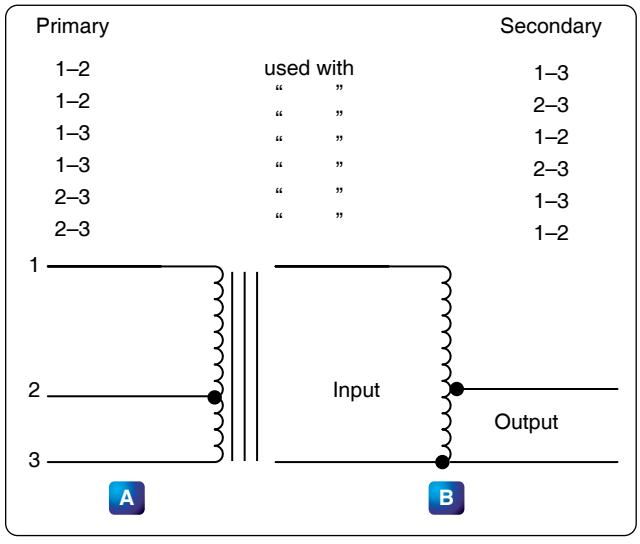


Figure 12-148. *Autotransformers.*

movement resistance is the actual DC resistance of the wire used to construct the meter coil.

In a standard d'Arsonval meter, movement may have a current sensitivity of 1 mA and a resistance of 50 Ω. If the meter is going to be used to measure more than 1 mA, then additional circuitry is required to accomplish the task. This additional circuitry is a simple shunt resistor. The purpose of the shunt resistor is to bypass current that exceeds the 1 mA limitation of the meter movement. To illustrate this, assume that the 1 mA meter in question is needed to measure 10 mA. The shunt resistor used should carry 9 mA while the remaining 1 mA is allowed to pass through the meter. *[Figure 12-149]*

To determine the proper shunt resistance for this situation:

R_{SH} = Shunt resistance
R_M = Meter resistance = 50 Ω

Because the shunt resistance and the 50 Ω meter resistance are in parallel, the voltage drop across both of them is the same.

$$E_{SH} = E_M$$

Using Ohm's Law, this relationship can be rewritten as:

$$E_{SH} = I_{SH} \times R_{SH}$$
$$E_M = I_M \times R_M$$
$$I_{SH} \times R_{SH} = I_M \times R_M$$

Simply solve for R_{SH}

$$R_{SH} = \frac{I_M \times R_M}{I_{SH}}$$

Substituting the values

$$R_{SH} = \frac{\text{ImA} \times 50\ \Omega}{9\ \text{mA}} = 5.56\ \Omega$$

Damping

To make meter readings quickly and accurately, it is desirable that the moving pointer overshoot its proper position only a small amount and come to rest after not more than one or two small oscillations. The term "damping" is applied to methods used to bring the pointer of an electrical meter to rest after it has been set in motion. Damping may be accomplished by electrical means, by mechanical means, or by a combination of both.

Electrical Damping

A common method of damping by electrical means is to wind the moving coil on an aluminum frame. As the coil moves in the field of the permanent magnet, eddy currents are set up in the aluminum frame. The magnetic field produced by the eddy currents opposes the motion of the coil. The pointer therefore swings more slowly to its proper position and comes to rest quickly with very little oscillation.

Mechanical Damping

Air damping is a common method of damping by mechanical means. As shown in *Figure 12-150*, a vane is attached to the shaft of the moving element and enclosed in an air chamber. The movement of the shaft is retarded because of the resistance that the air offers to the vane. Effective damping is achieved if the vane nearly touches the walls of the chamber.

A Basic Multirange Ammeter

Building upon the basic meter previously discussed is the more complex and useful multirange meter, which is more practical. The basic idea of a multirange ammeter is to make the meter usable over a wide range of voltages. In order to accomplish this, each range must utilize a different shunt resistance. The example given in this handbook is that of a two-range meter. However, once the basics of a two-range multirange ammeter are understood, the concepts can easily be transferred to the design of meters with many selectable ranges.

Figure 12-151 shows the schematic of an ammeter with two selectable ranges. This example builds upon the previous 10 mA range meter by adding a 100 mA range. With the switch selected to the 10 mA range, the meter indicates 10 mA when the needle is deflected to full-scale and likewise indicates 100 mA at full-scale when selected to 100 mA. The value of the 100 mA shunt resistor is determined the same way the 10 mA shunt resistor was determined. Recall that the meter movement can only carry 1 mA. This means that in a 100 mA range the remaining current of 99 mA must pass through the shunt resistor.

$$R_{SH} = \frac{I_M \times R_M}{I_{SH}}$$

Substituting the values:

$$R_{SH} = \frac{\text{ImA} \times 50\ \Omega}{99\ \text{mA}} = 0.51\ \Omega$$

Precautions

The precautions to observe when using an ammeter are summarized as follows:

1. Always connect ammeter in series with the element

through which the current flow is to be measured.

2. Never connect an ammeter across a source of voltage, such as a battery or generator. Remember that the resistance of an ammeter, particularly on the higher ranges, is extremely low and that any voltage, even a volt or so, can cause very high current to flow through the meter, causing damage to it.
3. Use a range large enough to keep the deflection less than full-scale. Before measuring a current, form some idea of its magnitude. Then switch to a large enough scale or start with the highest range and work down until the appropriate scale is reached. The most accurate readings are obtained at approximately half-scale deflection. Many milliammeters have been ruined by attempts to measure amperes. Therefore, be sure to read the lettering either on the dial or on the switch positions and choose proper scale before connecting the meter in the circuit.
4. Observe proper polarity in connecting the meter in the circuit. Current must flow through the coil in a definite direction in order to move the indicator needle up scale. Current reversal because of incorrect connection in the circuit results in a reversed meter deflection and frequently causes bending of the meter needle. Avoid improper meter connections by observing the polarity markings on the meter.

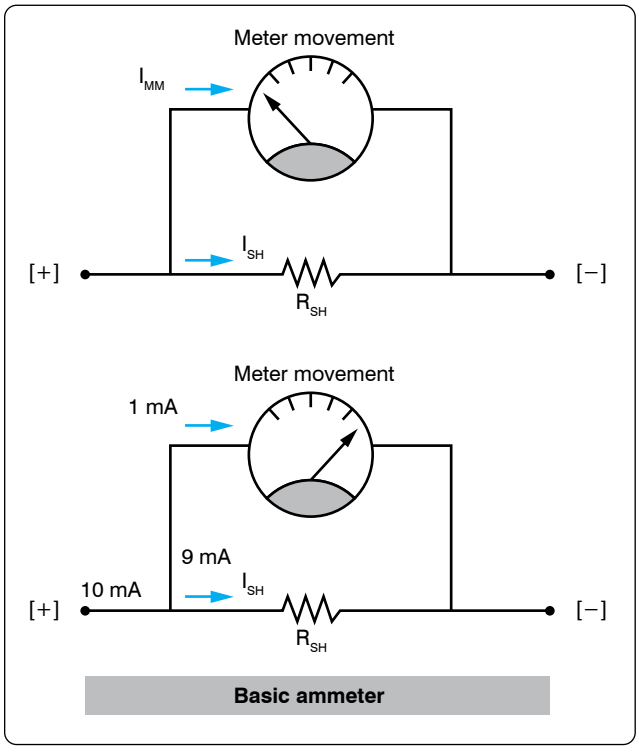


Figure 12-149. *Basic meter drawing.*

The Voltmeter

The voltmeter uses the same type of meter movement as the ammeter but employs a different circuit external to the meter movement.

As shown before, the voltage drop across the meter coil is a function of current and the coil resistance. In another example, 50 µA × 1,000 Ω = 50 mV. In order for the meter to be used to measure voltages greater than 50 mV, there must be added a series resistance to drop any excess voltage greater than that which the meter movement requires for a full-scale deflection. The case of the voltmeter, this resistance is called multiplier resistance and is designated as R_M. [*Figure 12-152]* The voltmeter only has one multiplier resistor for use in one range. In this example, the full-scale reading is 1 volt. R_M is determined in the following way:

The meter movement drops 50 mV at a full-scale deflection of 50 µA. The multiplying resistor R_M must drop the remaining voltage of 1 V − 50 mV = 950 mV. Since R_M is in series with the movement, it also carries 50 µA at full scale.

$$RM = \frac{950\ mV}{50\ \mu A} = 19k\ \Omega$$

Therefore, for 1 volt full-scale deflection, the total resistance of the voltmeter is 20k Ω. That is, the multiplier resistance and the coil resistance.

Voltmeter Sensitivity

Voltmeter sensitivity is defined in terms of resistance per volt (Ω/V). The meter used in the previous example has a sensitivity of 20k Ω and a full-scale deflection of 1 volt.

Multiple Range Voltmeters

The simplified voltmeter in *Figure 12-152* has only one range (1 volt), which means that it can measure voltages from 0 volts to 1 volt. In order for the meter to be more useful, additional multiplier resistors must be used. One resistor must be used for each desired range.

For a 50 µA movement, the total resistance required is 20k Ω for each volt of full-scale reading. In other words, the sensitivity for a 50 µA movement is always 20k Ω regardless of the selected range. The full-scale meter current is 50 µA at any range selection. To find the total meter resistance, multiply the sensitivity by the full-scale voltage for that particular range.

For example for a 10 volt range, $R_T = (20k\ \Omega/V)(10V) = 200k\ \Omega$. The total resistance for the 1 volt range is 20k Ω, so R_M for a 10 V range is 200k Ω − 20k Ω = 180k Ω. *[Figure 12-153]*

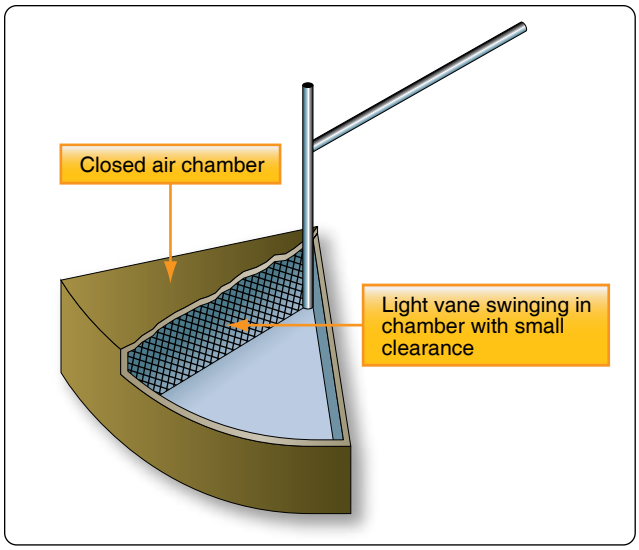


Figure 12-150. *Air damping.*

Voltmeter Circuit Connections

When voltmeters are used, they are connected in parallel with a circuit. If unsure about the voltage to be measured, take the first reading at the high value on the meter and then progressively move down through the range until a suitable read is obtained. Observe that the polarity is correct before connecting the meter to the circuit or damage occurs by driving the movement backwards.

Influence of the Voltmeter in the Circuit

When a voltmeter is connected across two points in a circuit, current is shunted. If the voltmeter has low resistance, it draws off a significant amount of current. This lowers the effective resistance of the circuit and change the voltage readings. When making a voltage measurement, use a high resistance voltmeter to prevent shunting of the circuit.

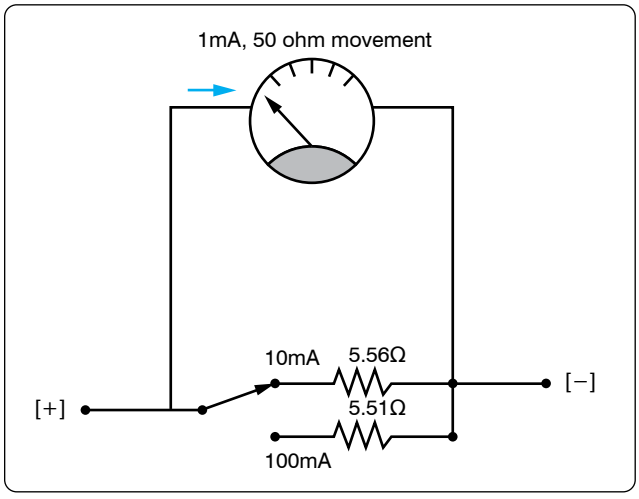


Figure 12-151. *Ammeter with two ranges.*

The Ohmmeter

The meter movement used for the ammeter and the voltmeter can also be used for the ohmmeter. The function of the ohmmeter is to measure resistance. A simplified one-stage ohmmeter is illustrated in *Figure 12-154*, which shows that the basic ohmmeter contains a battery and a variable resistor in series with the meter movement. To measure resistance, the leads of the meter are connected across an external resistance, which is to be measured. By doing this, the ohmmeter circuit is completed. This connection allows the internal battery to produce a current through the movement coil, causing a deflection of the pointer proportional to the value of the external resistance being measured.

Zero Adjustment

When the ohmmeter leads are open, the meter is at a full-scale deflection, indicating an infinite (∞) resistance or an open circuit. *[Figure 12-155]* When the leads are shorted as shown in figure "zero adjust," the pointer is at the full right-hand position, indicating a short circuit or zero resistance. The purpose of the variable resistor in this figure is to adjust the current so that the pointer is at exactly zero when the leads are shorted. This is used to compensate for changes in the internal battery voltage due to aging.

Ohmmeter Scale

Figure 12-156 shows a typical analog ohmmeter scale. Between zero and infinity (∞), the scale is marked to indicate various resistor values. Because the values decrease from left to right, this scale is often called a back-off scale.

In the case of the example given, assume that a certain ohmmeter uses a 50 µA, 1,000 Ω meter movement and has an internal 1.5 volt battery. A current of 50 µA produces a full-scale deflection when the test leads are shorted. To have 50 µA, the total ohmmeter resistance is 1.5 V/50 µA = 30k Ω. Therefore, since the coil resistance is 1k Ω, the variable zero adjustment resistor must be set to 30k Ω – 1k Ω = 29k Ω.

Now consider that a 120k Ω resistor is connected to the ohmmeter leads. Combined with the 30k Ω internal resistance, the total R is 150k Ω. The current is 1.5 V/150k Ω = 10 µA, which is 20 percent of the full-scale current and appears on the scale shown in *Figure 12-156*.

Now consider further that a 120k Ω resistor is connected to the ohmmeter leads. This results in a current of 1.5 V/75k Ω = 10 µA, which is 40 percent of the full-scale current and marked on the scale. Additional calculations of this type show that the scale is nonlinear. It is more compressed toward the left side than the right side. The center scale point corresponds to the internal meter resistance of 30k Ω. The reason is as follows:

With 30k Ω connected to the leads, the current is 1.5 V/60k Ω = 25 μA, which is half of the full-scale current of 50 μA.

The Multirange Ohmmeter

A practical ohmmeter has several operational ranges. These typically are indicated by R × 1, R × 10, R × 100, R × 1k, R × 100k and R × 1M. These range selections are interpreted in a different manner than that of an ammeter or voltmeter. The reading on the ohmmeter scale is multiplied by the factor indicated by the range setting. For example, if the pointer is set on the scale and the range switch is set at R × 100, the actual resistance measurement is 20 × 100 or 2k Ω.

To measure small resistance values, the technician must use a higher ohmmeter current than is needed for measuring large resistance values. Shunt resistors are needed to provide multiple ranges on the ohmmeter to measure a range of resistance values from the very small to very large. For each range, a different value of shunt resistance is switched in. The shunt resistance increases for higher ohm ranges and is always equal to the center scale reading on any selected range. In some meters, a higher battery voltage is used for the highest ohm range. *[Figure 12-157]*

Megger (Megohmmeter)

The megger, or megohmmeter, is a high range ohmmeter containing a hand-operated generator. It is used to measure insulation resistance and other high-resistance values. It is also used for ground, continuity, and short-circuit testing of electrical power systems. The chief advantage of the megger over an ohmmeter is its capacity to measure resistance with a high potential, or "breakdown" voltage. This type of testing ensures that insulation or a dielectric material will not short or leak under potential electrical stress.

The megger consists of two primary elements, both of which are provided with individual magnetic fields from a common permanent magnet: a hand-driven DC generator, G, which supplies the necessary current for making the measurement; and the instrument portion, which indicates the value of the resistance being measured. The instrument portion is of the opposed coil type. Coils A and B are mounted on the movable member with a fixed angular relationship to each other and are free to turn as a unit in a magnetic field. Coil B tends to move the pointer counterclockwise and coil A, clockwise. The coils are mounted on a light, movable frame that is pivoted in jewel bearings and free to move about axis 0. *[Figure 12-158]*

Coil A is connected in series with R3 and the unknown resistance, R_X, to be measured. The series combination of coil A, R3, and R_X is connected between the + and − brushes of the DC generator. Coil B is connected in series with R2, and this combination is also connected across the generator. There are no restraining springs on the movable member of the instrument portion of the megger. When the generator is not in operation, the pointer floats freely and may come to rest at any position on the scale.

If the terminals are open circuited, no current flows in coil A, and the current in coil B alone controls the movement of the moving element. Coil B takes a position opposite the gap in the core (since the core cannot move and coil B can), and the pointer indicates infinity on the scale. When a resistance is connected between the terminals, current flows in coil A, tending to move the pointer clockwise. At the same time, coil B tends to move the pointer counterclockwise. Therefore, the moving element, composed of both coils and the pointer, comes to rest at a position at which the two forces are balanced. This position depends upon the value of the external resistance, which controls the relative magnitude of current of coil A. Because changes in voltage affect both coils A and B in the same proportion, the position of the moving element is independent of the voltage. If the terminals are short circuited, the pointer rests at zero because the current in A is relatively large. The instrument is not damaged under these circumstances because the current is limited by R3.

There are two types of hand-driven meggers: the variable type and the constant pressure type. The speed of the variable pressure megger is dependent on how fast the hand crank

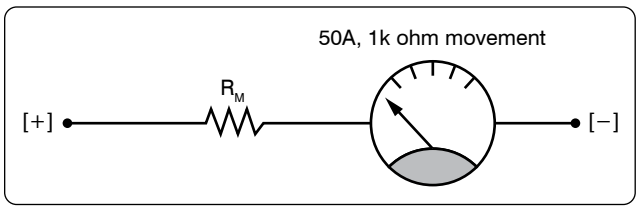


Figure 12-152. *Basic voltmeter.*

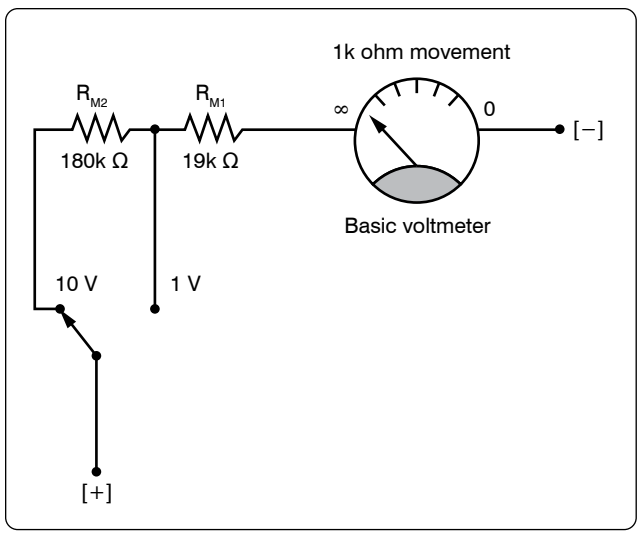


Figure 12-153. *Two range voltmeter.*

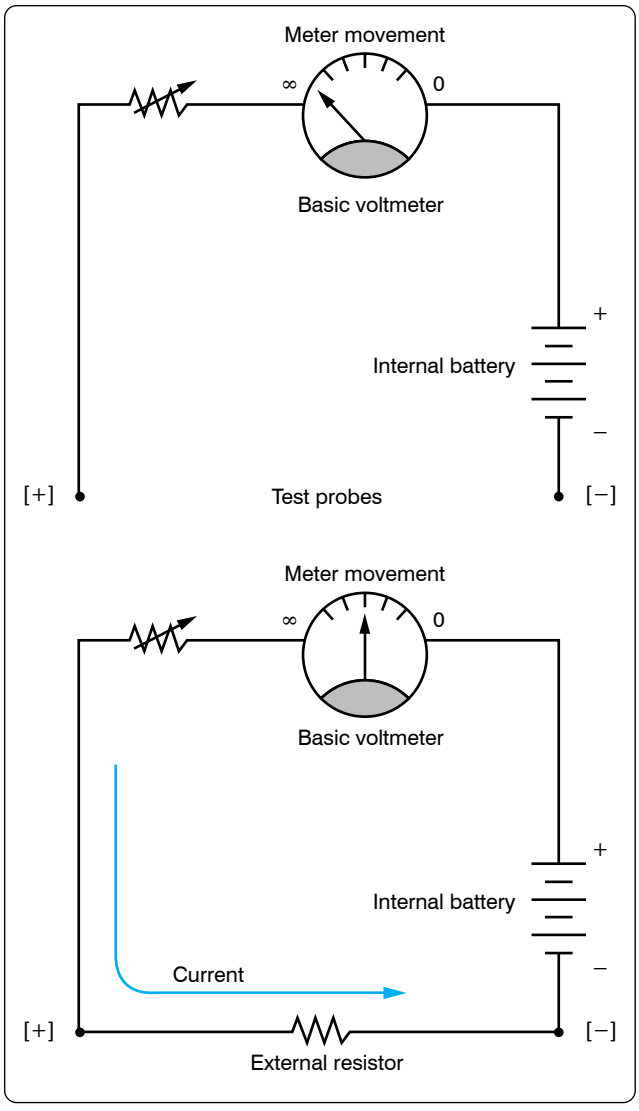


Figure 12-154. *Basic ohmmeter.*

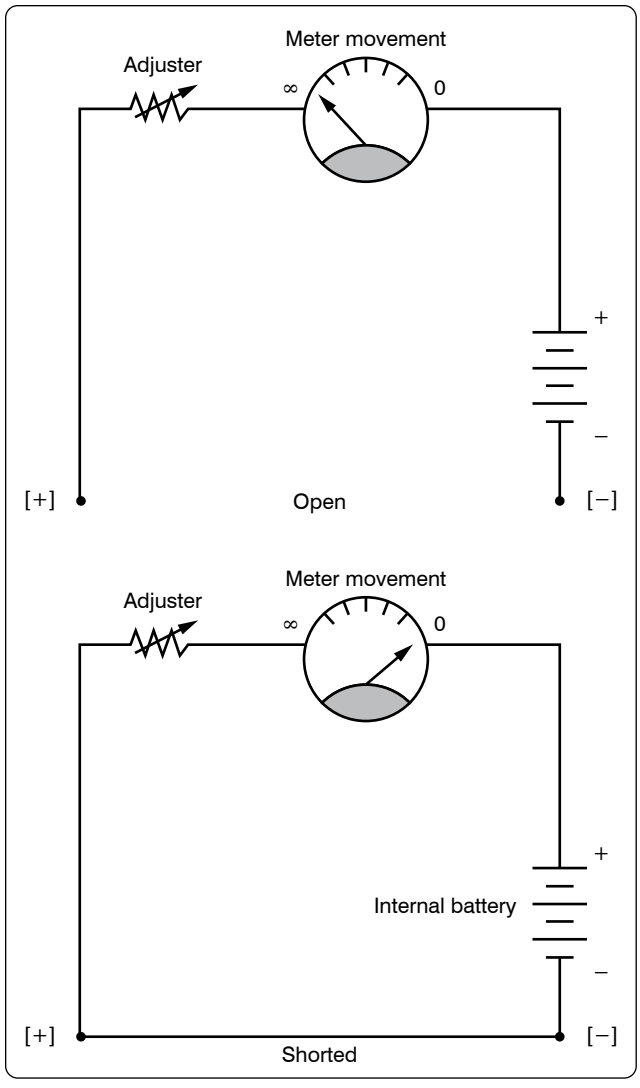


Figure 12-155. *Zero adjustment.*

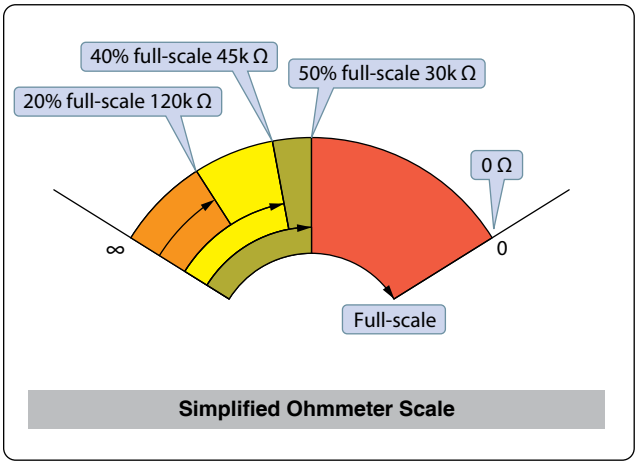


Figure 12-156. *Ohm scale.*

is turned. The constant pressure megger uses a centrifugal governor, or slip clutch. The governor becomes effective only when the megger is operated at a speed above its slip speed, at which speed its voltage remains constant.

AC Measuring Instruments

A DC meter, such as an ammeter, connected in an AC circuit indicates zero, because the meter movements used in a d'Arsonval type movement is restricted to DC. Since the field of a permanent magnet in the d'Arsonval type meter remains constant and in the same direction at all times, the moving coil follows the polarity of the current. The coil attempts to move in one direction during half of the AC cycle and in the reverse direction during the other half when the current reverses.

The current reverses direction too rapidly for the coil to follow, causing the coil to assume an average position. Since

the current is equal and opposite during each half of the AC cycle, the DC meter indicates zero, which is the average value. Thus, a meter with a permanent magnet cannot be used to measure alternating voltage and current. For AC measurements of current and voltage, additional circuitry is required. The additional circuitry has a rectifier, which converts AC to DC. There are two basic types of rectifiers: the half-wave rectifier and the full-wave rectifier. *[Figure 12-159]*

Figure 12-159 also shows a simplified block diagram of an AC meter. In this depiction, the full-wave rectifier precedes the meter movement. The movement responds to the average value of the pulsating DC. The scale can then be calibrated to show anything the designer wants. In most cases, it is the root mean square (RMS) value or peak value.

Electrodynamometer Meter Movement

The electrodynamometer can be used to measure alternating or direct voltage and current. It operates on the same principles as the permanent magnet moving coil meter, except that the permanent magnet is replaced by an air core electromagnet. The field of the electrodynamometer is developed by the same current that flows through the moving coil. *[Figure 12-160]*

Because this movement contains no iron, the electrodynamometer can be used as a movement for both AC and DC instruments. AC can be measured by connecting the stationary and moving coils in series. Whenever the current in the moving coil reverses, the magnetic field produced by the stationary coil reverses. Regardless of the direction of the current, the needle moves in a clockwise direction. However, for either voltmeter or ammeter applications, the electrodynamometer is too expensive to economically compete with the d'Arsonval-type movement.

Moving Iron Vane Meter

The moving iron vane meter is another basic type of meter. It can be used to measure either AC or DC. Unlike the d'Arsonval meter, which employs permanent magnets, it depends on induced magnetism for its operation. It utilizes the principle of repulsion between two concentric iron vanes, one fixed and one movable, placed inside a solenoid. A pointer is attached to the movable vane. *[Figure 12-161]*

When current flows through the coil, the two iron vanes become magnetized with North poles at their upper ends and South poles at their lower ends for one direction of current through the coil. Because like poles repel, the unbalanced component of force, tangent to the movable element, causes it to turn against the force exerted by the springs.

The movable vane is rectangular in shape and the fixed vane is tapered. This design permits the use of a relatively uniform scale.

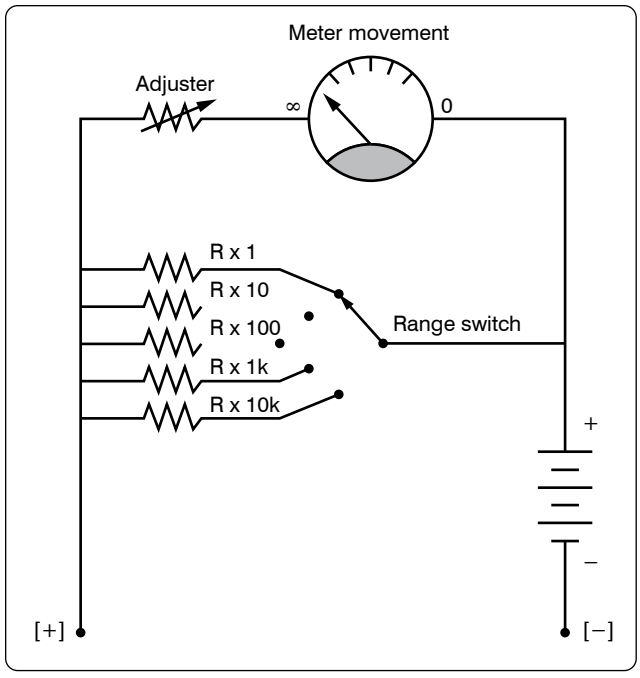


Figure 12-157. *Multirange ohmmeter.*

When no current flows through the coil, the movable vane is positioned so that it is opposite the larger portion of the tapered fixed vane, and the scale reading is zero. The amount of magnetization of the vanes depends on the strength of the field, which, in turn, depends on the amount of current flowing through the coil.

The force of repulsion is greater opposite the larger end of the fixed vane than it is nearer the smaller end. Therefore, the movable vane moves toward the smaller end through an angle that is proportional to the magnitude of the coil current. The movement ceases when the force of repulsion is balanced by the restraining force of the spring.

Because the repulsion is always in the same direction (toward the smaller end of the fixed vane), regardless of the direction of current flow through the coil, the moving iron vane instrument operates on either DC or AC circuits.

Mechanical damping in this type of instrument can be obtained by the use of an aluminum vane attached to the shaft so that, as the shaft moves, the vane moves in a restricted air space.

When the moving iron vane meter is used as an ammeter, the coil is wound with relatively few turns of large wire in order to carry the rated current. When the moving iron vane meter is used as a voltmeter, the solenoid is wound with many turns of small wire. Portable voltmeters are made with self-contained series resistance for ranges up to 750 volts. Higher ranges are obtained by the use of additional external multipliers.

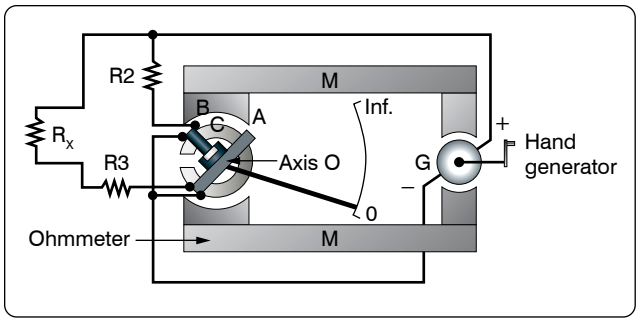


Figure 12-158. *Simplified megger circuit.*

The moving iron vane instrument may be used to measure DC but has an error due to residual magnetism in the vanes. Reversing the meter connections and averaging the readings may minimize the error. When used on AC circuits, the instrument has an accuracy of 0.5 percent. Because of its simplicity, relatively low cost, and the fact that no current is conducted to the moving element, this type of movement is used extensively to measure current and voltage in AC power circuits. However, because the reluctance of the magnetic circuit is high, the moving iron vane meter requires much more power to produce full-scale deflection than is required by a d'Arsonval meter of the same range. Therefore, the moving iron vane meter is seldom used in high-resistance low-power circuits.

Inclined Coil Iron Vane Meter

The principle of the moving iron vane mechanism is applied to the inclined coil type of meter, which can be used to measure both AC and DC. The inclined coil, iron vane meter has a coil mounted at an angle to the shaft. Attached obliquely to the shaft, and located inside the coil, are two soft iron vanes. When no current flows through the coil, a control spring holds the pointer at zero, and the iron vanes lie in planes parallel to the plane of the coil. When current flows through the coil, the vanes tend to line up with magnetic lines passing through the center of the coil at right angles to the plane of the coil. Thus, the vanes rotate against the spring action to move the pointer over the scale.

The iron vanes tend to line up with the magnetic lines regardless of the direction of current flow through the coil. Therefore, the inclined coil, iron vane meter can be used to measure either AC or DC. The aluminum disk and the drag magnets provide electromagnetic damping.

Like the moving iron vane meter, the inclined coil type requires a relatively large amount of current for full-scale deflection and is seldom used in high-resistance low-power circuits.

As in the moving iron vane instruments, the inclined coil instrument is wound with few turns of relatively large wire when used as an ammeter and with many turns of small wire when used as a voltmeter.

Varmeters

Multiplying the volts by the amperes in an AC circuit gives the apparent power: the combination of the true power (which does the work) and the reactive power (which does no work and is returned to the line). Reactive power is measured in units of vars (volt-amperes reactive) or kilovars (kilovolt-amperes reactive (kVAR). When properly connected, wattmeters measure the reactive power. As such, they are called varmeters. *[Figure 12-162]*

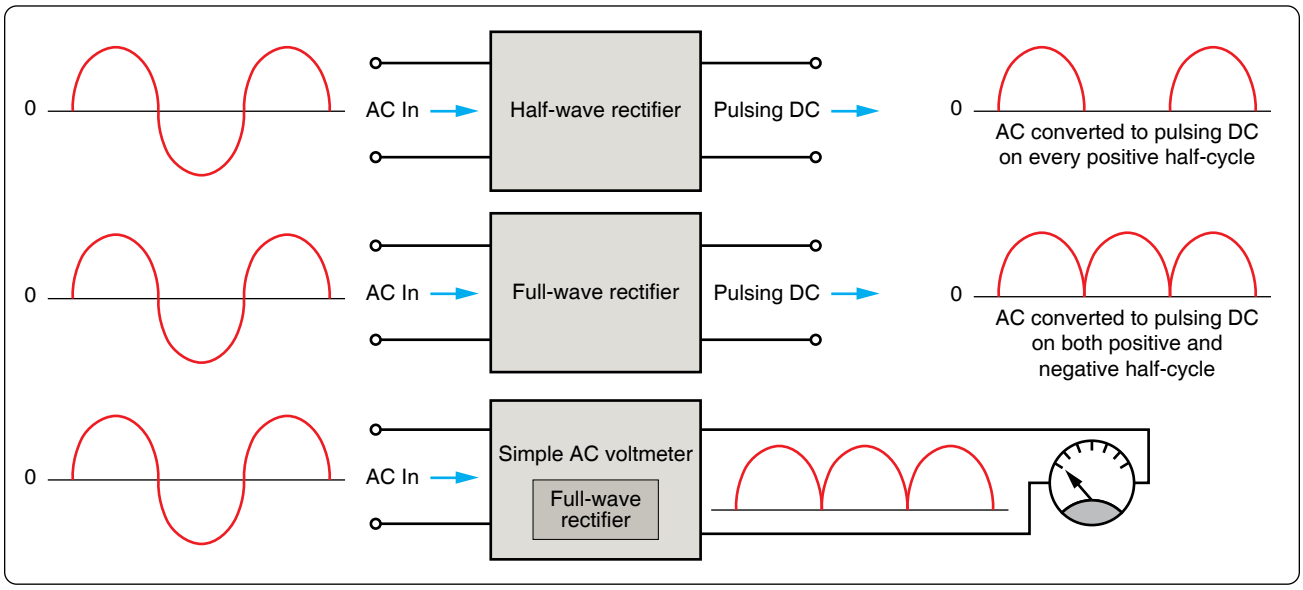


Figure 12-159. *Simplified block diagram of AC meter.*

Wattmeter

Electric power is measured by means of a wattmeter. Because electric power is the product of current and voltage, a wattmeter must have two elements, one for current and the other for voltage. For this reason, wattmeters are usually of the electrodynamometer type. *[Figure 12-163]*

The movable coil with a series resistance forms the voltage element, and the stationary coils constitute the current element. The strength of the field around the potential coil depends on the amount of current that flows through it. The current, in turn, depends on the load voltage applied across the coil and the high resistance in series with it. The strength of the field around the current coils depends on the amount of current flowing through the load. Thus, the meter deflection is proportional to the product of the voltage across the potential coil and the current through the current coils. The effect is almost the same (if the scale is properly calibrated) as if the voltage applied across the load and the current through the load were multiplied together.

If the current in the line is reversed, the direction of current in both coils and the potential coil is reversed, the net result is that the pointer continues to read up scale. Therefore, this type of wattmeter can be used to measure either AC or DC power.

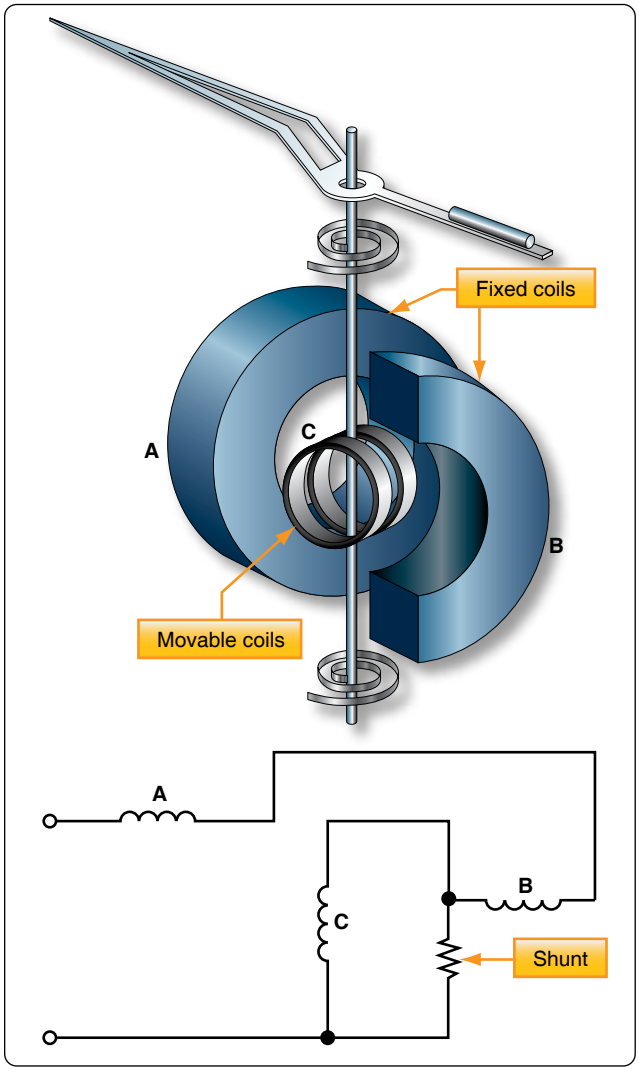


Figure 12-160. *Simplified diagram of an electrodynamometer movement.*

Frequency Measurement/Oscilloscope

The oscilloscope is by far one of the more useful electronic measurements available. The viewing capabilities of the oscilloscope make it possible to see and quantify various waveform characteristics, such as phase relationships, amplitudes, and durations. While oscilloscopes come in a variety of configurations and presentations, the basic operation is typically the same. Most oscilloscopes in general bench or shop applications use a cathode-ray tube (CRT), which is the device or screen that displays the waveforms.

The CRT is a vacuum instrument that contains an electron gun, which emits a very narrow and focused beam of electrons. A phosphorescent coat applied to the back of the screen forms the screen. The beam is electronically aimed and accelerated so that the electron beam strikes the screen. When the electron beam strikes the screen, light is emitted at the point of impact.

Figure 12-164 shows the basic components of the CRT with a block diagram. The heated cathode emits electrons. The magnitude of voltage on the control grid determines the actual flow of electrons and thus controls the intensity of the electron beam. The acceleration anodes increase the speed of the electrons, and the focusing anode narrows the beam down to a fine point. The surface of the screen is also an anode and assists in the acceleration of the electron beam.

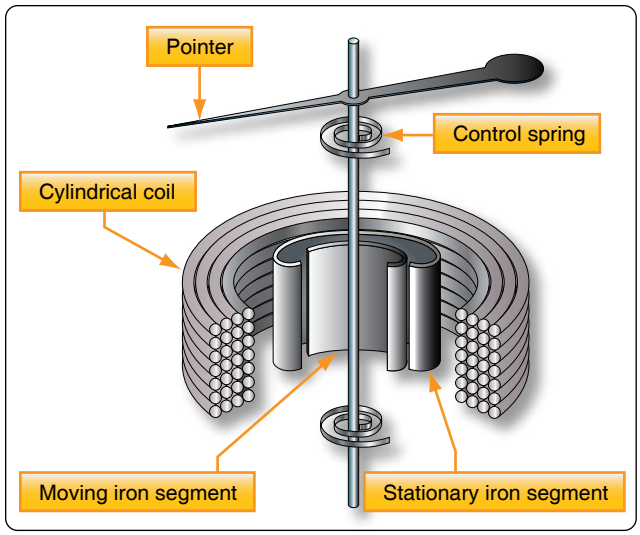


Figure 12-161. *Moving iron vane meter.*

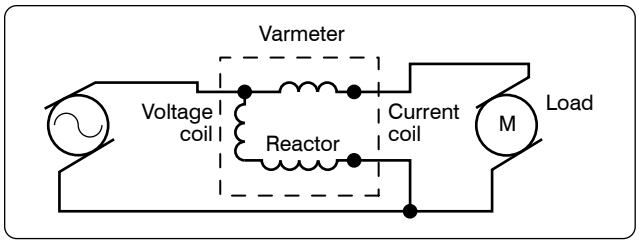


Figure 12-162. *A varmeter connected in an AC circuit.*

The purpose of the vertical and horizontal deflection plates is to bend the electron beam and position it to a specific point of the screen. *[Figure 12-165]* By providing a neutral or zero voltage to a deflection plate, the electron beam is unaffected. By applying a negative voltage to a plate, the electron beam is repelled and driven away from the plate. Finally, by applying a positive voltage, the electron beam is drawing to the plate. *Figure 12-165* provides a few possible plate voltage combinations and the resultant beam position.

Horizontal Deflection

To get a visual representation of the input signal, an internally generated saw-tooth voltage is generated and then applied to the horizontal deflection plates. *Figure 12-166* illustrates that the saw-tooth is a pattern of voltage applied, which begins at a negative voltage and increases at a constant rate to a positive voltage. This applied varying voltage draws or traces the electron beam from the far left of the screen to the far right side of the screen. The resulting display is a straight line, if the sweep rate is fast enough. This saw-tooth applied voltage is a repetitive signal so that the beam is repeatedly swept across the tube. The rate at which the saw-tooth voltage goes from negative to positive is determined by the frequency. This rate then establishes the sweep rate of the beam. When the saw-tooth reaches the end of its sweep from left to right, the beam then rapidly returns to the left side and is ready to make another sweep. During this time, the electron beam is stopped or blanked out and does not produce any kind of a trace. This period of time is called flyback.

Vertical Deflection

If this same signal were applied to the vertical plates, it would also produce a vertical line by causing the beam to trace from the down position to the up position.

Tracing a Sine Wave

Reproducing the sine wave on the oscilloscope combines both the vertical and horizontal deflection patterns. *[Figure 12-167]* If the sine wave voltage signal is applied across the vertical deflection plates, the result will be the vertical beam oscillation up and down on the screen. The amount that the beam moves above the centerline depends on the peak value of the voltage.

While the beam is being swept from the left to the right by the horizontal plates, the sine wave voltage is being applied to the vertical plates, causing the form of the input signal to be traced out on the screen.

Control Features on an Oscilloscope

While there are many different styles of oscilloscopes, which range from the simple to the complex, they all have some controls in common. Apart from the screen and the ON/OFF switch, some of these controls are listed as follows:

- Horizontal Position—allows for the adjustment of the neutral horizontal position of the beam. Use this control to reposition the waveform display in order to have a better view of the wave or to take measurements.
- Vertical Position—moves the traced image up or down allowing better observations and measurements.
- Focus—controls the electron beam as it is aimed and converges on the screen. When the beam is in sharp focus, it is narrowed down to a very fine point and does not have a fuzzy appearance.
- Intensity—essentially the brightness of the trace. Controlling the flow of electrons onto the screen varies the intensity. Do not keep the intensity too high for extended testing or when the beam is motionless and forms a dot on the screen. This can damage the screen.
- Seconds/Division—a time-based control that sets the horizontal sweep rate. Basically, the switch is used to select the time interval that each division on the horizontal scale represents. These divisions can be seconds, milliseconds, or even microseconds. A simple example would be if the technician had the seconds/division control set to 10 µS. If this technician is viewing a waveform that has a period of 4 divisions on the screen, then the period would be 40 µS. The frequency of this waveform can then be determined by taking the inverse of the period. In this case, $\frac{1}{40}$ µS equals a frequency of 25 kHz.
- Volts/Division—used to select the voltage interval that each division on the vertical scale represents. For example, suppose each vertical division was set to equal 10 mV. If a waveform was measured and had a peak value of 4 divisions, then the peak value in voltage would be 40 mV.
- Trigger—The trigger control provides synchronization between the saw-tooth horizontal sweep and the applied signal on the vertical plates. The benefit is that the waveform on the screen appears to be stationary and fixed and not drifting across the screen. A triggering circuit is used to initiate the start of a sweep rather than

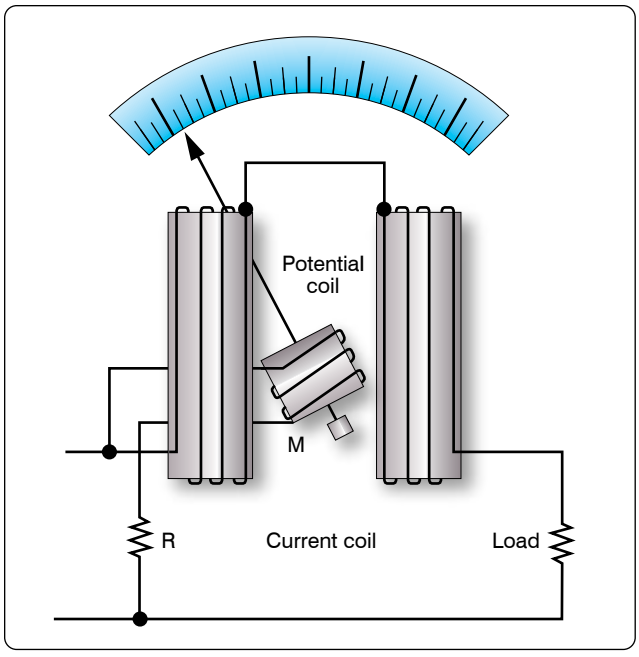


Figure 12-163. *Simplified electrodynamometer wattmeter circuit.*

the fixed saw-tooth sweep rate. In a typical oscilloscope, this triggering signal comes from the input signal itself at a selected point during the signal's cycle. The horizontal signal goes through one sweep, retraces back to the left side and waits there until it is triggered again by the input signal to start another sweep.

Flat Panel Color Displays for Oscilloscopes

While the standard CRT design of oscilloscope is still in service, the technology of display and control has evolved into use of the flat panel monitors. Furthermore, the newer oscilloscopes can even be integrated with the common personal computer (PC). *[Figure 12-168]* Some of the features of this technology include easy data capture, data transfer, documentation, and data analysis. Hand-held oscilloscopes are now available that can perform the functions of larger bench type equipment but are mobile and great tools for trouble shooting. *[Figure 12-169]*

Digital Multimeter

Traditionally, the meters that technicians have used have been the analog voltmeter, ammeter, and the ohmmeter. These have usually been combined into the same instrument and called a multimeter or a VOM (volt-ohm-milliammeter). This approach has been both convenient and economical. Digital multimeters (DMM) and digital voltmeters (DVM) are more common due to their ease of use. These meters are easier to read and provide greater accuracy when compared to the older analog units with needle movement. The multimeter's single-coil movement requires a number of scales, which are not always easy to read accurately. In addition, the loading characteristics due to the internal resistance sometimes affect the circuit and the measurements. Not only does the DVM offer greater accuracy and less ambiguity, but also higher input resistance, which has less of a loading effect and influence on a circuit.

Basic Circuit Analysis & Troubleshooting

Troubleshooting is the systematic process of recognizing the symptoms of a problem, identifying the possible cause, and locating the failed component or conductor in the circuit. To be proficient at troubleshooting, the technician must understand how the circuit operates and know how to properly use the test equipment. There are many ways in which a system can fail and to cover all of the possibilities is beyond the scope of this handbook. However, there are some basic concepts that enable the technician to handle many of the common faults encountered in the aircraft.

Before starting a discussion on basic circuits and troubleshooting, the following definitions are given.

- Short circuit—an unintentional low resistance path between two components in a circuit or between a

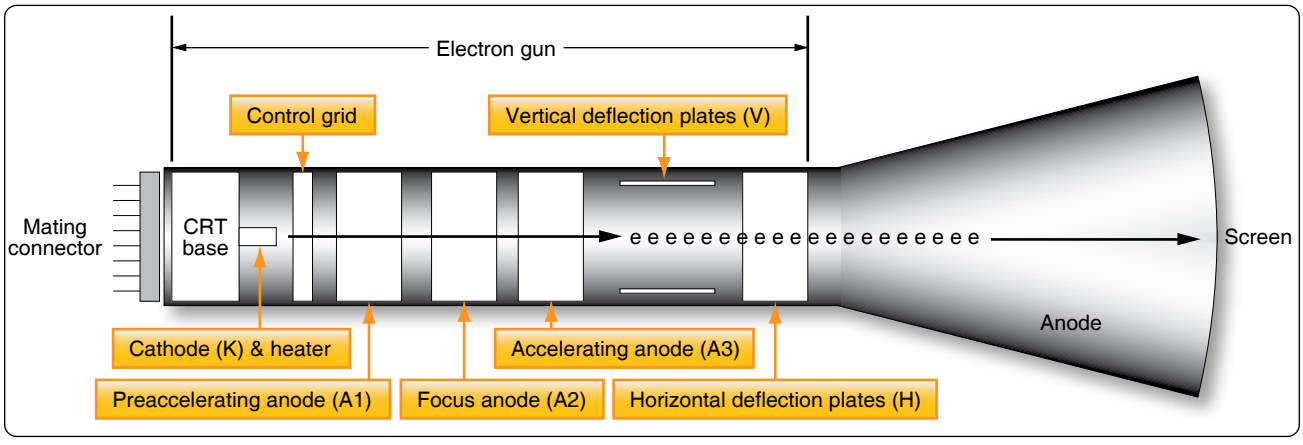


Figure 12-164. *Basic components of the CRT with a block diagram.*

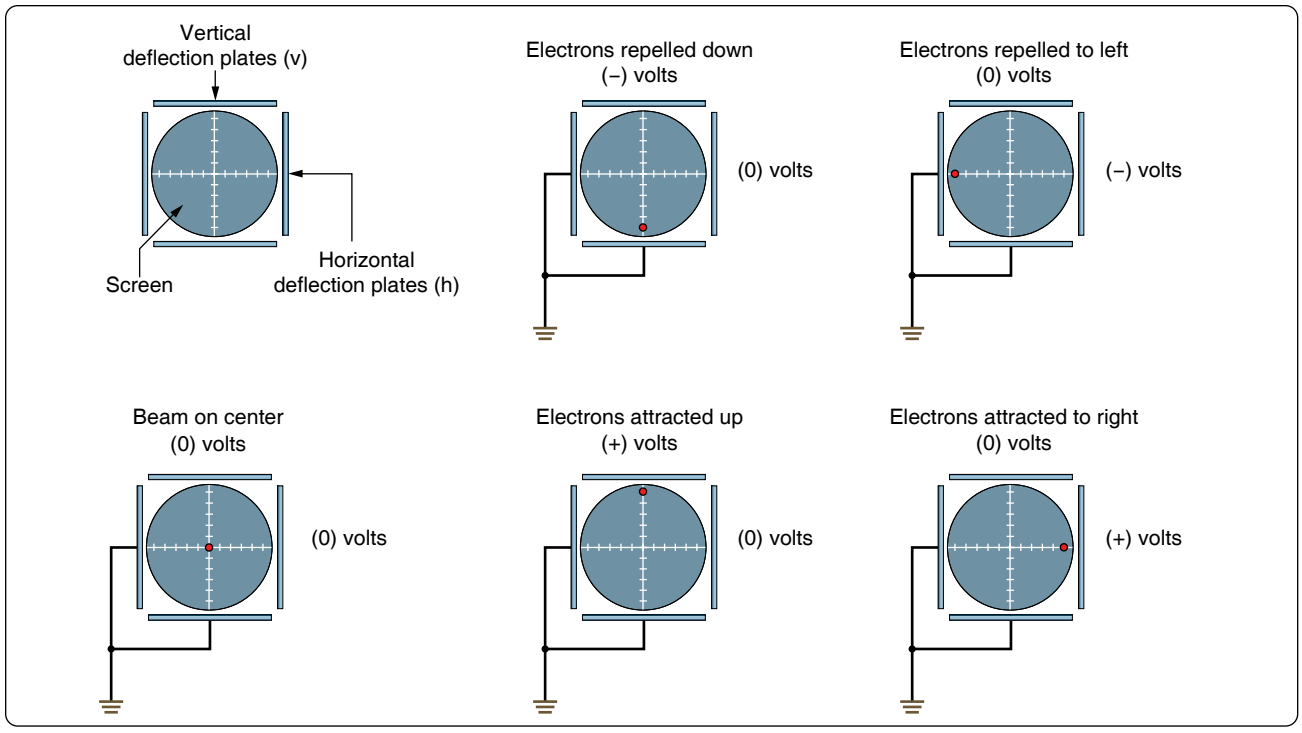


Figure 12-165. *Possible plate voltage combinations and the resultant beam position.*

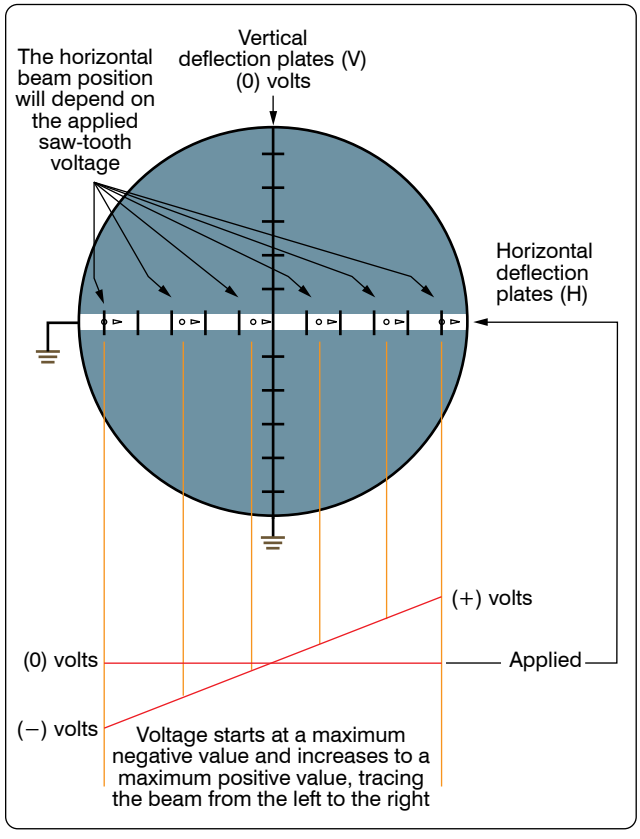


Figure 12-166. *Saw-tooth applied voltage.*

component/conductor and ground. It usually creates high current flow, which burns out or causes damage to the circuit conductor or components.

- Open circuit—a circuit that is not a complete or continuous path. An open circuit represents an infinitely large resistance. Switches are common devices used to open and close a circuit. Sometimes a circuit opens due to a component failure, such as a light bulb or a burned out resistor.
- Continuity—the state of being continuous, uninterrupted or connected together; the opposite of a circuit that is not broken or does not have an open.
- Discontinuity—the opposite of continuity, indicating that a circuit is broken or not continuous.

Voltage Measurement

Voltage is measured across a component with a voltmeter or the voltmeter position on a multimeter. Usually, there is a DC and an AC selection on the meter. Before the meter is used for measurements, make sure that the meter is selected for the correct type of voltage. When placing the probes across a component to take a measurement, take care to ensure that the polarity is correct. *[Figure 12-170]* Standard practice is for the red meter lead to be installed in the positive (+) jack and the black meter lead to be installed in the negative meter jack (–). Then when placing the probes across or in parallel with a component to measure the voltage, the leads should match the polarity of the component. The red lead is on the positive

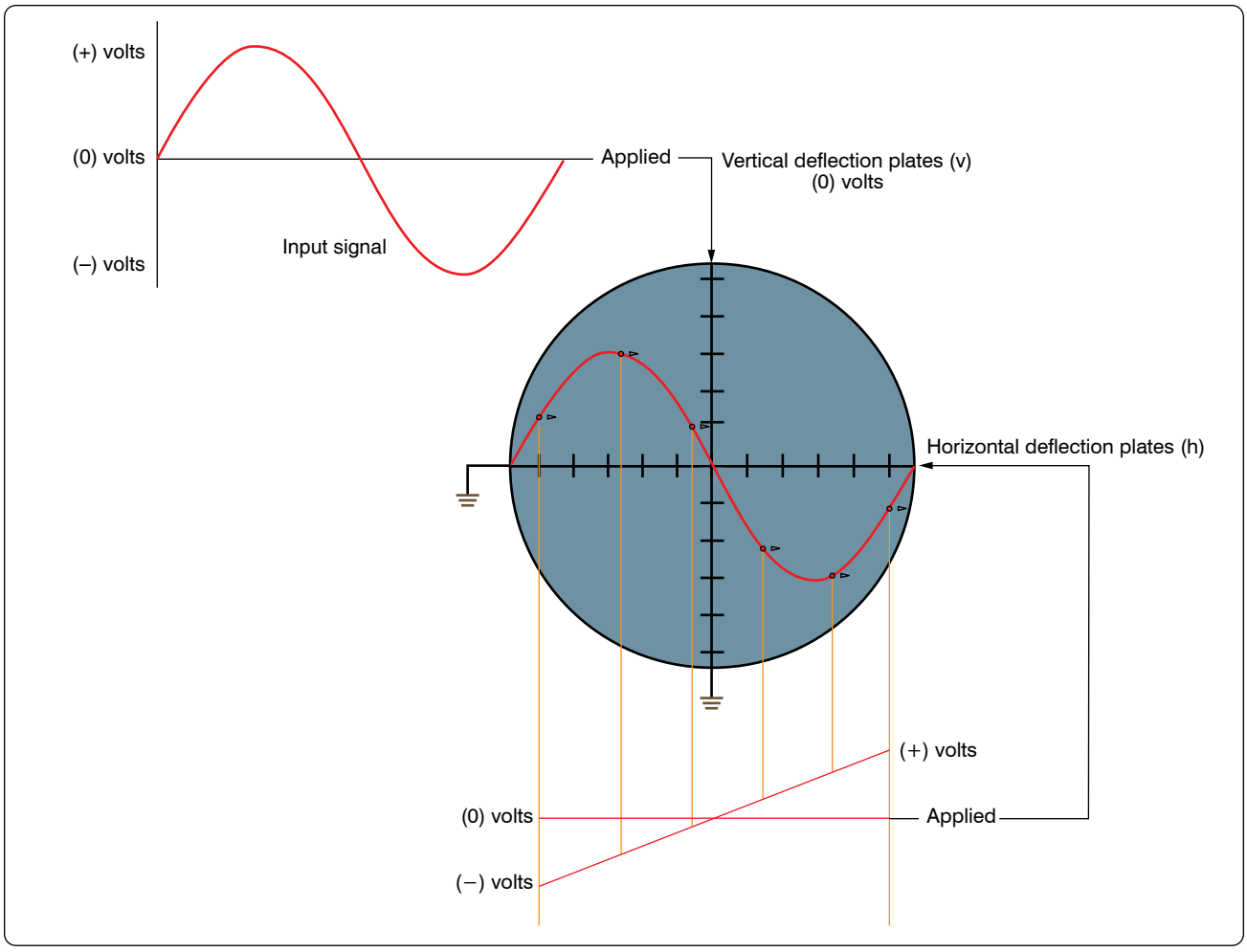


Figure 12-167. *Sine wave voltage signal.*

side of the component and the black is on the negative side, which prevents damage to the meter or incorrect readings.

All meters have some resistance and will shunt some of the current. This has the effect of changing the characteristic of the circuit because of this change in current. This is typically more of a concern with older analog type meters. If there are any questions about the magnitude of the voltage across a component, then the meter should be set to measure on the highest voltage range. This prevents the meter from "pegging" and possible damage. The range should then be selected to low values until the measured voltage is read at the mid-scale deflection. Readings taken at mid-scale are the most accurate.

Current Measurement

Current is measured with the ammeter connected in the current path by opening or breaking the circuit and inserting the meter in series. *[Figure 12-170]* Standard practice is for the red meter lead to be installed in the positive (+) jack and the black meter lead to be installed in the negative meter jack (−). The positive side of the meter is connected towards the positive voltage source. Ideally, the meter should not alter the current and influence the circuit and the measurements. However, the meter does have some effect because of its internal resistance that is connected with the rest of the circuit in series. The resistance is rather small and for most practical purposes, this can be neglected.

Checking Resistance in a Circuit

The ohmmeter is used to measure the resistance. In its more basic form, the ohmmeter consists of a variable resistor in series with a meter movement and a voltage source. The meter must first be adjusted before use.

Refer to *Figure 12-171* for meter configurations during adjustments. When the meter leads are not connected (open), the needle points to the full left-hand position, indicating infinite resistance or and open circuit. With the lead placed together, the circuit is shorted as shown with the meter needle to the full right-hand position. When a connection is made, the internal battery is allowed to produce a current through the movement coil, causing a deflection of the

needle in proportion to the value of the external resistance. In this case, the resistance is zero because the leads are shorted.

The purpose of the variable resistor in the meter is to adjust the current so that the pointer reads exactly zero when the leads are shorted. This is needed because as the battery continues to be used, the voltage changes, thus requiring an adjustment. The meter should be "zeroed" before each use.

To check the value of a resistor, the resistor must be disconnected from the circuit. This prevents any possible damage to the ohmmeter, and it prevents the possibility of any inaccurate readings due to the circuit being in parallel with the resistor in question. *[Figure 12-172]*

Continuity Checks

In many cases, the ohmmeter is not used for measuring the resistance of a component but to simply check the integrity of a connection from one portion of a circuit to another. If there is a good connection, then the ohmmeter reads a near zero resistance or a short. If the circuit is open or has a very poor connection at some point like an over-crimped pin in a connector, then the ohmmeter reads infinity or some very high resistance. Keep in mind that while any measurement is being taken, contact with the circuit or probes should be avoided. Contact can introduce another parallel path and provide misleading indications.

Capacitance Measurement

Figure 12-173 illustrates a basic test of a capacitor with an ohmmeter. There are usually two common modes of failure for a capacitor. One is a complete failure characterized by short circuit through the capacitor due to the dielectric breaking down or an open circuit. The more insidious failure occurs due to degradation, which is a gradual deterioration of the capacitor's characteristics.

If a problem is suspected, remove the capacitor from the circuit and check with an ohmmeter. The first step is to short the two leads of the capacitor to ensure that it is entirely discharged. Next, connect the two leads as shown in *Figure 12-173* across the capacitor and observe the needle movement. At first, the needle should indicate a short circuit. Then as the capacitor begins to charge, the needle should move to the left or infinity and eventually indicate an open circuit. The capacitor takes its charge from the internal battery of the ohmmeter. The greater the capacitance, the longer it takes to charge. If the capacitor is shorted, then the needle remains at a very low or shorted resistance. If there is some internal deterioration of the dielectric, then the needle never reaches a high resistance but some intermediate value, indicating a current.

Figure 12-168. *Oscilloscope with flat panel display.*

Figure 12-169. *Handheld oscilloscope.*

Inductance Measurement

The common mode of failure in an inductor is an open. To check the integrity of an inductor, it must be removed from the circuit and tested as an isolated component just like the capacitor. If there is an open in the inductor, a simple check with an ohmmeter shows it as an open circuit with infinite resistance. If in fact the inductor is in good condition, then the ohmmeter indicates the resistance of the coil.

On occasions, the inductor fails due to overheating. When the inductor is overheated, it is possible for the insulation covering the wire in the coil to melt, causing a short. The effects of a shorted coil are that of reducing the number of turns. At this point, further testing of the inductor must be

done with test equipment not covered in this handbook.

Troubleshooting Open Faults in a Series Circuit

One of the most common modes of failure is the "open" circuit. A component, such as a resistor, can overheat due to the power rating being exceeded. Other more frustrating problems can happen when a "cold" solder joint cracks leaving a wire disconnected from a relay or connector. This type of damage can occur during routine maintenance after a technician has accessed an area for inspections. In many cases, there is no visual indication that a failure has occurred, and the soon-to-be-frustrated technician is unaware that there is a problem until power is reapplied to the aircraft in the final days leading up to aircraft delivery and scheduled operations.

The first example is a simplified diagram shown in *Figures 12-174* through *12-176*. The circuit depicted in *Figure 12-174* is designed to cause current to flow through a lamp, but because of the open resistor, the lamp will not light. To locate this open, a voltmeter or an ohmmeter should be used.

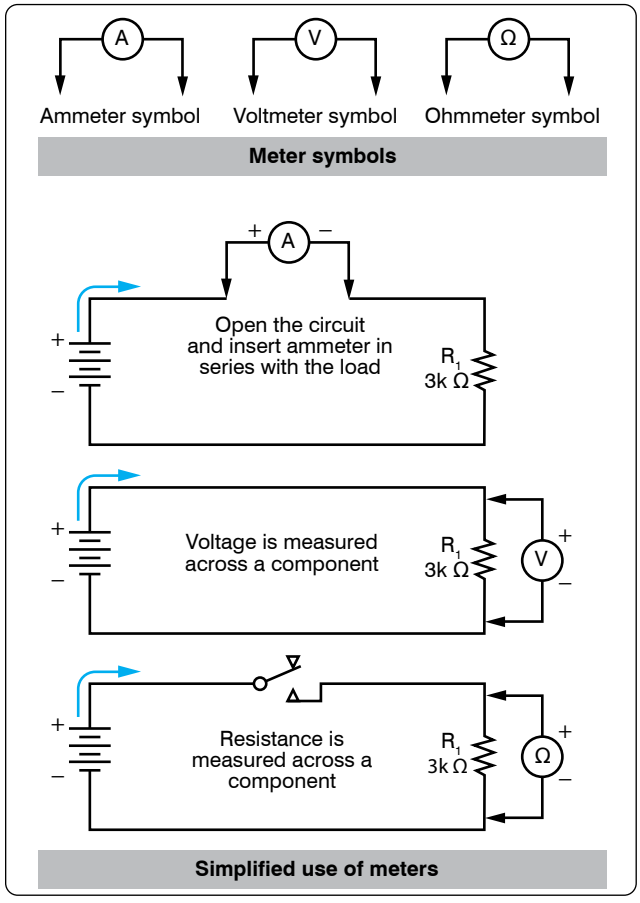


Figure 12-170. *Current, voltage, and resistance measurement.*

Tracing Opens with the Voltmeter

A general procedure to follow in this case is to measure the voltage drop across each component in the circuit, keeping in mind the following points. If there is an open in a series circuit, then the voltage drops on sides of the component. In this case, the total voltage must appear across the open resistor as per Kirchhoff's Voltage Law.

If a voltmeter is connected across the lamp, as shown in *Figure 12-175*, the voltmeter reads zero. Since no current can flow in the circuit because of the open resistor, there is no voltage drop across the lamp indicating that the lamp is good.

Next, the voltmeter is connected across the open resistor, as shown in *Figure 12-176*. The voltmeter has closed the circuit by shunting (paralleling) the burned out resistor, allowing current to flow. Current flows from the negative terminal of the battery, through the switch, through the voltmeter and the lamp, back to the positive terminal of the battery. However, the resistance of the voltmeter is so high that only a very small current flows in the circuit. The current is too small to light the lamp, but the voltmeter reads the battery voltage.

Tracing Opens with the Ohmmeter

A simplified circuit, as shown in *Figures 12-177* and *12-178,* illustrates how to locate an open in a series circuit using the ohmmeter. A general rule to keep in mind when troubleshooting with an ohmmeter is: when an ohmmeter is properly connected across a circuit component and a resistance reading is obtained, the component has continuity and is not open.

When an ohmmeter is used, the circuit component to be tested must be isolated and the power source removed from the circuit. In this case, these requirements can be met by opening the circuit switch as shown in *Figure 12-177*. The ohmmeter is zeroed and across all good components is zero. The voltage drop across the open component equals the total voltage across the series combination. This condition happens because the open component prevents current to pass through the series circuit. With there being no current, there can be no voltage drop across any of the good components. Because the current is zero, it can be determined by Ohm's Law that $E = IR = 0$ volts across a component. The voltage is the same on both places across (in parallel with) the lamp. In this testing configuration, some value of resistance is read indicating that the lamp is in good condition and is not the source of the open in the circuit.

Now the technician should move to the resistor and place the ohmmeter probe across it as shown in *Figure 12-178*. When the ohmmeter is connected across the open resistor, it indicates infinite resistance, or a discontinuity. Thus, the circuit open has now been located.

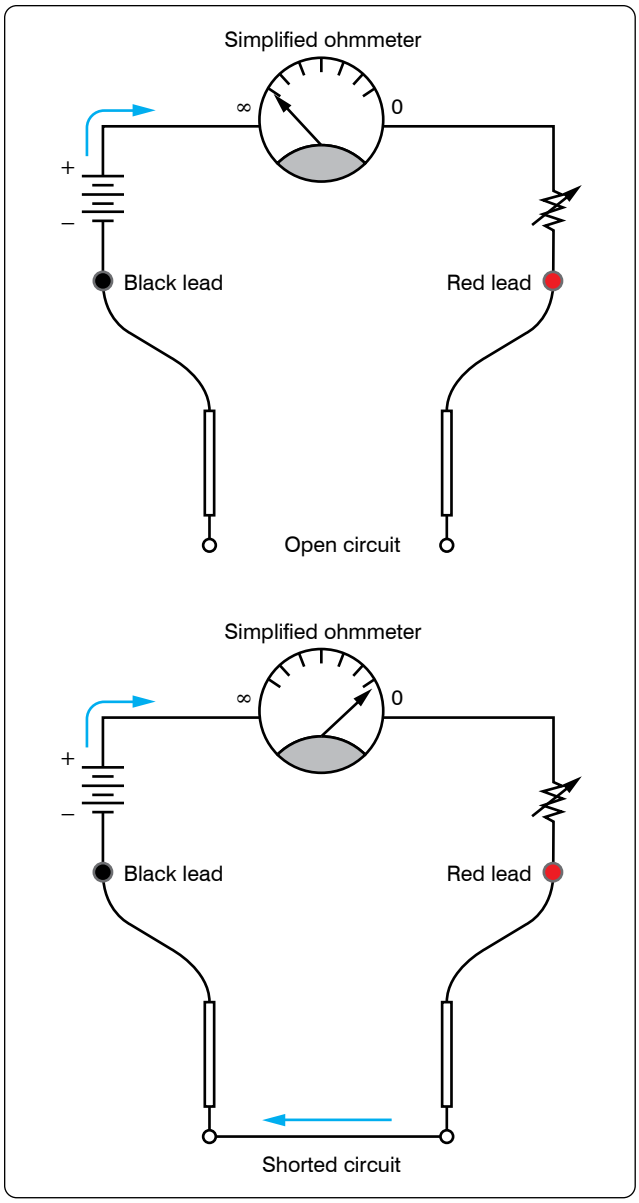


Figure 12-171. *Meter configurations during adjustments.*

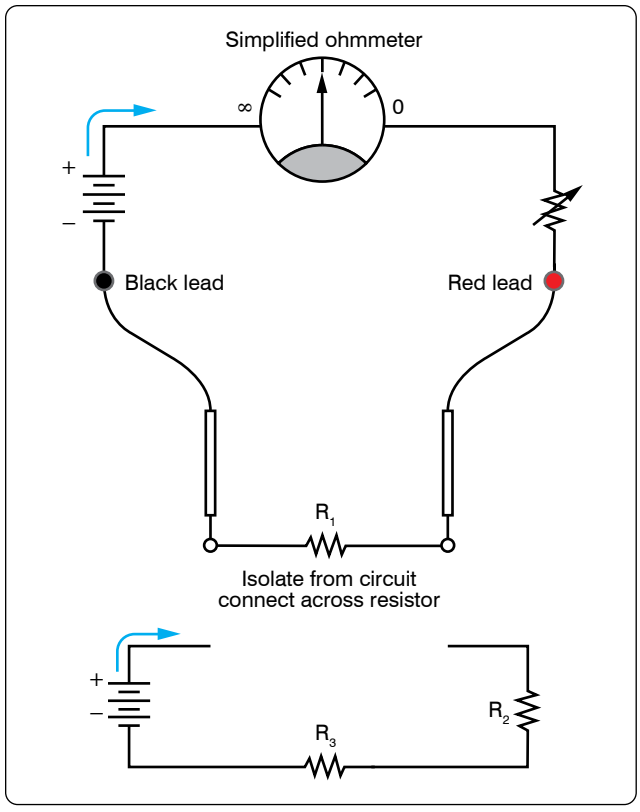


Figure 12-172. *Meter adjustment.*

Troubleshooting Shorting Faults in a Series Circuit

An open fault can cause a component or system not to work, which can be critical and hazardous. A shorting fault can potentially be more of a severe nature than the open type of fault. A short circuit, or "short," causes the opposite effect. A short across a series circuit produces a greater than normal current flow. Faults of this type can develop slowly when a wire bundle is not properly secured and is allowed to chafe against the airframe structure or other systems, such as hydraulic lines. Shorts can also occur due to a careless technician using incorrect hardware when installing an interior. If screws that are too long are used to install trim, it is possible to penetrate a wire bundle immediately causing numerous shorts. Worse yet, are the shorts that are not immediately seen but "latent" and do not show symptoms until the aircraft is in service. Another point to keep in mind is when closing panels. Wires can become pinched between the panel and the airframe causing either a short or a latent, intermittent short. The simplified circuit, shown in *Figures 12-179* through *12-182* is used to illustrate troubleshooting a short in a series circuit.

In *Figure 12-179*, a circuit is designed to light a lamp. A resistor is connected in the circuit to limit current flow. If the resistor is shorted, as shown in the illustration, the current flow increases and the lamp becomes brighter. If the applied voltage were high enough, the lamp would burn out, but in this case the fuse would protect the lamp by opening first. Usually a short circuit produces an open circuit by either blowing (opening) the fuse or burning out a circuit component. But in some circuits, there may be additional resistors which do not allow one shorted resistor to increase the current flow enough to blow the fuse or burn out a component. *[Figure 12-180]* Thus, with one resistor shorted out, the circuit still functions since the power dissipated by the other resistors does not exceed the rating of the fuse.

Tracing Shorts with the Ohmmeter

The shorted resistor can be located with an ohmmeter. *[Figure 12-181]* First the switch is opened to isolate the

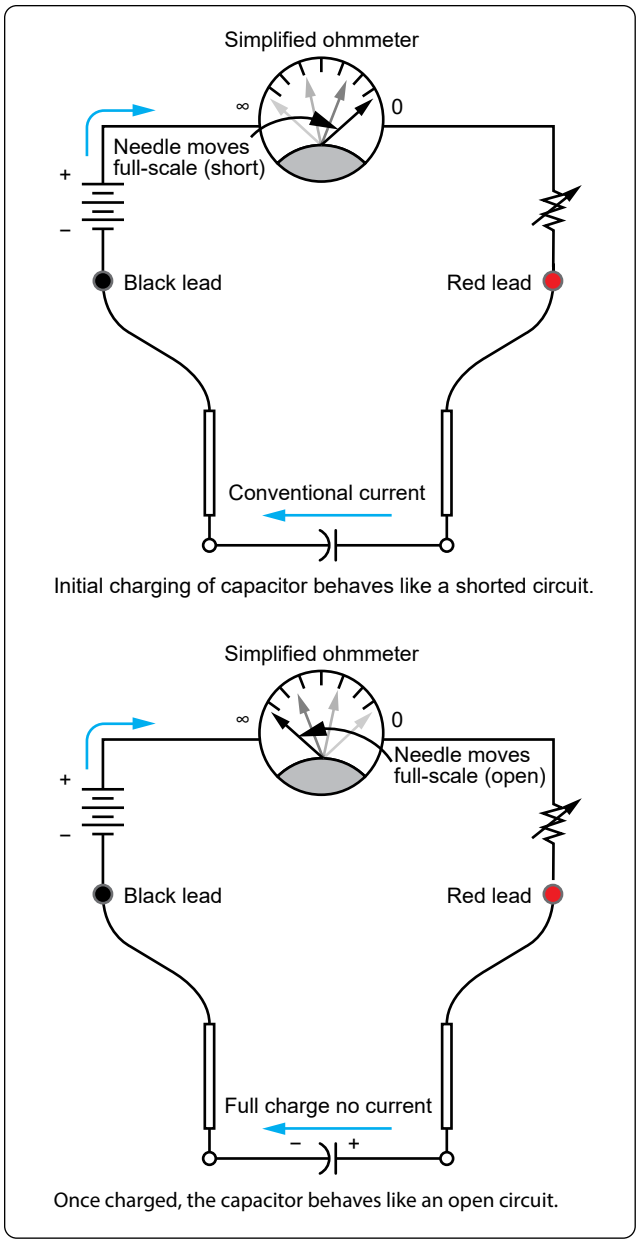


Figure 12-173. *Basic test of a capacitor with an ohmmeter.*

circuit components. In *Figure 12-181*, this circuit is shown with an ohmmeter connected across each of the resistors. Only the ohmmeter connected across the shorted resistor shows a zero reading, indicating that this resistor is shorted.

Tracing Shorts with the Voltmeter

To locate the shorted resistor while the circuit is functioning, a voltmeter can be used. *Figure 12-182* illustrates that when a voltmeter is connected across any of the resistors that are not shorted, a portion of the applied voltage is indicated on the voltmeter scale. When it is connected across the shorted resistor, the voltmeter reads zero.

Troubleshooting Open Faults in a Parallel Circuit

The procedures used in troubleshooting a parallel circuit are sometimes different from those used in a series circuit. Unlike a series circuit, a parallel circuit has more than one path in which current flows. A voltmeter cannot be used, since, when it is placed across an open resistor, it reads the voltage drop in a parallel branch. But an ammeter or the modified use of an ohmmeter can be employed to detect an open branch in a parallel circuit.

If the open resistor shown in *Figure 12-183* was not visually apparent, the circuit might appear to be functioning properly, because current would continue to flow in the other two branches of the circuit. To determine that the circuit is not operating properly, a determination must be made as to how the circuit should behave when working properly. First, the total resistance, total current, and the branch currents of the circuit should be calculated as if there were no open in the circuit. In this case, the total resistance can be simply determined by:

$$R_T = \frac{R}{N}$$

Where R_T is the total circuit resistance
N is the number of resistors
R is the resistor value

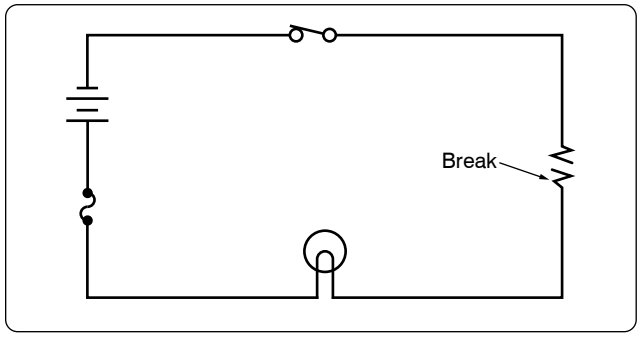


Figure 12-174. *An open circuit.*

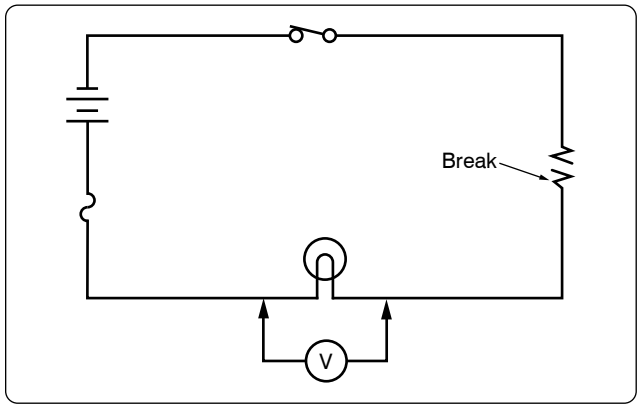


Figure 12-175. *Voltmeter across a lamp in an open circuit.*

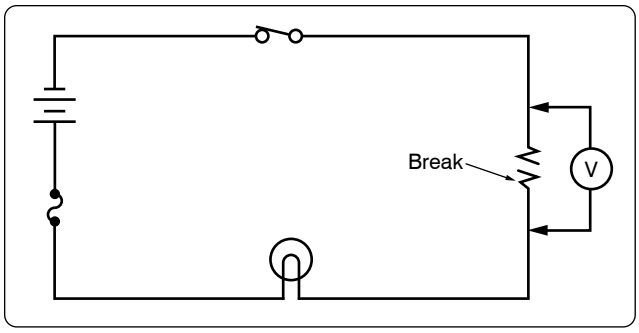


Figure 12-176. *Voltmeter across a resistor in an open circuit.*

$$R_T = \frac{30\ \Omega}{3} = 10\ \Omega$$

The total current of the circuit can now be determined by using Ohm's Law:

$$I_T = \frac{E_S}{R_T}$$

Where I_T is the total current
E_S is the source voltage across the parallel branch
R_T is the total resistance of the parallel branch

$$I_T = \frac{30\ v}{10\ \Omega} = 3 \text{ amperes (total current)}$$

Each branch current should be determined in a similar manner. For the first branch, the current is:

$$I_1 = \frac{E_S}{R_1}$$

Where I_1 is the current in the first branch
E_S is the source voltage across the parallel branch
R_1 is the resistance of the first branch

$$I_1 = \frac{30\ v}{30\ \Omega} = 1 \text{ ampere}$$

Because the other two branches are of the same resistive value, then the current in each of those branches is 1 ampere also. Adding up the amperes in each branch confirms the initial calculation of total current being 3 amperes.

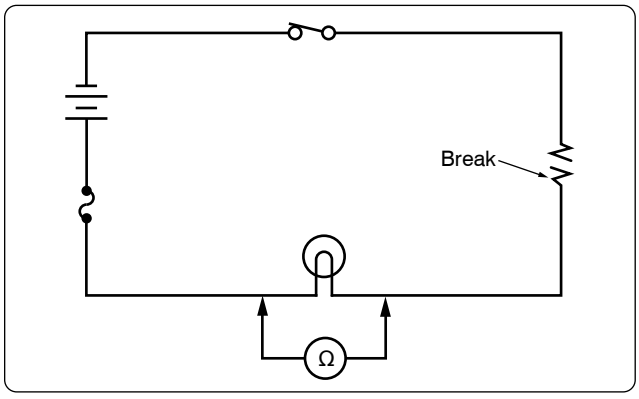


Figure 12-177. *Using an ohmmeter to check a circuit component.*

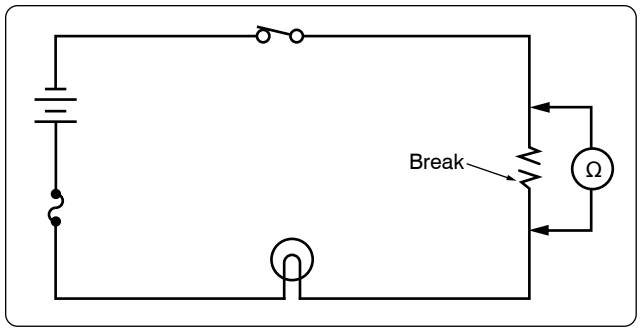


Figure 12-178. *Using an ohmmeter to locate an open in a circuit component.*

Tracing an Open with an Ammeter

If the technician now places an ammeter in the circuit, the total current would be indicated as 2 amperes as shown in *Figure 12-183* instead of the calculated 3 amperes. Since 1 ampere of current should be flowing through each branch, it is obvious that one branch is open. If the ammeter is then connected into the branches, one after another, the open branch is eventually located by a zero ammeter reading.

Tracing an Open with an Ohmmeter

A modified use of the ohmmeter can also locate this type of open. If the ohmmeter is connected across the open resistor, as shown in *Figure 12-184*, an erroneous reading of continuity would be obtained. Even though the circuit switch is open, the open resistor is still in parallel with R_1 and R_2, and the ohmmeter would indicate the open resistor had a resistance of 15 ohms, the equivalent resistance of the parallel combination of R_1 and R_2.

Therefore, it is necessary to open the circuit as shown in *Figure 12-185* in order to check the resistance of R_3. In this way, the resistor is not shunted (paralleled) by R_1 and R_2. The reading on the ohmmeter now indicates infinite resistance, which means the open component has been isolated.

Troubleshooting Shorting Faults in Parallel Circuits

As in a series circuit, a short in a parallel circuit usually causes an open circuit by blowing the fuse. But, unlike a series circuit, one shorted component in a parallel circuit stops current flow by causing the fuse to open. Refer to the circuit in *Figure 12-186*. If resistor R_3 is shorted, a path of almost zero resistance is offered the current, and all the circuit current flows through the branch containing the shorted resistor. Since this is practically the same as connecting a wire between the terminals of the battery, the current rises to an excessive value, and the fuse opens. Since the fuse opens almost as soon as a resistor shorts out, there is no time to

perform a current or voltage check. Thus, troubleshooting a parallel DC circuit for a shorted component should be accomplished with an ohmmeter. But, as in the case of checking for an open resistor in a parallel circuit, a shorted resistor can be detected with an ohmmeter only if one end of the shorted resistor is disconnected and isolated from the rest of the circuit.

Troubleshooting Shorting Faults in Series-Parallel Circuits

Logic in Tracing an Open

Troubleshooting a series-parallel resistive circuit involves locating malfunctions similar to those found in a series or a parallel circuit. *Figures 12-187* through *12-189* illustrate three points of failure in a series-parallel circuit and their generalized effects.

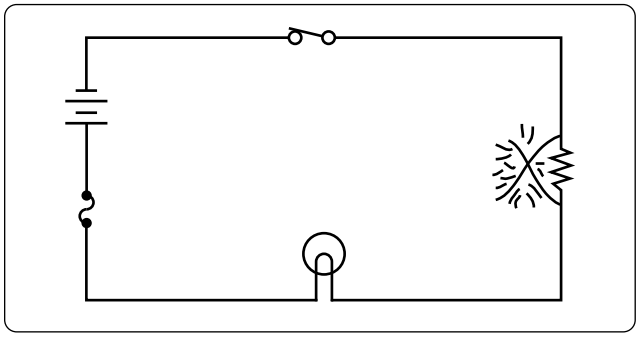


Figure 12-179. *A shorted resistor.*

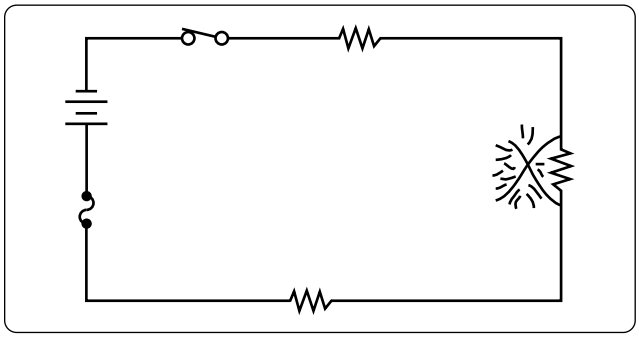


Figure 12-180. *A short that does not open the circuit.*

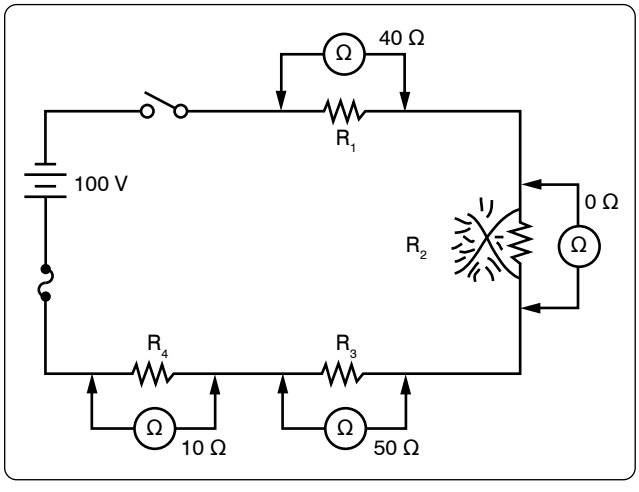


Figure 12-181. *Using an ohmmeter to locate a shorted resistor.*

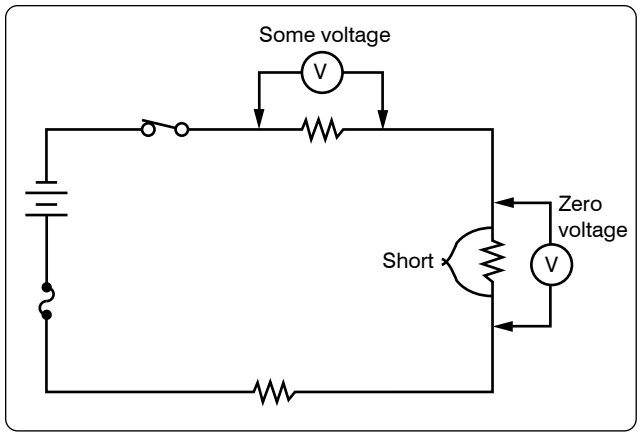


Figure 12-182. *Voltmeter connected across resistors.*

1. In the circuit shown in *Figure 12-187*, an open has occurred in the series portion of the circuit. When the open occurs anywhere in the series portion of a series-parallel circuit, current flow in the entire circuit stops. In this case, the circuit does not function, and the lamp, L_1, is not lit.
2. If the open occurs in the parallel portion of a series-parallel circuit, as shown in *Figure 12-188*, part of the circuit continues to function. In this case, the lamp continues to burn, but its brightness diminishes, since the total resistance of the circuit has increased and the total current has decreased.
3. If the open occurs in the branch containing the lamp, as shown in *Figure 12-189*, the circuit continues to function with increased resistance and decreased current, but the lamp does not light.

Tracing Opens with the Voltmeter

To explain how the voltmeter and ohmmeter can be used to troubleshoot series-parallel circuits, the circuit shown in *Figure 12-190* has been labeled at various points. A point-to-point description is listed below with expected results:

1. By connecting a voltmeter between points A and D, the battery and switch can be checked for opens.
2. By connecting the voltmeter between points A and B, the voltage drop across R_1 can be checked. This voltage drop is a portion of the applied voltage.
3. If R_1 is open, the reading between B and D is zero.
4. By connecting a voltmeter between A and E, the continuity of the conductor between the positive terminal of the battery and point E, as well as the fuse,

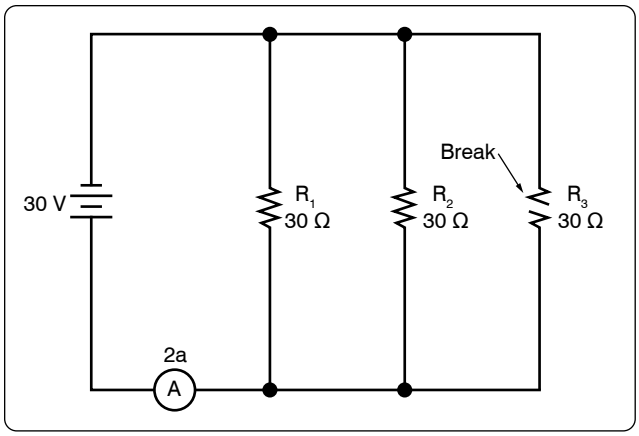


Figure 12-183. *Finding an open branch in a parallel circuit.*

can be checked. If the conductor or fuse is open, the voltmeter reads zero.

5. If the lamp is burning, it is obvious that no open exists in the branch containing the lamp, and the voltmeter could be used to detect an open in the branch containing R_2 by removing lamp, L_1, from the circuit.

Troubleshooting the series portion of a series-parallel circuit presents no difficulties, but in the parallel portion of the circuit, misleading readings can be obtained.

Batteries

Primary Cell

The dry cell is the most common type of primary-cell battery and is similar in its characteristics to that of an electrolytic cell. This type of a battery is basically designed with a metal electrode or graphite rod acting as the cathode (+) terminal, immersed in an electrolytic paste. This electrode/electrolytic build-up is then encased in a metal container, usually made of zinc, which itself acts as the anode (−) terminal. When the battery is in a discharge condition an electrochemical reaction takes place resulting in one of the metals being consumed. Because of this consumption, the charging process is not reversible. Attempting to reverse the chemical reaction in a primary cell by way of recharging is usually dangerous and can lead to a battery explosion.

These batteries are commonly used to power items such as flashlights. The most common primary cells today are found in alkaline batteries, silver-oxide, and lithium batteries. The earlier carbon-zinc cells, with a carbon post as cathode and a zinc shell as anode were once prevalent but are not as common.

Secondary Cell

A secondary cell is any kind of electrolytic cell in which the electrochemical reaction that releases energy is reversible. The lead-acid car battery is a secondary-cell battery. The electrolyte is sulfuric acid (battery acid), the positive electrode is lead peroxide, and the negative electrode is lead. A typical lead-acid battery consists of six lead-acid cells in a case. Each cell produces 2 volts, so the whole battery produces a total of 12 volts.

Other commonly used secondary cell chemistry types are nickel-cadmium (Ni-Cad), nickel-metal hydride (NiMH), lithium-ion (Li-ion), and Lithium-ion polymer (Li-ion polymer).

Lead-acid batteries used in aircraft are similar to automobile batteries. The lead acid battery is made up of a series of identical cells each containing sets of positive and negative plates. *Figure 12-191* illustrates each cell contains positive plates of lead dioxide (PbO_2), negative plates of spongy lead, and electrolyte (sulfuric acid and water). A practical cell is constructed with many more plates than just two in order to get the required current output. All positive plates are connected together as well as all the negatives. Because each positive plate is always positioned between two negative plates, there are always one or more negative plates than positive plates.

Between the plates are porous separators that keep the positive and negative plates from touching each other and shorting out the cell. The separators have vertical ribs on the side facing the positive plate. This construction permits the electrolyte to circulate freely around the plates. In addition, it provides a path for sediment to settle to the bottom of the cell.

Each cell is seated in a hard rubber casing through the top of which are terminal posts and a hole into which a nonspill vent cap is screwed. The hole provides access for testing the strength of the electrolyte and adding water. The vent plug permits gases to escape from the cell with a minimum of leakage of electrolyte, regardless of the position the airplane might assume. *[Figure 12-192]* In level flight, the lead weight

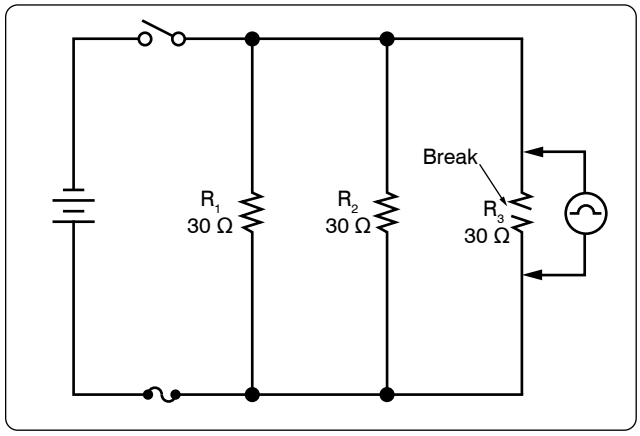


Figure 12-184. *A misleading ohmmeter indication.*

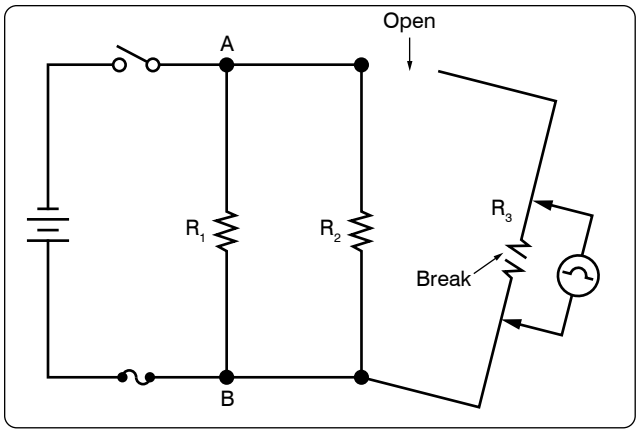


Figure 12-185. *Opening a branch circuit to obtain an accurate ohmmeter reading.*

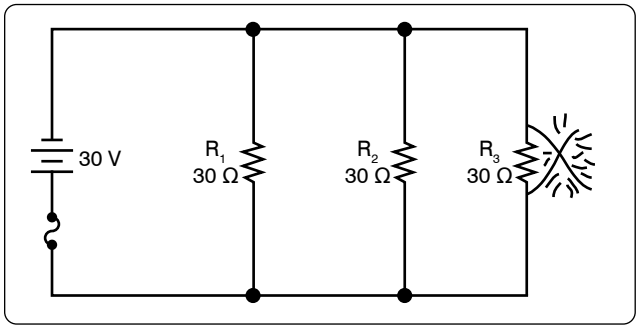


Figure 12-186. *A shorted component causes the fuse to open.*

permits venting of gases through a small hole. In inverted flight, this hole is covered by the lead weight.

The individual cells of the battery are connected in series by means of cell straps. *[Figure 12-193]* The complete assembly is enclosed in an acid resisting metal container (battery box), which serves as electrical shielding and mechanical protection. The battery box has a removable top. It also has a vent tube nipple at each end. When the battery is installed in an airplane, a vent tube is attached to each nipple. One tube is the intake tube and is exposed to the slipstream. The other is the exhaust vent tube and is attached to the battery drain sump, which is a glass jar containing a felt pad moistened with a concentrated solution of sodium bicarbonate (baking soda). With this arrangement, the airstream is directed through the battery case where battery gases are picked up, neutralized in the sump, and then expelled overboard without damage to the airplane.

To facilitate installation and removal of the battery in some aircraft, a quick disconnect assembly is used to connect the power leads to the battery. This assembly attaches the battery leads in the aircraft to a receptacle mounted on the side of the battery. *[Figure 12-194]* The receptacle covers the battery terminal posts and prevents accidental shorting during the

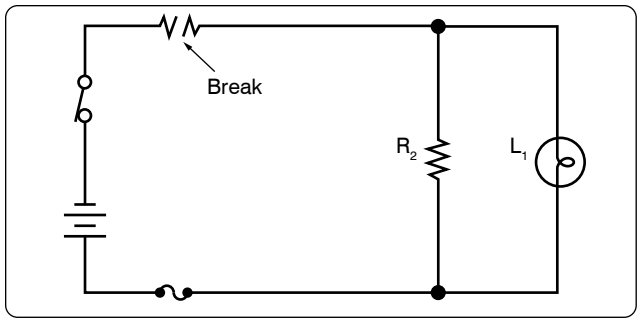


Figure 12-187. *An open in the series portion of a series-parallel circuit.*

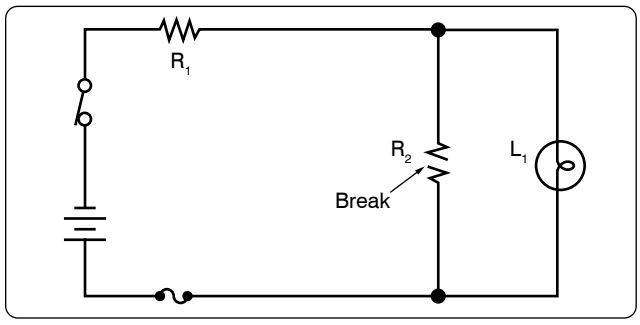


Figure 12-188. *An open in the parallel portion of a series-parallel circuit.*

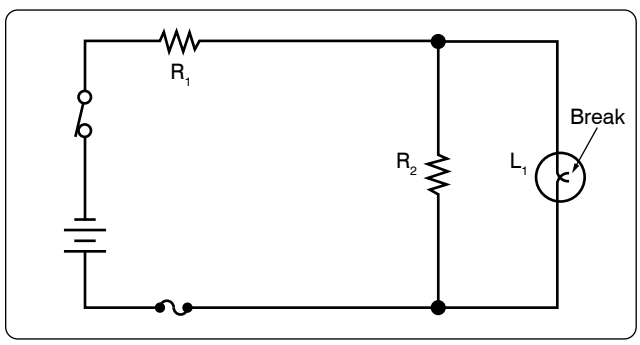


Figure 12-189. *An open lamp in a series-parallel circuit.*

installation and removal of the battery. The plug consists of a socket and a handwheel with a course pitch thread. It can be readily connected to the receptacle by the handwheel. Another advantage of this assembly is that the plug can be installed in only one position, eliminating the possibility of reversing the battery leads.

The voltage of lead acid cell is approximately two volts in order to attain the voltage required for the application. Each cell is then connected in series with heavy gauge metal straps to form a battery. In a typical battery, such as that used in an aircraft for starting, the voltage required is 12 or 24 volts. This voltage is achieved by connecting six cells or twelve cells respectively together in series and enclosing them in one plastic box.

Each cell containing the plates are filled with an electrolyte composed of sulfuric acid and distilled water with a specific gravity of 1.270 at 60 °F. This solution contains positive hydrogen ions and negative sulfate (SO_4) ions that are free to combine with other ions and form a new chemical compound. When the cell is discharged, electrons leave the negative plate and flow to the positive plates where they cause the lead dioxide (PbO_2) to break down into negative oxygen ions and positive lead ions. The negative oxygen ions join with positive hydrogen ions from the sulfuric acid and form water (H_2O). The negative sulfate ions join with the lead ions in both plates and form lead sulfate ($PbSO_4$). After the discharge, the specific gravity changes to about 1.150.

Battery Ratings

The voltage of a battery is determined by the number of cells connected in series to form the battery. Although the voltage of one lead-acid cell just removed from a charger is approximately 2.2 volts, a lead-acid cell is normally rated at approximately 2 volts. A battery rated at 12 volts consists of 6 lead-acid cells connected in series, and a battery rated at 24 volts is composed of 12 cells.

The most common battery rating is the ampere-hour rating. This is a unit of measurement for battery capacity. It is determined by multiplying a current flow in amperes by the time in hours that the battery is being discharged.

A battery with a capacity of 1 ampere-hour should be able to continuously supply a current of 1 amp to a load for exactly 1 hour, or 2 amps for ½ hour, or ⅓ amp for 3 hours, etc., before becoming completely discharged. Actually, the ampere-hour output of a particular battery depends on the rate at which it is discharged. Heavy discharge current heats the battery and decreases its efficiency and total ampere-hour output. For airplane batteries, a period of 5 hours has been established as the discharge time in rating battery capacity. However, this time of 5 hours is only a basis for rating and does not necessarily mean the length of time during which the battery is expected to furnish current. Under actual service conditions, the battery can be completely discharged within a few minutes, or it may never be discharged if the generator provides sufficient charge.

The ampere-hour capacity of a battery depends upon its total effective plate area. Connecting batteries in parallel increases ampere-hour capacity. Connecting batteries in series increases the total voltage but not the ampere-hour capacity.

Life Cycle of a Battery

Battery life cycle is defined as the number of complete charge/discharge cycles a battery can perform before its normal charge capacity falls below 80 percent of its initial rated capacity. Battery life can vary anywhere from 500 to 1,300 cycles. Various factors can cause deterioration of a battery and shorten its service life. The first is over-discharging, which causes excess sulfation; second, too-rapid charging or discharging that results in overheating of the plates and shedding of active material. The accumulation of shed material, in turn, causes shorting of the plates and results in internal discharge. A battery that remains in a low or discharged condition for a long period of time may be permanently damaged. The deterioration can continue to a point where cell capacity can drop to 80 percent after 1,000 cycles. In many cases, the cell can continue working to nearly 2,000 cycles but with a diminished capacity of 60 percent of its original state.

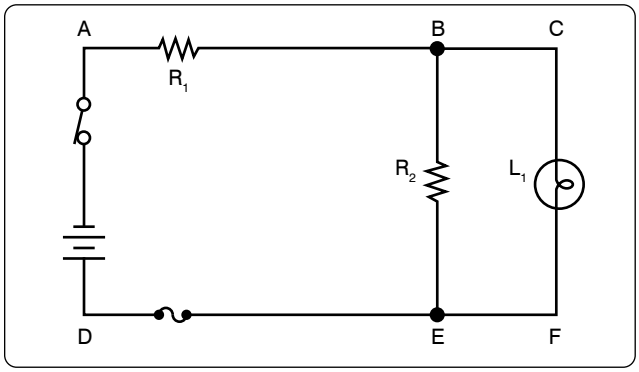


Figure 12-190. *Using the voltmeter to troubleshoot a series-parallel circuit.*

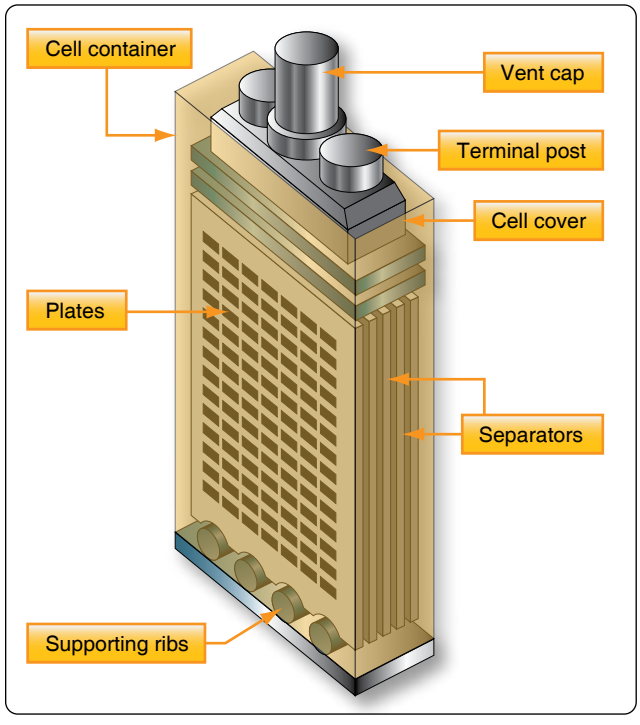


Figure 12-191. *Lead-acid cell construction.*

Lead-Acid Battery Testing Methods

The state of charge of a storage battery depends upon the condition of its active materials, primarily the plates. However, the state of charge of a battery is indicated by the density of the electrolyte and is checked by a hydrometer, an instrument that measures the specific gravity (weight as compared with water) of liquids.

The most commonly used hydrometer consists of a small sealed glass tube weighted at its lower end so it floats upright. *[Figure 12-195]* Within the narrow stem of the tube is a paper scale with a range of 1.100 to 1.300. When a hydrometer is used, a quantity of electrolyte sufficient to float the hydrometer is drawn up into the syringe. The depth to which the hydrometer sinks into the electrolyte is determined by the density of the electrolyte, and the scale value indicated at the level of the electrolyte is its specific gravity. The more dense the electrolyte, the higher the hydrometer floats; therefore, the highest number on the scale (1.300) is at the lower end of the hydrometer scale.

In a new, fully-charged aircraft storage battery, the electrolyte is approximately 30 percent acid and 70 percent water (by volume) and is 1.300 times as heavy as pure water. During discharge, the solution (electrolyte) becomes less dense and its specific gravity drops below 1.300. A specific gravity reading between 1.300 and 1.275 indicates a high state of charge; between 1.275 and 1.240, a medium state of charge; and between 1.240 and 1.200, a low state of charge. Aircraft batteries are generally of small capacity but are subject to heavy loads. The values specified for state of charge are therefore rather high. Hydrometer tests are made periodically on all storage batteries installed in aircraft. An aircraft battery in a low state of charge may have perhaps 50 percent charge remaining, but is nevertheless considered low in the face of heavy demands that would soon exhaust it. A battery in such a state of charge is considered in need of immediate recharging.

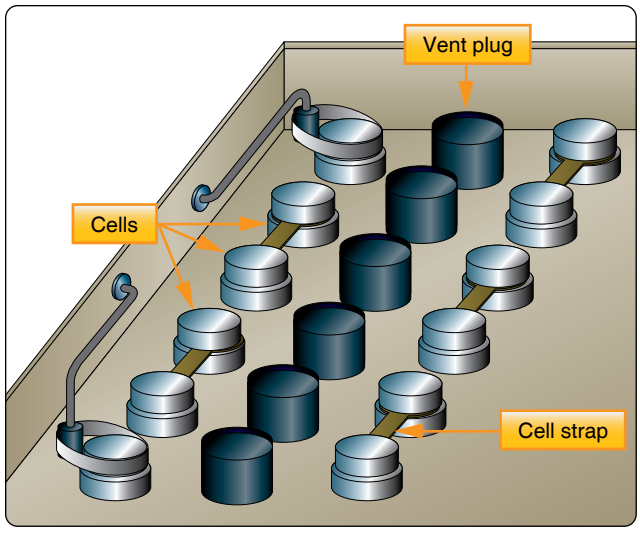


Figure 12-193. *Connection of storage battery.*

When a battery is tested using a hydrometer, the temperature of the electrolyte must be taken into consideration. The specific gravity readings on the hydrometer vary from the actual specific gravity as the temperature changes. No correction is necessary when the temperature is between 70 °F and 90 °F, since the variation is not great enough to consider. When temperatures are greater than 90 °F or less than 70 °F, it is necessary to apply a correction factor. Some hydrometers are equipped with a correction scale inside the tube. With other hydrometers, it is necessary to refer to a chart provided by the manufacturer. In both cases, the corrections should be added to, or subtracted from the reading shown on the hydrometer.

The specific gravity of a cell is reliable only if nothing has

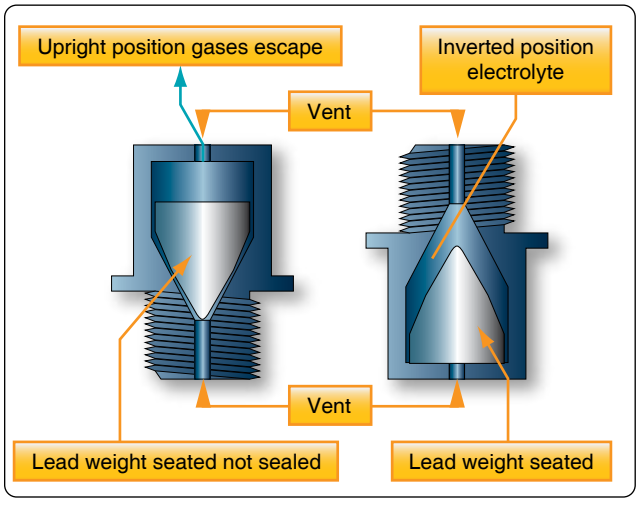


Figure 12-192. *Nonspill battery vent plug.*

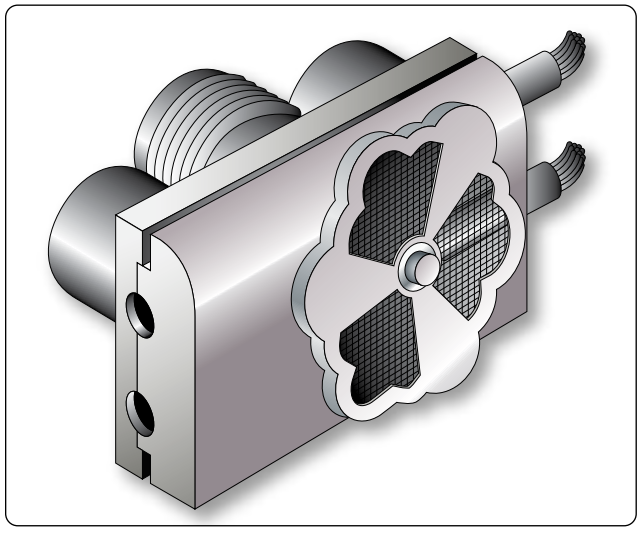


Figure 12-194. *A battery quick-disconnect assembly.*

been added to the electrolyte except occasional small amounts of distilled water to replace that lost as a result of normal evaporation. Always take hydrometer readings before adding distilled water, never after. This is necessary to allow time for the water to mix thoroughly with the electrolyte and to avoid drawing up into the hydrometer syringe a sample that does not represent the true strength of the solution.

Exercise extreme care when making the hydrometer test of a lead-acid cell. Handle the electrolyte carefully because sulfuric acid burns clothing and skin. If the acid does contact the skin, wash the area thoroughly with water and then apply bicarbonate of soda.

Lead-Acid Battery Charging Methods

Passing direct current through the battery in a direction opposite to that of the discharge current may charge a storage battery. Because of the internal resistance (IR) in the battery, the voltage of the external charging source must be greater than the open circuit voltage. For example, the open circuit voltage of a fully charged 12 cell, lead-acid battery is approximately 26.4 volts (12 × 2.2 volts), but approximately 28 volts are required to charge it. This larger voltage is needed for charging because of the voltage drop in the battery caused by the internal resistance. Hence, the charging voltage of a lead-acid battery must equal the open circuit voltage plus the IR drop within the battery (product of the charging current and the internal resistance).

Batteries are charged by either the constant voltage or constant current method. In the constant voltage method *[Figure 12-196A]*, a motor generator set with a constant, regulated voltage forces the current through the battery. In this method, the current at the start of the process is high but automatically tapers off, reaching a value of approximately 1 ampere when the battery is fully charged. The constant voltage method requires less time and supervision than does the constant current method.

In the constant current method *[Figure 12-196B]*, the current remains almost constant during the entire charging process. This method requires a longer time to charge a battery fully and, toward the end of the process, presents the danger of overcharging, if care is not exercised.

In the aircraft, the storage battery is charged by direct current from the aircraft generator system. This method of charging is the constant voltage method, since the generator voltage is held constant by use of a voltage regulator.

When a storage battery is being charged, it generates a certain amount of hydrogen and oxygen. Since this is an explosive mixture, it is important to take steps to prevent ignition of the gas mixture. Loosen the vent caps and leave in place. Do not permit open flames, sparks, or other sources of ignition in the vicinity. Before disconnecting or connecting a battery to the charge, always turn off the power by means of a remote switch. *Figure 12-197* shows battery charging equipment.

Nickel-Cadmium Batteries

Chemistry and Construction

Active materials in nickel-cadmium cells (Ni-Cad) are nickel hydroxide (NiOOH) in the charged positive plate (Anode) and sponge cadmium (Cd) in the charged negative plate (Cathode). The electrolyte is a potassium hydroxide (KOH) solution in concentration of 20–34 percent by weight pure KOH in distilled water.

Sintered nickel-cadmium cells have relatively thin sintered nickel matrices forming a plate grid structure. The grid structure is highly porous and is impregnated with the active positive material (nickel-hydroxide) and the negative material (cadmium-hydroxide). The plates are then formed by sintering nickel powder to fine-mesh wire screen. In other variations of the process, the active material in the sintered matrix is converted chemically, or thermally, to an active state and then formed. In general, there are many steps to these cycles of impregnation and formation. Thin sintered plate cells are ideally suited for very high rate charge and discharge service. Pocket plate nickel-cadmium cells have the positive or negative active material, pressed into pockets of perforated nickel-plated steel plates or into tubes. The active material is trapped securely in contact with a metal current collector so active material shedding is largely eliminated. Plate designs vary in thickness

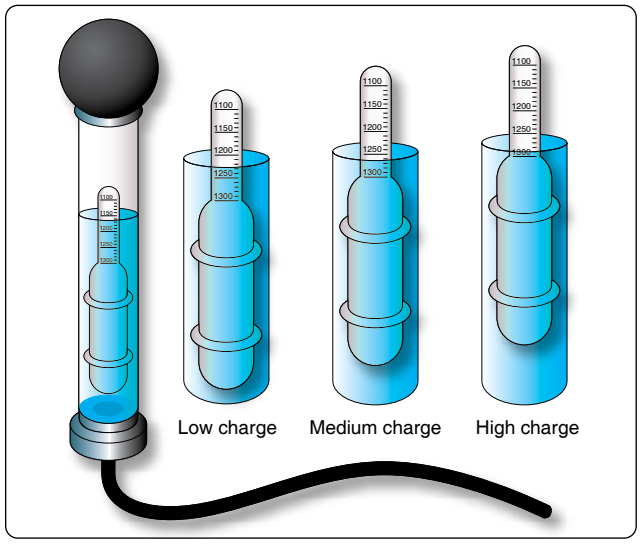


Figure 12-195. *Hydrometer (specific gravity readings).*

depending upon cycling service requirements. The typical open circuit cell voltage of a nickel-cadmium battery is about 1.25 volts. *Figure 12-198* shows a nickel-cadmium aircraft battery.

Operation of Nickel-Cadmium Cells

When a charging current is applied to a nickel-cadmium battery, the negative plates lose oxygen and begin forming metallic cadmium. The active material of the positive plates, nickel-hydroxide, becomes more highly oxidized. This process continues while the charging current is applied or until all the oxygen is removed from the negative plates and only cadmium remains.

Toward the end of the charging cycle, the cells emit gas. This also occurs if the cells are overcharged. This gas is caused by decomposition of the water in the electrolyte into hydrogen at the negative plates and oxygen at the positive plates. The voltage used during charging, as well as the temperature, determines when gassing occurs. To completely charge a nickel-cadmium battery, some gassing, however slight, must take place; thus, some water is used.

The chemical action is reversed during discharge. The positive plates slowly give up oxygen, which is regained by the negative plates. This process results in the conversion of the chemical energy into electrical energy. During discharge, the plates absorb a quantity of the electrolyte. On recharge, the level of the electrolyte rises and, at full charge, the electrolyte is at its highest level. Therefore, water should be added only when the battery is fully charged.
The nickel-cadmium battery is usually interchangeable with the lead-acid type. When replacing a lead-acid battery with a nickel-cadmium battery, the battery compartment must be clean, dry, and free of all traces of acid from the old battery. The compartment must be washed out and neutralized with ammonia or boric acid solution, allowed to dry thoroughly, and then painted with an alkali resisting varnish.

The pad in the battery sump jar should be saturated with a three percent (by weight) solution of boric acid and water before connecting the battery vent system.

General Maintenance and Safety Precautions

Refer to the battery manufacturer for detailed service instructions. Below are general recommendations for maintenance and safety precautions. For vented nickel-cadmium cells, the general maintenance requirements are:

1. Hydrate cells to supply water lost during overcharging.
2. Maintain inter-cell connectors at proper torque values.
3. Keep cell tops and exposed sides clean and dry.

Electrolyte spillage can form grounding paths. White moss around vent cap seals is potassium carbonate (K_2CO_3). Clean up these surfaces with distilled water and dry. While handling the caustic potassium hydroxide electrolyte, wear safety goggles to protect the eyes. The technician should also wear plastic gloves and an apron to protect skin and clothes. In case of spillage on hands or clothes, neutralize the alkali immediately with vinegar or dilute boric acid solution (one pound per gallon of water); then rinse with clear water.

During overcharging conditions, explosive mixtures of hydrogen and oxygen develop in nickel-cadmium cells. When this occurs, the cell relief valves vent these gases to the atmosphere, creating a potentially explosive hazard. Additionally, room ventilation should be such as to prevent a hydrogen build up in closed spaces from exceeding one percent by volume. Explosions can occur at concentrations above four percent by volume in air.

Sealed Lead Acid (SLA) Batteries

In many applications, sealed lead acid (SLA) batteries are gaining in use over flooded lead acid and Ni-Cad batteries. One leading characteristic of Ni-Cad batteries is that they perform well in low voltage, full-discharge, high cycle applications. However, they do not perform as well

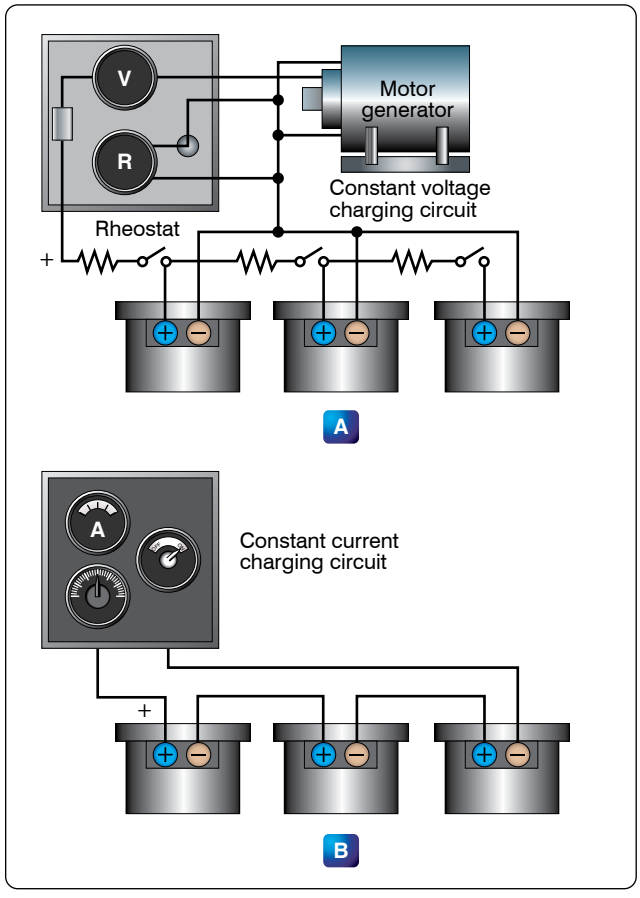


Figure 12-196. *Battery charging methods.*

in extended standby applications, such as auxiliary or as emergency battery packs used to power inertial reference units or stand-by equipment (attitude gyro).

It is typical during the servicing of a Ni-Cad battery to match as many as twenty individual cells in order to prevent unbalance and thus cell reversal during end of discharge. When a Ni-Cad does reverse, very high pressure and heat can result. The result is often pressure seal rupture, and in the worst case, a cell explosion. With SLA batteries, cell matching is inherent in each battery. Ni-Cads also have an undesirable characteristic caused by constant overcharge and infrequent discharges, as in standby applications. It is technically known as "voltage depression" and commonly but erroneously called "memory effect." This characteristic is only detectable when a full discharge is attempted. Thus, it is possible to believe a full charge exists, while in fact it does not. SLA batteries do not have this characteristic voltage depression (memory) phenomenon, and therefore do not require scheduled deep cycle maintenance as do Ni-Cads.

The Ni-Cad emergency battery pack requires relatively complicated test equipment due to the complex characteristics of the Ni-Cad. Sealed lead acid batteries do not have these temperamental characteristics and therefore it is not necessary to purchase special battery maintenance equipment. Some manufacturers of SLA batteries have included in the battery packs a means by which the battery can be tested while still installed on the aircraft. Ni-Cads must have a scheduled energy test performed on the bench due to the inability to measure their energy level on the aircraft, and because of their notable "memory" shortcoming.

The SLA battery can be designed to alert the technician if a battery is failing. Furthermore, it may be possible to test the failure detection circuits by activating a Built in Test (BITE) button. This practice significantly reduces FAA paperwork and maintenance workload. *Figure 12-199* shows a SLA battery.

Lithium-Ion Batteries

Lithium-ion batteries are the primary type of battery for many consumer type of equipment, such as cell phones, battery-powered tools, and computers, but now they are also being used in commercial and military aircraft. The FAA has certified lithium-ion batteries to be used on aircraft and one of the first aircraft to utilize the lithium-ion battery is the Boeing 787. The three primary functional components of a lithium-ion battery are the positive and negative electrodes and electrolyte. Generally, the negative electrode of a conventional lithium-ion cell is made from carbon. The positive electrode is a metal-oxide, and the electrolyte is a lithium salt in an organic solvent. The electrochemical roles of the electrodes reverse between anode and cathode, depending on the direction of current flow through the cell. Lithium-ion batteries can be dangerous under some conditions and can pose a safety hazard since they contain, unlike other rechargeable batteries, a flammable electrolyte and are also kept pressurized. Under certain conditions, they can overheat and a fire can occur. The Boeing 787 aircraft utilizes two large 32V 8 cell lithium-ion batteries. These batteries are much lighter and more powerful than Ni-Cad batteries used in similar-sized aircraft. These batteries can produce 150 A for airplane power up. *Figure 12-200* shows a B787 battery.

Inverters

An inverter is used in some aircraft systems to convert a portion of the aircraft's DC power to AC. This AC is used mainly for instruments, radio, radar, lighting, and other accessories. These inverters are usually built to supply current at a frequency of 400 cps, but some are designed to provide more than one voltage; for example, 26 volt AC in one winding and 115 volts in another.

There are two basic types of inverters: the rotary and the static. Either type can be single-phase or multiphase. The multiphase inverter is lighter for the same power rating than the single-phase, but there are complications in distributing multiphase power and in keeping the loads balanced.

Figure 12-197. *Battery charger.*

Figure 12-198. *Nickel-Cadmium aircraft battery.*

Rotary Inverters

There are many sizes, types, and configurations of rotary inverters. Such inverters are essentially AC generators and DC motors in one housing. The generator field, or armature, and the motor field, or armature, are mounted on a common shaft that rotates within the housing. One common type of rotary inverter is the permanent magnet inverter.

Permanent Magnet Rotary Inverter

A permanent magnet inverter is composed of a DC motor and a permanent magnet AC generator assembly. Each has a separate stator mounted within a common housing. The motor armature is mounted on a rotor and connected to the DC supply through a commutator and brush assembly. The motor field windings are mounted on the housing and connected directly to the DC supply. A permanent magnet rotor is mounted at the opposite end of the same shaft as the motor armature, and the stator windings are mounted on the housing, allowing AC to be taken from the inverter without the use of brushes. *Figure 12-201* shows an internal wiring diagram for this type of rotary inverter. The generator rotor has six poles, magnetized to provide alternate North and South poles about its circumference.

When the motor field and armature are excited, the rotor begins to turn. As the rotor turns, the permanent magnet rotates within the AC stator coils, and the magnetic flux developed by the permanent magnets are cut by the conductors in the AC stator coils. An AC voltage is produced in the windings whose polarity changes as each pole passes the windings.

This type inverter may be made multiphase by placing more AC stator coils in the housing in order to shift the phase the proper amount in each coil.

As the name of the rotary inverter indicates, it has a revolving armature in the AC generator section. *Figure 12-202* shows the diagram of a revolving armature, three phase inverter.

The DC motor in this inverter is a four pole, compound wound motor. The four field coils consist of many turns of fine wire, with a few turns of heavy wire placed on top. The fine wire is the shunt field, connected to the DC source through a filter and to ground through a centrifugal governor. The heavy wire is the series field, which is connected in series with the motor armature. The centrifugal governor controls the speed by shunting a resistor that is in series with the shunt field when the motor reaches a certain speed.

The alternator is a three-phase, four-pole, star-connected AC generator. The DC input is supplied to the generator field coils and connected to ground through a voltage regulator. The output is taken off the armature through three slip rings to provide three-phase power. The inverter would be a single-phase inverter if it had a single armature winding and one slip ring. The frequency of this type unit is determined by the speed of the motor and the number of generator poles.

Inductor-Type Rotary Inverter

Inductor-type inverters use a rotor made of soft iron laminations with grooves cut laterally across the surface to provide poles that correspond to the number of stator poles. *[Figure 12-203]* The field coils are wound on one set of stationary poles and the AC armature coils on the other set of stationary poles. When DC is applied to the field coils, a magnetic field is produced. The rotor turns within the field coils and, as the poles on the rotor align with the stationary poles, a low reluctance path for flux is established from the field pole through the rotor poles to the AC armature pole and through the housing back to the field pole. In this circumstance, there is a large amount of magnetic flux linking the AC coils.

When the rotor poles are between the stationary poles, there is a high reluctance path for flux, consisting mainly of air; then, there is a small amount of magnetic flux linking the AC coils. This increase and decrease in flux density in the stator induces an alternating current in the AC coils.

The number of poles and the speed of the motor determine the frequency of this type of inverter. The DC stator field current controls the voltage. A cutaway view of an inductor-type rotary inverter is shown in *Figure 12-204*.

Figure 12-205 is a simplified diagram of a typical aircraft AC power distribution system, utilizing a main and a standby rotary inverter system.

Static Inverters

In many applications where continuous DC voltage must be converted to alternating voltage, static inverters are used in place of rotary inverters or motor generator sets. The rapid progress made by the semiconductor industry is extending

Figure 12-199. *Sealed battery.*

the range of applications of such equipment into voltage and power ranges that would have been impractical a few years ago. Some such applications are power supplies for frequency sensitive military and commercial AC equipment, aircraft emergency AC systems, and conversion of wide frequency range power to precise frequency power. *[Figure 12-206]*

The use of static inverters in small aircraft also has increased rapidly in the last few years, and the technology has advanced to the point that static inverters are available for any requirement filled by rotary inverters. For example, 250 VA emergency AC supplies operated from aircraft batteries are in production, as are 2,500 VA main AC supplies operated from a varying frequency generator supply. This type of equipment has certain advantages for aircraft applications, particularly the absence of moving parts and the adaptability to conduction cooling.

Static inverters, referred to as solid-state inverters, are manufactured in a wide range of types and models that can be classified by the shape of the AC output waveform and the power output capabilities. One of the most commonly used static inverters produces a regulated sine wave output. A block diagram of a typical regulated sine wave static inverter is shown in *Figure 12-207*. This inverter converts a low DC voltage into higher AC voltage. The AC output voltage is held to a very small voltage tolerance, a typical variation of less than 1 percent with a full input load change. Output taps are normally provided to permit selection of various voltages; for example, taps may be provided for 105, 115, and 125 volt AC outputs. Frequency regulation is typically within a range of one cycle for a 0–100 percent load change. Variations of this type of static inverter are available, many of which provide a square wave output.

Since static inverters use solid-state components, they are considerably smaller, more compact, and much lighter in weight than rotary inverters. Depending on the output power rating required, static inverters that are no larger than a typical airspeed indicator can be used in aircraft systems. Some of the features of static inverters are:

1. High efficiency
2. Low maintenance, long life
3. No warmup period required
4. Capable of starting under load
5. Extremely quiet operation
6. Fast response to load changes

Static inverters are commonly used to provide power for such frequency sensitive instruments as the attitude gyro and directional gyro. They also provide power for autosyn and magnesyn indicators and transmitters, rate gyros, radar, and other airborne applications. *Figure 12-208* is a schematic of a typical small jet aircraft auxiliary battery system. It shows the battery as input to the inverter and the output inverter circuits to various subsystems.

Figure 12-200. *Boeing 787 lithium-ion battery.*

Semiconductors

To understand why solid-state devices function as they do, it is necessary to examine the composition and nature of semiconductors. The two most common materials used for semiconductors are germanium and silicon. The essential characteristic of these elements is that each atom has four valence electrons to share with adjacent atoms in forming bonds. While both elements are used in semiconductor construction, silicon is preferred in most modern applications due to its ability to operate over a wider range of temperatures. The nature of a bond between two silicon atoms is such that each atom provides one electron to share with the other. The two electrons shared are in fact shared equally between the two atoms. This form of sharing is known as a covalent bond. Such bonds are very stable and hold the two atoms together very tightly requiring much energy to break this bond. *[Figure 12-209]* In this case, all of the outer electrons are used to make covalent bonds with other silicon atoms. In this condition, because all of the outer shell atoms are used, silicon takes on the characteristic of a good insulator, due to the fact that there are no open positions available for electrons to migrate through the orbits.

For the silicon crystal to conduct electricity, there must be some means available to allow some electrons to move from place to place within the crystal, regardless of the covalent bonds present between the atoms. One way to accomplish this is to introduce an impurity, such as arsenic or phosphorus, into the crystal structure, which either provides an extra electron or create a vacant position in the outer shell for electrons

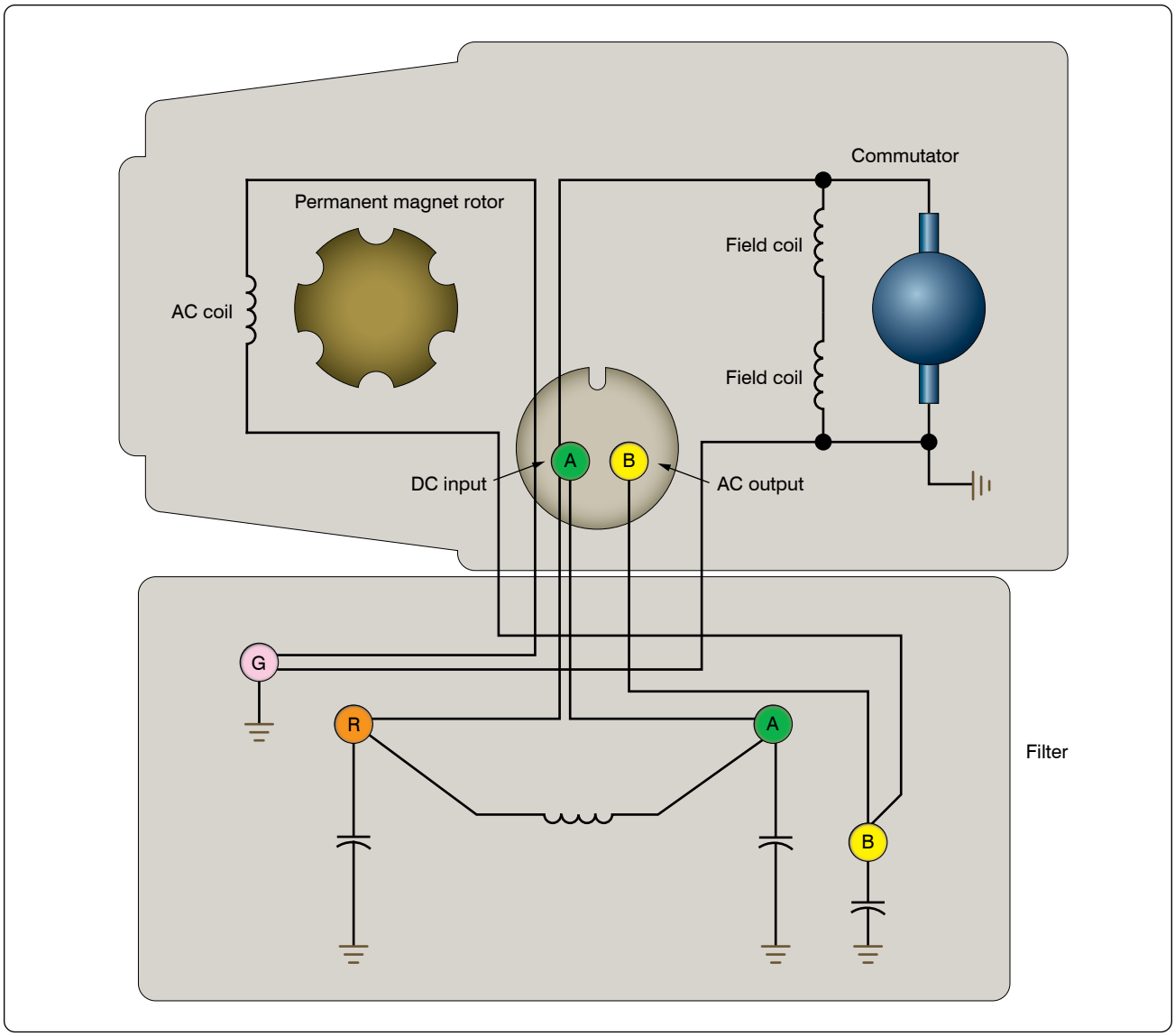


Figure 12-201. *Internal wiring diagram of single-phase permanent magnet rotary inverter.*

to pass though. The method used to create this condition is called doping.

Doping

Doping is the process by which small amounts of additives called impurities are added to the semiconductor material to increase their current flow by adding a few electrons or a few holes. Once the material is doped, it then falls into one of two categories: the N-type semiconductor and the P-type semiconductor.

An N-type semiconductor material is one that is doped with an N-type or a donor impurity. Elements such as phosphorus, arsenic, and antimony are added as impurities and have five outer electrons to share with other atoms. This causes the semiconductor material to have an excess electron. Due to the surplus of electrons, the electrons are then considered the majority current carriers. This electron can easily be moved with only a small applied electrical voltage. Current flow in an N-type silicon material is similar to conduction in a copper wire. That is, with voltage applied across the material, electrons will move through the crystal towards the positive terminal just like current flows in a copper wire.

A P-type semiconductor is one that is doped with a P-type or an acceptor impurity. Elements such as boron, aluminum, and gallium have only three electrons in the valence shell to share with the silicon atom. Those three electrons form covalent bonds with adjacent silicon atoms. However, the expected fourth bond cannot be formed and a complete connection is impossible here, leaving a “hole” in the structure of the crystal. There is an empty place where an electron would naturally go, and often an electron moves into that space.

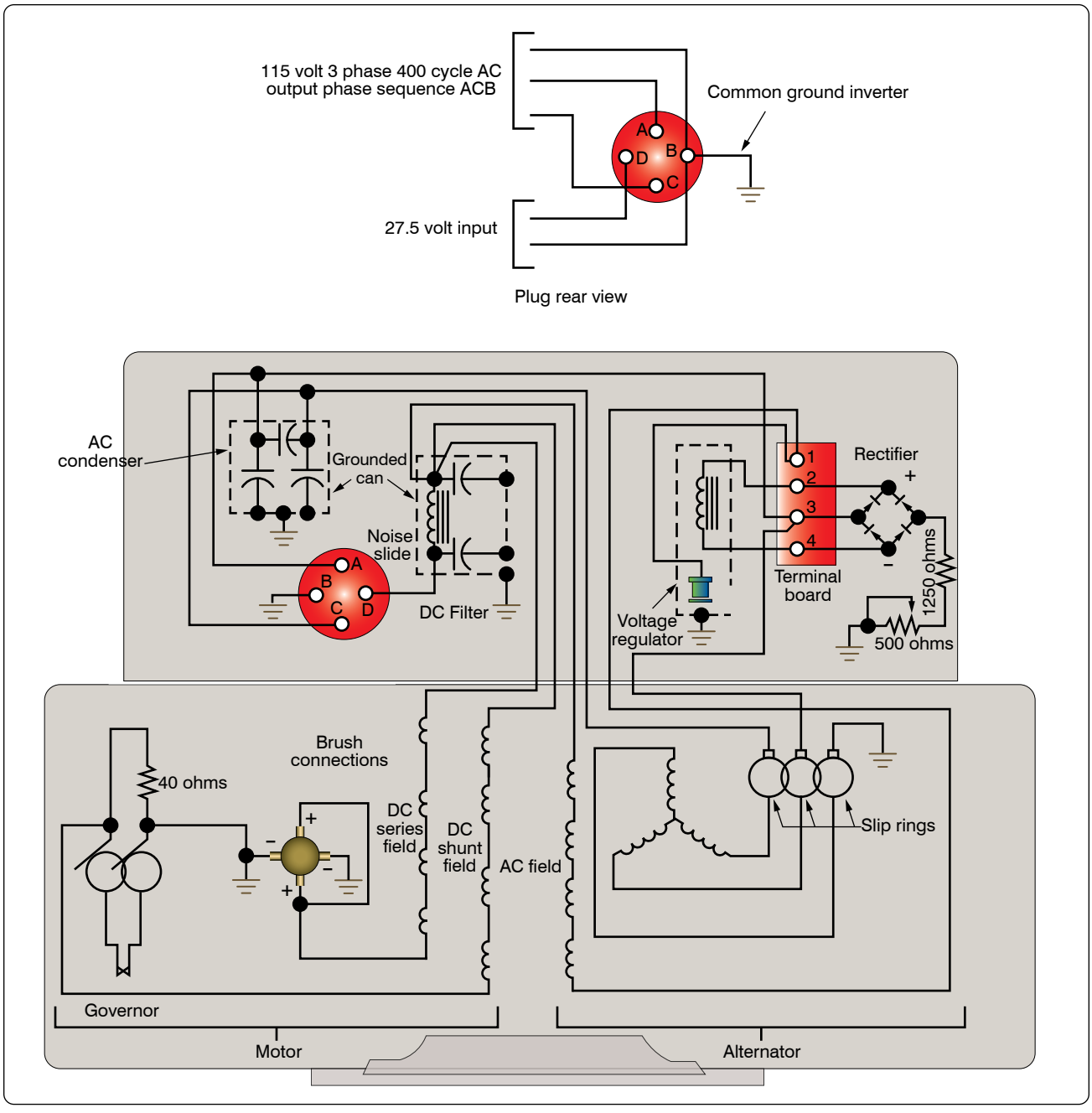


Figure 12-202. *Internal wiring diagram of three-phase, revolving armature.*

However, the electron filling the hole left a covalent bond behind to fill this empty space, which leaves another hole behind as it moves. Another electron may then move into that particular hole, leaving another hole behind. As this progression continues, holes appear to move as positive charges throughout the crystal. This type of semiconductor material is designated P-type silicon material. *Figure 12-210* shows the progression of a hole moving through a number of atoms. Notice that the hole illustrated at the far left of top depiction of *Figure 12-210* attracts the next valence electron into the vacancy, which then produces another vacancy called a hole in the next position to the right. Once again, this vacancy attracts the next valence electron. This exchange of holes and electrons continues to progress and can be viewed in one of two ways. The first way that this flow can be seen as that of electron movement. The electron is shown in *Figure 12-210* as moving from the right to the left through a series of holes. Likewise, the second depiction in *Figure 12-210* of the motion of the vacated hole can be seen as migrating from the left to the right. This view is often called hole movement. The valence electron in the structure progresses along a path detailed by the arrows. Holes,

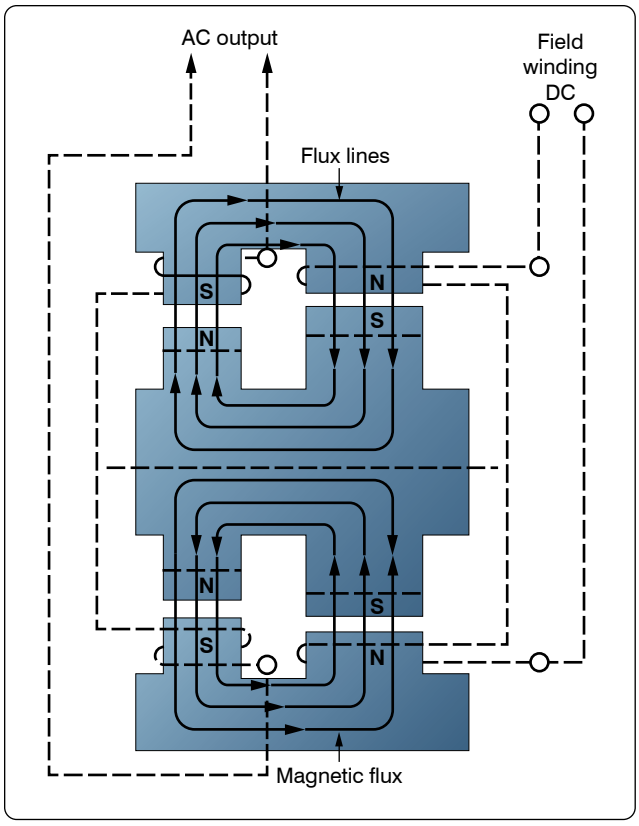


Figure 12-203. *Diagram of basic inductor-type inverter.*

however, move along a path opposite that of the electrons.

PN Junctions & the Basic Diode

A single type of semiconductor material by itself is not very useful. Useful applications are developed only when a single component contains both P-type and N-type materials. The semiconductor diode is also known as a PN junction diode. This is a two-element semiconductor device that makes use of the rectifying properties of a PN junction to convert alternating current into direct current by permitting current flow in one direction only.

Figure 12-211 illustrates the electrical characteristics of an unbiased diode, which means that no external voltage is applied. The P-side in the illustration is shown to have many holes, while the N-side shows many electrons. The electrons on the N-side tend to diffuse out in all directions. When an electron enters the P region, it becomes a minority carrier. By definition, a minority carrier is an electron or hole, whichever is the less dominant carrier in a semiconductor device. In P-type materials, electrons are the minority carrier and in N-type material, the hole is considered the minority carrier. With so many holes around the electron, the electron soon drops into a hole. When this occurs, the hole then disappears, and the conduction band electron becomes a valence electron.

Each time an electron crosses the PN junction, it creates a pair of ions. *Figure 12-211* shows this area outlined by dashed lines. The circled plus signs and the circled negative signs are the positive and negative ions, respectively. These ions are fixed in the crystal and do not move around like electrons or holes in the conduction band. Thus, the depletion zone constitutes a layer of a fixed charge. An electrostatic field, represented by a small battery in *Figure 12-211*, is established across the junction between the oppositely charged ions.

The junction barrier is an electrostatic field, which has been created by the joining of a section of N-type and P-type material. Because holes and electrons must overcome this field to cross the junction, the electrostatic field is usually called a barrier. Because there is a lack or depletion of free electrons and holes in the area around the barrier, this area is called the depletion region. *[Figure 12-211]* As the diffusion of electrons and holes across the junction continue, the strength of the electrostatic field increases until it is strong enough to prevent electrons or holes from crossing over. At this point, a state of equilibrium exists, and there is no further movement across the junction. The electrostatic field created at the junction by the ions in the depletion zone is called a barrier.

Forward Biased Diode

Figure 12-212 illustrates a forward biased PN junction. When an external voltage is applied to a PN junction, it is called bias. In a forward biased PN junction or diode, the negative voltage source is connected to the N-type material and the positive voltage source is connected to the P-type material. In this configuration, the current can easily flow. If a battery is used to bias the PN junction and it is connected in such a way that the applied voltage opposes the junction field, it has the effect of reducing the junction barrier and consequently aids in the current flow through the junction.

The electrons move toward the junction and the right end of the diode becomes slightly positive. This occurs because electrons at the right end of the diode move toward the junction and leave positively charged atoms behind. The positively charged atoms then pull electrons into the diode from the negative terminal of the battery.

When electrons on the N-type side approach the junction, they recombine with holes. Basically, electrons are flowing into the right end of the diode, while the bulk of the electrons in the N-type material move toward the junctions. The left edge of this moving front of electrons disappears by dropping into holes at the junction. In this way, there is a continuous current of electrons from the battery moving toward the junction.

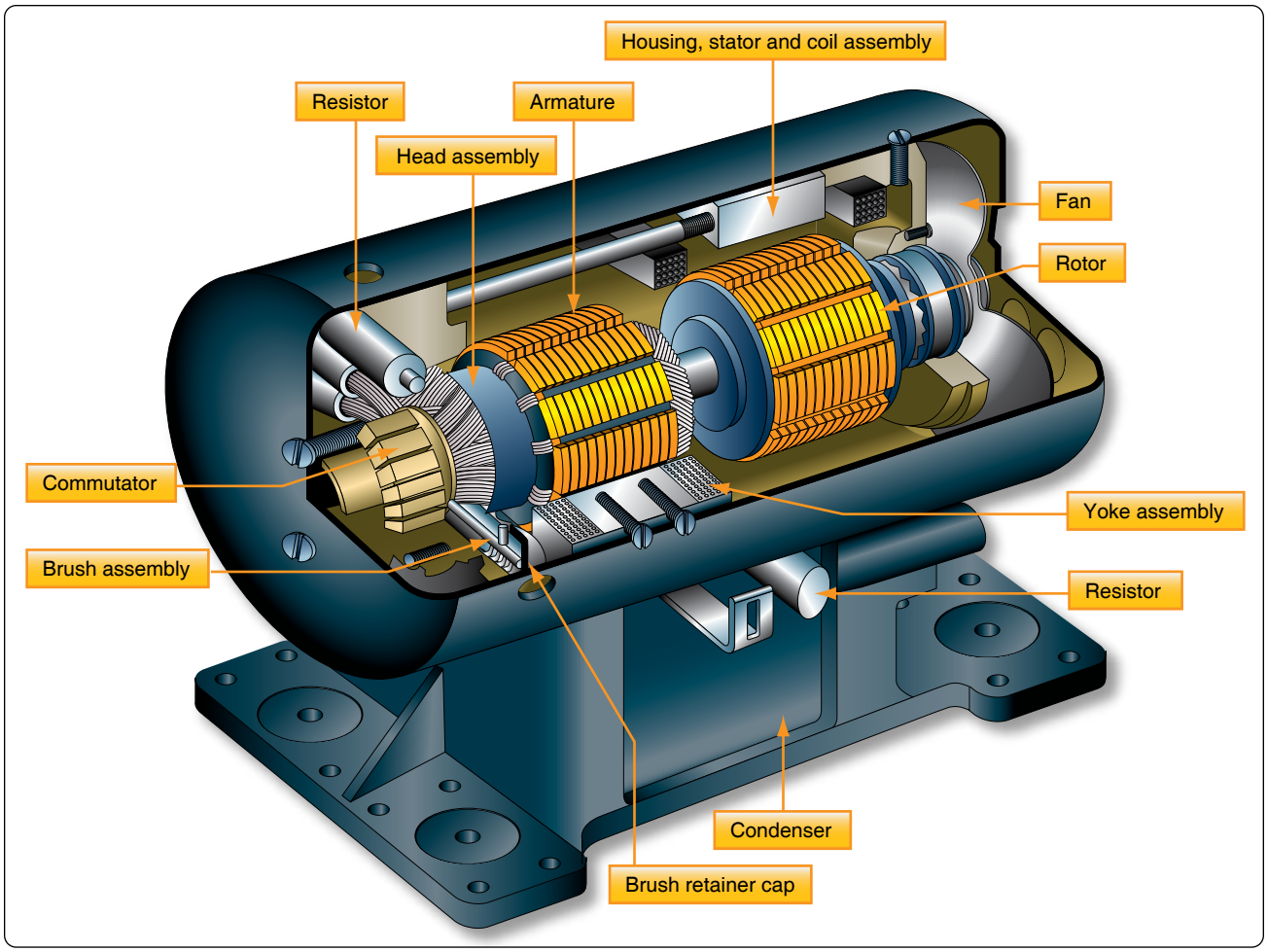


Figure 12-204. *Cutaway view of inductor-type rotary inverter.*

When the electrons hit the junction, they then become valence electrons. Once a valence electron, they can then move through the holes in the P-type material. When the valence electrons move through the P-type material from the right to the left, a similar movement is occurring with the holes by moving from the left side of the P-type material to the right. Once the valence electron reaches the end of the diode, it then flows back into the positive terminal of the battery.

In summary:

1. Electron leaves negative terminal of the battery and enters the right end (N-type material) of the diode.
2. Electron then travels through the N-type material.
3. The electron nears the junction and recombines and becomes a valence electron.
4. The electron now travels through the P-type material as a valence electron.
5. The electron then leaves the diode and flows back to the positive terminal of the battery.

Reverse Biased Diode

When the battery is turned around as shown in *Figure 12-213*, then the diode is reverse biased and current does not flow. The most noticeable effect seen is the widened depletion zone.

The applied battery voltage is in the same direction as the depletion zone field. Because of this, holes and electrons tend to move away from the junction. Simply stated, the negative terminal attracts the holes away from the junction, and the positive terminal attracts the electrons away from the barrier. Therefore, the result is a wider depletion zone. This action increases the barrier width because there are more negative ions on the P-side of the junction and more positive ions on the N-side of the junction. This increase in the number of ions at the junction prevents current flow across the barrier by the majority carriers.

To summarize, the important thing to remember is that these PN junction diodes offer very little resistance to current when the diode is forward biased. Maximum resistance happens

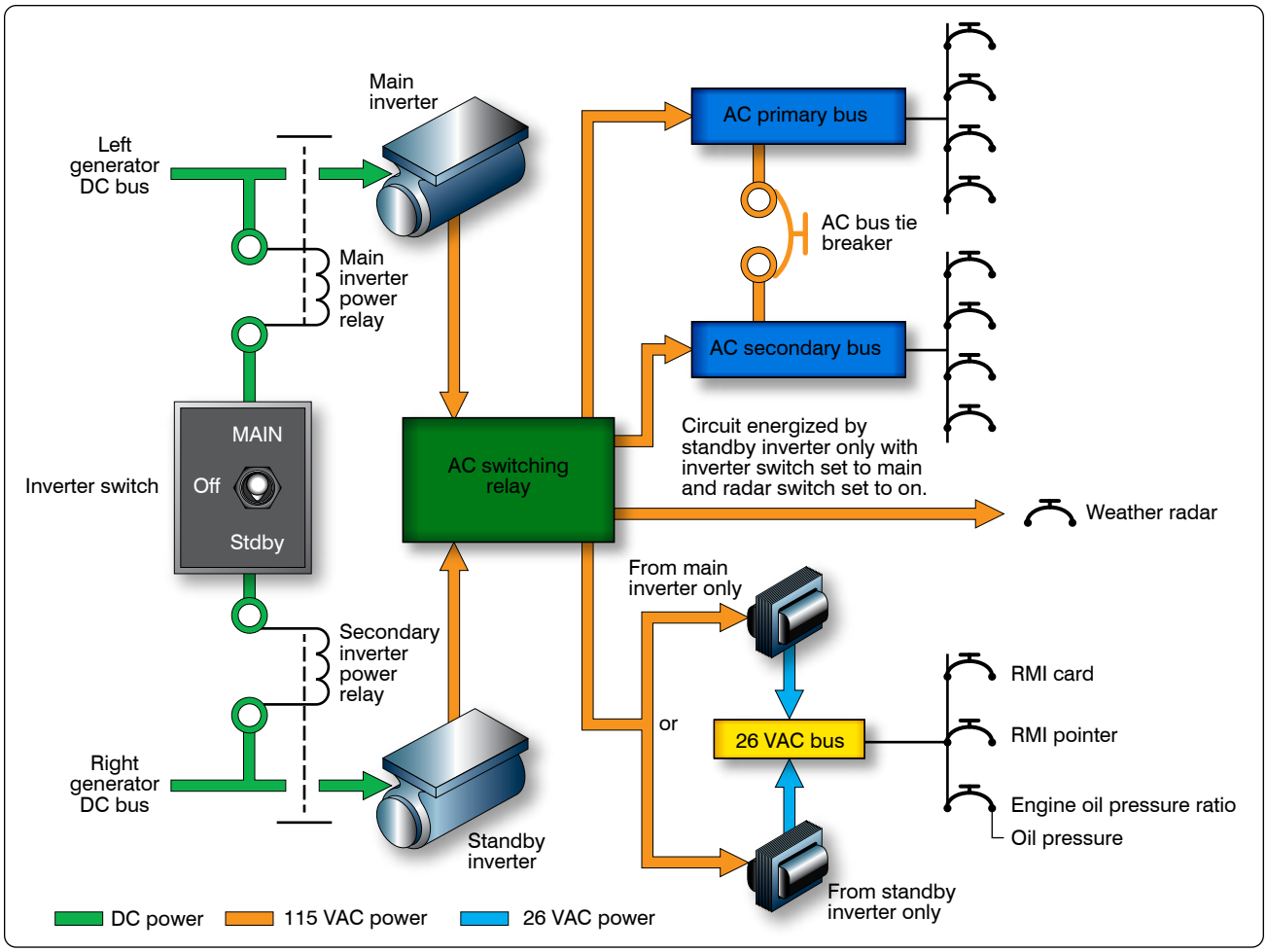


Figure 12-205. *A typical aircraft AC power distribution system using main and standby rotary inverters.*

Figure 12-206. *Static inverter.*

when the diode is reversed biased. *Figure 12-214* shows a graph of the current characteristics of a diode that is biased in both directions.

Rectifiers

Many devices in an aircraft require high amperage, low voltage DC for operation. This power may be furnished by DC engine-driven generators, motor generator sets, vacuum tube rectifiers, or dry disk or solid-state rectifiers.

In aircraft with AC systems, a special DC generator is not desirable since it would be necessary for the engine accessory section to drive an additional piece of equipment. Motor generator sets, consisting of air-cooled AC motors that drive DC generators, eliminate this objection because they operate directly off the AC power system. Vacuum tube or various types of solid-state rectifiers provide a simple and efficient method of obtaining high voltage DC at low amperage. Dry disk and solid-state rectifiers, on the other hand, are an excellent source of high amperage at low voltage.

A rectifier is a device that transforms AC into DC by limiting or regulating the direction of current flow. The principal types of rectifiers are dry disk and solid state. Solid-state, or semiconductor, rectifiers have replaced virtually all other types; and, since dry disk and motor generators are largely limited to older model aircraft, the major part of the study of rectifiers is devoted to solid-state devices used for rectification. The two methods discussed in this handbook

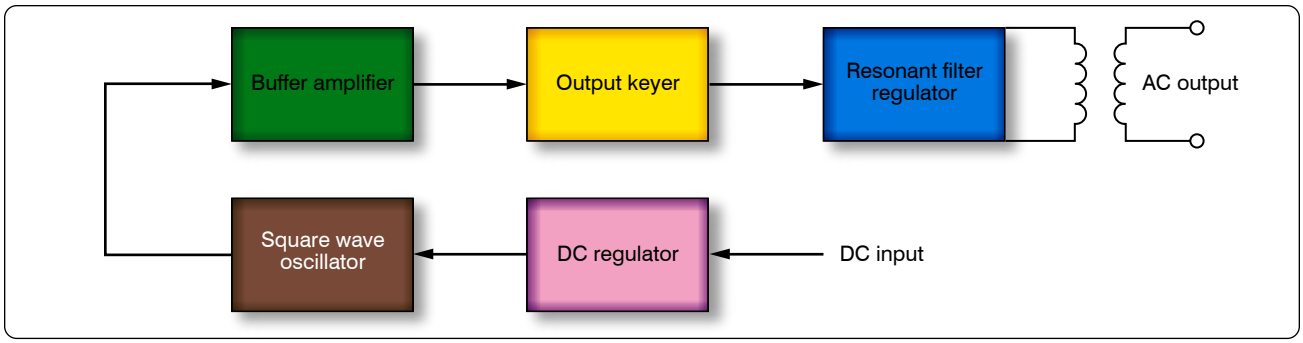


Figure 12-207. *Regulated sine wave static inverter.*

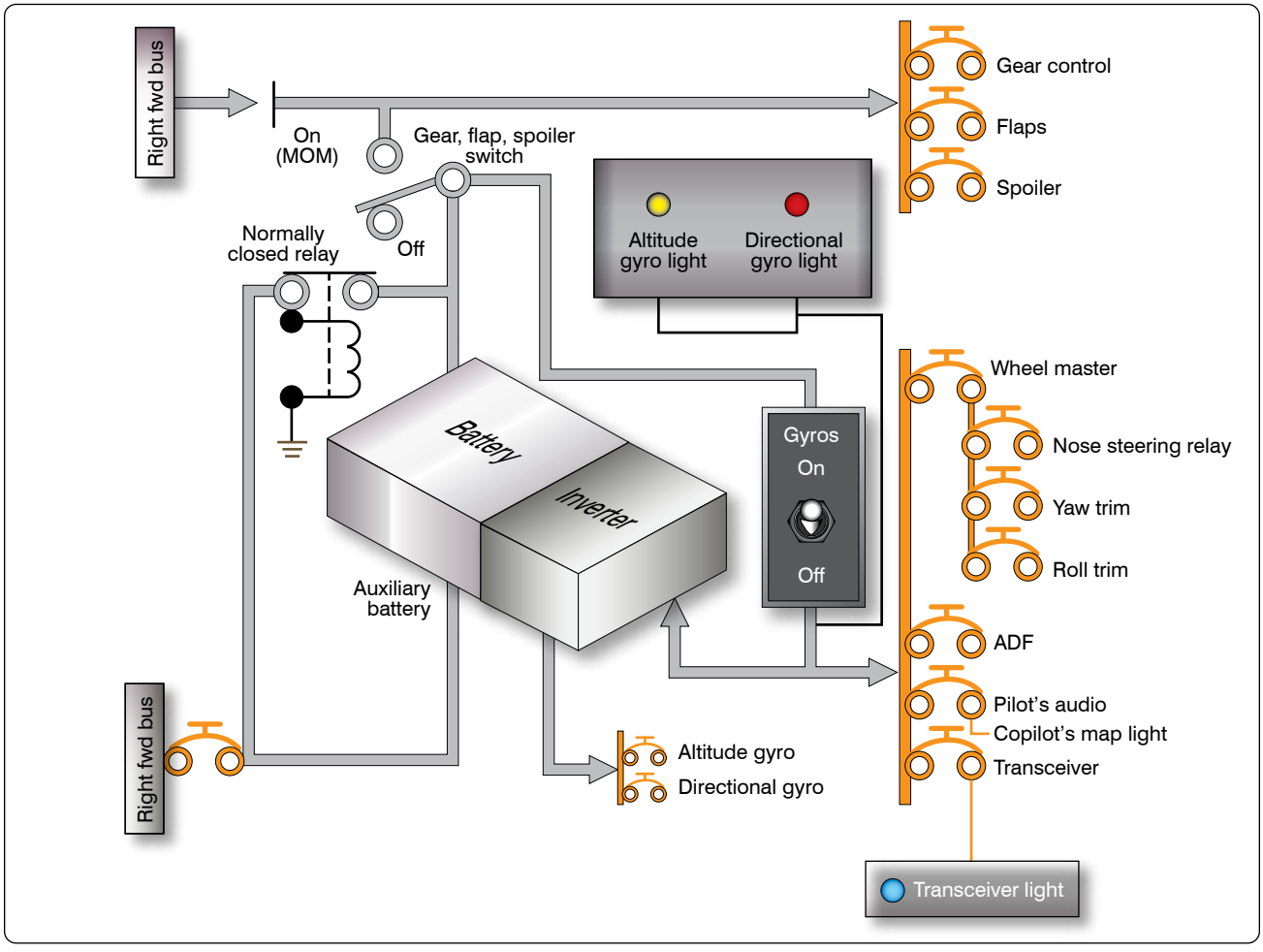


Figure 12-208. *Auxiliary battery system using static inverter.*

are the half-wave rectifier and the full-wave rectifier.

Half-Wave Rectifier

Figure 12-215 illustrates the basic concept of a half-wave rectifier. When an AC signal is on a positive swing as shown in *Figure 12-215A*, the polarities across the diode and the load resistor are also positive. In this case, the diode is forward biased and can be replaced with a short circuit as shown in the figure. The positive portion of the input signal appears across the load resistor with no loss in potential across the series diode.

Figure 12-215B now shows the input signal being reversed. Note that the polarities across the diode and the load resistor are also reversed. In this case, the diode is now reverse biased and can be replaced with an equivalent open circuit. The current in the circuit is now 0 amperes and the voltage drop over the load resistor is 0 volts. The resulting waveform for a complete

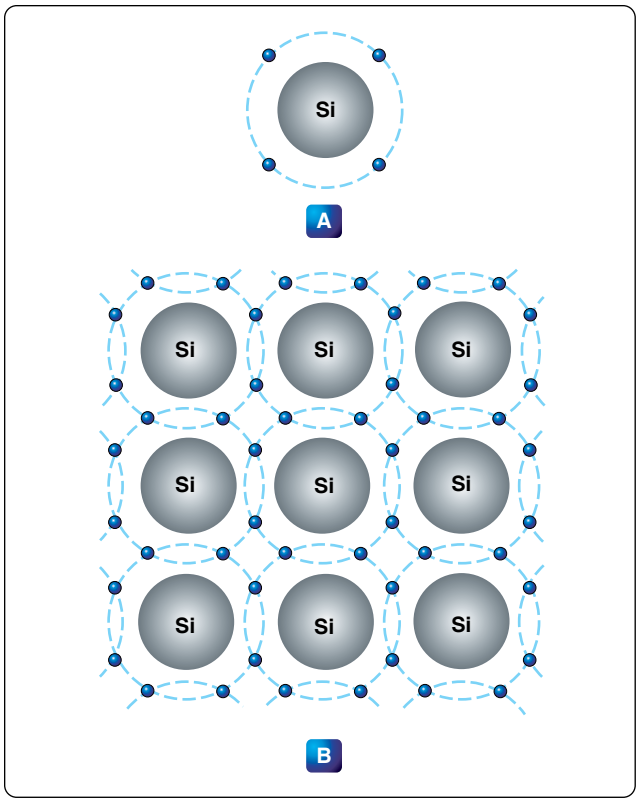


Figure 12-209. *Valence electrons.*

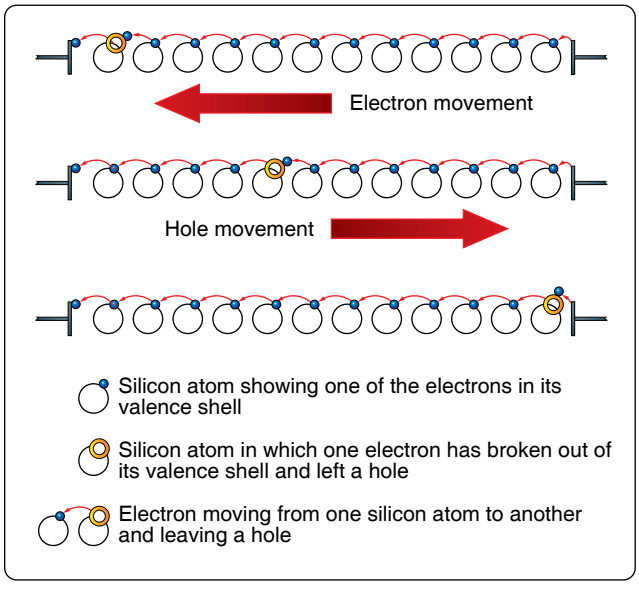


Figure 12-210. *A hole moving through atoms.*

sinusoidal input can be seen at the far right of *Figure 12-215*. The output waveform is a reproduction of the input waveform minus the negative voltage swing of the wave. For this reason, this type of rectifier is called a half-wave rectifier.

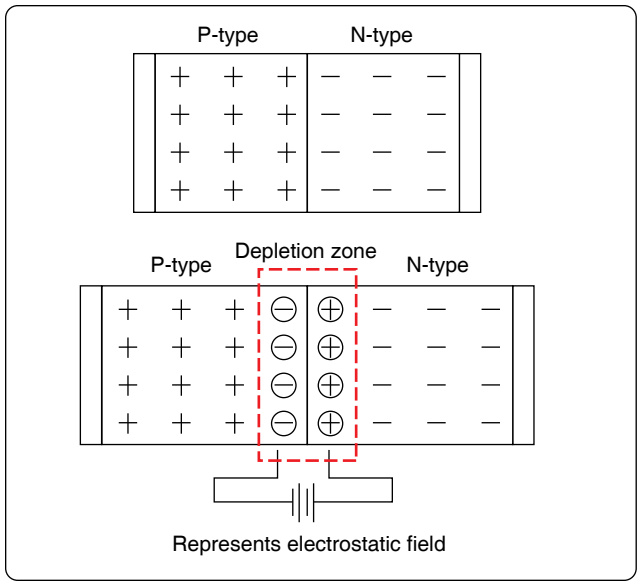


Figure 12-211. *Depletion region.*

Full-Wave Rectifier

Figure 12-216 illustrates a more common use of the diode as a rectifier. This type of a rectifier is called a full-wave bridge rectifier. The term "full-wave" indicates that the output is a continuous sequence of pulses rather than having gaps that appear in the half-wave rectifier.

Figure 12-216C shows the initial condition, during which, a positive portion of the input signal is applied to the network. Note the polarities across the diodes. Diodes D2 and D4 are reverse biased and can be replaced with an open circuit. Diodes D1 and D3 are forward biased and act as an open circuit. The current path through the diodes is clear to see, and the resulting waveform is developed across the load resistor.

During the negative portion of the applied signal, the diodes reverse their polarity and bias states. The result is a network shown in *Figure 12-216D*. Current now passes through diodes D4 and D2, which are forward biased, while diodes D1 and D3 are essentially open circuits due to being reverse biased. Note that during both alternations of the input waveform, the current passes through the load resistor in the same direction. This results in the negative swing of the waveform being flipped up to the positive side of the time line.

Dry Disk

Dry disk rectifiers operate on the principle that electric current flows through a junction of two dissimilar conducting materials more readily in one direction than it does in the opposite direction. This is true because the resistance to current flow in one direction is low, while in the other direction it is high. Depending on the materials used, several amperes may flow in the direction of low resistance but only

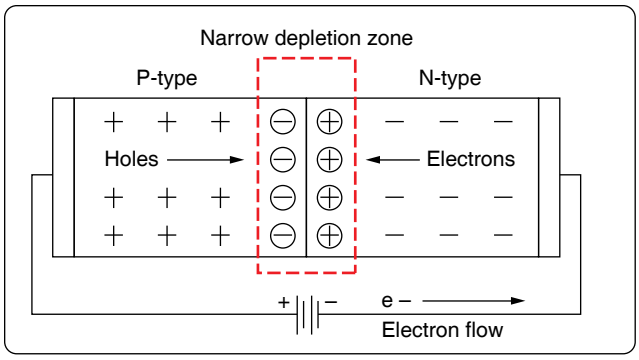


Figure 12-212. *Forward biased PN junction.*

a few milliamperes in the direction of high resistance.

Three types of dry disk rectifiers may be found in aircraft: the copper oxide rectifier, the selenium rectifier, and the magnesium copper-sulfide rectifier. The copper oxide rectifier consists of a copper disk upon which a layer of copper oxide has been formed by heating. *[Figure 12-217]* It may also consist of a chemical copper oxide preparation spread evenly over the copper surface. Metal plates, usually lead plates, are pressed against the two opposite faces of the disk to form a good contact. Current flow is from the copper to the copper oxide.

The selenium rectifier consists of an iron disk, similar to a washer, with one side coated with selenium. Its operation is similar to that of the copper oxide rectifier. Current flows from the selenium to the iron.

The magnesium copper sulfide rectifier is made of washer-shaped magnesium disks coated with a layer of copper sulfide. The disks are arranged similarly to the other types. Current flows from the magnesium to the copper sulfide.

Types of Diodes

Today, there are many varieties of diodes that can be grouped into one of several basic categories.

Power Rectifier Diodes

The rectifier diode is usually used in applications that require high current, such as power supplies. The range in which the diode can handle current can vary anywhere from one ampere to hundreds of amperes. One common example of diodes is the series of diodes, part numbers 1N4001 to 1N4007. The "1N" indicates that there is only one PN junction, or that the device is a diode. The average current carrying range for these rectifier diodes is about one ampere with a peak inverse voltage between 50 volts to 1,000 volts. Larger rectifier diodes can carry currents up to 300 amperes when forward biased and have a peak inverse voltage of 600 volts. A recognizable feature of the larger rectifier diodes is that they are encased in metal in order to provide a heat sink. *[Figure 12-218]*

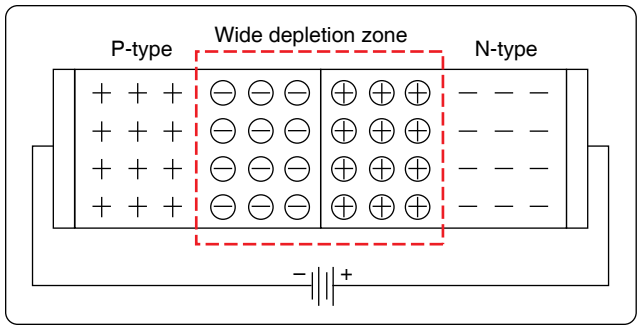


Figure 12-213. *Reversed diode.*

Zener Diodes

Zener diodes (sometimes called "breakdown diodes") are designed so that they break down (allow current to pass) when the circuit potential is equal to or in excess of the desired reverse bias voltage. The range of reverse bias breakdown-voltages commonly found can range from 2 volts to 200 volts depending on design. Once a specific reverse bias voltage has been reached, the diode conducts and behaves like a constant voltage source. Within the normal operating range, the zener functions as a voltage regulator, waveform clipper, and other related functions. Below the desired voltage, the zener blocks the circuit like any other diode biased in the reverse direction. Because the zener diode allows free flow in one direction when it is used in an AC circuit, two diodes connected in opposite directions must be used. This takes care of both alternations of current. Power ratings of these devices range from about 250 milliwatts to 50 watts.

Special Purpose Diodes

The unique characteristics of semiconductor material have allowed for the development of many specialized types of diodes. A short description of some of the more common diode types is given for general familiarization. *[Figure 12-219]*

Light-Emitting Diode (LED)

In a forward biased diode, electrons cross the junction and fall into holes. As the electrons fall into the valence band, they radiate energy. In a rectifier diode, this energy is dissipated as heat. However, in the light-emitting diode (LED), the energy is dissipated as light. By using elements such as gallium, arsenic, and phosphorous, an LED can be designed to radiate colors, such as red, green, yellow, blue, and infrared light. LEDs that are designed for the visible light portion of the spectrum are useful for instruments, indicators, and even cabin lighting. The advantages of the LED over the incandescent lamps are longer life, lower voltage, faster on and off operations, and less heat.

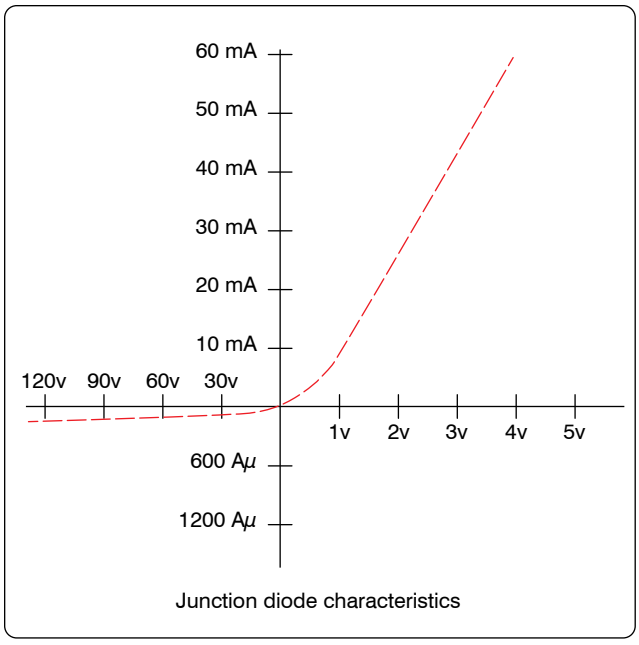


Figure 12-214. *Diode biased in both directions.*

Liquid Crystal Displays (LCD)

The liquid crystal display (LCD) has an advantage over the LED in that it requires less power to operate. Where LEDs commonly operate in the milliwatt range, the LCD operates in the microwatt range. The liquid crystal is encapsulated between two glass plates. When voltage is not applied to the LCD, the display is clear. However, when a voltage is applied, the result is a change in the orientation of the atoms of the crystals. The incident light is then reflected in a different direction. A frosted appearance results in the regions that have voltage applied and permits distinguishing of numeric values.

Photodiode

Thermal energy produces minority carriers in a diode. The higher the temperature, the greater the current in a reverse current diode. Light energy can also produce minority carriers. By using a small window to expose the PN junction, a photodiode can be built. When light falls upon the junction of a reverse-biased photodiode, electrons-hole pairs are created inside the depletion layer. The stronger the light, the greater the number of light-produced carriers, which in turn causes a greater magnitude of reverse-current. Because of this characteristic, the photodiode can be used in light detecting circuits.

Varactors

The varactor is simply a variable-capacitance diode. The reverse voltage applied controls the variable-capacitance of the diode. The transitional capacitance decreases as the reverse voltage is increasingly applied. In many

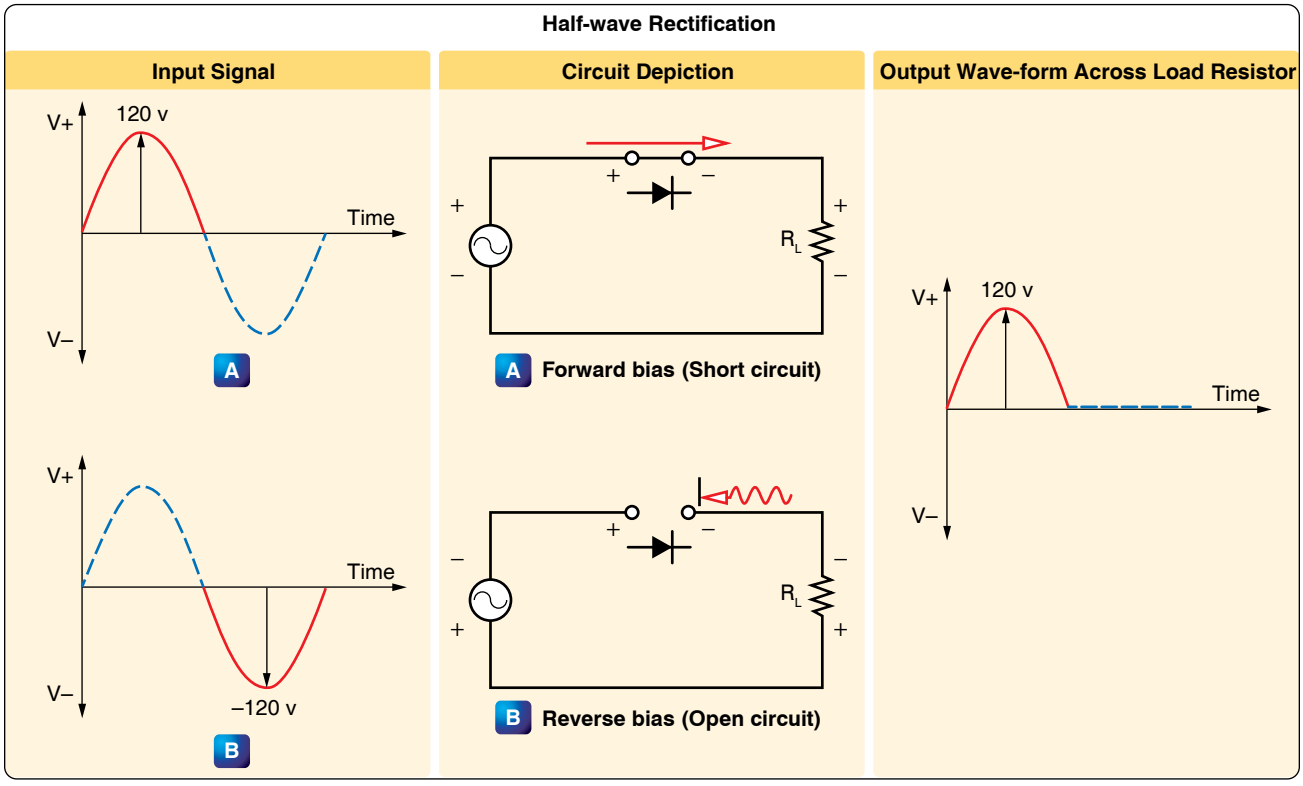


Figure 12-215. *Basic concept of half-wave rectifier.*

applications, the varactor has replaced the old mechanically tuned capacitors. Varactors can be placed in parallel with an inductor and provide a resonant tank circuit for a tuning circuit. By simply varying the reverse voltage across the varactor, the resonant frequency of the circuit can be adjusted.

Schottky Diodes

Schottky diodes are designed to have metal, such as gold, silver, or platinum, on one side of the junction and doped silicon, usually an N-type, on the other side of the junction. This type of a diode is considered a unipolar device because free electrons are the majority carrier on both sides of the junction. The Schottky diode has no depletion zone or charge storage, which means that the switching time can be as high as 300 MHz. This characteristic exceeds that of the bipolar diode.

Diode Identification

Figure 12-218 illustrates a number of methods employed for identifying diodes. Typically manufacturers place some form of an identifier on the diode to indicate which end is the anode and which end is the cathode. Dots, bands, colored bands, the letter 'k' or unusual shapes indicate the cathode end of the diode.

Introduction to Transistors

The transistor is a three-terminal device primarily used to amplify signals and control current within a circuit. *[Figure 12-220]* The basic two-junction semiconductor must have one type of region sandwiched between two of the other type. The three regions in a transistor are the collector (C), which is moderately doped, the emitter (E), which is heavily doped, and the base (B), which is significantly less doped. The alternating layers of semiconductor material type provide the common commercial name for each type of transistor. The interface between the layers is called a junction. Selenium and germanium diodes previously discussed are examples of junction diodes. Note that the sandwiched layer or base is significantly thinner than the collector or the emitter. In general, this permits a "punching through" action for the carriers passing between the collector and emitter terminals.

Classification

The transistors are classified as either NPN or PNP according to the arrangement of their N and P-materials. The NPN transistor is formed by introducing a thin region of P-material between two regions of N-type material. The opposite is true for the PNP configuration.

The two basic types of transistors along with their circuit

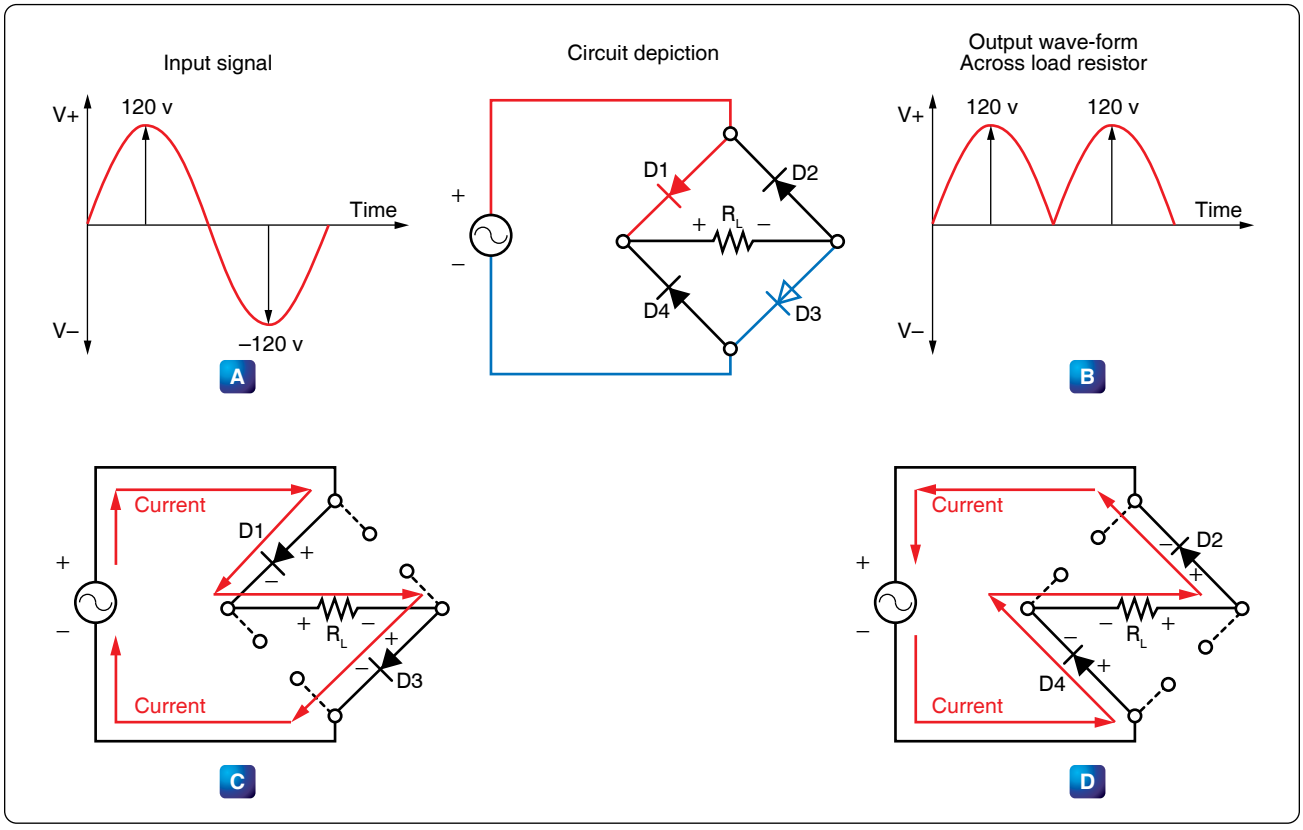


Figure 12-216. *Full-wave bridge rectifier.*

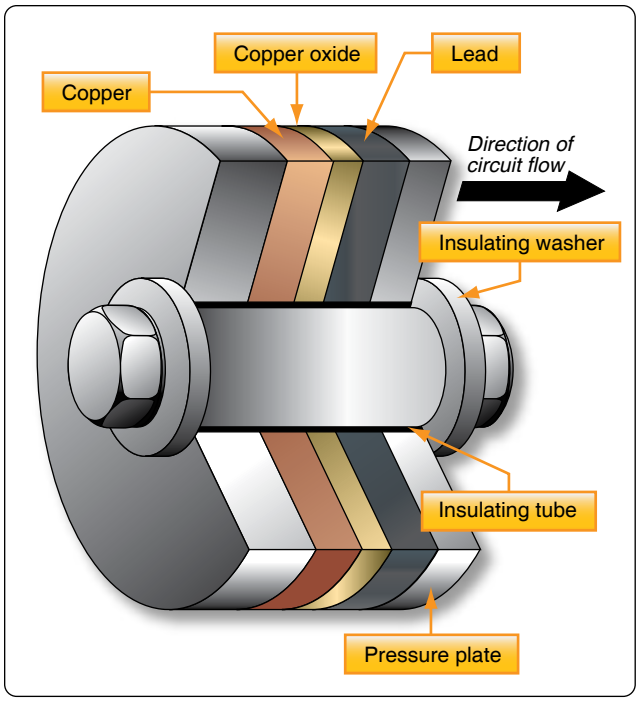


Figure 12-217. *Copper oxide dry disk rectifier.*

symbols are shown in *Figure 12-221*. Note that the two symbols are different. The horizontal line represents the base, and two angular lines represent the emitter and collector. The angular line with the arrow on it is the emitter, while the line without is the collector. The direction of the arrow on the emitter determines whether or not the transistor is a PNP or an NPN type. If the arrow is pointing in, the transistor is a PNP. On the other hand, if the arrow is pointing out, then it is an NPN type.

Transistor Theory

As discussed in the section on diodes, the movement of the electrons and holes can be considered current. Electron current moves in one direction, while hole current travels in the opposite direction. In transistors, both electrons and holes act as carriers of current.

A forward biased PN junction is comparable to a low-resistance circuit element, because it passes a high current for a given voltage. On the other hand, a reverse-biased PN junction is comparable to a high-resistance circuit element. By using Ohm's Law formula for power ($P = I^2R$) and assuming current is held constant through both junctions, it can be concluded that the power developed across the high resistance junction is greater than that developed across a low resistance junction. Therefore, if a crystal were to contain two PN junctions, one forward biased and the other reverse biased, and a low-power signal injected into the forward biased junction, a high-power signal could be produced at the reverse-biased junction.

To use the transistor as an amplifier, some sort of external bias voltage must modify each of the junctions. The first PN junction (emitter-base) is biased in the forward direction. This produces a low resistance. The second junction, which is the collector-base junction, is reverse biased to produce a high resistance. *[Figure 12-222]*

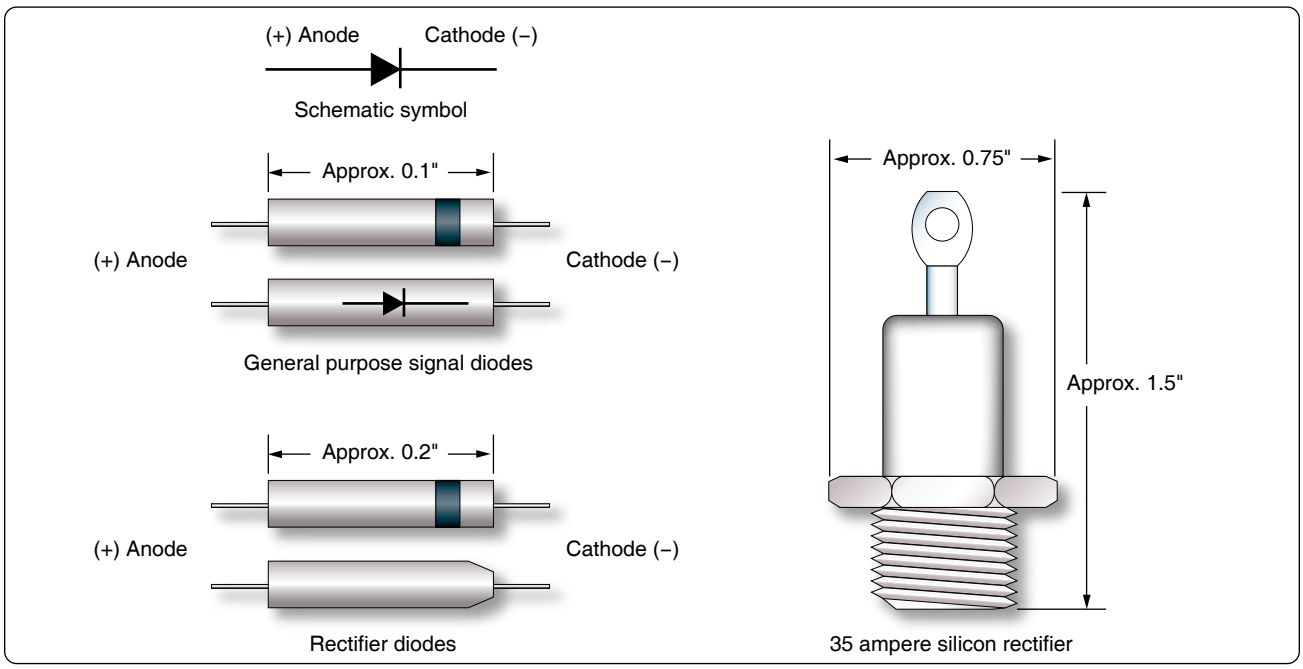


Figure 12-218. *General purpose diodes.*

With the emitter-base junction biased in the forward direction, electrons leave the negative terminal of the battery and enter the N-material. These electrons pass easily through the emitter, cross over the junction, and combine with the hole in the P-material in the base. For each electron that fills a hole in the P-material, another electron leaves the P-material, which creates a new hole and enters the positive terminal of the battery.

The second PN junction, which is the base-collector junction, is reverse biased. This prevents the majority carriers from crossing the junction, thus creating a high-resistance circuit. It is worth noting that there still is a small current passing through the reversed PN junction in the form of minority carriers—that is, electrons in the P-material and holes in the N-material. The minority carriers play a significant part in the operation of the NPN transistor.

Figure 12-223 illustrates the basic interaction of the NPN junction. There are two batteries in the circuit used to bias the NPN transistor. Vbb is considered the base voltage supply, rated in this illustration at 1 volt, and the battery voltage Vcc, rated at 6 volts, is called the collector voltage supply.

Current within the external circuit is simply the movement of free electrons originating at the negative terminal of the battery and flowing to the N-material. *[Figure 12-223]*
As the electrons enter the N-material, they become the majority carrier and move through the N-material to the emitter-base PN junction. This emitter-base junction is forward biased at about 0.65 to 0.7 volts positive with respect to the emitter and presents no resistance to the flow of electrons from the emitter into the base, which is composed of P-material. As these electrons move into the base, they drop into available holes. For every electron that drops into a hole, another electron exits the base by way of the base lead and becomes the base current or Ib. Of course, when one electron leaves the base, a new hole is formed. From the standpoint of the collector, these electrons that drop into holes are lost and of no use. To reduce this loss of electrons, the transistor is designed so that the base is very thin in relation to the emitter and collector, and the base is lightly doped.

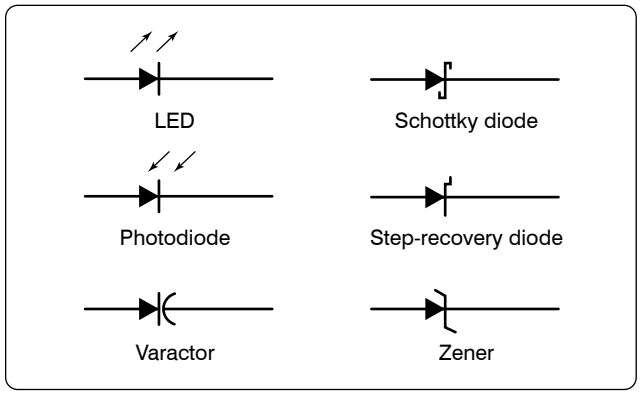


Figure 12-219. *Schematic symbols for special purpose diodes.*

Most of the electrons that move into the base fall under the influence of the reverse bias of the collector. While collector-base junction is reverse biased with respect to the majority carriers, it behaves as if it is forward biased to the electrons or minority carriers in this case. The electrons are accelerated through the collector-base junction and into the collector. The collector is comprised of the N-type material; therefore, the electrons once again become the majority carrier. Moving easily through the collector, the electrons return to the positive terminal of the collector supply battery Vcc, which is shown in *Figure 12-223* as Ic.

Because of the way this device operates to transfer current (and its internal resistances) from the original conduction path to another, its name is a combination of the words "transfer" and "resistor"—transistor.

PNP Transistor Operation

The PNP transistor generally works the same way as the NPN transistor. The primary difference is that the emitter, base, and collector materials are made of different material than the NPN. The majority and minority current carriers are the opposite in the PNP to that of the NPN. In the case of the PNP, the majority carriers are the holes instead of the electrons in the NPN transistor. To properly bias the PNP, the polarity of the bias network must be reversed.

Identification of Transistors

Figure 12-224 illustrates some of the more common transistor lead identifications. The methods of identifying leads vary due to a lack of a standard and require verification using manufacturer information to properly identify. However, a short description of the common methods is discussed below.

Figure 12-224D shows an oval-shaped transistor. The collector lead in this case is identified by the wide space between it and the lead for the base. The final lead at the far left is the emitter. In many cases, colored dots indicate the collector lead, and short leads relative to the other leads indicate the emitter. In a conventional power diode, as seen in *Figure 12-224E*, the collector lead is usually a part of the mounting bases, while the emitter and collector are leads or tines protruding from the mounting surface.

Field Effect Transistors

Another transistor design that has become more important than the bipolar transistor is the field-effect transistor (FET). The primary difference between the bipolar transistor and the FET is that the bipolar transistor has two PN junctions and

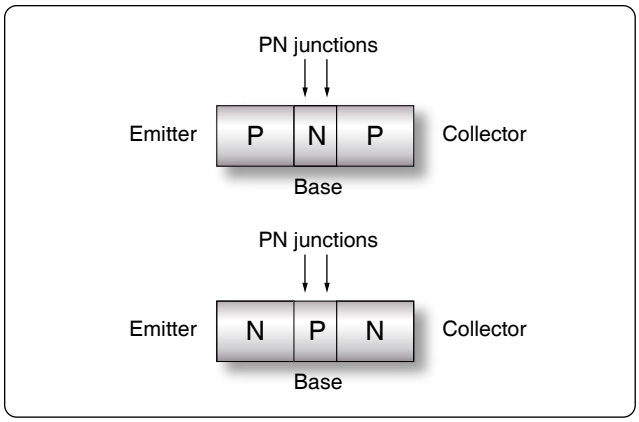


Figure 12-220. *Transistor.*

is a current-controlled device, while the FET has only one PN junction and is a voltage-controlled device. Within the FET family, there are two general categories of components. One category is called the junction FET (JFET), which has only one PN junction. The other category is known as the enhancement-type or metal-oxide JET (MOSFET).

Figure 12-225 shows the basic construction of the JFET and the schematic symbol. In this figure, it can be seen that the drain (D) and source (S) are connected to an N-type material, and the gate (G) is connected to the P-type material. With gate voltage Vgg set to 0 volts and drain voltage Vdd set to some positive voltage, a current flows between the source and the drain, through a narrow band of N-material. If then, Vgg is adjusted to some negative voltage, the PN junction is reverse biased, and a depletion zone (no charge carriers) is established at the PN junction. By reducing the region of noncarriers, it has the effect of reducing the dimensions of the N-channel, resulting in a reduction of source to drain current.

Because JFETs are voltage-controlled devices, they have some advantages over the bipolar transistor. One such advantage is that because the gate is reverse biased, the circuit that it is connected to sees the gate as a very high resistance. This means that the JFET has less of an insertion influence in the circuit. The high resistance also means that less current is used.

Like many other solid-state devices, careless handling and static electricity can damage the JFET. Technicians should take all precautions to prevent such damage.

Metal-Oxide-Semiconductor FET (MOSFET)

Figure 12-226 illustrates the general construction and the schematic symbol of the MOSFET transistor. The biasing arrangement for the MOSFET is essentially the same as that for the JFET. The term "enhancement" comes from the idea that when there is no bias voltage applied to the gate (G), then there is no channel for current conduction between the

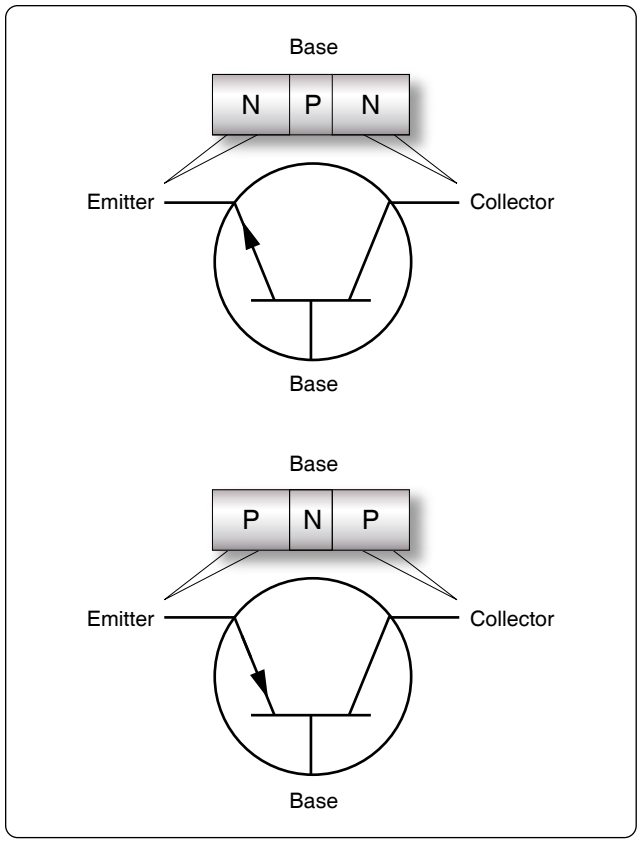


Figure 12-221. *Two basic transistors with circuit symbols.*

source (S) and the drain (D). By applying a greater voltage on the gate (G), the P-channel begins to materialize and grow in size. Once this occurs, the source (S) to drain (D) current Id increases. The schematic symbol reflects this characteristic by using a broken line to indicate that the channel does not exist without a gate bias.

Common Transistor Configurations

A transistor may be connected in one of three different configurations: common-emitter (CE), common-base (CB), and common-collector (CC). The term "common" is used to indicate which element of the transistor is common to both the input and the output. Each configuration has its own characteristics, which makes each configuration suitable for particular applications. A way to determine what configuration you may find in a circuit is to first determine which of the three transistor elements is used for the input signal. Then, determine the element used for the output signal. At that point, the remaining element, (base, emitter, or collector) is the common element to both the input and output, and thus you determine the configuration.

Common-Emitter (CE) Configuration

This is the configuration most commonly used in amplifier

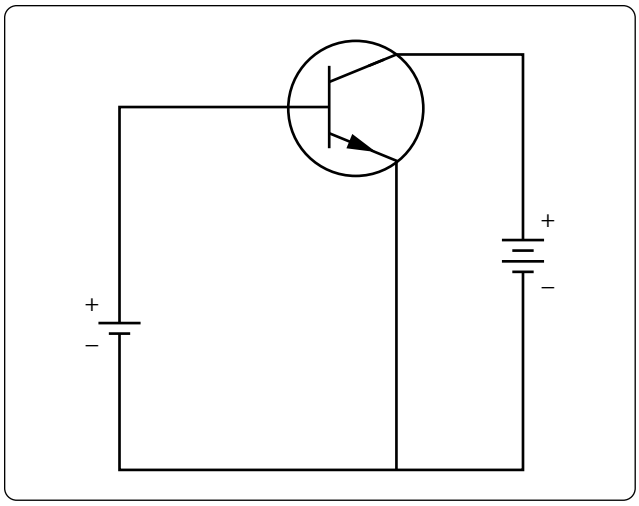


Figure 12-222. *NPN transistor.*

circuits because they provide good gains for voltage, current, and power. The input signal is applied to the base-emitter junction, which is forward biased (low resistance), and the output signal is taken off the collector-emitter junction, which is reverse biased (high resistance). Then the emitter is the common element to both input and output circuits. *[Figure 12-227]*

When the transistor is connected in a CE configuration, the input signal is injected between the base and emitter, which is a low-resistance, low-current circuit. As the input signal goes positive, it causes the base to go positive relative to the emitter. This causes a decrease in the forward bias, which in turn reduces the collector current I_C and increases the collector voltage (E_C being more negative). During the negative portion of the input signal, the voltage on the base is driven more negative relative to the emitter. This increases the forward bias and allows an increase in collector current

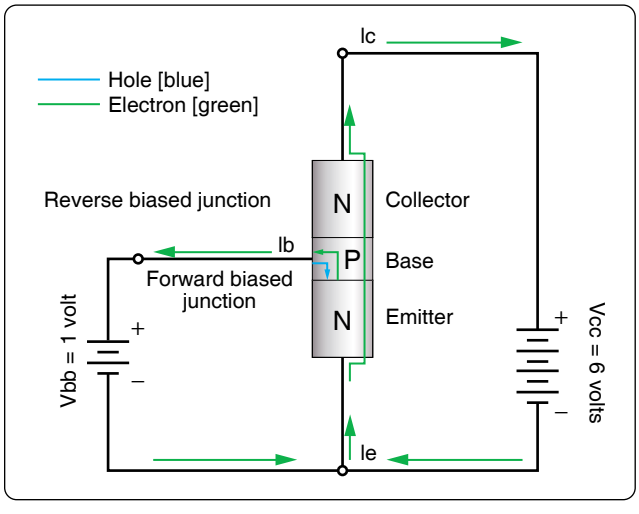


Figure 12-223. *NPN Junction.*

I_C and a decrease in collector voltage (E_C being less negative and going positive). The collector current, which flows through the reverse-biased junction, also flows through a high-resistance load resulting in a high level of amplification.

Because the input signal to the CE goes positive when the output goes negative, the two signals are 180° out of phase. This is the only configuration that provides a phase reversal. The CE is the most popular of the three configurations because it has the best combination of current and voltage gain. Gain is a term used to indicate the magnitude of amplification. Each transistor configuration has its unique gain characteristics even though the same transistors are used.

Common-Collector (CC) Configuration

This transistor configuration is usually used for impedance matching. It is also used as a current driver due to its high current gain. It is also very useful in switching circuits since it has the ability to pass signals in either direction. *[Figure 12-227]*

In the CC circuit, the input signal is applied to the base, and the output signal is taken from the emitter, leaving the collector as the common point between the input and the output. The input resistance of the CC circuit is high, while the output resistance is low. The current gain is higher than that in the CE, but it has a lower power gain than either the CE or CB configuration. Just like the CB configuration, the output signal of the CC circuit is in phase with the input signal. The CC is typically referred to as an emitter-follower because the output developed on the emitter follows the input signal applied to the base.

Common-Base (CB) Configuration

The primary use of this configuration is for impedance matching because it has low input impedance and high output resistance. Two factors, however, limit the usefulness of this circuit application. First is the low-input resistance and second is its lack of current, which is always below 1. Since the CB configuration gives voltage amplification, there are some applications for this circuit, such as microphone amplifiers. *[Figure 12-227]*

In the CB circuit, the input signal is applied to the emitter and the output signal is taken from the collector. In this case, both the input and the output have the base as a common element. When an input signal is applied to the emitter, it causes the emitter-base junction to react in the same manner as that in the CE circuit. When an input adds to the bias, it increases the transistor current; conversely, when the signal opposes the bias, the current in the transistor decreases.

The signal adds to the forward bias, since it is applied to the emitter, causing the collector current to increase. This increase in I_C results in a greater voltage drop across the

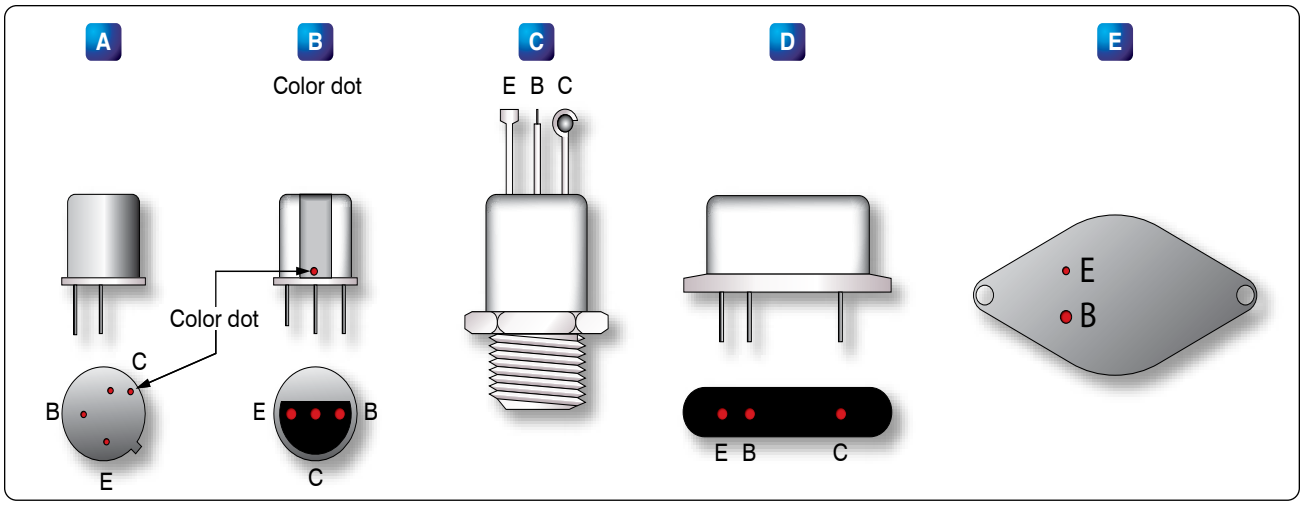


Figure 12-224. *Common transistor lead identifications.*

load resistor RL, thus lowering the collector voltage E_C. The collector voltage, in becoming less negative, swings in a positive direction and is therefore in phase with the incoming positive signal.

Vacuum Tubes

The use of vacuum tubes in aircraft electrical and electronic systems has rapidly declined due to the many advantages of using transistors. However, some systems still employ vacuum tubes in special applications, and possibly some older model aircraft still in service are equipped with devices that use vacuum tubes. While these components may still be in service, their infrequent occurrence does not warrant a detailed discussion.

Originally, vacuum tubes were developed for radio work. They are used in radio transmitters as amplifiers for controlling voltage and current, as oscillators for generating audio and radio frequency signals, and as rectifiers for converting AC into DC. While there are many types of vacuum tubes for a variety of applications, the most common types fall into one of the following families: (1) diode, (2) triode, (3) tetrode, and (4) pentode. Each of these vacuum tube types operates on the following fundamental principles.

When a piece of metal is heated, the speed of the electrons in the metal is increased. If the metal is heated to a high enough temperature, the electrons are accelerated to the point where some of them actually leave the surface of the metal. In a vacuum tube, electrons are supplied by a piece of metal called a cathode, which is heated by an electric current. Within limits, the hotter the cathode, the greater the number of electrons it gives off or emits.

To increase the number of electrons emitted, the cathode is usually coated with special chemical compounds. If an external field does not draw the emitted electrons away, they form about the cathode into a negatively-charged cloud called the space charge. The accumulation of negative electrons near the emitter repels others coming from the emitter. The emitter, if insulated, becomes positive because of the loss of electrons. This establishes an electrostatic field between the cloud of negative electrons and the now positive cathode. A balance is reached when only enough electrons flow from the cathode to the area surrounding it to supply the loss caused by diffusion of the space charge.

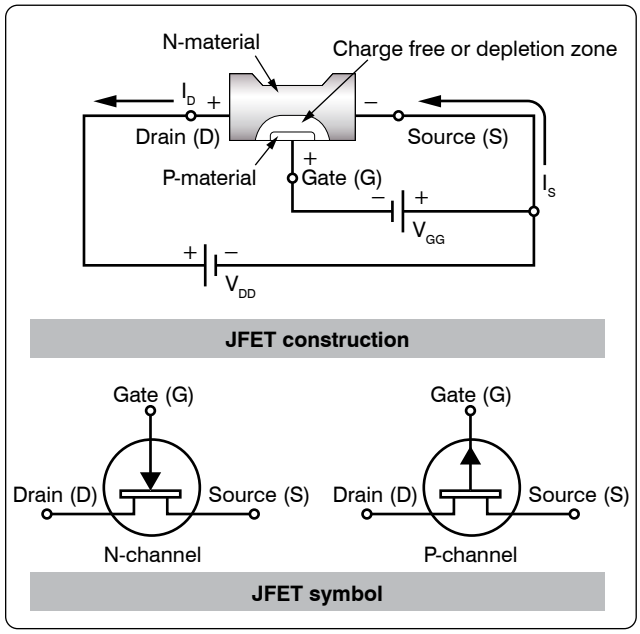


Figure 12-225. *JFET and the schematic symbol.*

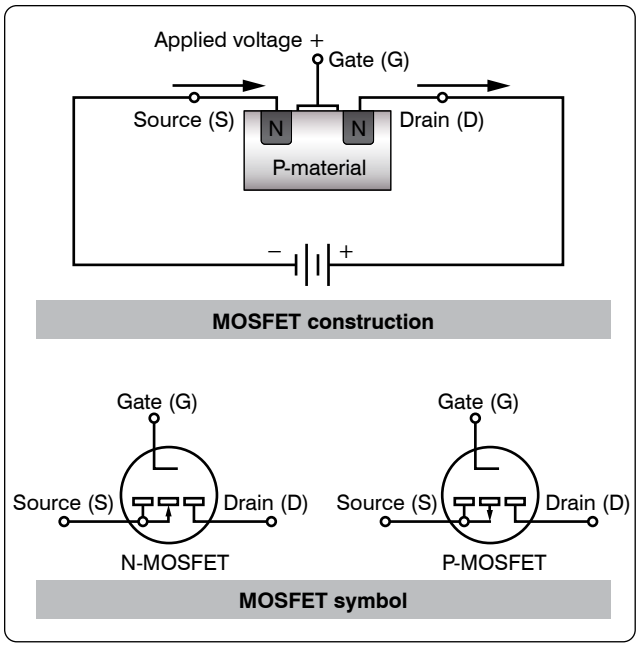


Figure 12-226. *General construction and schematic symbol of MOSFET transistor.*

Filtering

One of the more common uses of the capacitor and inductor that the technician may find in the field is that of the filter.

Filtering Characteristics of Capacitors

The nature of capacitance opposes a voltage change across its terminal by storing energy in its electrostatic field. Whenever the voltage tends to rise, the capacitor converts this voltage change to stored energy. When the voltage tends to fall, the capacitor converts this stored energy back to voltage. The use of a capacitor for filtering the output of a rectifier is illustrated in *Figure 12-228*. The rectifier is shown as a block, and the capacitor C_1 is connected in parallel with the load R_1.

The capacitor C_1 is chosen to offer very low impedance to the AC ripple frequency and very high impedance to the DC component. The ripple voltage is therefore bypassed to ground through the low impedance path of the capacitor, while the DC voltage is applied unchanged to the load. The effect of the capacitor on the output of the rectifier can be seen in the waveshapes shown in *Figure 12-229*. Dotted lines show the rectifier output, while the solid lines show the effect of the capacitor. In this example, full-wave rectifier outputs are shown. The capacitor C_1 charges when the rectifier voltage output tends to increase and discharges when the voltage output tends to decrease. In this manner, the voltage across the load R_1 is kept fairly constant.

Filtering Characteristics of Inductors

The inductance provided by an inductor may be used as a filter, because it opposes a change in current through it by storing energy in its electromagnetic field. Whenever the current increases, the stored energy in the electromagnetic field increases. When the current through the inductor decreases, the inductor supplies the energy back into the circuit in order to maintain the existing flow of current. The use of an inductor for filtering the output of a rectifier is shown in *Figure 12-230*. Note that in this network the inductor L_1 is in series with the load R_1.

The inductance L_1 is selected to offer high impedance to the AC ripple voltage and low impedance to the DC component. The result is a very large voltage drop across the inductor and a very small voltage drop across the load R_1. For the DC component, however, a very small voltage drop occurs across the inductor and a very large voltage drop across the load. The effect of an inductor on the output of a full-wave rectifier in the output waveshape is shown in *Figure 12-231*.

Common Filter Configurations

Capacitors and inductors are combined in various ways to provide more satisfactory filtering than can be obtained with a single capacitor or inductor. These are referred to collectively as LC filters. Several combinations are shown schematically in *Figure 12-232*. Note that the L, or inverted L-type, and the T-type filter sections resemble schematically

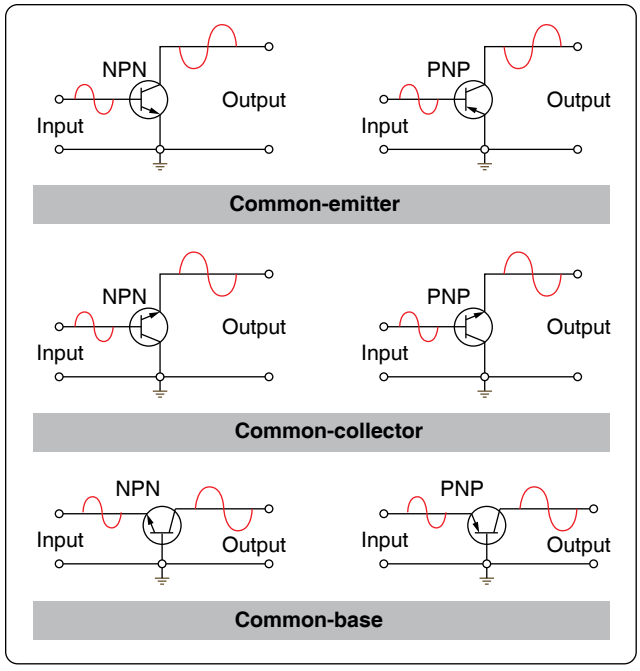


Figure 12-227. *Transistor configurations (common-emitter, common-collector, common-base).*

the corresponding letters of the alphabet. The pi-type filter section resembles the Greek letter pi (π) schematically.
All the filter sections shown are similar in that the inductances are in series and the capacitances are in parallel with the load. The inductances must, therefore, offer very high impedance and the capacitors very low impedance to the ripple frequency. Since the ripple frequency is comparatively low, the inductances are iron core coils having large values of inductance (several henries). Because they offer such high impedance to the ripple frequency, these coils are called chokes. The capacitors must also be large (several microfarads) to offer very little opposition to the ripple frequency. Because the voltage across the capacitor is DC, electrolytic capacitors are frequently used as filter capacitors. Always observe the correct polarity in connecting electrolytic capacitors.

LC filters are also classified according to the position of the capacitor and inductor. A capacitor input filter is one in which the capacitor is connected directly across the output terminals of the rectifier. A choke input filter is one in which a choke precedes the filter capacitor.

If it is necessary to increase the applied voltage to more than a single rectifier can tolerate, the usual solution is to stack them. These rectifiers are similar to resistors added in series. Each resistor drops a portion of the applied voltage rather than the total voltage. The same theory applies to rectifiers added in series or stacked. Series stacking increases the voltage rating. If, for example, a rectifier is destroyed with an applied voltage exceeding 50 volts, and it is to be used in a circuit with an applied voltage of 150 volts, stacking of diodes can be employed. The result is shown in *Figure 12-233*.

Basic LC Filters

Analog filters are circuits that perform signal processing functions, specifically intended to remove unwanted signal components, such as ripple, and enhance desired signals. The simplest analog filters are based on combinations of inductors and capacitors. The four basic categories of filters discussed are: low-pass, high-pass, band-pass and band-stop. All these types are collectively known as passive filters, because they do not depend on any external power source.

The operation of a filter relies on the characteristic of variable inductive and capacitive reactance based on the applied frequency. In review, the inductor blocks high-frequency signals (high reactance) and conducts low-frequency signals (low reactance), while capacitors do the reverse. A filter in which the signal passes through an inductor, or in which a capacitor provides a path to earth, presents less attenuation (reduction) to a low-frequency signal than to a high-frequency signal and is considered a low-pass filter. If the signal passes through a capacitor, or has a path to ground through an inductor, then the filter presents less attenuation to high-frequency signals than low-frequency signals and is then considered a high-pass filter. Typically after an AC signal is rectified, the pulses of voltage are changed to usable form of DC by way of filtering.

Low-Pass Filter

A low-pass filter is a filter that passes low frequencies well, but attenuates (reduces) higher frequencies. The so-called cutoff frequency divides the range of frequencies that are passed and the range of frequencies that are stopped. In other words, the frequency components higher than the cutoff frequency are stopped by a low-pass filter. The actual amount of attenuation for each frequency varies by filter design.

An inductive low-pass filter inserts an inductor in series with the load, where a capacitive low-pass filter inserts a resistor in series and a capacitor in parallel with the load. The former filter design tries to block the unwanted frequency signal while the latter tries to short it out. *Figure 12-234* illustrates this type of circuit and the frequency/current flow response.

High-Pass Filter (HPF)

A high-pass filter (HPF) is a filter that passes high frequencies well, but attenuates (reduces) frequencies lower than the cutoff frequency. The actual amount of attenuation for each frequency varies once again depending on filter design. In some cases, it is called a low-cut filter. A HPF is essentially the opposite of a low-pass filter.

It is useful as a filter to block any unwanted low frequency components of a signal while passing the desired higher frequencies. *Figure 12-235* illustrates this type of circuit and the frequency/current flow response.

Band-Pass Filter

A band-pass filter is basically a combination of a high-pass and a low-pass. There are some applications where a particular range of frequencies need to be singled out or filtered from a wider range of frequencies. Band-pass filter circuits are designed to accomplish this task by combining the properties of low-pass and high-pass into a single filter. *Figure 12-236* illustrates this type of circuit and the frequency/current flow response.

Band-Stop Filter

In signal processing, a band-stop filter or band-rejection filter is a filter that passes most frequencies unaltered, but attenuates those in a range to very low levels. It is the opposite of a band-pass filter. A notch filter is a band-stop filter with a narrow stopband (high Q factor). Notch filters are used in live sound reproduction (public address (PA) systems) and in instrument

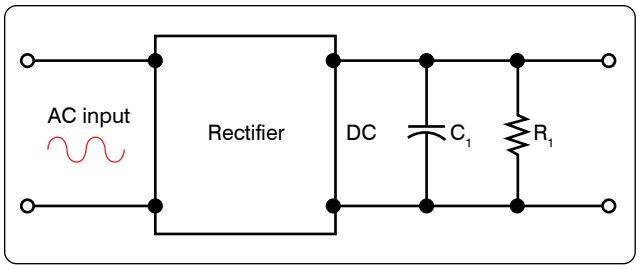


Figure 12-228. *A capacitor used as a filter.*

amplifier (especially amplifiers or preamplifiers for acoustic instruments, such as acoustic guitar, mandolin, bass instrument amplifier, etc.) to reduce or prevent feedback, while having little noticeable effect on the rest of the frequency spectrum. Other names include "band limit filter," "T-notch filter," "band-elimination filter," and "band-rejection filter."

Typically, the width of the stop-band is less than 1 to 2 decades (that is, the highest frequency attenuated is less than 10 to 100 times the lowest frequency attenuated). In the audio band, a notch filter uses high and low frequencies that may be only semitones apart.

A band-stop filter is the general case. A notch filter is a specific type of band-stop filter with a very narrow range. Also called band-elimination, band-reject, or notch filters, this kind of filter passes all frequencies above and below a particular range set by the component values. Not surprisingly, it can be made out of a low-pass and a high-pass filter, just like the band-pass design, except that this time we connect the two filter sections in parallel with each other instead of in series. *Figure 12-237* illustrates this type of circuit and the

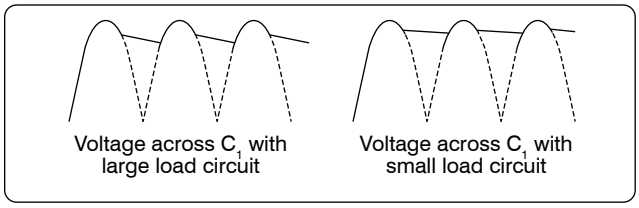


Figure 12-229. *Half-wave and full-wave rectifier outputs using capacitor filter.*

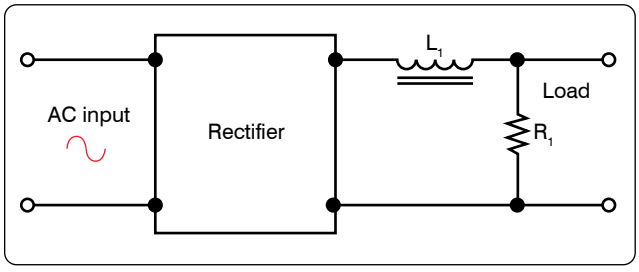


Figure 12-230. *An inductor used as a filter.*

frequency/current flow response.

Amplifier Circuits

An amplifier is a device that enables an input signal to control an output signal. The output signal has some or all of the characteristics of the input signal but generally is a greater magnitude than the input signal in terms of voltage, current, or power. Gain is the basic function of all amplifiers. Because of this gain, we can expect the output signal to be greater than the input signal. For example, if we have an input signal of 1 volt and an output signal of 10 volts, then the gain factor can be determined by:

Gain = Signal out /signal in
Gain = 10 V/1 V = 10

Voltage gain is usually used to describe the operation of a small gain amplifier. In this type of an amplifier, the output signal voltage is larger than the input signal voltage. Power gain, on the other hand, is usually used to describe the operation of large signal amplifiers. In the case of power gain amplifiers, the gain is not based on voltage but on watts. A power amplifier is an amplifier in which the output signal power is greater than the input signal power. Most power amplifiers are used as the final stage of amplification and drive the output device. The output device could be a flight deck or cabin speaker, an indicator, or antenna. Whatever the device, the power to make it work comes from the final stage of amplification. Drivers for autopilot servos are sometimes contained in line replaceable units (LRUs) called autopilot amplifiers. These units take the low signal commands from the flight guidance system and amplify the signals to a level usable for driving the servo motors.

Classification

The classification of a transistor amplifier circuit is determined by the percentage of the time that the current flows through the output circuit in relation to the input signal. There are four classifications of operation: A, AB, B, and C. Each class of operation has a certain use and characteristic. No individual class of amplifiers is considered the best. The best use of an amplifier is a matter of proper selection for the particular operation desired.

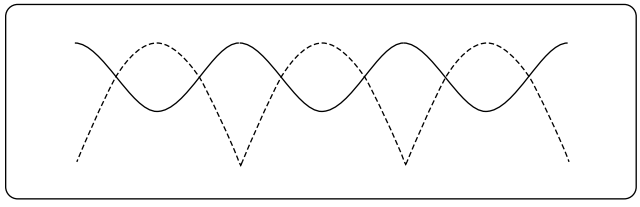


Figure 12-231. *Output of an inductor filter rectifier.*

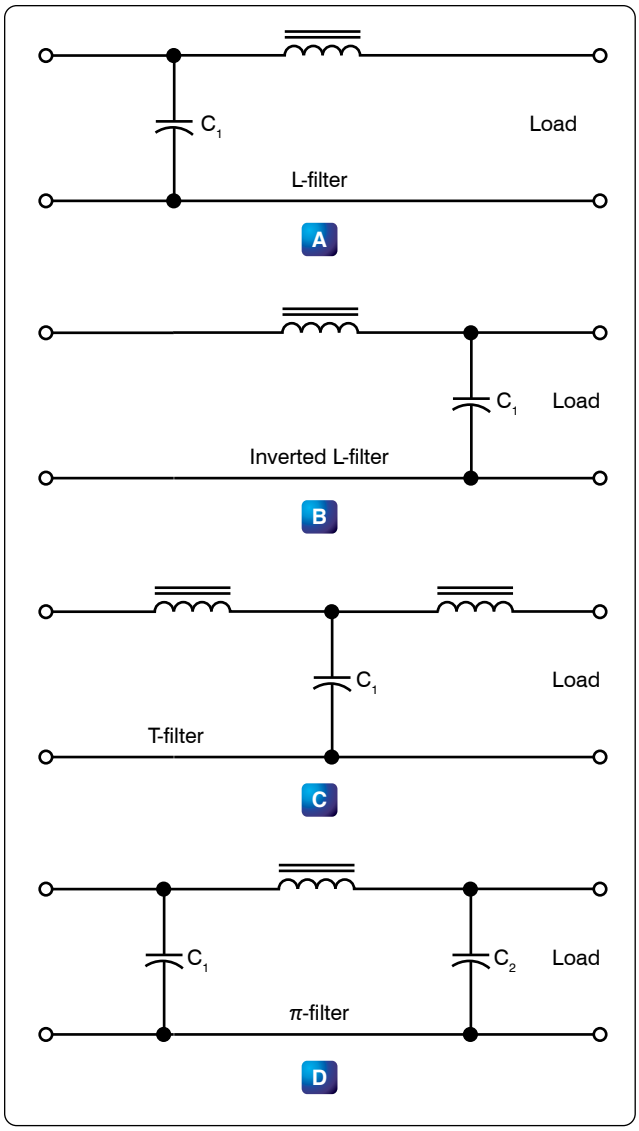


Figure 12-232. *LC filters.*

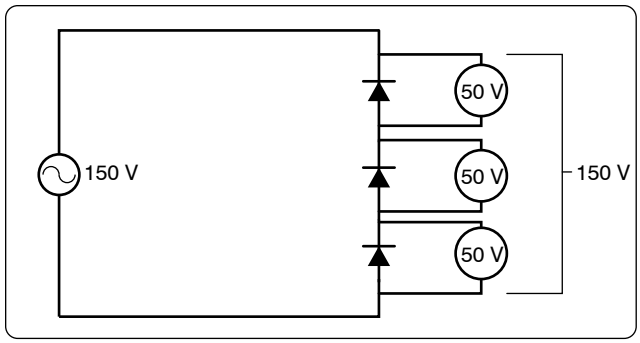


Figure 12-233. *Stacking diodes in a circuit.*

Class A

In the Class A operation, the current in the transistor flows for 100 percent or 360° of the input signal. *[Figure 12-238]* Class A operation is the least efficient class of operation but provides the best fidelity. Fidelity simply means that the output signal is a good reproduction of the input signal in all respects other than the amplitude, which is amplified. In some cases, there may be some phase shifting between the input signal and the output signal. Typically, the phase difference is 180°. If the output signal is not a good reproduction of the input signal, then the signal is said to be distorted. Distortion is any undesired change to the signal from the input to the output.

The efficiency of an amplifier refers to the amount of power delivered to the output compared to the power supplied to the circuit. Every device in the circuit consumes power in order to operate. If the amplifier operates for 360° of input signal, then it is using more power than if it was using only 180° of input signal. The more power consumed by the amplifier, the less there is available for the output signal. Usually the Class A amplifier is used where efficiency is of little concern and where fidelity in reproduction is desired.

Class AB

In the Class AB operation, the transistor current flows for more than 50 percent but less than 100 percent of the input signal. *[Figure 12-239]* Unlike the Class A amplifier, the output signal is distorted. A portion of the output circuit appears to be truncated. This is due to the lack of current through the transistor during this point of operation. When the emitter in this case becomes positive enough, the transistor cannot conduct because the base to emitter junction is no longer forward biased. The input signal going positive beyond this point does not produce any further output and the output remains level. The Class AB amplifier has a better efficiency and a poorer fidelity than the Class A amplifier. These amplifiers are used when an exact reproduction of the input is not required but both the positive and negative portions of the input signals need to be available on the output.

Class B

In Class B operation, the transistor current flows for only 50 percent of the input signal. *[Figure 12-240]* In this illustration, the base-emitter bias does not allow the transistor to conduct whenever the input signal is greater than zero. In this case, only the negative portion of the input signal is reproduced. Unlike the rectifier, the Class B amplifier does not only reproduce half of the input signal, but it also amplifies it. Class B amplifiers are twice as efficient as the Class A amplifier because the amplifying device only uses power for half of the input signal.

Class C

In Class C operations, transistor current flows for less than 50 percent of the input signal. *[Figure 12-241]* This class of

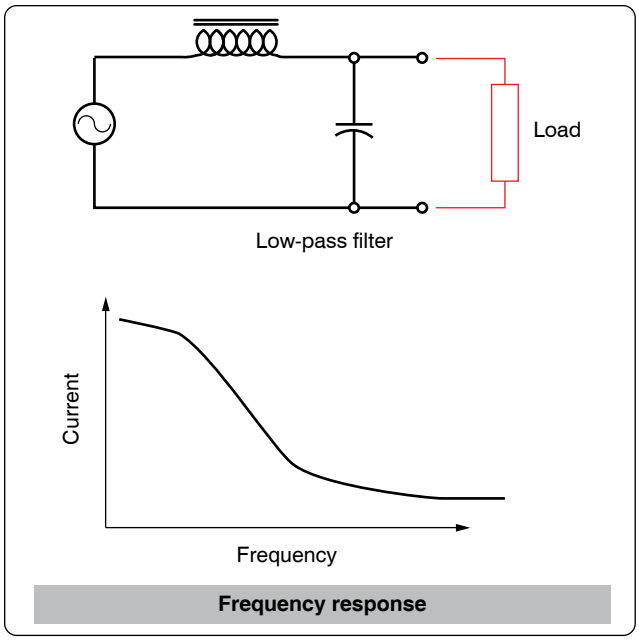


Figure 12-234. *Low-pass filter.*

operation is the most efficient. Because the transistor does not conduct except during a small portion of the input signal, this is the most efficient class of amplifier. The distortion of the Class C amplifier is greater (poor fidelity) than the Class A, AB, and B amplifiers because a small portion of the input signal is reproduced on the output. Class C amplifiers are used when the output signal is used for only small portions of time.

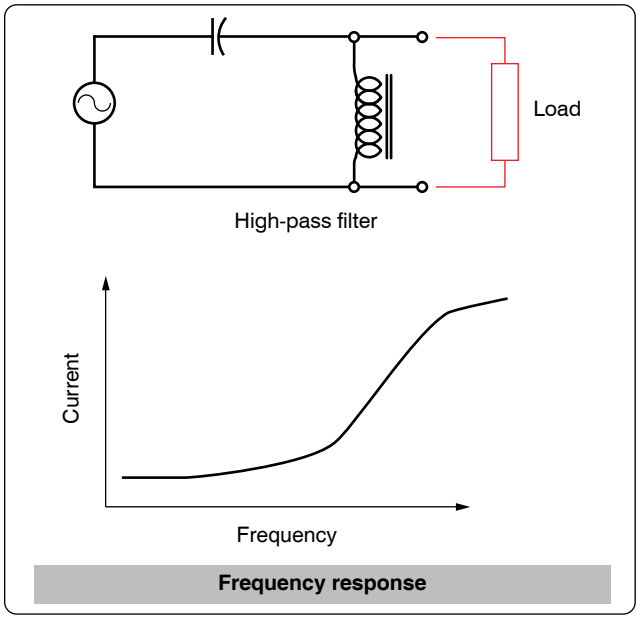


Figure 12-235. *High-pass filter.*

Methods of Coupling

Coupling is used to transfer a signal from one stage on an amplifier to another stage. Regardless of whether an amplifier is a single stage or one in a series of stages, there must be a method for the signal to enter and leave the circuit. Coupling is the process of transferring the energy between circuits. There are a number of ways for making this transfer and to discuss these methods in detail goes beyond the scope of this handbook. However, four methods are listed below with a brief description of their operation.

Direct Coupling

Direct coupling is the connection of the output of one stage directly to the input of the next stage. Direct coupling provides a good frequency response because no frequency-sensitive components, such as capacitors and inductors, are used. Yet this method is not used very often due to the complex power supply requirements and the impedance matching problems.

RC Coupling

RC coupling is the most common method of coupling and

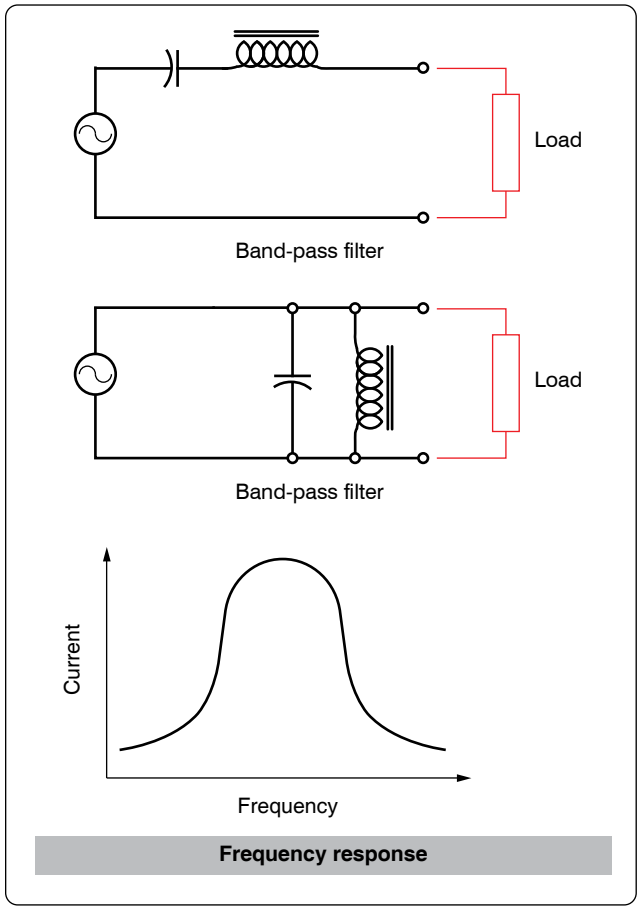


Figure 12-236. *Band-pass filter.*

uses a coupling capacitor and signal developing resistors. *[Figure 12-242]* In this circuit, R1 acts as a load resistor for Q1 and develops the output signal for that stage. The capacitor C1 blocks the DC bias signal and passes the AC output signal. R2 then becomes the load over which the passes AC signal is developed as an input to the base of Q2. This arrangement allows for the bias voltage of each stage to be blocked, while the AC signal is passed to the next stage.

Impedance Coupling

Impedance coupling uses a coil as a load for the first stage but otherwise functions just as an RC coupling. *[Figure 12-243]* This method is similar to the RC coupling method. The difference is that R1 is replaced with inductor L1 as the output load. The amount of signal developed on the output load depends on the inductive reactance of the coil. In order for the inductive reactance to be high, the inductance must be large; the frequency must be high or both. Therefore, load inductors should have relatively large amounts of inductance and are most effective at high frequencies.

Transformer Coupling

Transformer coupling uses a transformer to couple the signal from one stage to the next. *[Figure 12-244]* The transformer action of T1 couples the signal from the first stage to the second stage. The primary coil of T1 acts as a load for the output of the first stage while the secondary coil acts as the developing impedance for the second stage Q2. Transformer coupling is very efficient and the transformer can aid in impedance matching.

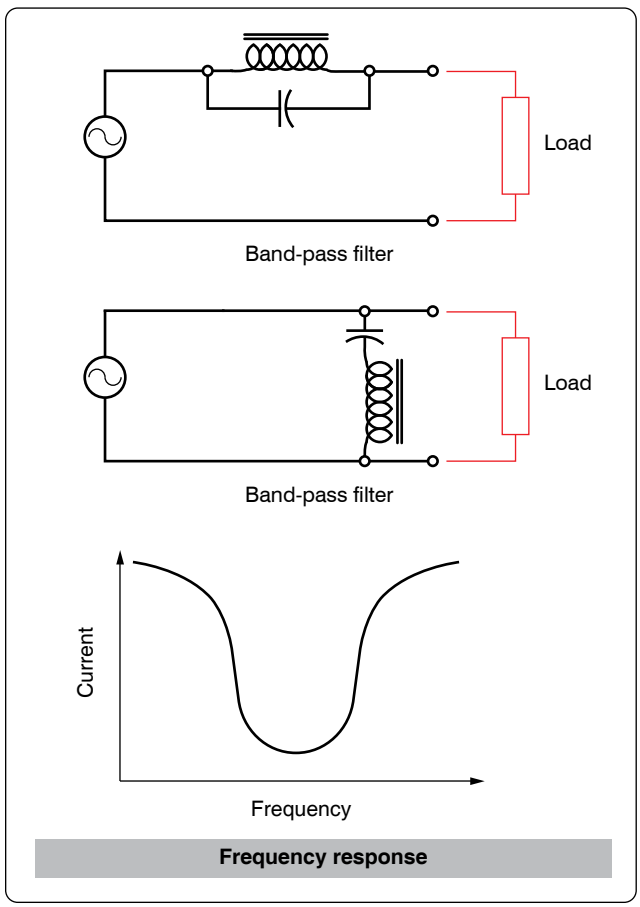


Figure 12-237. *Band-stop filter.*

Feedback

Feedback occurs when a small portion of the output signal is sent back to the input signal to the amplifier. There are two types of feedback in amplifiers:

1. Positive (regenerative)
2. Negative (degenerative)

The main difference between these two signals is whether the feedback signal adds to the input signal or if the feedback signal diminishes the input signal.

When the feedback is positive, the signal being returned to the input is in phase with the input signal and thus interferes constructively. *Figure 12-245* illustrates this concept applied in the amplified circuit through a block diagram. Notice that the feedback signal is in phase with the input signal, which regenerates the input signal. This results in an output signal with amplitude greater than would have been without the constructive, positive feedback. This type of positive feedback is what causes an audio system to squeal.

Figure 12-245 also illustrates with a block diagram how negative or degenerative feedback occurs. In this case, the feedback signal is out of phase with the input signal. This causes destructive interference and degenerates the input signal. The result is a lower amplitude output signal than would have occurred without the feedback.

Operational Amplifiers (OP AMP)

An operational amplifier (OP AMP) is designed to be used with other circuit components and performs either computing functions or filtering. *[Figure 12-246]* Operational amplifiers are usually high-gain amplifiers with the amount of gain governed by the amount of feedback.

Operational amplifiers were originally developed for analog computers and used to perform mathematical functions. Today many devices use the operational amplifier for DC amplifiers, AC amplifiers, comparators, oscillators, and filter circuits. The widespread use is due to the fact that the OP AMP is a versatile device, small, and inexpensive. Built into the integrated chip, the operational amp is used as a basic building block of larger circuits.

There are two inputs to the operational amplifier, inverting (−) and non-inverting (+), and there is one output. The polarity

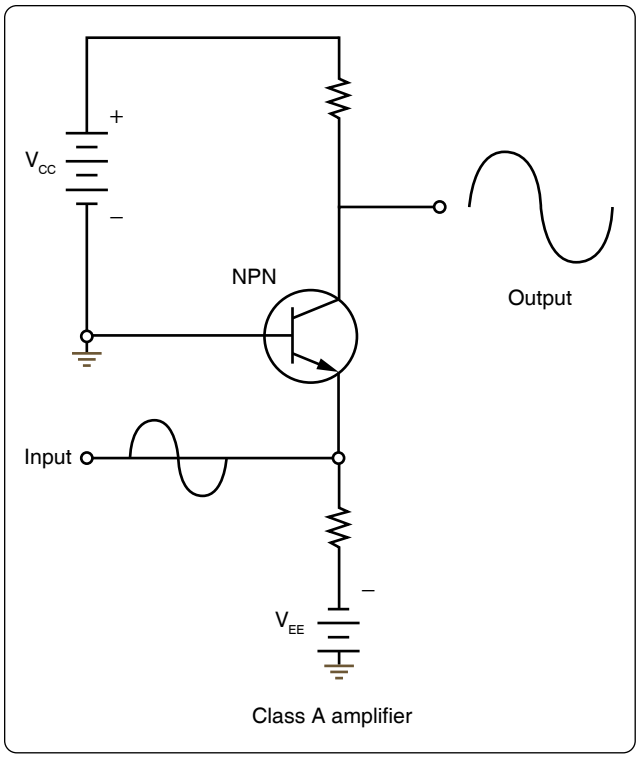


Figure 12-238. *Simplified Class A amplifier circuit.*

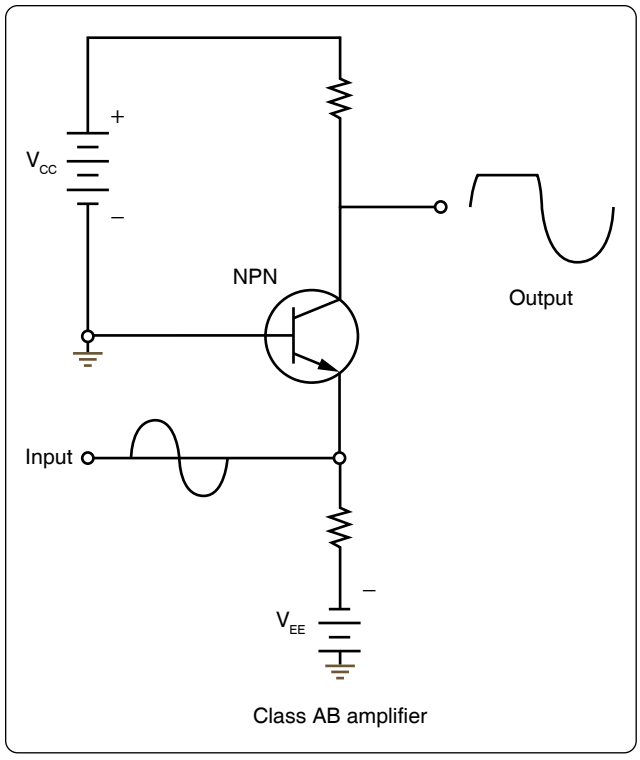


Figure 12-239. *Simplified Class AB amplifier circuit.*

of a signal applied to the inverting input (−) is reversed at the output. A signal applied to the non-inverting (+) input retains its polarity on the output. To be classified as an operational amplifier, the circuit must have certain characteristics:

1. Very high gain
2. Very high input impedance
3. Very high output impedance

This type of a circuit can be made up of discrete components, such as resistors and transistors. However, the most common form of an operational amplifier is found in the integrated circuit. This integrated circuit or chip contains the various stages of the operational amplifier and can be treated as if it were a single stage.

Applications

The number of applications for OP AMPs is too numerous to detail in this handbook. However, the technician occasionally comes across these devices in modern aircraft and should be able to recognize their general purpose in a circuit. Some of the basic applications are:

1. Go/no-go detectors
2. Square wave circuits
3. Non-inverting amplifier
4. Inverting amplifier

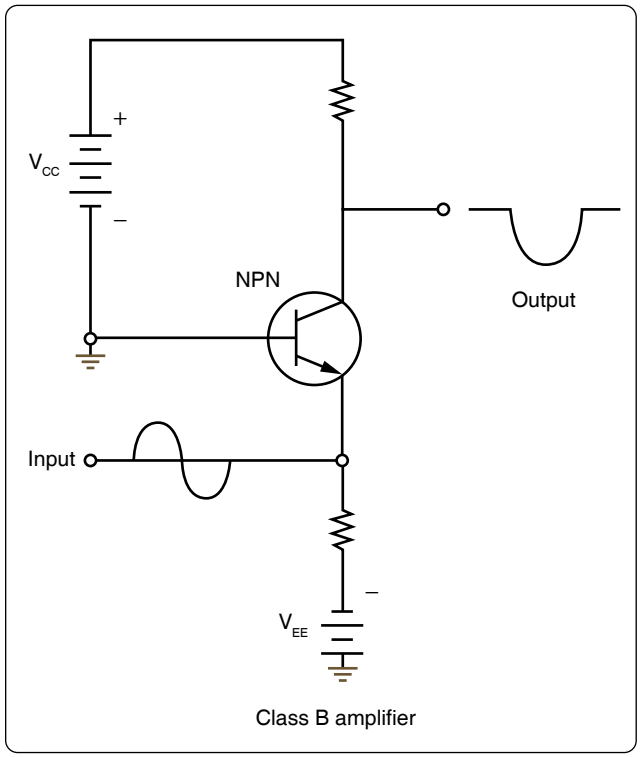


Figure 12-240. *Simplified Class B amplifier circuit.*

5. Half-wave rectifier

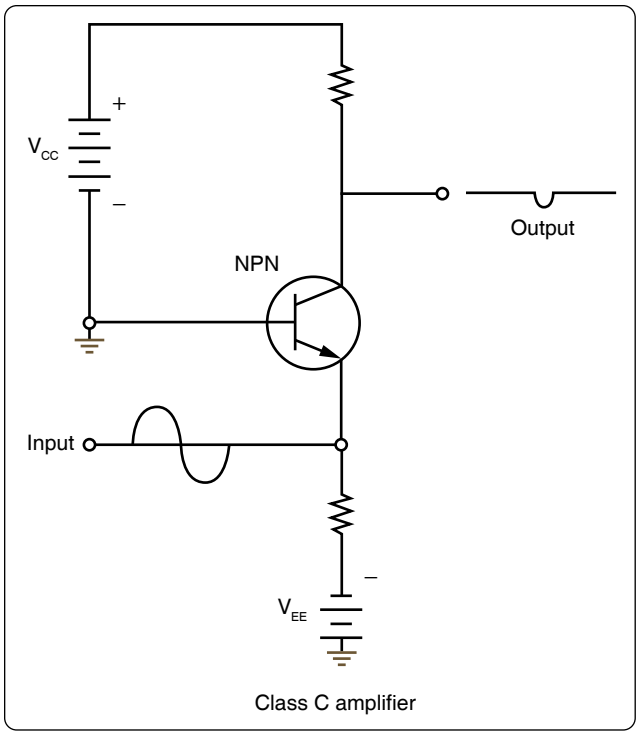


Figure 12-241. *Simplified Class C amplifier circuit.*

Magnetic Amplifiers

Magnetic amplifiers do not amplify magnetism but use electromagnetism to amplify a signal. Essentially, the magnetic amplifier is a power amplifier with a very limited frequency response. The frequency range most commonly associated with the magnetic amplifier is 100 Hz and less, which places it in the audio range. As a technical point, the magnetic amplifier is a low-frequency amplifier.

Advantages of the magnetic amplifier are:

1. Very high efficiency, on the order of approximately 90 percent
2. High reliability
3. Very rugged, able to withstand vibrations, moisture, and overloads
4. No warm-up time

Some of the disadvantages of the magnetic amplifier are:

1. Incapacity to handle low-voltage signals
2. Not usable in high-frequency applications
3. Time delay associated with magnetic affects
4. Poor fidelity

The basic operating principles of the magnetic amplifier are fairly simple. Keep in mind that all amplifiers are current control devices. In this particular case, power that is delivered

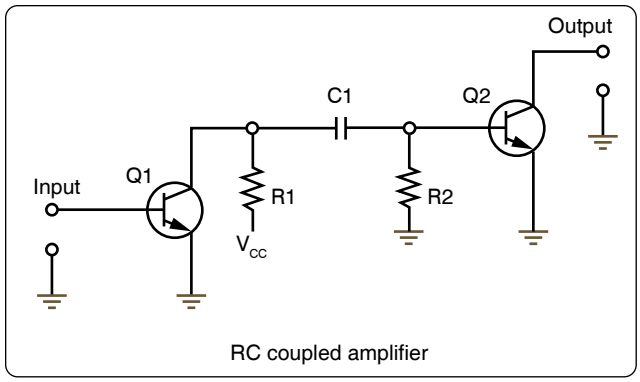


Figure 12-242. *Simplified RC coupling circuit.*

to the load is controlled by a variable inductance.

If an AC voltage is applied to the primary winding of an iron core transformer, the iron core is magnetized and demagnetized at the same frequency as that of the applied voltage. This, in turn, induces a voltage in the transformers secondary winding. The output voltage across the terminals of the secondary depends on the relationship of the number of turns in the primary and the secondary of the transformer.

The iron core of the transformer has a saturation point after which the application of a greater magnetic force produces no change in the intensity of magnetization. Hence, there is no change in transformer output, even if the input is greatly increased. The magnetic amplifier circuit in *Figure 12-247* is used to explain how a simple magnetic amplifier functions.

1. Assume that there is 1 ampere of current in coil A, which has 10 turns of wire. If coil B has 10 turns of wire, an output of 1 ampere is obtained if coil B is properly loaded.
2. By applying direct current to coil C, the core of the magnetic amplifier coil can be further magnetized. Assume that coil C has the proper number of turns and, upon the application of 30 milliamperes, that the core

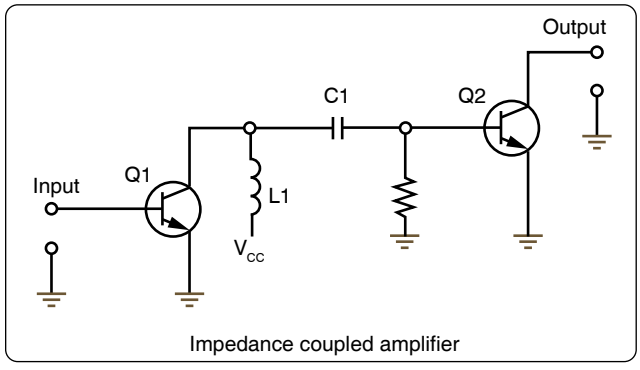


Figure 12-243. *Simplified impedance coupling circuit.*

is magnetized to the point where 1 ampere on coil A results in only 0.24 ampere output from coil B.

3. By making the DC input to coil C a continuous variable from 0 to 30 milliamperes and by maintaining an input of 1 ampere on coil A, it is possible to control the output of coil B to any point between 0.24 ampere and 1 ampere in this example.

The term "amplifier" is used for this arrangement because, by use of a few milliamperes, control of an output of 1 or more amperes is obtained.

Saturable-Core Reactor

The same procedure can be used with the circuit shown in *Figure 12-248*. A saturable-core reactor is a magnetic-core coil whose reactance is controlled by changing the permeability of the core. Varying the unidirectional flux controls the permeability of the core.

By controlling the extent of magnetization of the iron ring, it is possible to control the amount of current flowing to the load, since the amount of magnetization controls the impedance of the AC input winding. This type of magnetic amplifier is called a simple saturable reactor circuit.

Adding a rectifier to such a circuit would remove half the cycle of the AC input and permit DC to flow to the load. The amount of DC flowing in the load circuit is controlled by a DC control winding (sometimes referred to as bias). This type of magnetic amplifier is referred to as being self-saturating.

To use the full AC input power, a circuit such as that shown in *Figure 12-249* may be used. This circuit uses a full-wave bridge rectifier. The load receives a controlled DC by using the full AC input. This type of circuit is known as a self-saturating, full-wave magnetic amplifier.

In *Figure 12-250,* it is assumed that the DC control winding is supplied by a variable source, such as a sensing circuit. To control such a source and use its variations to control the AC output, it is necessary to include another DC winding that has a constant value. This winding, referred to as the reference winding, magnetizes the magnetic core in one direction.

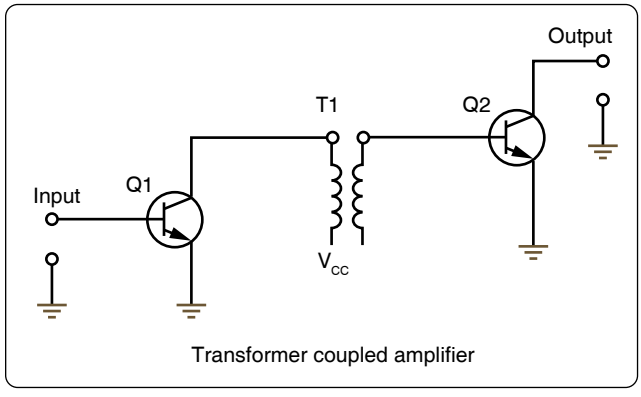


Figure 12-244. *Simplified transformer coupling circuit.*

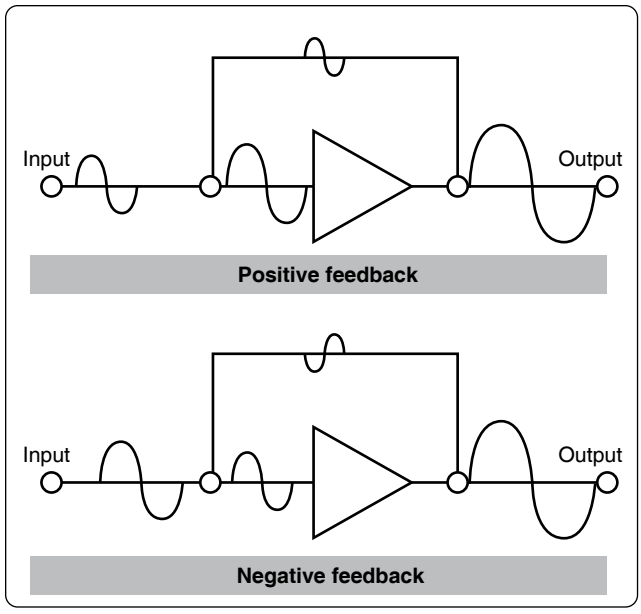


Figure 12-245. *Feedback.*

The DC control winding, acting in opposition to the reference winding, either increases (degenerative) or decreases (regenerative) the magnetization of the core to change the amount of current flowing through the load. This is essentially a basic preamplifier.

Logic Circuits

Logic is considered the science of reasoning—the development of a reasonable conclusion based on known information. Human reasoning tells us that certain propositions are true if certain conditions or premises are true. An annunciator being lit in the master warning panel is an example of a proposition, which is either true or false. For example, predetermined and designed conditions must be met in order for an annunciator in a master warning panel to be lit. A "LOW HYDRAULIC PRESS" annunciator may have a simple set of conditions that cause it to be illuminated. If the conditions are met, such as a hydraulic reservoir that is low on fluid causing the line press to be low, then the logic is true and the annunciator lights. Several propositions, when combined, form a logical function. In the example above, the "LOW HYDRAULIC PRESS" annunciator is on if the LED is not burned out and the hydraulic press is low or if the LED is not burned out and the annunciator test is being asserted.

This section on logic circuits only serves as an introduction to the basic concepts. The technician encounters many

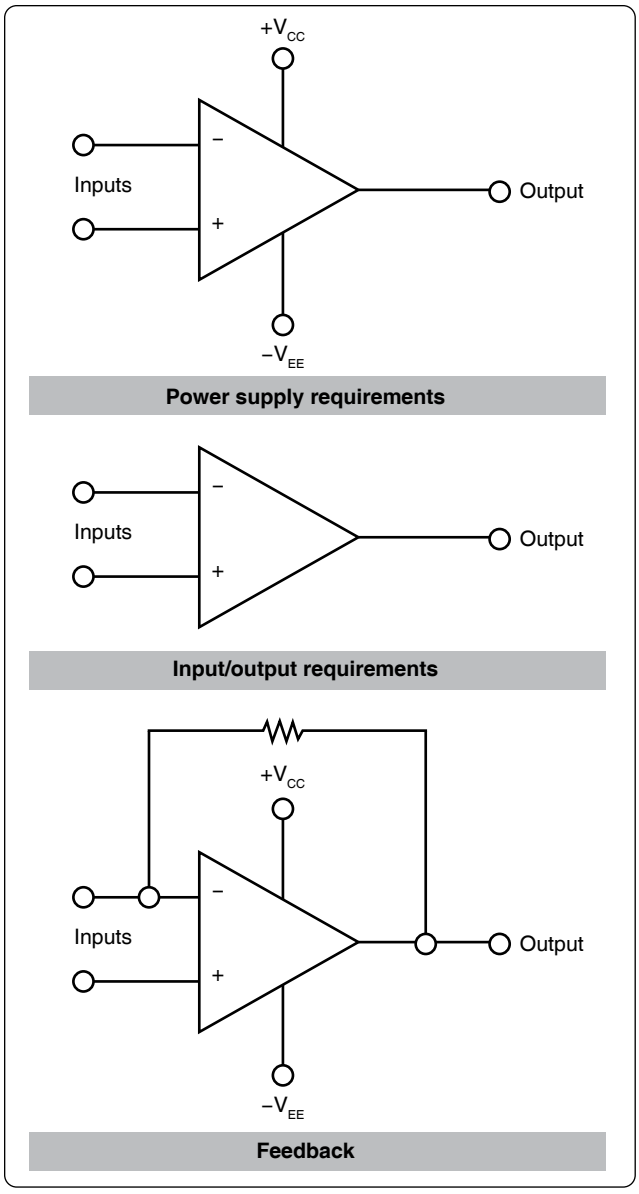


Figure 12-246. *Schematic symbol for the operational amplifier.*

situations or problems in everyday life that can be expressed in some form of a logical function. Many problems and situations can be condensed down to simple yes/no or true/false statements that, if logically ordered, can filter a problem down to a reasonable answer. The digital logic circuits are well suited for this task and have been employed in today's integrated circuits found in virtually all of the devices that we take for granted in modern aircraft. These logical circuits are used to carry out the logical functions for such things as navigation and communications. There are several fundamental elements that form the building blocks of the complex digital systems found in line replaceable units (LRUs) and avionics card cages. The following is a very basic outline of what those elements are and what logic conditions they process. It is far beyond the scope of this handbook to cover digital logic systems due to the vast body of knowledge that it represents. However, this serves as an introduction and, in some limited cases, is useful in reading system block diagrams that use logic symbols to aid the technician in understanding how a given circuit operates.

Logic Polarity

Electrical pulses can represent two logic conditions and any two differing voltages can be used for this purpose. For example, a positive voltage pulse could represent a true or 1 condition and a negative voltage pulse could then represent a false or 0 logic condition. The condition in which the voltage changes to represent a true or 1 logic is known as the logic polarity. Logic circuits are usually divided into two broad classes: positive polarity and negative polarity. The voltage levels used and a statement indicating the use of positive or negative logic is usually specified in the logic diagrams provided by the original equipment manufacturers (OEMs).

Positive

When a signal that activates a circuit to a 1, true or high condition, has an electrical level that is relatively more positive than the other 0 or false condition, then the logic polarity is said to be positive. An example would be:

Active State: 1 or True = +5 volts direct current (VDC)
0 or False = −5 VDC

Negative

When the signal that actives a circuit to a 1, true or high condition, has an electrical level that is relatively more negative than the other 0 or false condition, then the logic polarity is said to be negative. An example would be:

Active State: 1 or True = 0 VDC
0 or False = +5 VDC

Pulse Structure

Figure 12-251 illustrates the positive and negative pulse in an idealized form. In both forms, the pulse is composed of two edges—one being the leading edge and the other the trailing edge. In the case of the positive pulse logic, the positive transition from a lower state to a higher state is the leading edge and the trailing edge is the opposite. In the case of the negative logic pulse, the negative transition from a higher state to a lower state is the leading edge while the rise from the lower state back to the higher state is the trailing edge. *Figure 12-251* is considered an ideal pulse because the rise and fall times are instantaneous. In reality, these changes take time, although in actual practice, the rise and fall can be assumed as instantaneous. *Figure 12-252* shows the non-ideal pulse and its characteristics. The time required for a pulse to go from a low state to a high state is called the rise

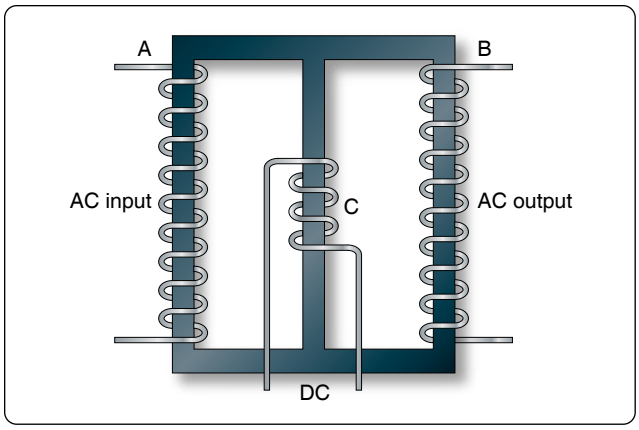


Figure 12-247. *Magnetic amplifier circuit.*

time, and the time required for the pulse to return to zero is called the fall time. It is common practice to measure the rise and fall time between 10 percent amplitude and 90 percent amplitude. The reason for taking the measurements in these points is due to the non-linear shape of the pulse in the first 10 percent and final 90 percent of the rise and fall amplitudes. The pulse width is defined as the duration of the pulse. To be more specific, it is the time between the 50 percent amplitude point on both the pulse rise and fall.

Basic Logic Circuits

Boolean logic is a symbolic system used in representing the truth value of statements. It is employed in the binary system used by digital computers primarily because the only truth values (true and false) can be represented by the binary digits 1 and 0. A circuit in computer memory can be open or closed, depending on the value assigned to it. The fundamental operations of Boolean logic, often called Boolean operators, are "and," "or," and "not;" combinations of these make up 13 other Boolean operators. Six of these operators are discussed.

The Inverter Logic

The inverter circuit performs a basic logic function called inversion. The purpose of the inverter is to convert one logic state into the opposite state. In terms of a binary digit, this would be like converting a 1 to a 0 or a 0 to a 1. When a high voltage is applied to the inverter input, low voltage is the output. When a low voltage is applied to the input, a high voltage is on the output. This operation can be put into what is known as a logic or truth table. The standard logic symbol is shown in *Figure 12-253*. *Figure 12-254* shows the possible logic states for this gate. This is the common symbol for an amplifier with a small circle on the output. This type of logic can also be considered a NOT gate.

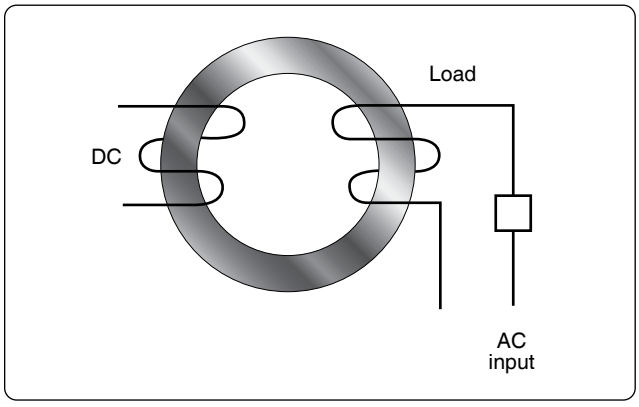


Figure 12-248. *Saturable CORE reactor circuit.*

The AND Gate

The AND gate is made up of two or more inputs and a single output. The logic symbol is shown in *Figure 12-255*. Inputs are on the left and the output is on the right in each of the depictions. Gates with two, three, and four inputs are shown; however, any number of inputs can be used in the AND logic as long as the number is greater than one. The operation of the AND gate is such that the output is high only when all of the inputs are high. If any of the inputs are low, the output is also low. Therefore, the basic purpose of an AND gate is to determine when certain conditions have been met at the same time. A high level on all inputs produces a high level on the output. *Figure 12-256* shows a simplified diagram of the AND logic with two switches and a light bulb. Notice that both switches need to be closed in order for the light bulb to turn on. Any other combination of switch positions is an open circuit and the light does not turn on. An example of AND logic could possibly be engage logic found in an autopilot. In this case, the autopilot would not be allowed to be engaged unless certain conditions are first met. Such conditions could be: Vertical gyro is valid AND directional gyro is valid AND all autopilot control knobs are in detents AND servo circuits are operational. Only when these conditions are met does the

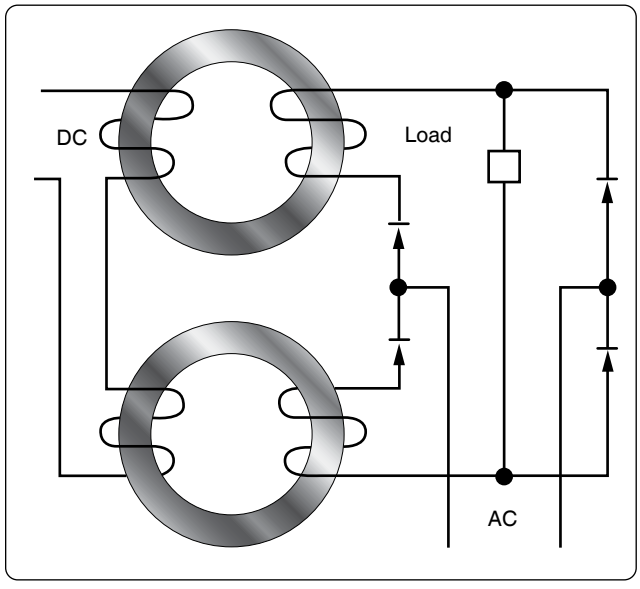


Figure 12-249. *Self-saturating, full-wave magnetic amplifier.*

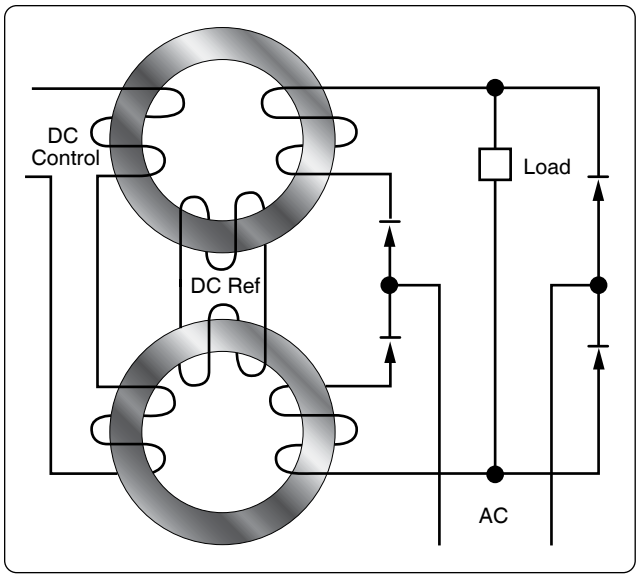


Figure 12-250. *Basic preamplifier circuit.*

autopilot engage. *[Figure 12-257]*

The OR Gate

The OR gate has two or more inputs and one output and is normally represented by the standard logic symbol and truth table. *[Figure 12-258]* Note that the OR gate can have any number of inputs as long as it is greater than one. The operation of the OR gate is such that a high on any one of the inputs produces a high on the output. The only time that a low is produced on the output is if there are no high levels on any input. *Figure 12-259* is a simplified circuit that illustrates the OR logic. The example used is a "DOOR UNSAFE" annunciator. Let's say in this case that the plane has one cabin door and a baggage door. In order for the annunciator light on the master warning panel to extinguish, both doors must be closed and locked. If any one of the doors is not secured properly, the baggage door OR the cabin door, then the "DOOR UNSAFE" annunciator illuminates. In this case, two switches are in parallel with each other. If either one of the two switches is closed, the light bulb lights up. The lamp is off only when both switches are open.

The NAND Gate

The term NAND is a combination of the NOT-AND gate and indicates an AND function with an inverted output. A standard logic symbol for a two input NAND gate is shown in *Figure 12-260*. Notice that an equivalent AND gate with an inverter is also shown. The logical operation of the NAND gate is such that a low output occurs only if all inputs are high. If any of the inputs are low, the output is high. An example of a two input NAND gate and its corresponding truth table are shown in *Figure 12-261*.

The NOR Gate

The term NOR is a combination of the NOT and OR and indicates an OR function with an inverted output. The standard logic symbol for a two-inputs NOR gate is shown in *Figure 12-263*. Notice that an equivalent AND gate with an inverter is also shown. The logical operation of the NOR gate is such that a low output happens when any of its inputs are high. Only when all of its inputs are low is the output high. The logic of this gate produces resultant outputs that are the opposite of the OR gate. In the NOR gate, the low output is the active output level. *Figure 12-263* illustrates the logical operation of a two-input NOR gate for all of its possible combinations and the truth table.

Exclusive OR Gate

The exclusive OR gate is a modified OR gate that produces a 1 output when only one of the inputs is a 1. The abbreviation often used is X-OR. It is different from the standard OR gate in that when both inputs are a 1, then the output remains at a 0. The standard symbol and truth table for the X-OR gate are shown in *Figure 12-264*.

Exclusive NOR Gate

The exclusive NOR (X-NOR) gate is nothing more than an X-OR gate with an inverted output. It produces a 1 output when all inputs are 1s and also when all inputs are 0s. The standard symbol is shown in *Figure 12-265*.

The Integrated Circuit

All of the logic functions so far discussed plus many other components are available in some form of an integrated circuit. The digital systems found in today's aircraft owe their existence to a large extent to the design of the integrated circuit (IC). In most cases, the IC has an advantage over the use of discrete components in that they are smaller, consume less power, are very reliable, and are inexpensive. The most noticeable characteristic of the IC is its size and in comparison to the discrete semiconductor component, can easily be on the order of thousands of times smaller. *[Figure 12-266]*\

A monolithic integrated circuit is an electronic circuit that is constructed entirely on a single chip or wafer of semiconductor material. All of the discrete components, such as resistors, transistors, diodes, and capacitors, can be constructed on these small pieces of semiconductor material and are an integral part of the chip. There are a number of levels of integration. Those levels are: small-scale integration, medium-scale integration, large-scale integration, and microprocessors. The small-scale integration is considered the least complex design of the digital ICs. These ICs contain the basic components, such as the AND, OR, NOT, NOR and NAND gates. *[Figure 12-267]* The medium-scale integration can contain

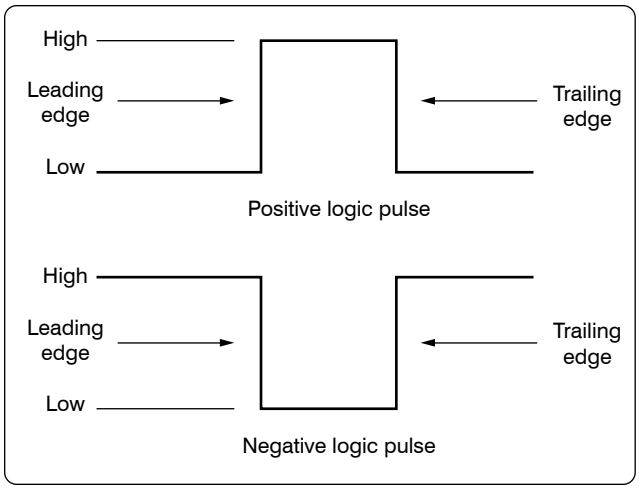


Figure 12-251. *Positive and negative pulse in an idealized form.*

the same components as found in the small-scale design but in larger numbers ranging from 12 to 100. The medium-scale designs are house circuits that are more complex, such as encoders, decoders, registers, counters, multiplexers, smaller memories, and arithmetic circuits. *[Figure 12-268]* The large-scale integrated circuits contain even more logic gates, larger memories than the medium-scale circuits, and in some cases microprocessors.

Microprocessors

The microprocessor is a device that can be programmed to perform arithmetic and logical operations and other functions in a preordered sequence. The microprocessor is usually used as the central processing unit (CPU) in today's computer systems when it is connected to other components, such as memory chips and input/output circuits. The basic arrangement and design of the circuits residing in the microprocessor is called the architecture.

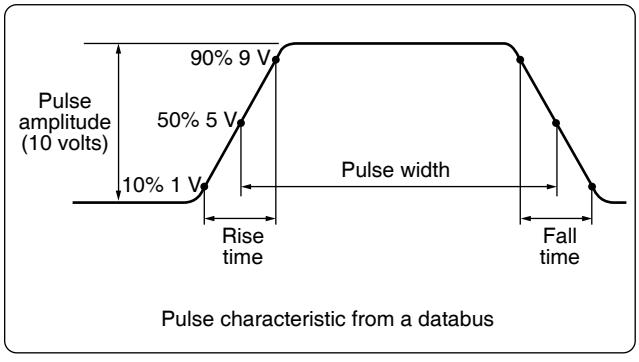


Figure 12-252. *Non-ideal pulse and its characteristics.*

DC Generators

Theory of Operation

In the study of alternating current, basic generator principles were introduced to explain the generation of an AC voltage by a coil rotating in a magnetic field. Since this is the basis for all generator operation, it is necessary to review the principles of generation of electrical energy.

When lines of magnetic force are cut by a conductor passing through them, voltage is induced in the conductor. The strength of the induced voltage is dependent upon the speed of the conductor and the strength of the magnetic field. If the ends of the conductor are connected to form a complete circuit, a current is induced in the conductor. The conductor and the magnetic field make up an elementary generator.

This simple generator is illustrated in *Figure 12-269*, together with the components of an external generator

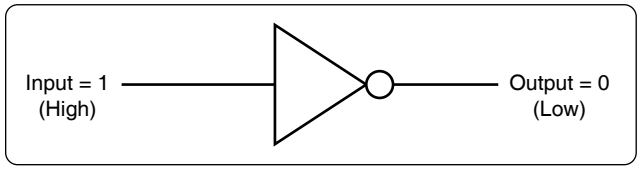


Figure 12-253. *Standard logic symbol.*

Input	Output
High	Low
Low	High

Figure 12-254. *Possible logic states.*

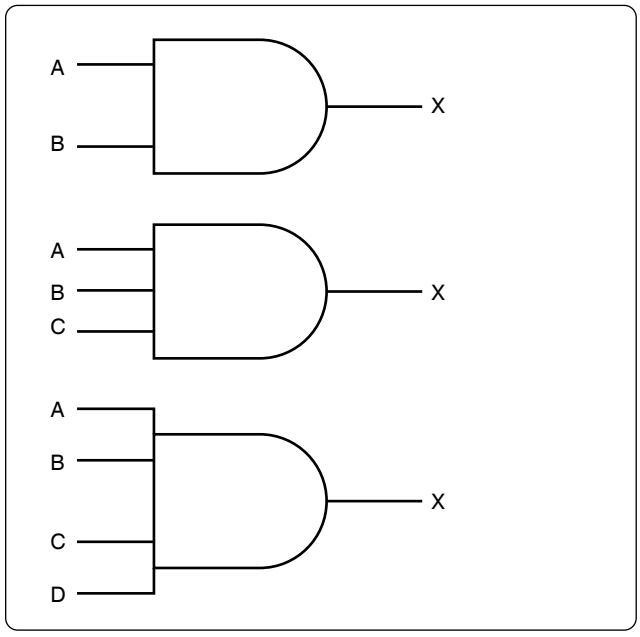


Figure 12-255. *AND gate logic symbol.*

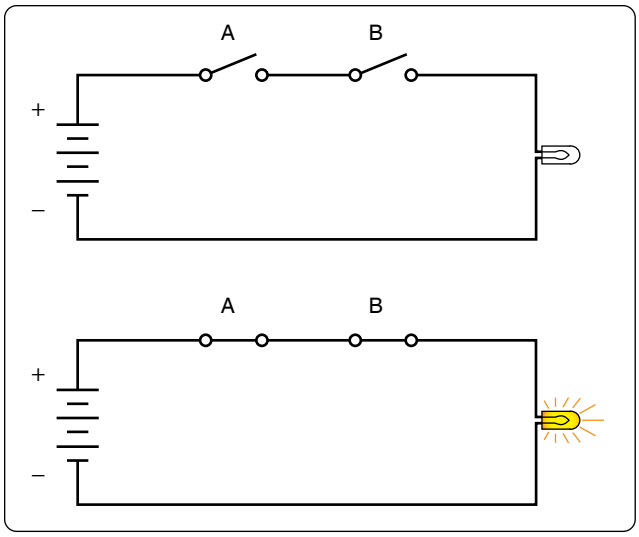


Figure 12-256. *Simplified diagram of the AND logic.*

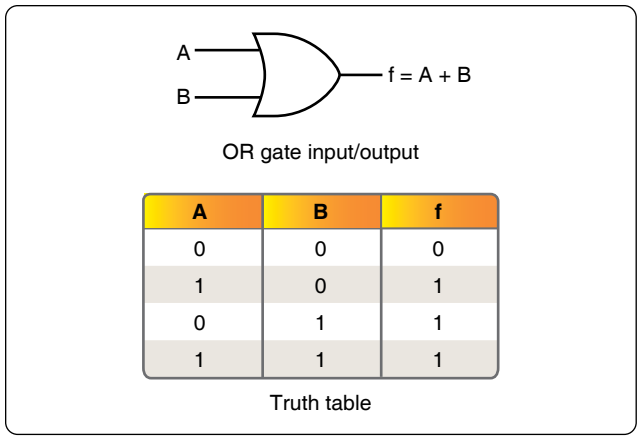


Figure 12-258. *OR gate.*

circuit which collect and use the energy produced by the simple generator. The loop of wire *[Figure 12-269A* and *B]* is arranged to rotate in a magnetic field. When the plane of the loop of wire is parallel to the magnetic lines of force, the voltage induced in the loop causes a current to flow in the direction indicated by the arrows in *Figure 12-269*. The voltage induced at this position is maximum, since the wires are cutting the lines of force at right angles, thus cutting more lines of force per second than in any other position relative to the magnetic field. As the loop approaches the vertical position shown in *Figure 12-270*, the induced voltage decreases because both sides of the loop (A and B) are approximately parallel to the lines of force and the rate of cutting is reduced. When the loop is vertical, no lines of force are cut since the wires are momentarily traveling parallel to the magnetic lines of force, and there is no induced voltage. As the rotation of the loop continues, the number of lines of force cut increases until the loop has rotated an additional 90° to a horizontal plane. As shown in *Figure 12-271*, the number of lines of force cut and the induced voltage once again are maximum. The direction of cutting, however, is in the opposite direction to that occurring in *Figures 12-269* and *12-270*, so the direction (polarity) of the induced voltage is reversed. As rotation of the loop continues, the number of lines of force having been cut again decreases, and the induced voltage becomes zero at the position shown in *Figure 12-272*, since the wires A and B are again parallel to the magnetic lines of force. If the voltage induced throughout the entire 360° of rotation

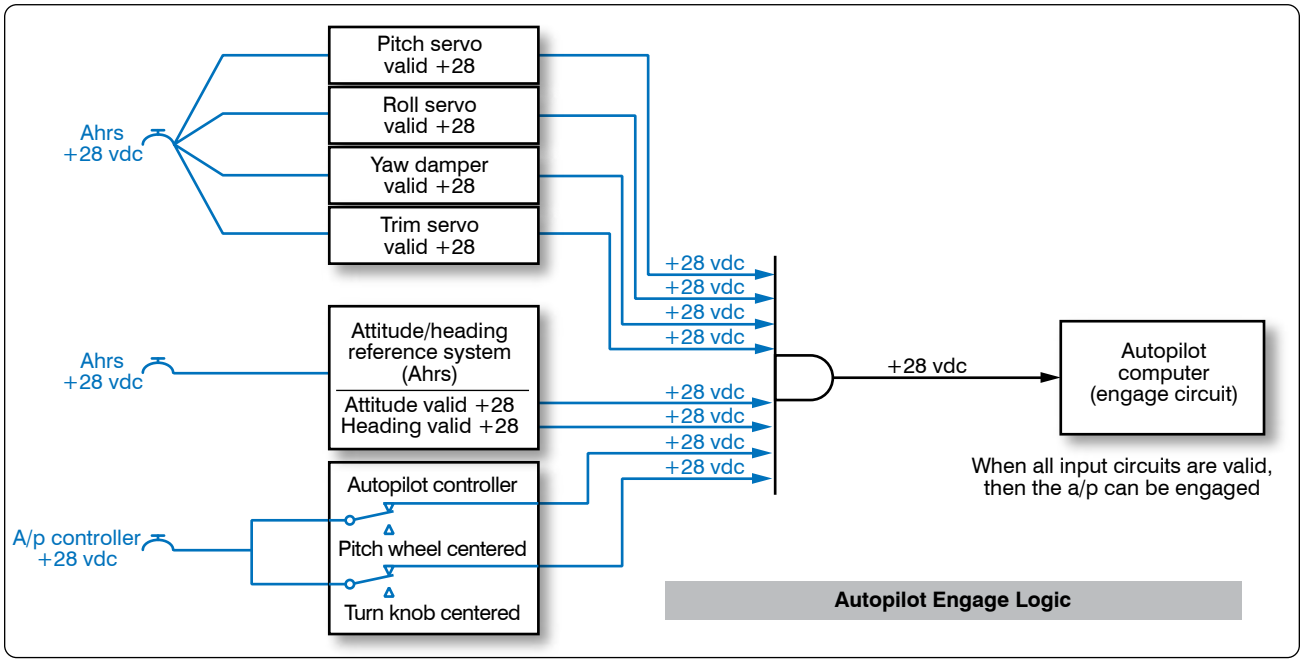


Figure 12-257. *AND logic of system found in the aircraft wiring diagrams.*

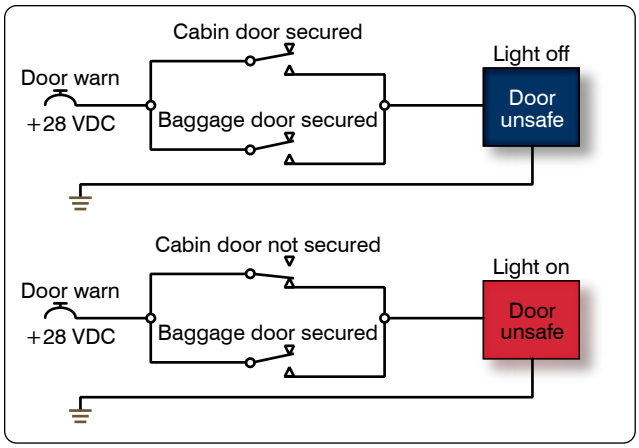


Figure 12-259. *Simplified circuit that illustrates OR logic.*

is plotted, the curve shown in *Figure 12-273* results. This voltage is called an alternating voltage because of its reversal from positive to negative value, first in one direction and then in the other.

To use the voltage generated in the loop for producing a current flow in an external circuit, some means must be provided to connect the loop of wire in series with the external circuit. Such an electrical connection can be effected by opening the loop of wire and connecting its two ends to two metal rings, called slip rings, against which two metal or carbon brushes ride. The brushes are connected to the external circuit. By replacing the slip rings of the basic AC generator with two half cylinders, called a commutator, a basic DC generator is obtained. *[Figure 12-274]* In this illustration, the black side of the coil is connected to the black segment, and the white side of the coil to the white segment. The segments are insulated from each other. The two stationary brushes are placed on opposite sides of the commutator and are so mounted that each brush contacts each segment of the commutator as the latter revolves simultaneously with the loop. The rotating parts of a

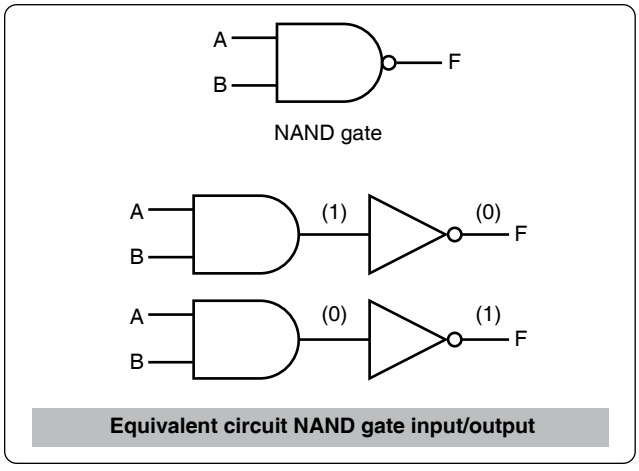


Figure 12-260. *Standard logic symbol for two input NAND gate.*

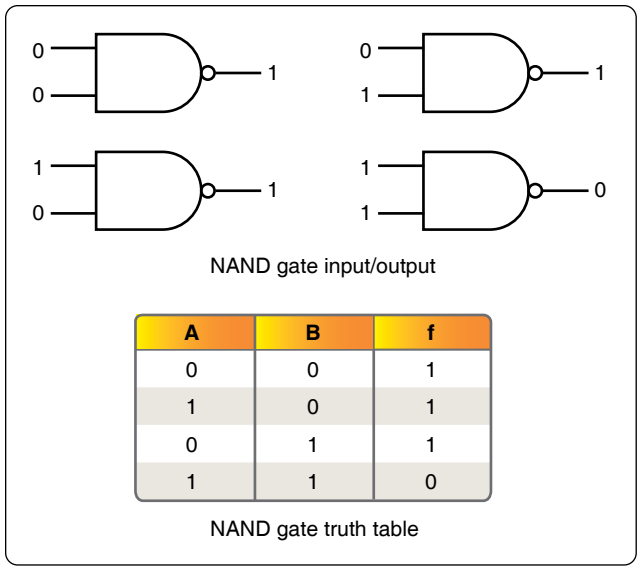


Figure 12-261. *Two input NAND gate and corresponding truth table.*

DC generator (coil and commutator) are called an armature. The generation of an emf by the loop rotating in the magnetic field is the same for both AC and DC generators, but the action of the commutator produces a DC voltage.

Generation of a DC Voltage

Figure 12-275 illustrates in an elementary, step-by-step manner, how a DC voltage is generated. This is accomplished by showing a single wire loop rotating through a series of positions within a magnetic field.

Position A

The loop starts in position A and is rotating clockwise. However, no lines of force are cut by the coil sides, which means that no emf is generated. The black brush is shown coming into contact with the black segment of the commutator, and the white brush is just coming into contact with the white segment.

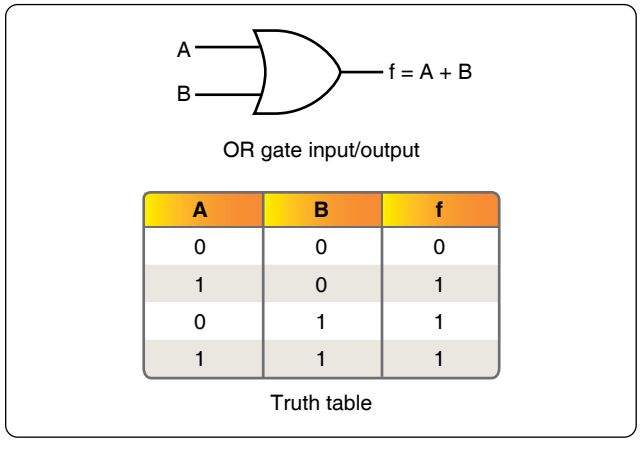
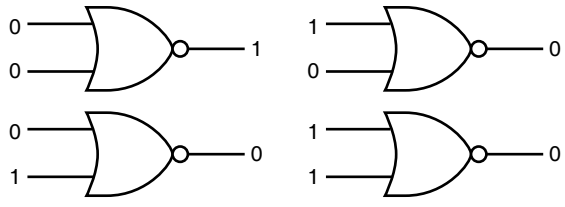


Figure 12-262. *Standard logic symbol for two inputs OR gate.*



A	B	f
0	0	1
0	1	0
1	0	0
1	1	0

NOR gate truth table

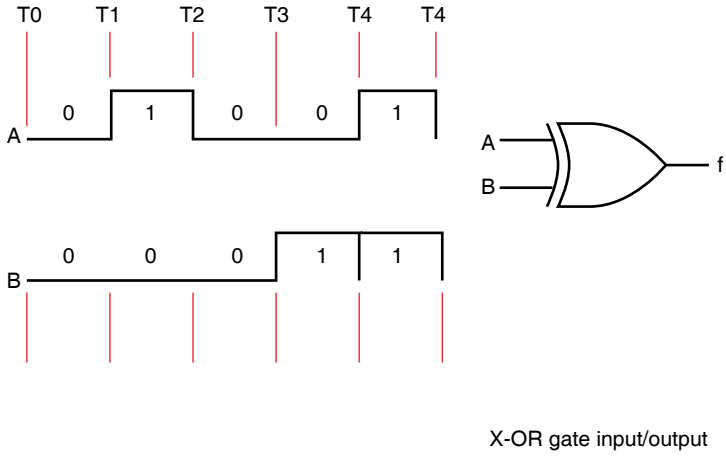
Figure 12-263. *Logical operation of two-input NOR gate and truth table.*

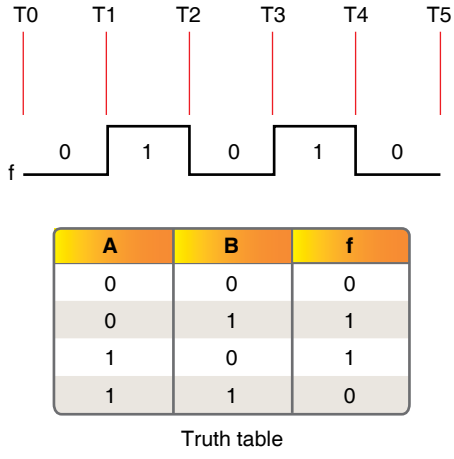
Position B

In position B, the flux is now being cut at a maximum rate, which means that the induced emf is maximum. At this time, the black brush is contacting the black segment, and the white brush is contacting the white segment. The deflection of the meter is toward the right, indicating the polarity of the output voltage.

Position C

At position C, the loop has completed 180° of rotation. Like position A, no flux lines are being cut and the output voltage is zero. The important condition to observe at position C is the action of the segments and brushes. The black brush at the 180° angle is contacting both black and white segments on one side of the commutator, and the white brush is contacting both segments on the other side of the commutator. After the

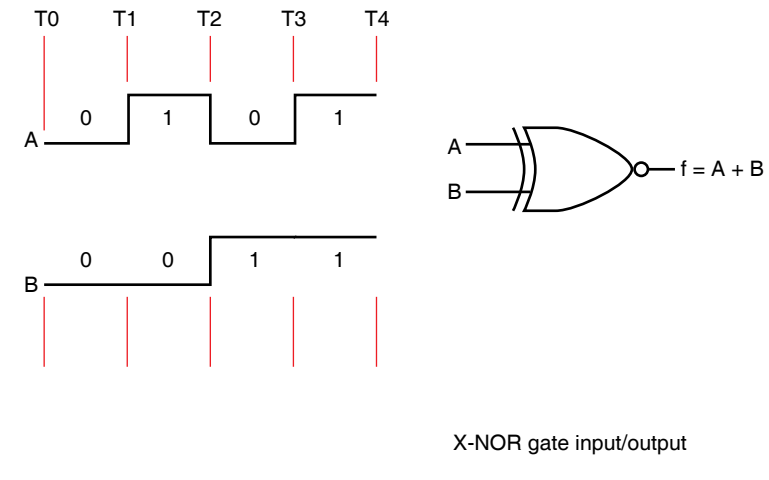


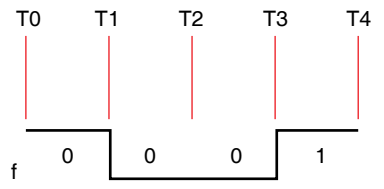


A	B	f
0	0	0
0	1	1
1	0	1
1	1	0

Truth table

Figure 12-264. *Standard symbol and truth table for X-OR gate.*





A	B	f
0	0	1
0	1	0
1	0	0
1	1	1

Truth table

Figure 12-265. *Standard Symbol for X-NOR gate.*

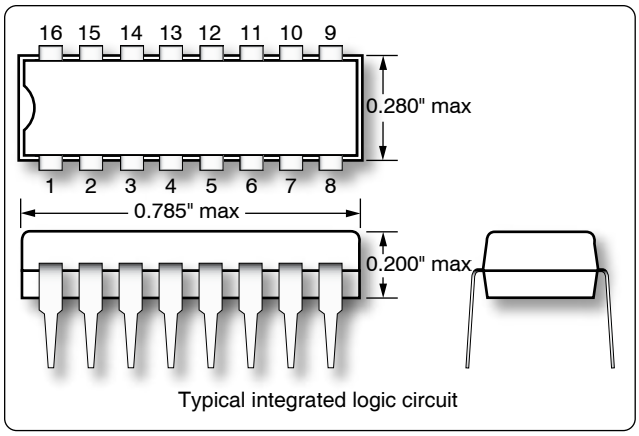


Figure 12-266. *Integrated circuit.*

loop rotates slightly past the 180° point, the black brush is contacting only the white segment, and the white brush is contacting only the black segment.

Because of this switching of commutator elements, the black brush is always in contact with the coil side moving downward, and the white brush is always in contact with the coil side moving upward. Though the current actually reverses its direction in the loop in exactly the same way as in the AC generator, commutator action causes the current to flow always in the same direction through the external circuit or meter.

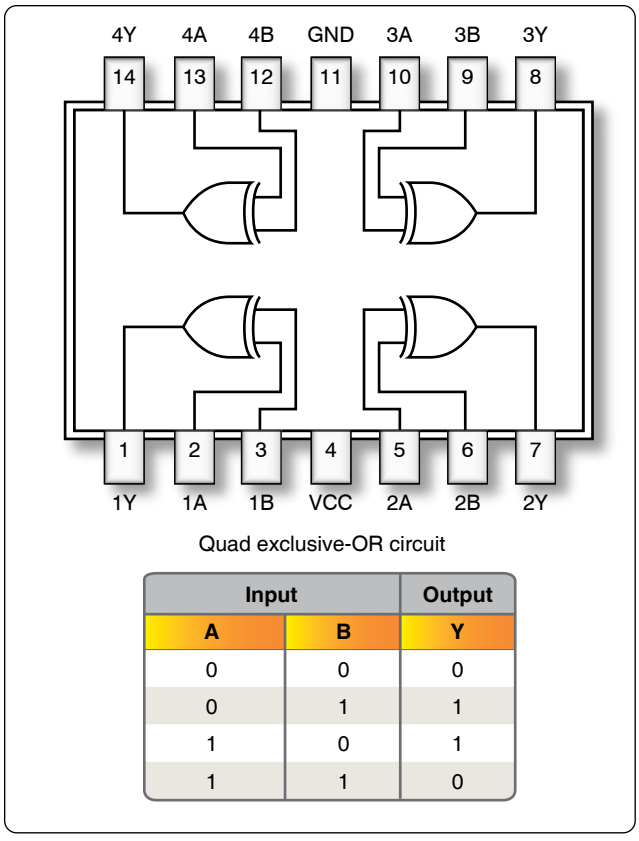
Position D

At position D, commutator action reverses the current in the external circuit, and the second half cycle has the same waveform as the first half cycle. The process of commutation is sometimes called rectification, since rectification is the converting of AC voltage to DC voltage.

The Neutral Plane

At the instant that each brush is contacting two segments on the commutator [*Figure 12-275A, C,* and *E*], a direct short circuit is produced. If an emf were generated in the loop at this time, a high current would flow in the circuit, causing an arc and thus damaging the commutator. For this reason, the brushes must be placed in the exact position where the short occurs when the generated emf is zero. This position is called the neutral plane. If the brushes are installed properly, no sparking occurs between the brushes and the commutator. Sparking is an indication of improper brush placement, which is the main cause of improper commutation.

The voltage generated by the basic DC generator in *Figure 12-275* varies from zero to its maximum value twice for each revolution of the loop. This variation of DC voltage is called "ripple," and may be reduced by using more loops, or coils, as shown in *Figure 12-276A*. As the number of loops



Input		Output
A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

Figure 12-267. *Small-scale integration schematic form.*

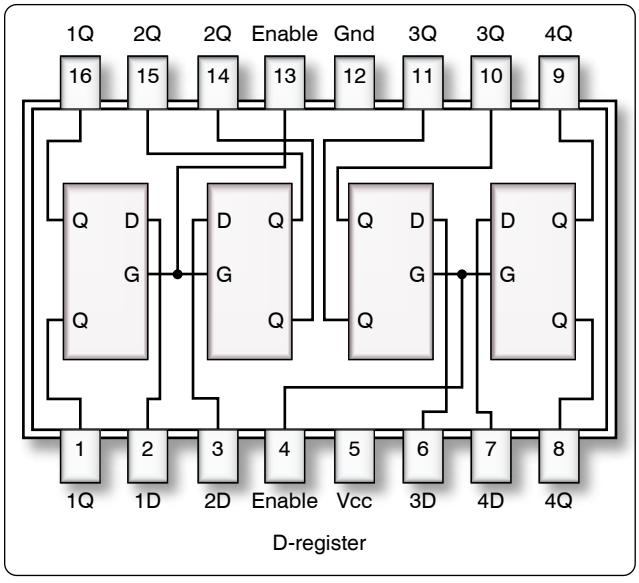


Figure 12-268. *Medium-scale integration schematic form.*

is increased, the variation between maximum and minimum values of voltage is reduced *[Figure 12-276B]*, and the output voltage of the generator approaches a steady DC value. In *Figure 12-276A,* the number of commutator segments is increased in direct proportion to the number of loops; that

is, there are two segments for one loop, four segments for two loops, and eight segments for four loops.

The voltage induced in a single turn loop is small. Increasing the number of loops does not increase the maximum value of generated voltage, but increasing the number of turns in each loop increases this value. Within narrow limits, the output voltage of a DC generator is determined by the product of the number of turns per loop, the total flux per pair of poles in the machine, and the speed of rotation of the armature.

An AC generator, or alternator, and a DC generator are identical as far as the method of generating voltage in the rotating loop is concerned. However, if the current is taken from the loop by slip rings, it is an alternating current, and the generator is called an AC generator, or alternator. If the current is collected by a commutator, it is direct current, and the generator is called a DC generator.

Construction Features of DC Generators

Generators used on aircraft may differ somewhat in design, since various manufacturers make them. All, however, are of the same general construction and operate similarly. The major parts, or assemblies, of a DC generator are a field frame (or yoke), a rotating armature, and a brush assembly. The parts of a typical aircraft generator are shown in *Figure 12-277*.

Field Frame

The field frame is also called the yoke, which is the foundation or frame for the generator. The frame has two functions: It completes the magnetic circuit between the poles and acts as a mechanical support for the other parts of the generator. In *Figure 12-278A*, the frame for a two-pole generator is shown in a cross-sectional view. A four-pole generator frame is shown in *Figure 12-278B*.

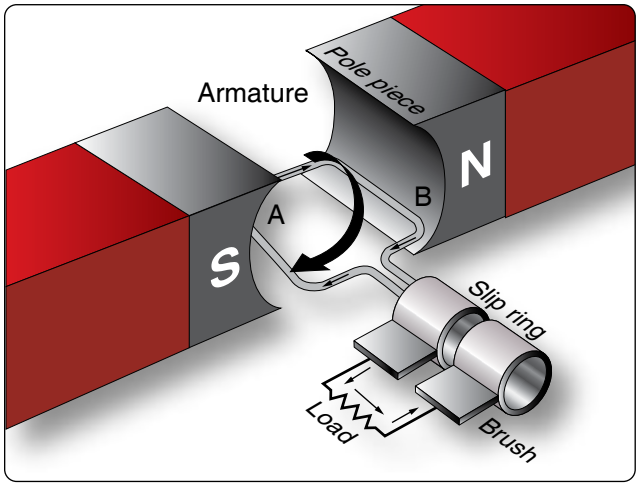


Figure 12-269. *Inducing maximum voltage in an elementary generator.*

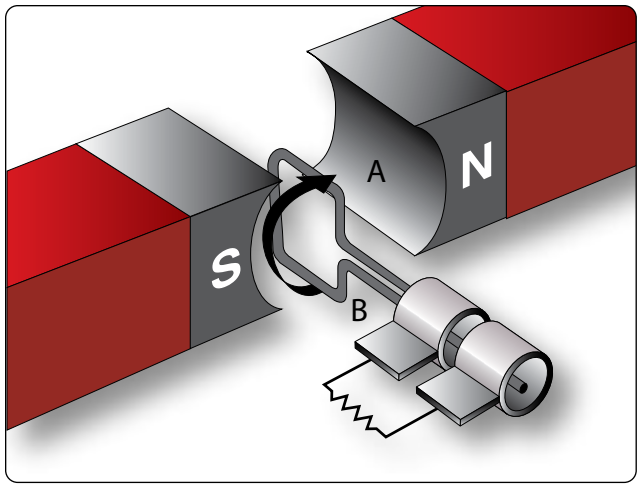


Figure 12-270. *Inducing minimum voltage in an elementary generator.*

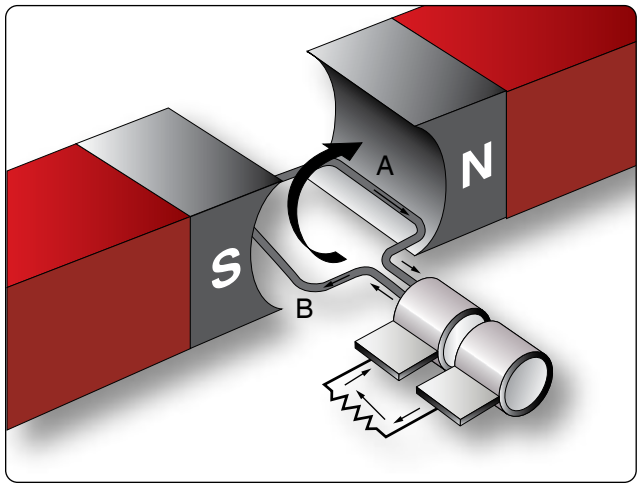


Figure 12-271. *Inducing maximum voltage in the opposite direction.*

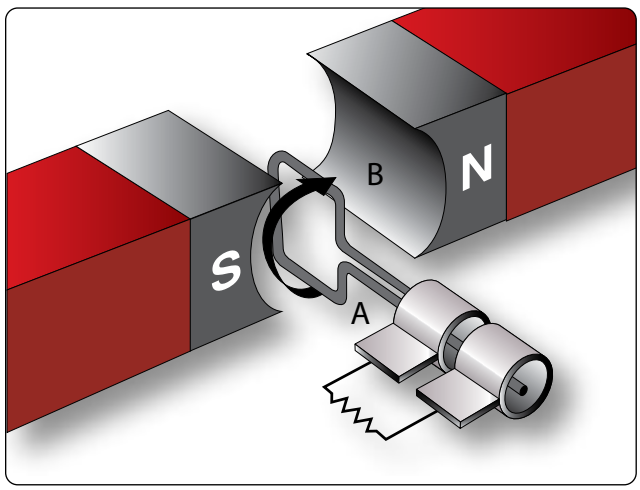


Figure 12-272. *Inducing a minimum voltage in the opposite direction.*

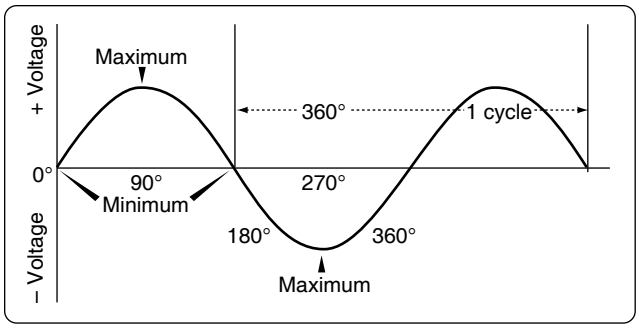


Figure 12-273. *Output of an elementary generator.*

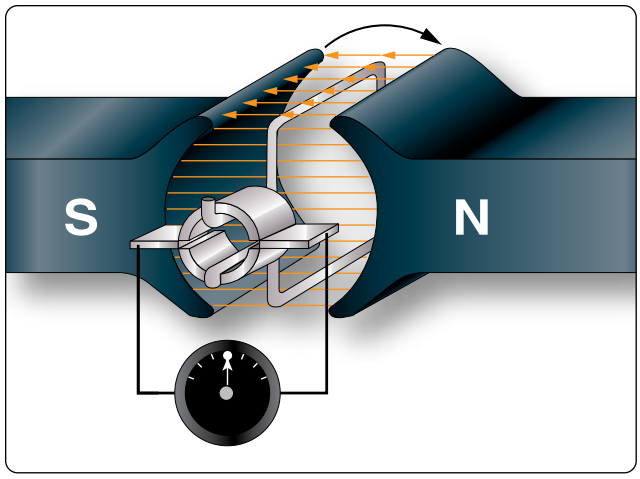


Figure 12-274. *Basic DC generator.*

In small generators, the frame is made of one piece of iron, but in larger generators, it is usually made up of two parts bolted together. The frame has high magnetic properties and, together with the pole pieces, forms the major part of the magnetic circuit. The field poles are bolted to the inside of the frame and form a core on which the field coil windings are mounted. *[Figure 12-278]*

The poles are usually laminated to reduce eddy current losses and serve the same purpose as the iron core of an electromagnet; that is, they concentrate the lines of force produced by the field coils. The entire frame, including field poles, is made from high-quality magnetic iron or sheet steel.

A practical DC generator uses electromagnets instead of permanent magnets. To produce a magnetic field of the necessary strength with permanent magnets would greatly increase the physical size of the generator.

The field coils are made up of many turns of insulated wire and are usually wound on a form that fits over the iron core of the pole to which it is securely fastened. *[Figure 12-279]* The exciting current, which is used to produce the magnetic field and which flows through the field coils, is obtained from an external source or from the generated DC of the machine. No electrical connection exists between the windings of the field coils and the pole pieces.

Most field coils are connected so that the poles show alternate polarity. Since there is always one North pole for each South pole, there must always be an even number of poles in any generator.

Note that the pole pieces in *Figure 12-278* project from the frame. Because air offers a great amount of reluctance to the magnetic field, this design reduces the length of the air gap between the poles and the rotating armature and increases the efficiency of the generator. When the pole pieces are made to project they are called salient poles. *[Figure 12-278]*

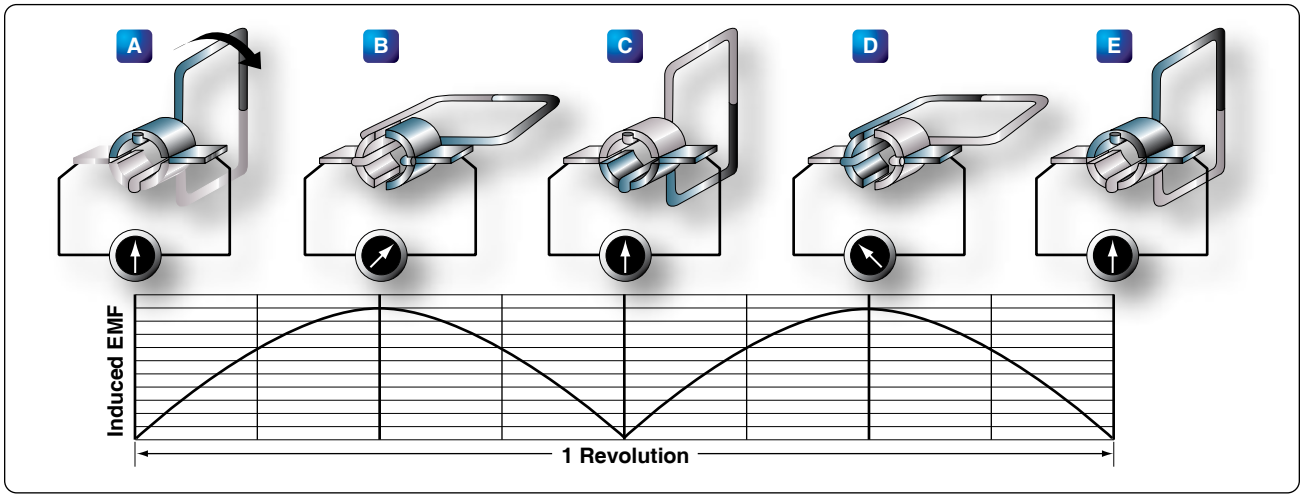


Figure 12-275. *Operation of a basic DC generator.*

Armature

The armature assembly of a generator consists of many armature coils wound on an iron core, a commutator, and associated mechanical parts. These additional loops of wire are actually called windings and are evenly spaced around the armature so that the distance between each winding is the same. Mounted on a shaft, it rotates through the magnetic field produced by the field coils. The core of the armature acts as an iron conductor in the magnetic field and, for this reason, is laminated to prevent the circulation of eddy currents.

Gramme-Ring Armature

There are two general kinds of armatures: the ring and the drum. *Figure 12-280* shows a ring-type armature made up of an iron core, an eight-section winding, and an eight-segment commutator. The disadvantage of this arrangement is that the windings, located on the inner side of the iron ring, cut few lines of flux. As a result, they have very little voltage induced in them. For this reason, the Gramme ring armature is not widely used.

Drum-Type Armature

A drum-type armature is shown in *Figure 12-281*. The armature core is in the shape of a drum and has slots cut into it where the armature windings are placed. The advantage is that each winding completely surrounds the core so that the entire length of the conductor cuts through the magnetic flux. The total induced voltage in this arrangement is far greater than that of the Gramme ring-type armature.

Drum-type armatures are usually constructed in one of two methods: lap winding and the wave winding. Each method having its own advantage. Lap windings are used in generators that are designed for high current. The windings are connected in parallel paths and for this reason require several brushes. The wave winding is used in generators that are designed for high voltage outputs. The two ends of each coil are connected to commutator segments separated by the distance between poles. This results in a series arrangement of the coils and is additive of all the induced voltages.

Commutators

Figure 12-282 shows a cross-sectional view of a typical commutator. The commutator is located at the end of an armature and consists of wedge shaped segments of hard drawn copper, insulated from each other by thin sheets of mica. The segments are held in place by steel V-rings or clamping flanges fitted with bolts. Rings of mica insulate the segments from the flanges. The raised portion of each segment is called a riser, and the leads from the armature coils are soldered to the risers. When the segments have no risers, the leads are soldered to short slits in the ends of the segments.

The brushes ride on the surface of the commutator, forming the electrical contact between the armature coils and the external circuit. A flexible, braided copper conductor, commonly called a pigtail, connects each brush to the external circuit. The brushes, usually made of high-grade carbon and held in place by brush holders insulated from the frame, are free to slide up and down in their holders in order to follow any irregularities in the surface of the commutator. The brushes are usually adjustable so that the pressure of the brushes on the commutator can be varied and the position of the brushes with respect to the segments can be adjusted.

The constant making and breaking of connections to the coils in which a voltage is being induced necessitates the use of material for brushes, which has a definite contact resistance. Also, this material must be such that the friction between the commutator and the brush is low, to prevent excessive wear. For these reasons, the material commonly used for brushes is high-grade carbon. The carbon must be soft enough to prevent undue wear of the commutator and yet hard enough to provide reasonable brush life. Since the contact resistance of carbon is fairly high, the brush must be quite large to provide a large area of contact. The commutator surface is highly polished to reduce friction as much as possible. Oil or grease must never be used on a commutator, and extreme care must be used when cleaning it to avoid marring or scratching the surface.

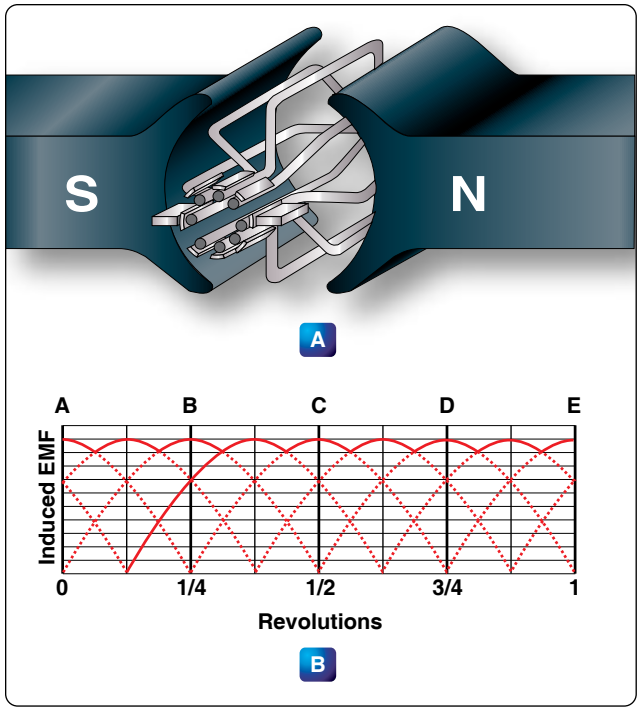


Figure 12-276. *Increasing the number of coils reduces the ripple in the voltage.*

Armature Reaction

Current flowing through the armature sets up electromagnetic fields in the windings. These new fields tend to distort or bend the magnetic flux between the poles of the generator from a straight-line path. Since armature current increases with load, the distortion becomes greater with an increase in load. This distortion of the magnetic field is called armature reaction. *[Figure 12-283]*

Armature windings of a generator are spaced so that, during rotation of the armature, there are certain positions when the brushes contact two adjacent segments, thereby shorting the armature windings to these segments. When the magnetic field is not distorted, there is usually no voltage being induced in the shorted windings, and therefore no harmful results occur from the shorting of the windings. However, when the field is distorted, a voltage is induced in these shorted windings, and sparking takes place between the brushes and the commutator segments. Consequently, the commutator becomes pitted, the wear on the brushes becomes excessive, and the output of the generator is reduced. To correct this condition, the brushes are set so that the plane of the coils, which are shorted by the brushes, is perpendicular to the distorted magnetic field, which is accomplished by moving the brushes forward in the direction of rotation. This operation is called shifting the brushes to the neutral plane or plane of commutation. The neutral plane is the position where the plane of the two opposite coils is perpendicular to the magnetic field in the generator. On a few generators, the brushes can be shifted manually ahead of the normal neutral plane to the neutral plane caused by field distortion. On nonadjustable brush generators, the manufacturer sets the brushes for minimum sparking.

Compensating windings or interpoles may be used to counteract some of the effects of field distortion, since shifting the brushes is inconvenient and unsatisfactory, especially when the speed and load of the generator are changing constantly.

Compensating Windings

The compensating windings consist of a series of coils embedded in slots in the pole faces. These coils are also connected in series with the armature. Consequently, this series connection with the armature produces a magnetic field in the compensating windings that varies directly with the armature current. The compensating windings are wound in such a manner that the magnetic field produced by them counteracts the magnetic field produced by the armature. As a result, the neutral plane remains stationary any magnitude of armature current. With this design, once the brushes are set correctly, they do not need to be moved again. *Figure 12-284A* illustrates how the windings are set into the pole faces.

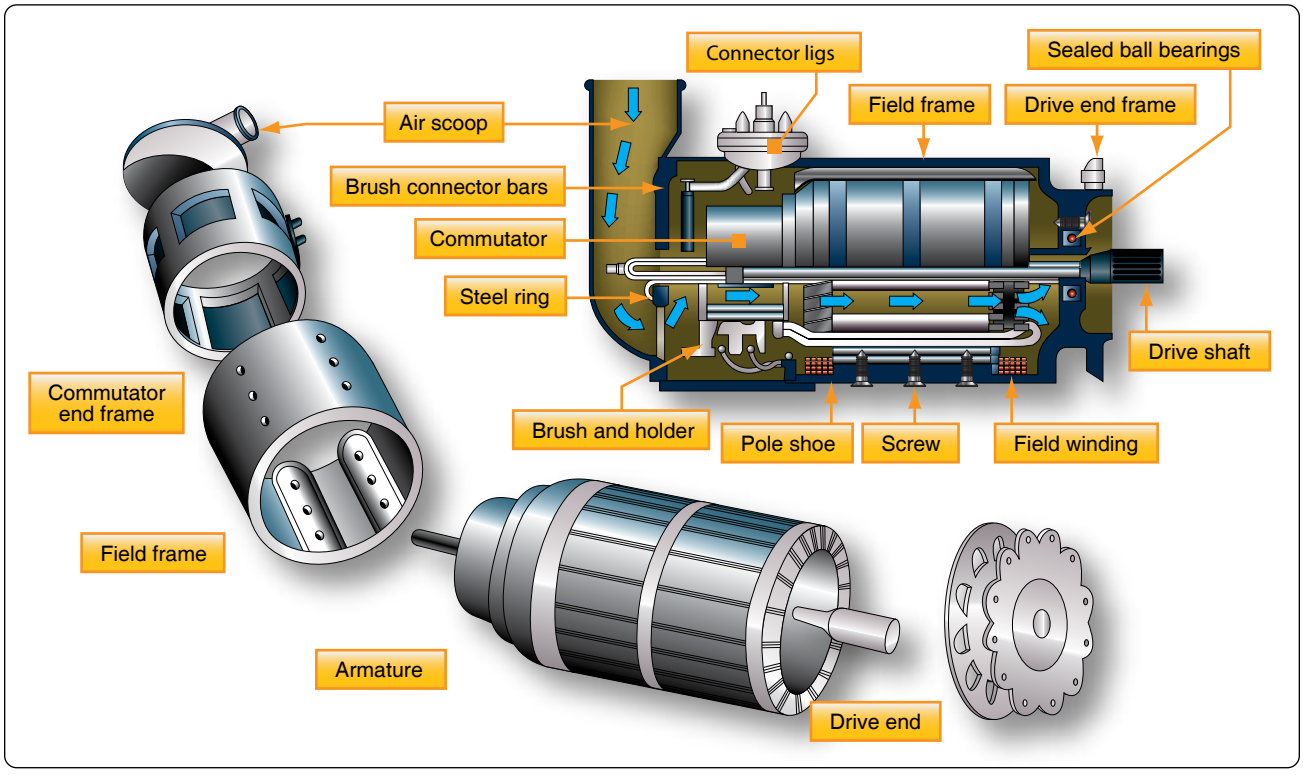


Figure 12-277. *Typical 24-volt aircraft generator.*

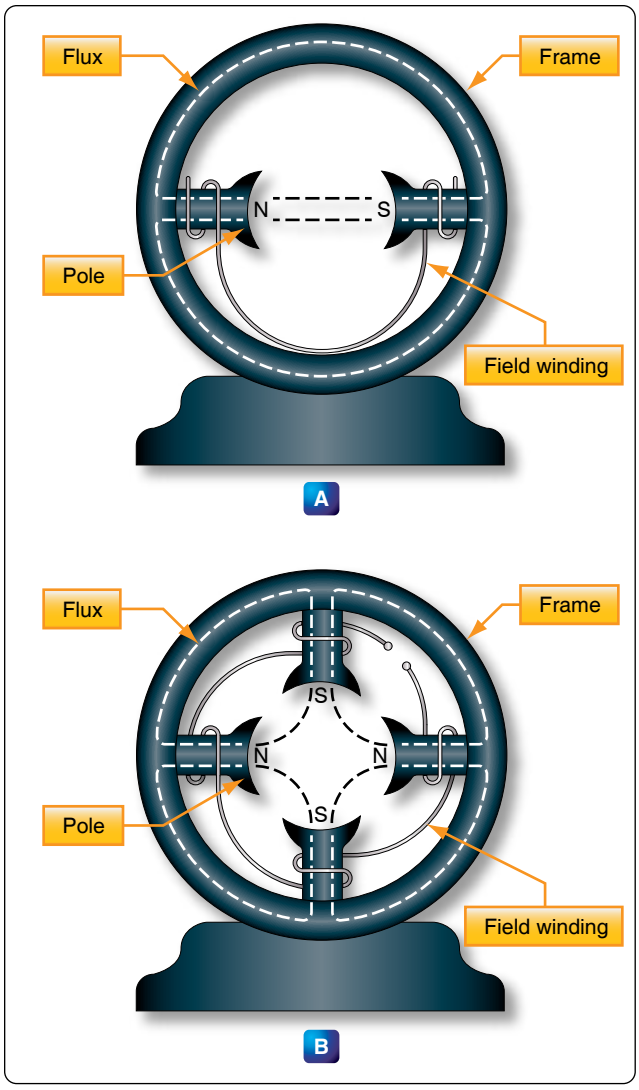


Figure 12-278. *A two-pole and a four-pole frame assembly.*

Interpoles

An interpole is a pole placed between the main poles of a generator. An example of interpole placement is shown in *Figure 12-284B*. This is a simple two-pole generator with two interpoles.

An interpole has the same polarity as the next main pole in the direction of rotation. The magnetic flux produced by an interpole causes the current in the armature to change direction as an armature winding passes under it. This cancels the electromagnetic fields about the armature windings. The magnetic strength of the interpoles varies with the load on the generator; and since field distortion varies with the load, the magnetic field of the interpoles counteracts the effects of the field set up around the armature windings and minimizes field distortion. Thus, the interpole tends to keep the neutral plane in the same position for all loads on the generator; therefore, field distortion is reduced by the interpoles, and the efficiency, output, and service life of the brushes are improved.

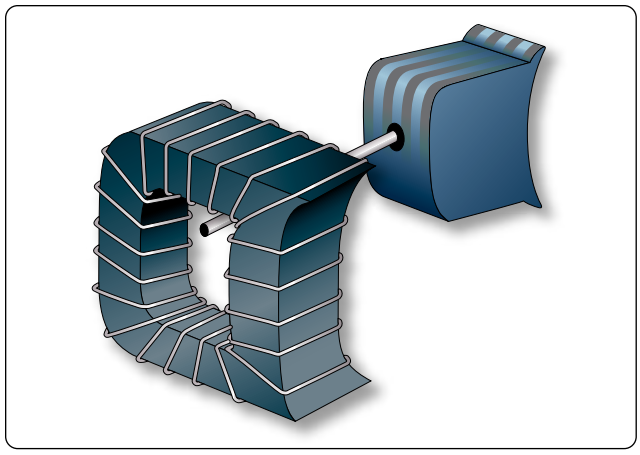


Figure 12-279. *A field coil removed from a field pole.*

Types of DC Generators

There are three types of DC generators: series wound, shunt wound, and shunt series or compound wound. The difference in type depends on the relationship of the field winding to the external circuit.

Series Wound DC Generators

The field winding of a series generator is connected in series with the external circuit called the load. *[Figure 12-285]* The field coils are composed of a few turns of large wire; the magnetic field strength depends more on the current flow rather than the number of turns in the coil. Series generators have very poor voltage regulation under changing load, since the greater the current through the field coils to the external circuit, the greater the induced emf and the greater the terminal or output voltage. Therefore, when the load is increased, the voltage increases; likewise, when the load is decreased, the voltage

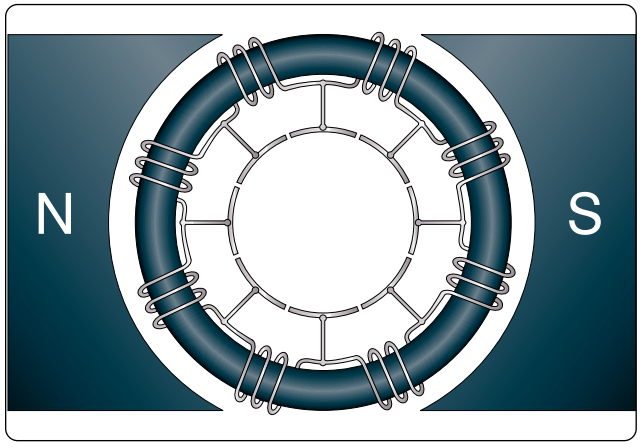


Figure 12-280. *An eight-section, ring-type armature.*

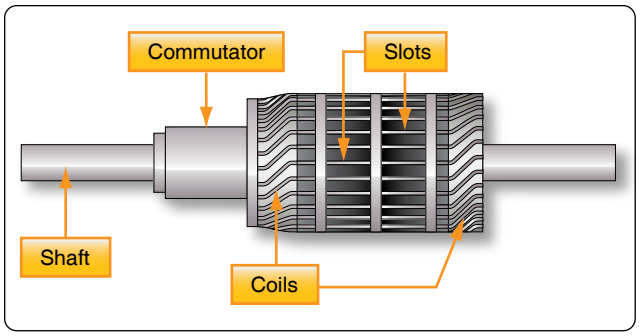


Figure 12-281. *A drum-type armature.*

decreases. The output voltage of a series wound generator may be controlled by a rheostat in parallel with the field windings. *[Figure 12-285A]* Since the series wound generator has such poor regulation, it is never employed as an airplane generator. Generators in airplanes have field windings, which are connected either in shunt or in compound.

Shunt Wound DC Generators

A generator having a field winding connected in parallel with the external circuit is called a shunt generator. *[Figure 12-286A* and *B]* The field coils of a shunt generator contain many turns of small wire; the magnetic strength is derived from the large number of turns rather than the current strength through the coils. If a constant voltage is desired, the shunt wound generator is not suitable for rapidly fluctuating loads. Any increase in load causes a decrease in the terminal or output voltage, and any decrease in load causes an increase in terminal voltage; since the armature and the load are connected in series, all current flowing in the external circuit passes through the armature winding. Because of the resistance in the armature winding, there is a voltage drop (IR drop = current × resistance). As the load increases, the armature current increases and the IR drop in the armature increases. The voltage delivered to the terminals is the difference between the induced voltage and the voltage drop; therefore, there is a decrease in terminal voltage. This decrease in voltage causes a decrease in field strength, because the current in the field coils decreases in proportion to the decrease in terminal voltage; with a weaker field, the voltage is further decreased. When the load decreases, the output voltage increases accordingly, and a larger current flows in the windings. This action is cumulative, so the output voltage continues to rise to a point called field saturation, after which there is no further increase in output voltage.

The terminal voltage of a shunt generator can be controlled by means of a rheostat inserted in series with the field windings. *[Figure 12-286A]* As the resistance is increased, the field current is reduced; consequently, the generated voltage is reduced also. For a given setting of the field rheostat, the terminal voltage at the armature brushes is approximately equal to the generated voltage minus the IR

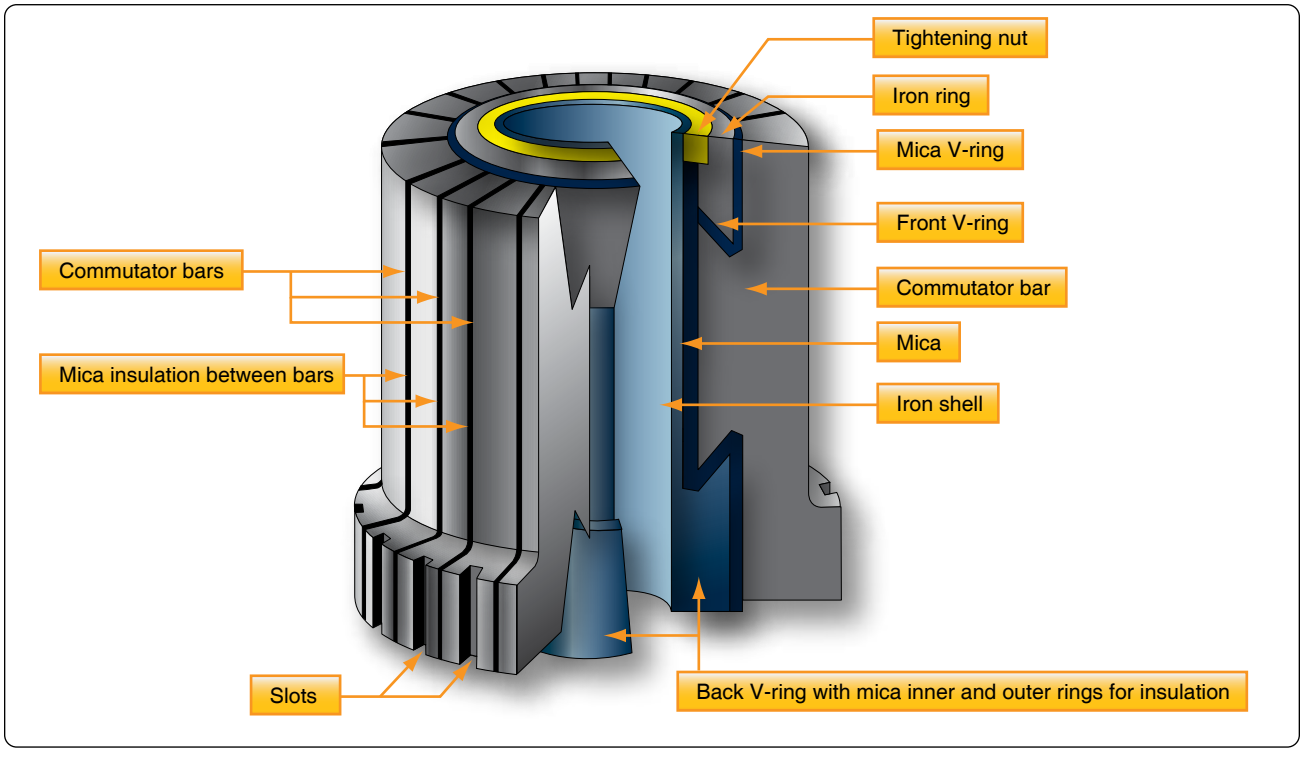


Figure 12-282. *Commutator with portion removed to show construction.*

drop produced by the load current in the armature; thus, the voltage at the terminals of the generator drops as the load is applied. Certain voltage sensitive devices are available that automatically adjust the field rheostat to compensate for variations in load. When these devices are used, the terminal voltage remains essentially constant.

Compound Wound DC Generators

A compound wound generator combines a series winding and a shunt winding in such a way that the characteristics of each are used to advantage. The series field coils are made of a relatively small number of turns of large copper conductor, either circular or rectangular in cross section, and are connected in series with the armature circuit. These coils are mounted on the same poles on which the shunt field coils are mounted and, therefore, contribute a magnetomotive force which influences the main field flux of the generator. A diagrammatic and a schematic illustration of a compound wound generator is shown in *Figure 12-287A* and *B*.

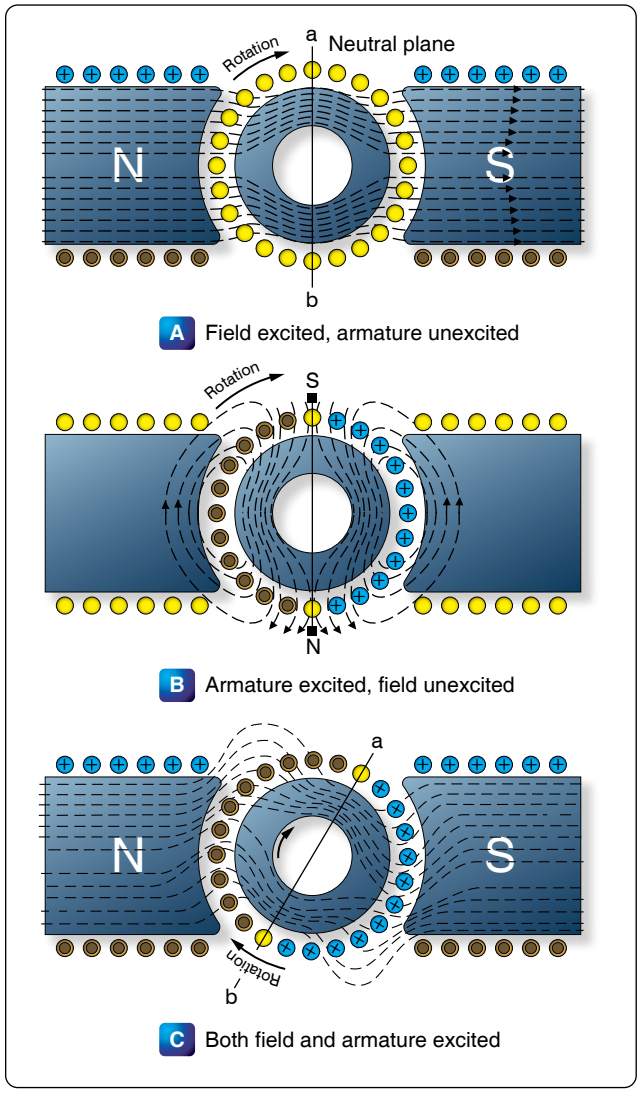


Figure 12-283. *Armature reaction.*

If the ampere turns of the series field act in the same direction as those of the shunt field, the combined magnetomotive force is equal to the sum of the series and shunt field components. Load is added to a compound generator in the same manner in which load is added to a shunt generator, by increasing the number of parallel paths across the generator terminals. Thus, the decrease in total load resistance with added load is accompanied by an increase in armature circuit and series field circuit current. The effect of the additive series field is that of increased field flux with increased load. The extent of the increased field flux depends on the degree of saturation of the field as determined by the shunt field current. Thus, the terminal voltage of the generator may increase or decrease with load, depending on the influence of the series field coils. This influence is referred to as the degree of compounding. A flat compound generator is one in which the no load and full load voltages have the same value; whereas an under compound generator has a full load voltage less than the no load value, and an over compound generator has a full load voltage which is higher than the no load value. Changes in terminal voltage with increasing load depend upon the degree of compounding.

If the series field aids the shunt field, the generator is said to be cumulative compounded. If the series field opposes the shunt field, the machine is said to be differentially compounded or is called a differential generator. Compound generators are usually designed to be overcompounded. This feature permits varied degrees of compounding by connecting a variable shunt across the series field. Such a shunt is sometimes called a diverter. Compound generators are used where voltage regulation is of prime importance.

Differential generators have somewhat the same characteristics as series generators in that they are essentially constant current generators. However, they generate rated voltage at no load, the voltage dropping materially as the load current increases. Constant current generators are ideally suited as power sources for electric arc welders and are used almost universally in electric arc welding.

If the shunt field of a compound generator is connected across both the armature and the series field, it is known as a long shunt connection, but if the shunt field is connected across the armature alone, it is called a short shunt connection. These connections produce essentially the same generator characteristics. A summary of the characteristics of the various types of generators discussed is shown in *Figure 12-288*.

Generator Ratings

A generator is rated in power output. Since a generator is designed to operate at a specified voltage, the rating usually is given as the number of amperes the generator can safely

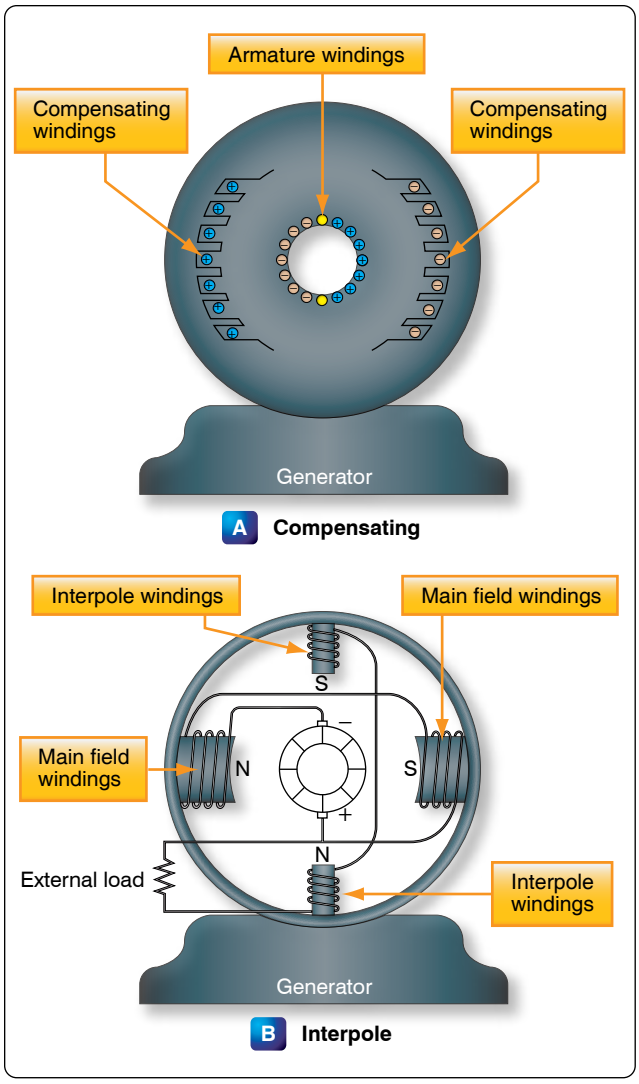


Figure 12-284. *Simple two-pole generator with two interpoles.*

supply at its rated voltage. Generator rating and performance data are stamped on the nameplate attached to the generator. When replacing a generator, it is important to choose one of the proper rating.

The rotation of generators is termed either clockwise or counterclockwise, as viewed from the driven end. Usually, the direction of rotation is stamped on the data plate. If no direction is stamped on the plate, the rotation may be marked by an arrow on the cover plate of the brush housing. It is important that a generator with the correct direction of rotation be used; otherwise, the voltage is reversed.

The speed of an aircraft engine varies from idle rpm to takeoff rpm; however, during the major portion of a flight, it is at a constant cruising speed. The generator drive is usually geared to revolve the generator between 1⅛ and 1½ times the engine crankshaft speed. Most aircraft generators have a speed at

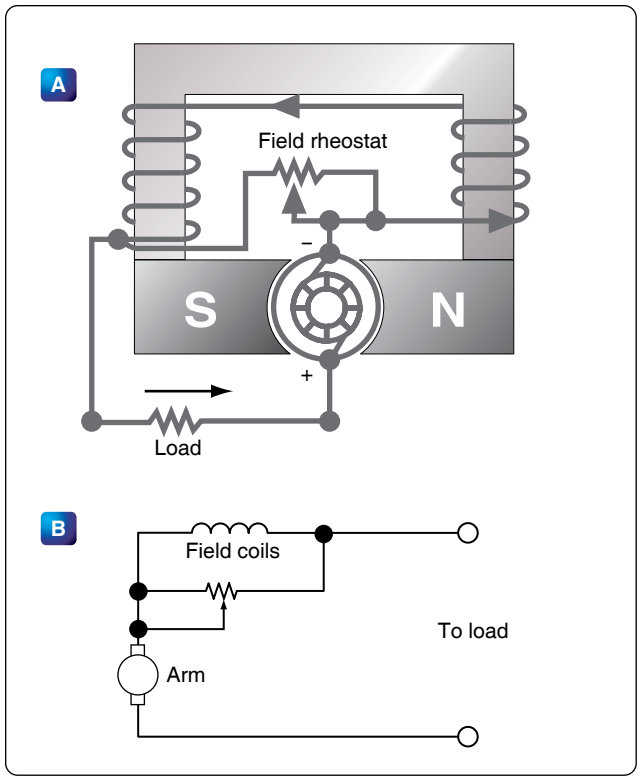


Figure 12-285. *Diagram and schematic of a series wound generator.*

which they begin to produce their normal voltage. Termed the "coming in" speed, it is usually about 1,500 rpm.

Generator Terminals

On most large 24-volt generators, electrical connections are made to terminals marked B, A, and E. The positive armature lead in the generator connects to the B terminal. The negative armature lead connects to the E terminal. The positive end of the shunt field winding connects to terminal A, and the opposite end connects to the negative terminal brush. Terminal A receives current from the negative generator brush through the shunt field winding. This current passes through the voltage regulator and back to the armature through the positive brush. Load current, which leaves the armature through the negative brushes, comes out of the E lead and passes through the load before returning to the armature through the positive brushes.

DC Generator Maintenance

Inspection

The following information about the inspection and maintenance of DC generator systems is general in nature because of the large number of differing aircraft generator systems. These procedures are for familiarization only. Always follow the applicable manufacturer's instructions for a given generator system.

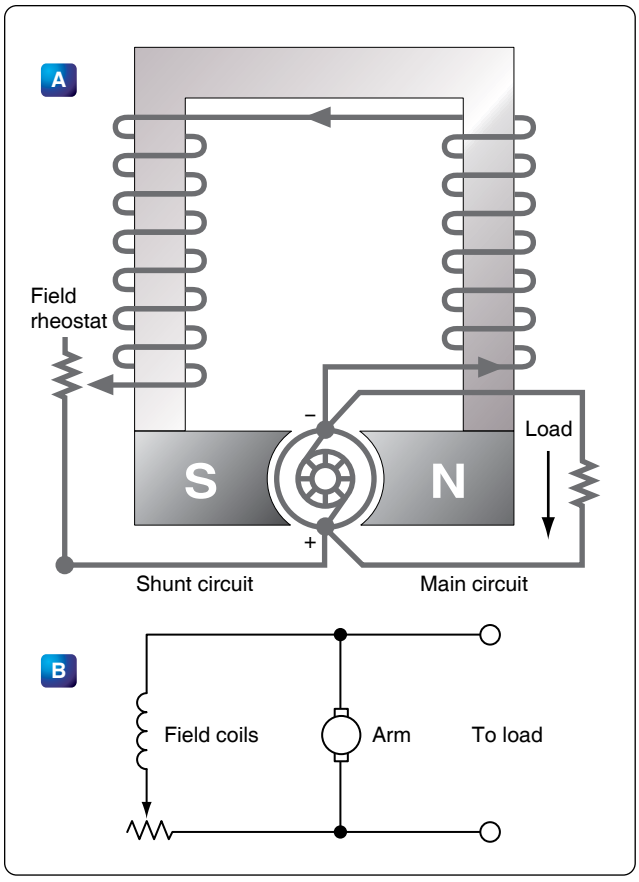


Figure 12-286. *Shunt wound generator.*

In general, the inspection of the generator installed in the aircraft should include the following items:

1. Security of generator mounting
2. Condition of electrical connections
3. Dirt and oil in the generator—if oil is present, check engine oil seal. Blow out dirt with compressed air.
4. Condition of generator brushes
5. Generator operation
6. Voltage regulator operation

Condition of Generator Brushes

Sparking of brushes quickly reduces the effective brush area in contact with the commutator bars. The degree of such sparking should be determined. Excessive wear warrants a detailed inspection.

The following information pertains to brush seating, brush pressure, high mica condition, and brush wear. Manufacturers usually recommend the following procedures to seat brushes that do not make good contact with slip rings or commutators.

Lift the brush sufficiently to permit the insertion of a strip of No. 000, or finer, sandpaper under the brush, rough side out. *[Figure 12-289]* Pull the sandpaper in the direction of armature rotation, being careful to keep the ends of the sandpaper as close to the slip ring or commutator surface as possible in order to avoid rounding the edges of the brush.

When pulling the sandpaper back to the starting point, raise the brush so it does not ride on the sandpaper. Sand the brush only in the direction of rotation.

After the generator has run for a short period, brushes should be inspected to make sure that pieces of sand have not become embedded in the brush and are collecting copper.

Under no circumstances should emery cloth or similar abrasives be used for seating brushes (or smoothing commutators), since they contain conductive materials that cause arcing between brushes and commutator bars.

Excessive pressure causes rapid wear of brushes. Too little pressure, however, allows "bouncing" of the brushes, resulting in burned and pitted surfaces.

A carbon, graphite, or light metalized brush should exert a pressure of 1½ to 2½ psi on the commutator. The pressure recommended by the manufacturer should be checked by the use of a spring scale graduated in ounces. Brush spring

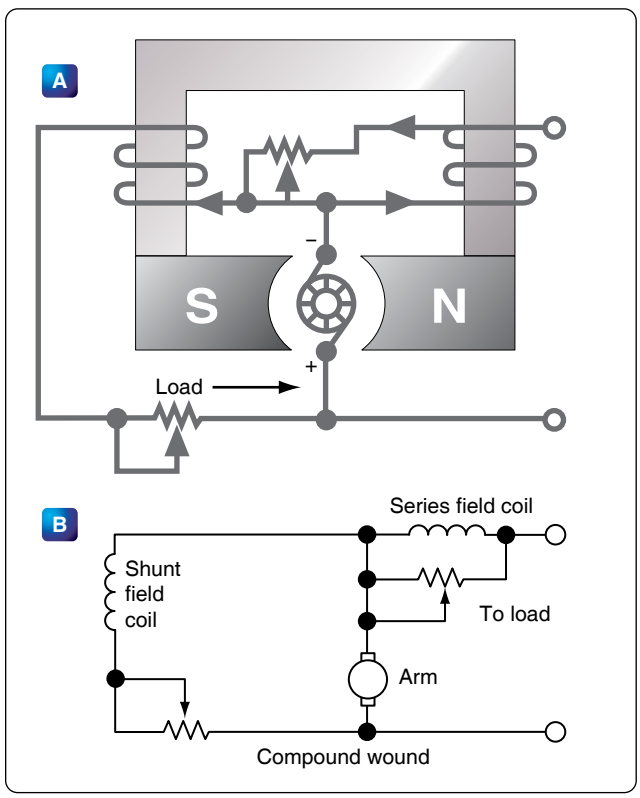


Figure 12-287. *Compound wound generator.*

tension is usually adjusted between 32 to 36 ounces; however, the tension may differ slightly for each specific generator.

When a spring scale is used, the measurement of the pressure that a brush exerts on the commutator is read directly on the scale. The scale is applied at the point of contact between the spring arm and the top of the brush, with the brush installed in the guide. The scale is drawn up until the arm just lifts off the brush surface. At this instant, the force on the scale should be read.

Flexible low resistance pigtails are provided on most heavy current carrying brushes, and their connections should be securely made and checked at frequent intervals. The pigtails should never be permitted to alter or restrict the free motion of the brush.

The purpose of the pigtail is to conduct the current, rather than subjecting the brush spring to currents that would alter its spring action by overheating. The pigtails also eliminate any possible sparking to the brush guides caused by the movement of the brushes within the holder, thus minimizing side wear of the brush.

Carbon dust resulting from brush sanding should be thoroughly cleaned from all parts of the generators after a sanding operation. Such carbon dust has been the cause of several serious fires, as well as costly damage to the generator.

Operation over extended periods of time often results in the mica insulation between commutator bars protruding above the surface of the bars. This condition is called "high mica" and interferes with the contact of the brushes to the commutator. Whenever this condition exists, or if the armature has been turned on a lathe, carefully undercut the mica insulation to a depth equal to the width of the mica, or approximately 1.20 inch.

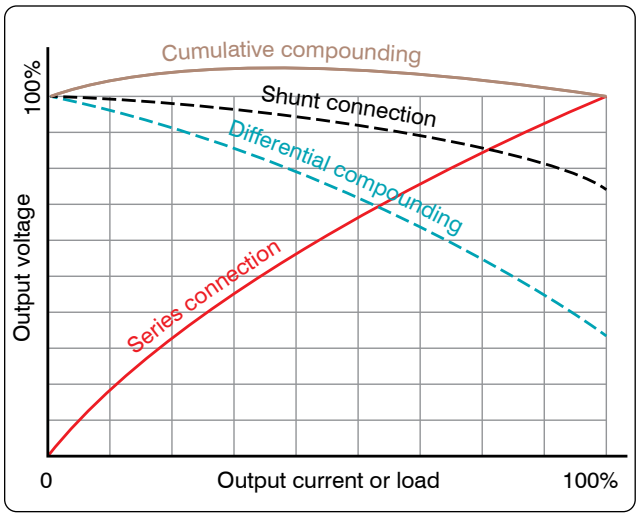


Figure 12-288. *Generator characteristics.*

Each brush should be a specified length to work properly. If a brush is too short, the contact it makes with the commutator will be faulty, which can also reduce the spring force holding the brush in place. Most manufacturers specify the amount of wear permissible from a new brush length. When a brush has worn to the minimum length permissible, it must be replaced.

Some special generator brushes should not be replaced because of a slight grooving on the face of the brush. These grooves are normal and will appear in AC and DC generator brushes which are installed in some models of aircraft generators. These brushes have two cores made of a harder material with a higher expansion rate than the material used in the main body of the brush. Usually, the main body of the brush face rides on the commutator. However, at certain temperatures, the cores extend and wear through any film on the commutator.

DC Motors

Most devices in an airplane, from the starter *[Figure 12-290]* to the automatic pilot, depend upon mechanical energy furnished by DC motors. A DC motor is a rotating machine, which transforms DC energy into mechanical energy. It consists of two principal parts—a field assembly and an armature assembly. The armature is the rotating part in which current carrying wires are acted upon by the magnetic field.

Whenever a current carrying wire is placed in the field of a magnet, a force acts on the wire. The force is not one of attraction or repulsion; however, it is at right angles to the wire and also at right angles to the magnetic field set up by the magnet. The action of the force upon a current carrying wire placed in a magnetic field is shown in *Figure 12-291*. A wire is located between two permanent magnets. The lines of force in the magnetic field are from the North pole to the South pole. When no current flows, no force is exerted on the wire, but when current flows through the wire, a magnetic field is set up about it. *[Figure 12-291]* The direction of the field depends on the direction of current flow. Current in one direction creates a clockwise field about the wire, and current in the other direction, a counterclockwise field.

Since the current carrying wire produces a magnetic field, a reaction occurs between the field about the wire and the magnetic field between the magnets. When the current flows

in a direction to create a counterclockwise magnetic field about the wire, this field and the field between the magnets add or reinforce at the bottom of the wire because the lines of force are in the same direction. At the top of the wire, they subtract or neutralize, since the lines of force in the two fields are opposite in direction. Thus, the resulting field at the bottom is strong and the one at the top is weak. Consequently, the wire is pushed upward. *[Figure 12-291C]* The wire is always pushed away from the side where the field is strongest. If current flow through the wire were reversed in direction, the two fields would add at the top and subtract at the bottom. Since a wire is always pushed away from the strong field, the wire would be pushed down.

Force Between Parallel Conductors

Two wires carrying current in the vicinity of one another exert a force on each other because of their magnetic fields. An end view of two conductors is shown in *Figure 12-292*. In *Figure 12-292A*, electron flow in both conductors is toward the reader, and the magnetic fields are clockwise around the conductors. Between the wires, the fields cancel because the directions of the two fields oppose each other. The wires are forced in the direction of the weaker field, toward each other. This force is one of attraction. In *Figure 12-292B*, the electron flow in the two wires is in opposite directions.

The magnetic fields are, therefore, clockwise in one and counterclockwise in the other, as shown. The fields reinforce each other between the wires, and the wires are forced in the direction of the weaker field, away from each other. This force is one of repulsion.

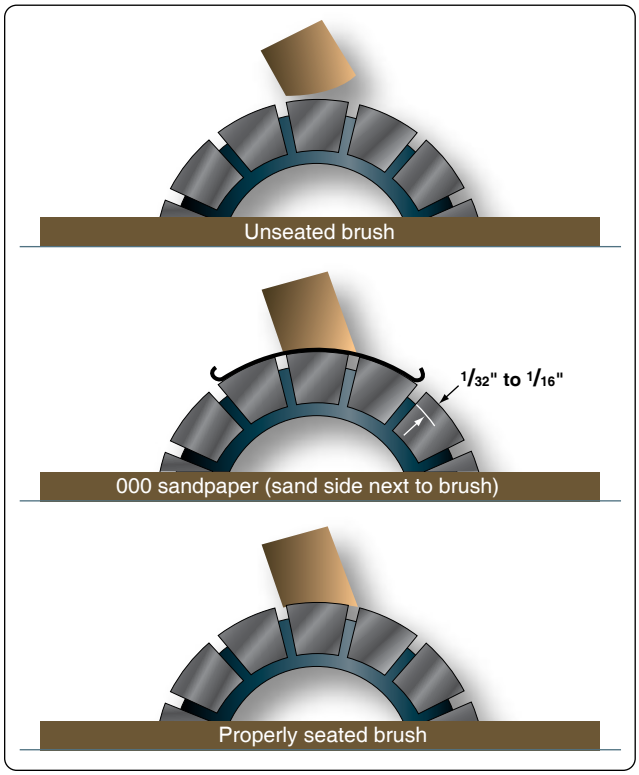


Figure 12-289. *Seating brushes with sandpaper.*

To summarize: conductors carrying current in the same direction tend to be drawn together; conductors carrying current in opposite directions tend to be repelled from each other.

Developing Torque

If a coil in which current is flowing is placed in a magnetic field, a force is produced which causes the coil to rotate. In the coil shown in *Figure 12-293*, current flows inward on side A and outward on side B. The magnetic field about B is clockwise and that about A, counterclockwise. As previously explained, a force develops which pushes side B downward. At the same time, the field of the magnets and the field about A, in which the current is inward, adds at the bottom and subtracts at the top. Therefore, A moves upward. The coil rotates until its plane is perpendicular to the magnetic lines between the North and South poles of the magnet, as indicated in *Figure 12-293* by the white coil at right angles to the black coil.

The tendency of a force to produce rotation is called torque. When the steering wheel of a car is turned, torque is applied. The engine of an airplane gives torque to the propeller. Torque is developed also by the reacting magnetic fields about the current carrying coil just described. This is the torque, which turns the coil.

The right-hand motor rule can be used to determine the direction a current carrying wire moves in a magnetic field. As illustrated in *Figure 12-294*, if the index finger of the right hand is pointed in the direction of the magnetic field and the second finger in the direction of current flow, the thumb indicates the direction the current carrying wire moves. The amount of torque developed in a coil depends upon several factors: the strength of the magnetic field, the number of turns in the coil, and the position of the coil in the field. Magnets are made of special steel that produces a strong field. Since there is torque acting on each turn, the greater the number of turns on the coil, the greater the torque. In a coil carrying a steady current located in a uniform magnetic field, the torque varies at successive positions of rotation. *[Figure 12-295]* When the plane of the coil is parallel to the lines of force, the torque is zero. When its plane cuts the lines of force at right angles, the torque is 100 percent. At intermediate positions, the torque ranges between 0 and 100 percent.

Basic DC Motor

A coil of wire through which the current flows rotates when placed in a magnetic field. This is the technical basis governing the construction of a DC motor. *[Figure 12-296]* However,

if the connecting wires from the battery were permanently fastened to the terminals of the coil and there was a flow of current, the coil would rotate only until it lined itself up with the magnetic field. Then, it would stop, because the torque at that point would be 0.

A motor, of course, must continue rotating. It is therefore necessary to design a device that reverses the current in the coil just at the time the coil becomes parallel to the lines of force. This creates torque again and causes the coil to rotate. If the current reversing device is set up to reverse the current each time the coil is about to stop, the coil can be made to continue rotating as long as desired.

One method of doing this is to connect the circuit so that, as the coil rotates, each contact slides off the terminal to which it connects and slides onto the terminal of opposite polarity. In other words, the coil contacts switch terminals continuously as the coil rotates, preserving the torque and keeping the coil rotating. In *Figure 12-296*, the coil terminal segments are labeled A and B. As the coil rotates, the segments slide onto and past the fixed terminals or brushes. With this arrangement, the direction of current in the side of the coil next to the North-seeking pole flows toward the reader, and the force acting on that side of the coil turns it downward. The part of the motor that changes the current from one wire to another is called the commutator.

Position A

When the coil is positioned as shown in *Figure 12-296A*, current flows from the negative terminal of the battery to the negative (−) brush, to segment B of the commutator, through the loop to segment A of the commutator, to the positive (+) brush, and then back to the positive terminal of the battery. By using the right-hand motor rule, it is seen that the coil rotates counterclockwise. The torque at this position of the coil is maximum, since the greatest number of lines of force is being cut by the coil.

Figure 12-290. *DC series starter motor.*

Position B

When the coil has rotated 90° to the position shown in *Figure 12-296B*, segments A and B of the commutator no longer make contact with the battery circuit and no current can flow through the coil. At this position, the torque has reached a minimum value, since a minimum number of lines of force are being cut. However, the momentum of the coil carries it beyond this position until the segments again make contact with the brushes, and current again enters the coil; this time, though, it enters through segment A and leaves through segment B. However, since the positions of segments A and B have also been reversed, the effect of the current is as before, the torque acts in the same direction, and the coil continues its counterclockwise rotation.

Position C

On passing through the position shown in *Figure 12-296C*, the torque again reaches maximum.

Position D

Continued rotation carries the coil again to a position of minimum torque as in *Figure 12-296D*. At this position, the brushes no longer carry current, but once more the momentum rotates the coil to the point where current enters through

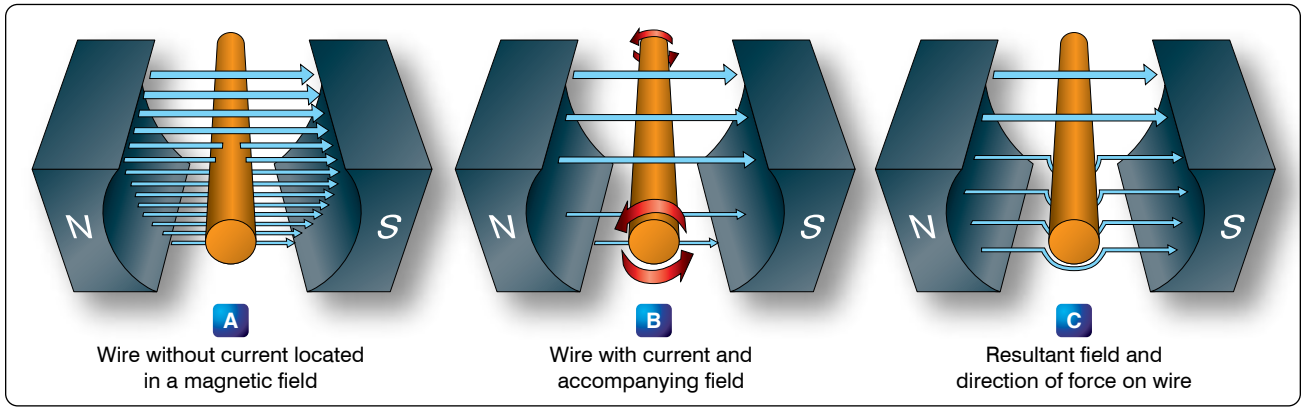


Figure 12-291. *Force on a current carrying wire.*

segment B and leaves through A. Further rotation brings the coil to the starting point and, thus, one revolution is completed.

The switching of the coil terminals from the positive to the negative brushes occurs twice per revolution of the coil.

The torque in a motor containing only a single coil is neither continuous nor very effective, for there are two positions where there is actually no torque at all. To overcome this, a practical DC motor contains a large number of coils wound on the armature. These coils are so spaced that, for any position of the armature, there are coils near the poles of the magnet. This makes the torque both continuous and strong. The commutator, likewise, contains a large number of segments instead of only two.

The armature in a practical motor is not placed between the poles of a permanent magnet but between those of an electromagnet, since a much stronger magnetic field can be furnished. The core is usually made of a mild or annealed steel, which can be magnetized strongly by induction. The current magnetizing the electromagnet is from the same source that supplies the current to the armature.

DC Motor Construction

The major parts in a practical motor are the armature assembly, the field assembly, the brush assembly, and the end frame. *[Figure 12-297]*

Armature Assembly

The armature assembly contains a laminated, soft-iron core, coils, and a commutator, all mounted on a rotatable steel shaft. Laminations made of stacks of soft iron, insulated from each other, form the armature core. Solid iron is not used, since a solid iron core revolving in the magnetic field would heat and use energy needlessly. The armature windings are insulated copper wire, which are inserted in slots insulated with fiber paper (fish paper) to protect the windings. The ends of the windings are connected to the commutator segments. Wedges or steel bands hold the windings in place to prevent them from flying out of the slots when the armature is rotating at high speeds. The commutator consists of a large number of copper segments insulated from each other and the armature shaft by pieces of mica. Insulated wedge rings hold the segments in place.

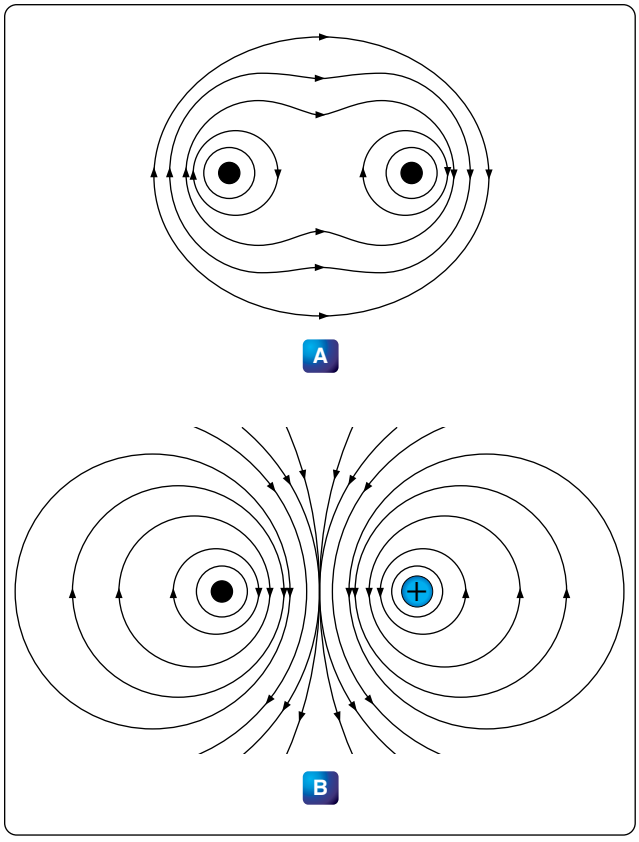


Figure 12-292. *Fields surrounding parallel conductors.*

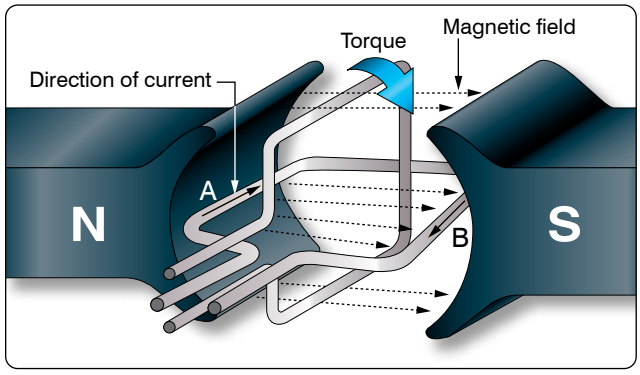


Figure 12-293. *Developing a torque.*

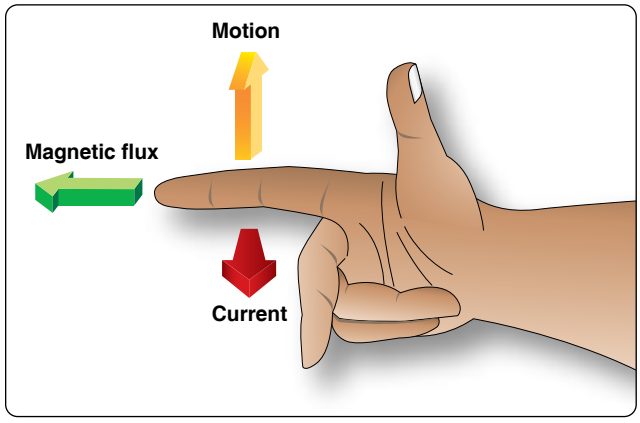


Figure 12-294. *Right-hand motor rule.*

Field Assembly

The field assembly consists of the field frame, the pole pieces, and the field coils. The field frame is located along the inner wall of the motor housing. It contains laminated, soft-steel pole pieces on which the field coils are wound. A coil, consisting of several turns of insulated wire, fits over each pole piece and, together with the pole, constitutes a field pole. Some motors have as few as two poles, others as many as eight.

Brush Assembly

The brush assembly consists of the brushes and their holders. The brushes are usually small blocks of graphitic carbon, since this material has a long service life and also causes minimum wear to the commutator. The holders permit some play in the brushes so they can follow any irregularities in the surface of the commutator and make good contact. Springs hold the brushes firmly against the commutator. *[Figure 12-298]*

End Frame

The end frame is the part of the motor opposite the commutator. Usually, the end frame is designed so that it can be connected to the unit to be driven. The bearing for the drive end is also located in the end frame. Sometimes the end frame is made a part of the unit driven by the motor. When this is done, the bearing on the drive end may be located in any one of a number of places.

Types of DC Motors

There are three basic types of DC motors: series motors, shunt motors, and compound motors. They differ largely in the method in which their field and armature coils are connected.

Series DC Motor

In the series motor, the field windings, consisting of a relatively few turns of heavy wire, are connected in series with the armature winding. Both a diagrammatic and a schematic illustration of a series motor are shown in *Figure 12-299*. The same current flowing through the field winding also flows through the armature winding. Any increase in current, therefore, strengthens the magnetism of

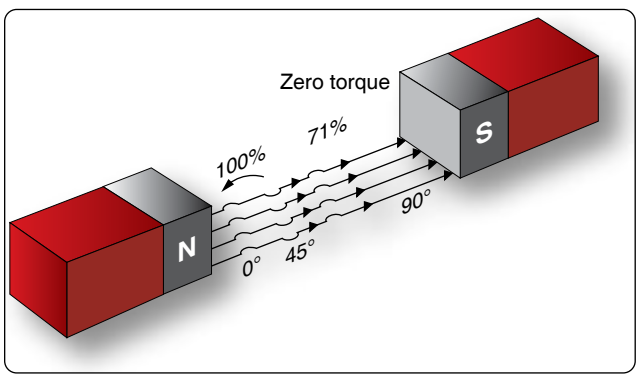


Figure 12-295. *Torque on a coil at various angles of rotation.*

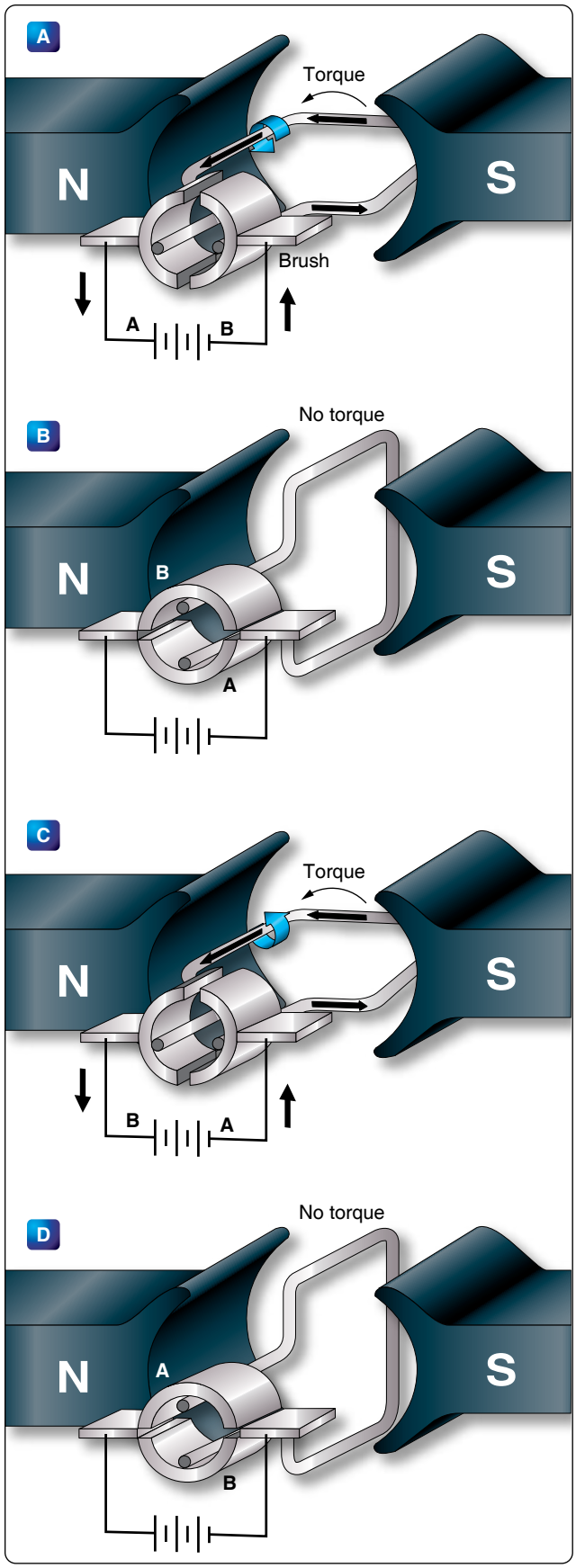


Figure 12-296. *Basic DC motor operation.*

both the field and the armature.

Because of the low resistance in the windings, the series motor is able to draw a large current in starting. This starting current, in passing through both the field and armature windings, produces a high starting torque, which is the series motor's principal advantage.

The speed of a series motor is dependent upon the load. Any change in load is accompanied by a substantial change in speed. A series motor runs at high speed when it has a light load and at low speed with a heavy load. If the load is removed entirely, the motor may operate at such a high speed that the armature falls apart. If high starting torque is needed under heavy load conditions, series motors have many applications. Series motors are often used in aircraft as engine starters and for raising and lowering landing gears, cowl flaps, and wing flaps.

Shunt DC Motor

In the shunt motor, the field winding is connected in parallel or in shunt with the armature winding. *[Figure 12-300]* The resistance in the field winding is high. Since the field winding is connected directly across the power supply, the current through the field is constant. The field current does not vary with motor speed, as in the series motor and, therefore, the torque of the shunt motor varies only with the current through the armature. The torque developed at starting is less than that developed by a series motor of equal size.

The speed of the shunt motor varies very little with changes in load. When all load is removed, it assumes a speed slightly higher than the loaded speed. This motor is particularly suitable for use when constant speed is desired and when high starting torque is not needed.

Compound DC Motor

The compound motor is a combination of the series and shunt motors. There are two windings in the field: a shunt winding and a series winding. *[Figure 12-301]* The shunt winding is composed of many turns of fine wire and is connected in parallel with the armature winding. The series winding consists of a few turns of large wire and is connected in series with the armature winding. The starting torque is higher than in the shunt motor but lower than in the series motor. Variation of speed with load is less than in a series wound motor but greater than in a shunt motor. The compound motor is used whenever the combined characteristics of the series and shunt motors are desired.

Like the compound generator, the compound motor has both series and shunt field windings. The series winding may either aid the shunt wind (cumulative compound) or oppose the shunt winding (differential compound). The starting and load characteristics of the cumulative compound motor are somewhere between those of the series and those of the shunt motor.

Because of the series field, the cumulative compound motor has a higher starting torque than a shunt motor. Cumulative compound motors are used in driving machines, which are subject to sudden changes in load. They are also used where a high starting torque is desired, but a series motor cannot be used easily.

In the differential compound motor, an increase in load creates an increase in current and a decrease in total flux in this type of motor. These two tend to offset each other and the result is a practically constant speed. However, since an increase in load tends to decrease the field strength, the speed characteristic becomes unstable. Rarely is this type of motor used in aircraft systems.

A graph of the variation in speed with changes of load of the various types of DC motors is shown in *Figure 12-302*.

Counter Electromotive Force (emf)

The armature resistance of a small, 28-volt DC motor is extremely low, about 0.1 ohm. When the armature is connected across the 28-volt source, current through the armature is apparently:

$$I = \frac{E}{R} = \frac{28}{0.1} = 280 \text{ amperes}$$

This high value of current flow is not only impracticable but also unreasonable, especially when the current drain, during normal operation of a motor, is found to be about 4 amperes. This is because the current through a motor armature during operation is determined by more factors than ohmic resistance.

When the armature in a motor rotates in a magnetic field, a voltage is induced in its windings. This voltage is called the back or counter emf and is opposite in direction to the voltage applied to the motor from the external source.

Counter emf opposes the current, which causes the armature to rotate. The current flowing through the armature, therefore, decreases as the counter emf increases. The faster the armature rotates, the greater the counter emf. For this reason, a motor connected to a battery may draw a fairly high current on starting, but as the armature speed increases, the current flowing through the armature decreases. At rated speed, the counter emf may be only a few volts less than the battery voltage. Then, if the load on the motor is increased, the motor slows down, less counter emf is generated, and the current drawn from the external source increases. In a shunt motor, the counter emf affects only the current in the armature, since

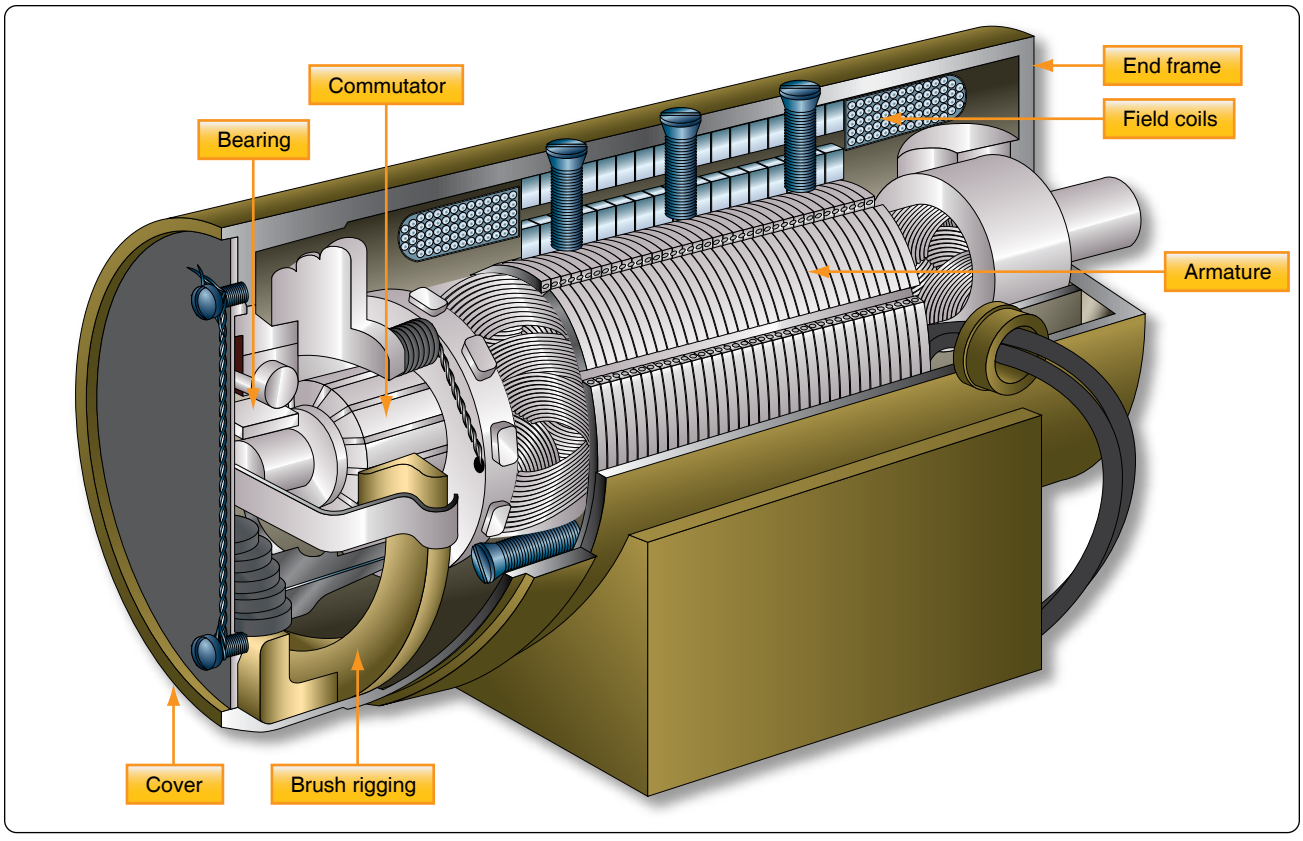


Figure 12-297. *Cutaway view of practical DC motor.*

the field is connected in parallel across the power source. As the motor slows down and the counter emf decreases, more current flows through the armature, but the magnetism in the field is unchanged. When the series motor slows down, the counter EMF decreases and more current flows through the field and the armature, thereby strengthening their magnetic fields. Because of these characteristics, it is more difficult to stall a series motor than a shunt motor.

Types of Duty

Electric motors are called upon to operate under various conditions. Some motors are used for intermittent operation;

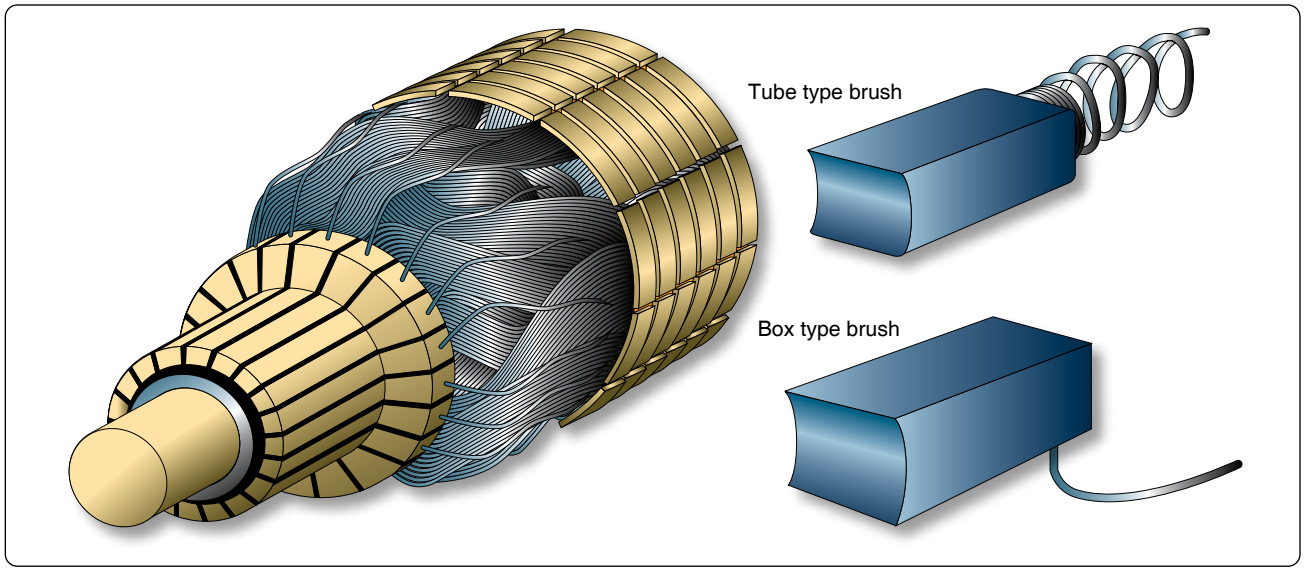


Figure 12-298. *Commutator and brushes.*

others operate continuously. Motors built for intermittent duty can be operated for short periods only and, then, must be allowed to cool before being operated again. If such a motor is operated for long periods under full load, the motor becomes overheated. Motors built for continuous duty may be operated at rated power for long periods.

Reversing Motor Direction

By reversing the direction of current flow in either the armature or the field windings, the direction of a motor's rotation may be reversed. This reverses the magnetism of either the armature or the magnetic field in which the armature rotates. If the wires connecting the motor to an external source are interchanged, the direction of rotation is not reversed, since changing these wires reverses the magnetism of both field and armature and leaves the torque in the same direction as before.

One method for reversing direction of rotation employs two field windings wound in opposite directions on the same pole. This type of motor is called a split field motor. *[Figure 12-303]* The single pole, double throw switch makes it possible to direct current through either of the two windings. When the switch is placed in the lower position, current flows through the lower field winding, creating a North pole at the lower field winding and at the lower pole piece, and a South pole at the upper pole piece. When the switch is placed in the upper position, current flows through the upper field winding, the magnetism of the field is reversed, and the armature rotates in the opposite direction. Some split field motors are built with two separate field windings wound on alternate poles. The armature in such a motor, a four pole reversible motor, rotates in one direction when current flows through the windings of one set of opposite pole pieces, and in the opposite direction when current flows through the other set of windings.

Another method of direction reversal, called the switch method, employs a double pole, double throw switch which changes the direction of current flow in either the armature or the field. In the illustration of the switch method shown in *Figure 12-304*, current direction may be reversed through the field but not through the armature. When the switch is thrown to the "up" position, current flows through the field winding to establish a North pole at the right side of the motor and a South pole at the left side of the motor. When the switch is thrown to the "down" position, this polarity is reversed and the armature rotates in the opposite direction.

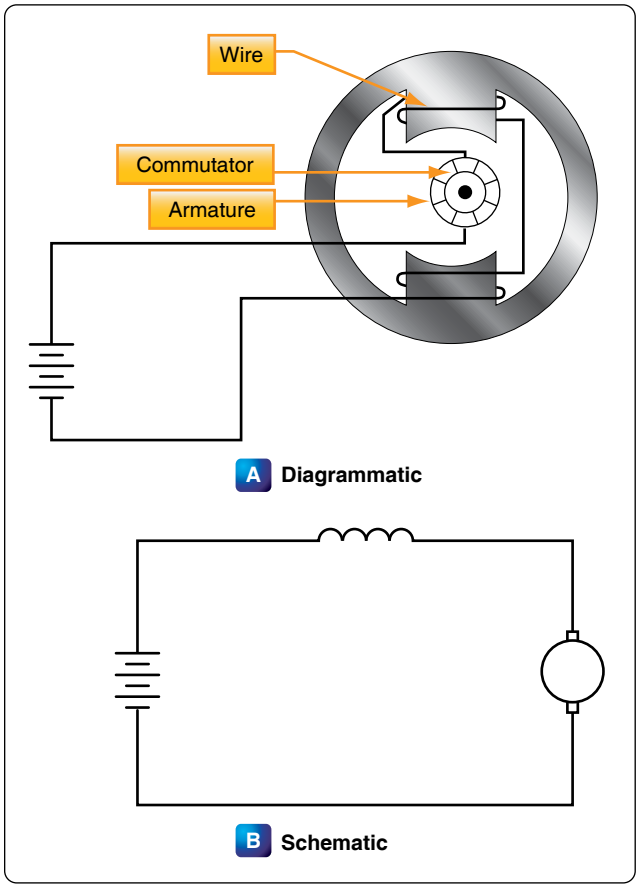


Figure 12-299. *Series motor.*

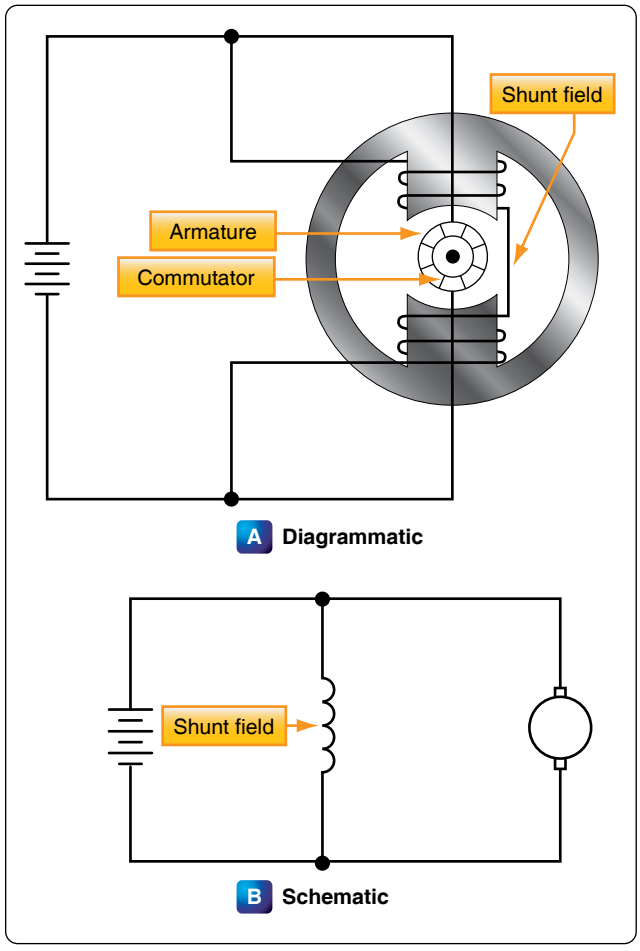


Figure 12-300. *Shunt motor.*

Motor Speed

Motor speed can be controlled by varying the current in the field windings. When the amount of current flowing through the field windings is increased, the field strength increases, but the motor slows down since a greater amount of counter EMF is generated in the armature windings. When the field current is decreased, the field strength decreases, and the motor speeds up because the counter EMF is reduced. A motor in which speed can be controlled is called a variable speed motor. It may be either a shunt or series motor.

In the shunt motor, speed is controlled by a rheostat in series with the field windings. *[Figure 12-305]* The speed depends on the amount of current that flows through the rheostat to the field windings. To increase the motor speed, the resistance in the rheostat is increased, which decreases the field current. As a result, there is a decrease in the strength of the magnetic field and in the counter EMF. This momentarily increases the armature current and the torque. The motor then automatically speeds up until the counter EMF increases and causes the armature current to decrease to its former value. When this occurs, the motor operates at a higher fixed speed than before.

To decrease the motor speed, the resistance of the rheostat is decreased. More current flows through the field windings and increases the strength of the field; then, the counter EMF increases momentarily and decreases the armature current. As a result, the torque decreases and the motor slows down until the counter EMF decreases to its former value; then the motor operates at a lower fixed speed than before.

In the series motor, the rheostat speed control is connected either in parallel or in series with the motor field, or in parallel with the armature. When the rheostat is set for maximum resistance, the motor speed is increased in the parallel armature connection by a decrease in current. When the rheostat resistance is maximum in the series connection, motor speed is reduced by a reduction in voltage across the motor. For above normal speed operation, the rheostat is in

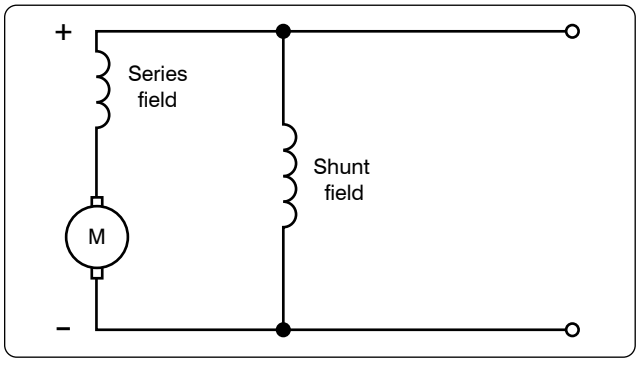


Figure 12-301. *Compound motor.*

parallel with the series field. Part of the series field current

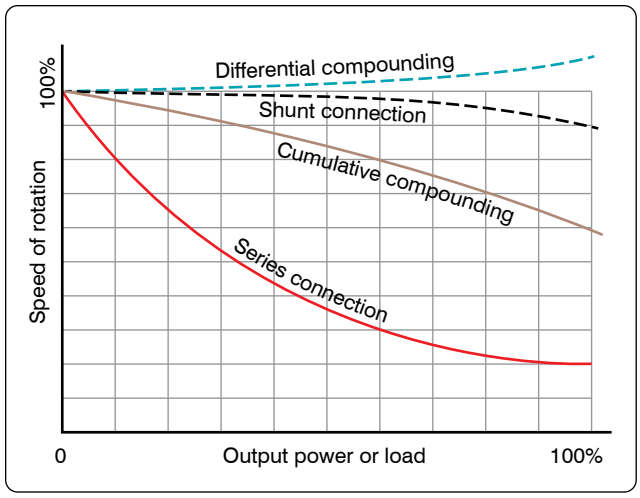


Figure 12-302. *Load characteristics of DC motors.*

is bypassed and the motor speeds up. *[Figure 12-306]*

Energy Losses in DC Motors

Losses occur when electrical energy is converted to mechanical energy (in the motor), or mechanical energy is converted to electrical energy (in the generator). For the machine to be efficient, these losses must be kept to a minimum. Some losses are electrical; others are mechanical. Electrical losses are classified as copper losses and iron losses; mechanical losses occur in overcoming the friction of various parts of the machine.

Copper losses occur when electrons are forced through the copper windings of the armature and the field. These losses are proportional to the square of the current. They are sometimes called I^2R losses, since they are due to the power dissipated in the form of heat in the resistance of the field and armature windings.

Iron losses are subdivided in hysteresis and eddy current losses. Hysteresis losses are caused by the armature revolving in an alternating magnetic field. It, therefore, becomes magnetized first in one direction and then in the other. The residual magnetism of the iron or steel of which the armature is made causes these losses. Since the field magnets are always magnetized in one direction (DC field), they have no hysteresis losses.

Eddy current losses occur because the iron core of the armature is a conductor revolving in a magnetic field. This sets up an EMF across portions of the core, causing currents to flow within the core. These currents heat the core and, if they become excessive, may damage the windings. As far as the output is concerned, the power consumed by eddy currents is a loss. To reduce eddy currents to a minimum, a laminated

core usually is used. A laminated core is made of thin sheets of iron electrically insulated from each other. The insulation between laminations reduces eddy currents, because it is "transverse" to the direction in which these currents tend to flow. However, it has no effect on the magnetic circuit. The thinner the laminations, the more effectively this method reduces eddy current losses.

Inspection and Maintenance of DC Motors

Use the following procedures to make inspection and maintenance checks:

1. Check the operation of the unit driven by the motor in accordance with the instructions covering the specific installation.
2. Check all wiring, connections, terminals, fuses, and switches for general condition and security.
3. Keep motors clean and mounting bolts tight.
4. Check brushes for condition, length, and spring tension. Minimum brush lengths, correct spring tension, and procedures for replacing brushes are given in the applicable manufacturer's instructions.
5. Inspect commutator for cleanness, pitting, scoring, roughness, corrosion, or burning. Check for high mica (IF the copper wears down below the mica, the mica insulates the brushes from the commutator.) Clean dirty commutators with a cloth moistened with the recommended cleaning solvent. Polish rough or corroded commutators with fine sandpaper (000 or finer) and blow out with compressed air. Never use emery paper since it contains metallic particles that may cause shorts. Replace the motor if the commutator is burned, badly pitted, grooved, or worn to the extent that the mica insulation is flush with the commutator surface.
6. Inspect all exposed wiring for evidence of overheating. Replace the motor if the insulation on leads or windings is burned, cracked, or brittle.
7. Lubricate only if called for by the manufacturer's instructions covering the motor. Most motors used in today's airplanes require no lubrication between overhauls.
8. Adjust and lubricate the gearbox, or unit which the motor drives, in accordance with the applicable manufacturer's instructions covering the unit.

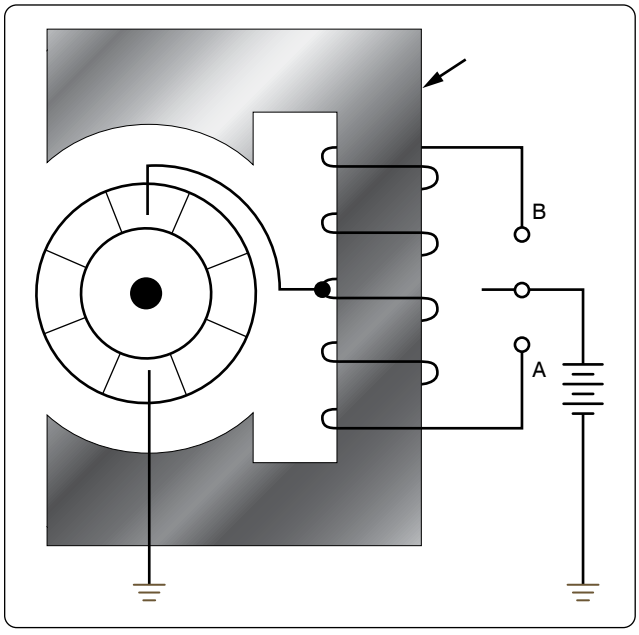


Figure 12-303. *Split field series motor.*

When trouble develops in a DC motor system, check first to determine the source of the trouble. Replace the motor only when the trouble is due to a defect in the motor itself. In most cases, the failure of a motor to operate is caused by a defect in the external electrical circuit or by mechanical failure in the mechanism driven by the motor.

Check the external electrical circuit for loose or dirty

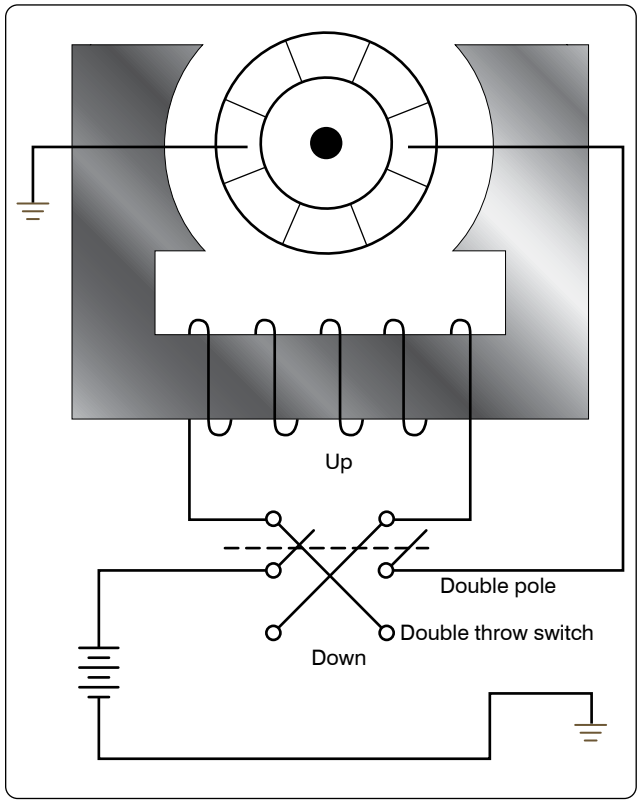


Figure 12-304. *Switch method of reversing motor direction.*

connections and for improper connection of wiring. Look for open circuits, grounds, and shorts by following the applicable manufacturer's circuit testing procedure. If the fuse is not blown, failure of the motor to operate is usually due to an open circuit. A blown fuse usually indicates an accidental ground or short circuit. A low battery usually causes the chattering of the relay switch, which controls the motor. When the battery is low, the open circuit voltage of the battery is sufficient to close the relay, but with the heavy current draw of the motor, the voltage drops below the level required to hold the relay closed. When the relay opens, the voltage in the battery increases enough to close the relay again. This cycle repeats and causes chattering, which is very harmful to the relay switch due to the heavy current causing an arc which burns the contacts.

Check the unit driven by the motor for failure of the unit or drive mechanism. If the motor has failed as a result of a failure in the driven unit, the fault must be corrected before installing a new motor. If it has been determined that the fault is in the motor itself (by checking for correct voltage at the motor terminals and for failure of the driven unit), inspect the commutator and brushes. A dirty commutator or defective or binding brushes may result in poor contact between brushes and commutator. Clean the commutator, brushes, and brush holders with a cloth moistened with the recommended cleaning solvent. If brushes are damaged or worn to the specified minimum length, install new brushes in accordance with the applicable manufacturer's instructions covering the motor. If the motor still fails to operate, replace it with a serviceable motor.

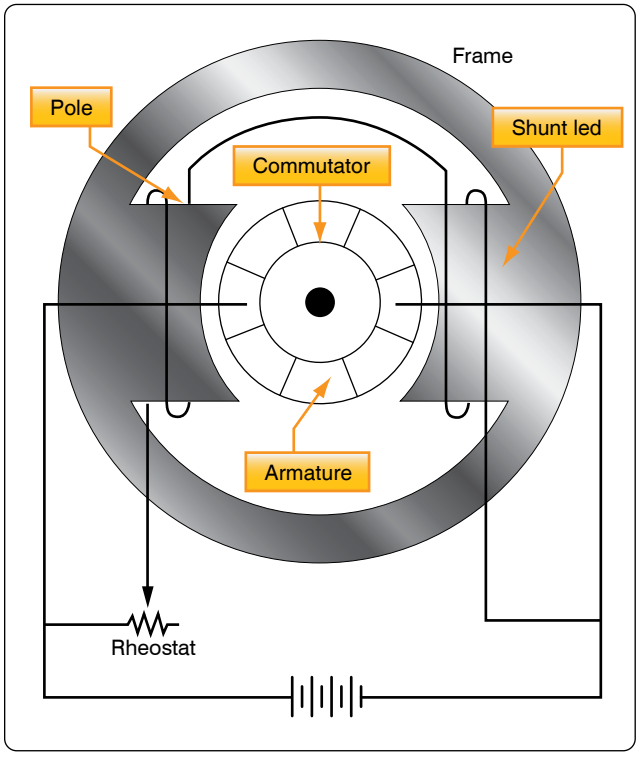


Figure 12-305. *Shunt motor with variable speed control.*

AC Motors

Because of their advantages, many types of aircraft motors are designed to operate on alternating current. In general, AC motors are less expensive than comparable DC motors. In many instances, AC motors do not use brushes and commutators so sparking at the brushes is avoided. AC motors are reliable and require little maintenance. They are also well suited for constant speed applications and certain types are manufactured that have, within limits, variable speed characteristics. Alternating current motors are designed to operate on polyphase or single-phase lines and at several voltage ratings.

The speed of rotation of an AC motor depends upon the number of poles and the frequency of the electrical source of power:

$$\text{rpm} = \frac{120 \times \text{Frequency}}{\text{Number of poles}}$$

Since airplane electrical systems typically operate at 400 cycles, an electric motor at this frequency operates at about seven times the speed of a 60-cycle commercial motor with the same number of poles. Because of this high speed of rotation, 400-cycle AC motors are suitable for operating small, high-speed rotors, through reduction gears, in lifting and moving heavy loads, such as the wing flaps, the retractable landing gear, and the starting of engines. The 400-cycle induction type motor operates at speeds ranging from 6,000 rpm to 24,000 rpm. Alternating current motors are rated in horsepower output, operating voltage, full load current, speed, number of phases, and frequency. Whether the motors operate continuously or intermittently (for short intervals) is also considered in the rating.

Types of AC Motors

There are two general types of AC motors used in aircraft systems: induction motors and synchronous motors. Either type may be single-phase, two-phase, or three-phase. Three-phase induction motors are used where large amounts of power are required. They operate such devices as starters, flaps, landing gears, and hydraulic pumps. Single-phase induction motors are used to operate devices such as surface locks, intercooler shutters, and oil shutoff valves in which the power requirement is low. Three-phase synchronous motors operate at constant synchronous speeds and are commonly used to operate flux gate compasses and propeller synchronizer systems. Single-phase synchronous motors are common sources of power to operate electric clocks and other small, precision equipment. They require some auxiliary method to bring them up to

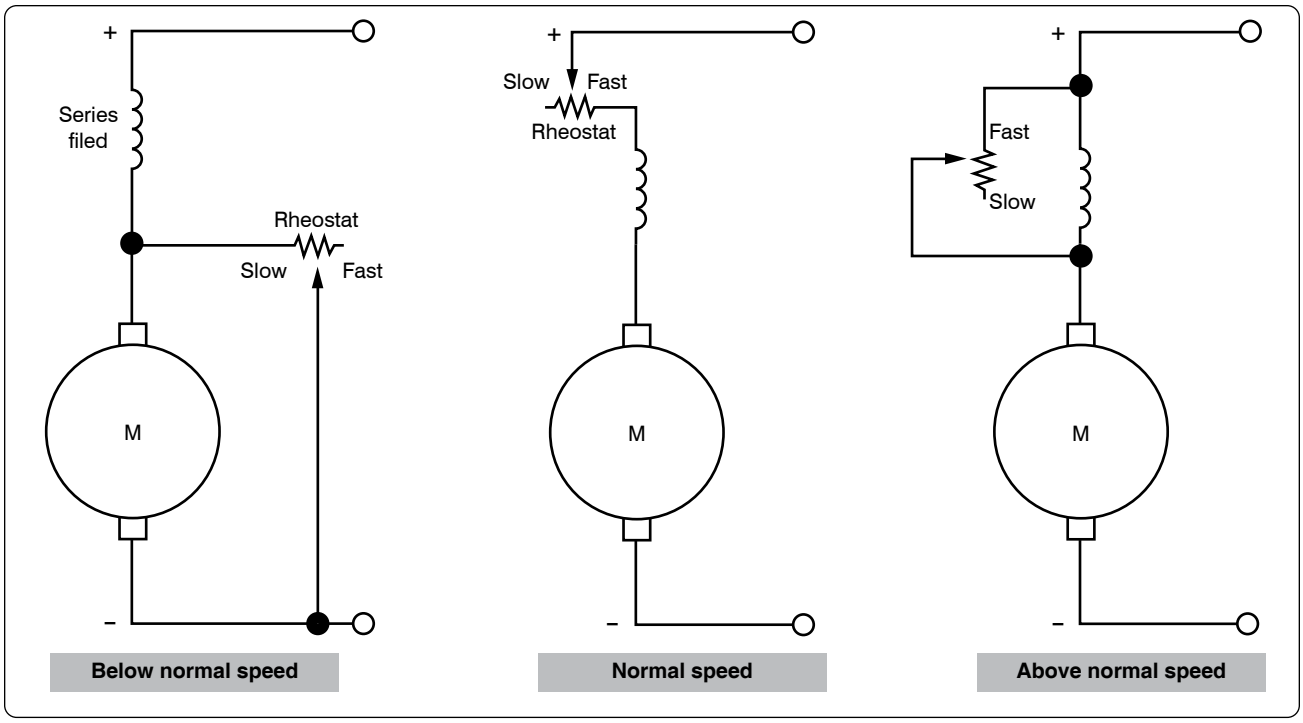


Figure 12-306. *Controlling the speed of a series DC motor.*

synchronous speeds; that is, to start them. Usually the starting winding consists of an auxiliary stator winding.

Three-Phase Induction Motor

The three-phase AC induction motor is also called a squirrel cage motor. Both single-phase and three-phase motors operate on the principle of a rotating magnetic field. A horseshoe magnet held over a compass needle is a simple illustration of the principle of the rotating field. The needle takes a position parallel to the magnetic flux passing between the two poles of the magnet. If the magnet is rotated, the compass needle follows. A rotating magnetic field can be produced by a two-or three-phase current flowing through two or more groups of coils wound on inwardly projecting poles of an iron frame. The coils on each group of poles are wound alternately in opposite directions to produce opposite polarity, and each group is connected to a separate phase of voltage. The operating principle depends on a revolving, or rotating, magnetic field to produce torque. The key to understanding the induction motor is a thorough understanding of the rotating magnetic field.

Rotating Magnetic Field

The field structure shown in *Figure 12-307A* has poles whose windings are energized by three AC voltages: a, b, and c. These voltages have equal magnitude but differ in phase. *[Figure 12-307B]* At the instant of time shown as 0, the resultant magnetic field produced by the application of the three voltages has its greatest intensity in a direction extending from pole 1 to pole 4. Under this condition, pole 1 can be considered as a North pole and pole 4 as a South pole. At the instant of time shown as 1, the resultant magnetic field has its greatest intensity in the direction extending from pole 2 to pole 5. In this case, pole 2 can be considered as a North pole and pole 5 as a South pole. Thus, between instant 0 and instant 1, the magnetic field has rotated clockwise. At instant 2, the resultant magnetic field has its greatest intensity in the direction from pole 3 to pole 6, and the resultant magnetic field has continued to rotate clockwise. At instant 3, poles 4 and 1 can be considered as North and South poles, respectively, and the field has rotated still farther. At later instants of time, the resultant magnetic field rotates to other positions while traveling in a clockwise direction, a single revolution of the field occurring in one cycle. If the exciting voltages have a frequency of 60 cps, the magnetic field makes 60 revolutions per second, or 3,600 rpm. This speed is known as the synchronous speed of the rotating field.

Construction of Induction Motor

The stationary portion of an induction motor is called a stator, and the rotating member is called a rotor. Instead of salient poles in the stator, as shown in *Figure 12-307A*, distributed windings are used. These windings are placed in slots around the periphery of the stator. It is usually impossible to determine the number of poles in an induction motor by visual inspection, but the information can be obtained from the nameplate of the motor. The nameplate usually gives the number of poles and the speed at which the motor is designed

to run. This rated, or nonsynchronous, speed is slightly less than the synchronous speed. To determine the number of poles per phase on the motor, divide 120 times the frequency by the rated speed. Written as an equation, it is:

$$P = \frac{120 \times f}{N}$$

Where: P is the number of poles per phase
f is the frequency in cps
N is the rated speed in rpm, and
120 is a constant

The result is nearly equal to the number of poles per phase. For example, consider a 60-cycle, three-phase motor with a rated speed of 1,750 rpm. In this case:

$$P = \frac{120 \times 60}{1,750} = \frac{7,200}{1,750} = 4.1$$

Therefore, the motor has four poles per phase. If the number of poles per phase is given on the nameplate, the synchronous speed can be determined by dividing 120 times the frequency by the number of poles per phase. In the example used above, the synchronous speed is equal to 7,200/4, or 1,800 rpm.

The rotor of an induction motor consists of an iron core having longitudinal slots around its circumference in which heavy copper or aluminum bars are embedded. These bars are welded to a heavy ring of high conductivity on either end. The composite structure is sometimes called a squirrel cage, and motors containing such a rotor are called squirrel cage induction motors. *[Figure 12-308]*

Induction Motor Slip

When the rotor of an induction motor is subjected to the revolving magnetic field produced by the stator windings, a voltage is induced in the longitudinal bars. The induced voltage causes a current to flow through the bars. This current, in turn, produces its own magnetic field, which combines with the revolving field so that the rotor assumes a position in which the induced voltage is minimized. As a result, the rotor revolves at very nearly the synchronous speed of the stator field, the difference in speed being just sufficient enough to induce the proper amount of current in the rotor to overcome the mechanical and electrical losses in the rotor. If the rotor were to turn at the same speed as the rotating field, the rotor conductors would not be cut by any magnetic lines of force, no EMF would be induced in them, no current could flow, and there would be no torque. The rotor would then slow down. For this reason, there must always be a difference in speed between the rotor and the rotating field. This difference in speed is called slip and is expressed as a percentage of the synchronous speed. For example, if the rotor turns at 1,750 rpm and the synchronous speed is 1,800 rpm, the difference in speed is 50 rpm. The slip is then equal to 50/1,800 or 2.78 percent.

Single-Phase Induction Motor

The previous discussion has applied only to polyphase motors. A single-phase motor has only one stator winding. This winding generates a field, which merely pulsates, instead of rotating. When the rotor is stationary, the expanding and collapsing stator field induces currents in the rotor. These currents generate a rotor field opposite in polarity to that of the stator. The opposition of the field exerts a turning force on the upper and lower parts of the rotor trying to turn it 180° from its position. Since these forces are exerted through the center of the rotor, the turning force is equal in each direction. As a result, the rotor does not turn. If the rotor starts turning, it continues to rotate in the direction in which it started, since the turning force in that direction is aided by the momentum of the rotor.

Shaded Pole Induction Motor

The first effort in the development of a self-starting, single-phase motor was the shaded pole induction motor. *[Figure 12-309]* This motor has salient poles, a portion of each pole being encircled by a heavy copper ring. The presence of the ring causes the magnetic field through the ringed portion of the pole face to lag appreciably behind that through the other part of the pole face. The net effect is the production of a slight component of rotation of the field, sufficient to cause the rotor to revolve. As the rotor accelerates, the torque increases until the rated speed is obtained. Such motors have low starting torque and find their greatest application in small fan motors where the initial torque required is low.

In *Figure 12-309*, a diagram of a pole and the rotor is shown. The poles of the shaded pole motor resemble those of a DC motor.

A low resistance, short-circuited coil or copper band is placed across one tip of each small pole from which the motor gets the name of shaded pole. The rotor of this motor is the squirrel cage type. As the current increases in the stator winding, the flux increases. A portion of this flux cuts the low resistance shading coil. This induces a current in the shading coil, and by Lenz's Law, the current sets up a flux that opposes the flux inducing the current. Hence, most of the flux passes through the unshaded portion of the poles. *[Figure 12-310]*

When the current in the winding and the main flux reaches a maximum, the rate of change is zero; thus, no emf is induced in the shading coil. A little later, the shading coil current, which causes the induced emf to lag, reaches zero, and there is no opposing flux. Therefore, the main field flux

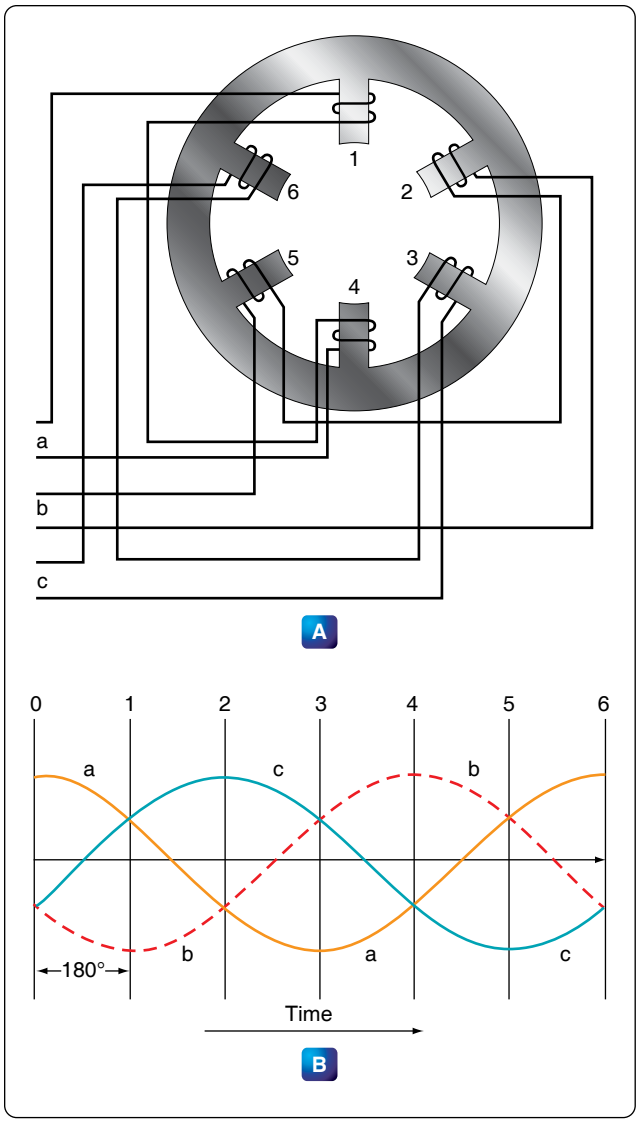


Figure 12-307. *Rotating magnetic field developed by application of three-phase voltages.*

passes through the shaded portion of the field pole. The main field flux, which is now decreasing, induces a current in the shading coil. This current sets up a flux that opposes the decrease of the main field flux in the shaded portion of the pole. The effect is to concentrate the lines of force in the shaded portion of the pole face. In effect, the shading coil retards, in time phase, the portion of the flux passing through the shaded part of the pole. This lag in time phase of the flux in the shaded tip causes the flux to produce the effect of sweeping across the face of the pole, from left to right in the direction of the shaded tip. This behaves like a very weak rotating magnetic field, and sufficient torque is produced to start a small motor. The starting torque of the shaded pole motor is exceedingly weak, and the power factor is low. Consequently, it is built in sizes suitable for driving such devices as small fans.

Split-Phase Motor

There are various types of self-starting motors, known as split-phase motors. Such motors have a starting winding displaced 90 electrical degrees from the main or running winding. In some types, the starting winding has a fairly high resistance, which causes the current in this winding to be out of phase with the current in the running winding. This condition produces, in effect, a rotating field and the rotor revolves. A centrifugal switch disconnects the starting winding automatically after the rotor has attained approximately 25 percent of its rated speed.

Capacitor Start Motor

With the development of high-capacity electrolytic capacitors, a variation of the split-phase motor known as the capacitor start motor, has been made. Nearly all fractional horsepower motors in use today on refrigerators and other similar appliances are of this type. *[Figure 12-311]* In this adaptation, the starting winding and running winding have the same size and resistance value. The phase shift between currents of the two windings is obtained by using capacitors connected in series with the starting winding.

Capacitor start motors have a starting torque comparable to their torque at rated speed and can be used in applications where the initial load is heavy. Again, a centrifugal switch is required for disconnecting the starting winding when the rotor speed is approximately 25 percent of the rated speed.

Although some single-phase induction motors are rated as high as 2 horsepower (hp), the major field of application is 1 hp, or less, at a voltage rating of 115 volts for the smaller sizes and 110 to 220 volts for one-fourth hp and up. For even larger power ratings, polyphase motors generally are used, since they have excellent starting torque characteristics.

Direction of Rotation of Induction Motors

The direction of rotation of a three-phase induction motor can be changed by simply reversing two of the leads to the motor. The same effect can be obtained in a two-phase motor by reversing connections to one phase. In a single-phase motor, reversing connections to the starting winding reverses the direction of rotation.

Most single-phase motors designed for general application have provision for readily reversing connections to the starting winding. Nothing can be done to a shaded pole motor to reverse the direction of rotation because the direction is determined by the physical location of the copper shading ring. If, after starting, one connection to a three-phase motor is broken, the motor continues to run but delivers only one-third the rated power. Also, a two-phase motor runs at one-half its rated power if one phase is disconnected. Neither

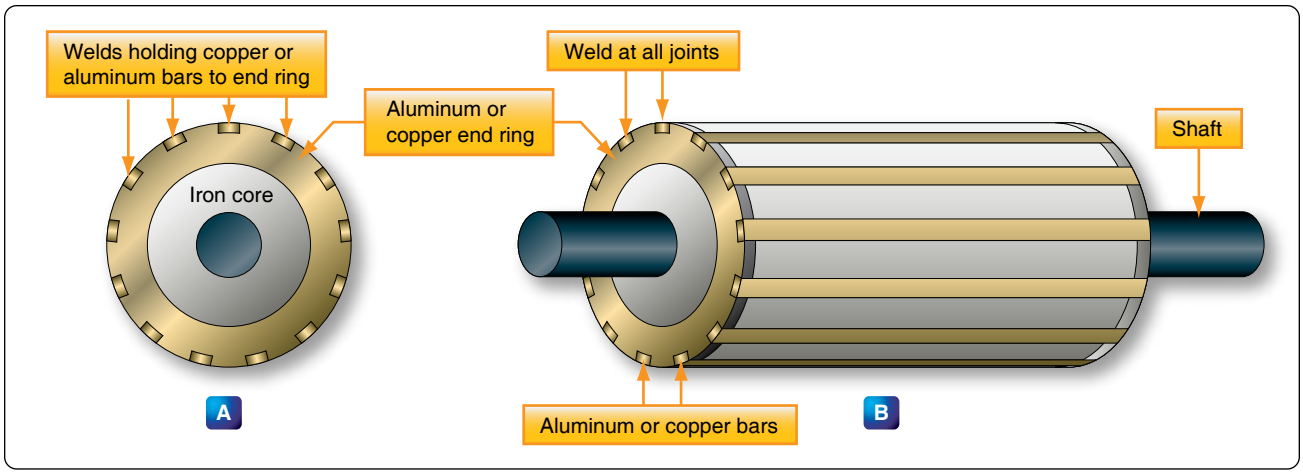


Figure 12-308. *Squirrel cage rotor for an AC induction motor.*

motor starts under these abnormal conditions.

Synchronous Motor

The synchronous motor is one of the principal types of AC motors. Like the induction motor, the synchronous motor makes use of a rotating magnetic field. Unlike the induction motor, however, the torque developed does not depend on the induction of currents in the rotor. Briefly, the principle of operation of the synchronous motor is as follows: A multiphase source of AC is applied to the stator windings, and a rotating magnetic field is produced. A direct current is applied to the rotor winding, and another magnetic field is produced. The synchronous motor is so designed and constructed that these two fields react to each other in such a manner that the rotor is dragged along and rotates at the same speed as the rotating magnetic field produced by the stator windings.

An understanding of the operation of the synchronous motor can be obtained by considering the simple motor of *Figure 12-312*. Assume that poles A and B are being rotated clockwise by some mechanical means in order to produce a rotating magnetic field. They induce poles of opposite polarity in the soft iron rotor, and forces of attraction exist between corresponding North and South poles.

Consequently, as poles A and B rotate, the rotor is dragged along at the same speed. However, if a load is applied to the rotor shaft, the rotor axis momentarily falls behind that of the rotating field but, thereafter, continues to rotate with the field at the same speed as long as the load remains constant. If the load is too large, the rotor pulls out of synchronism with the rotating field and, as a result, no longer rotates with the field at the same speed. Thus, the motor is said to be overloaded.

Such a simple motor as shown in *Figure 12-312* is never used. The idea of using some mechanical means of rotating the poles

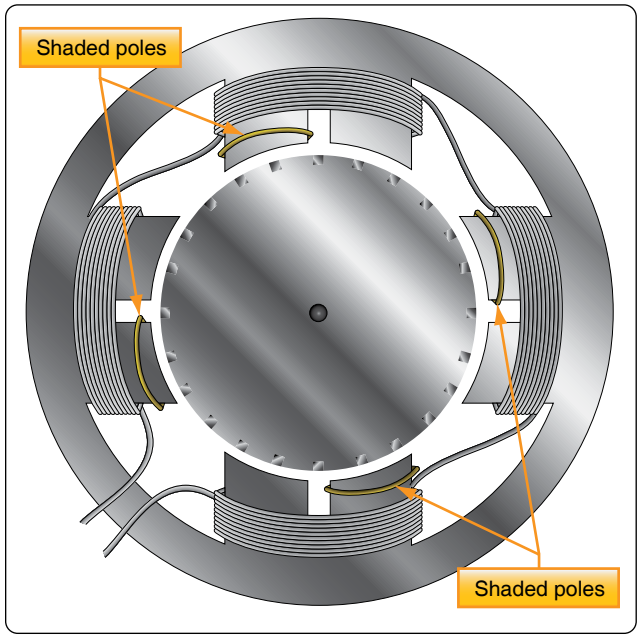


Figure 12-309. *Shaded pole induction motor.*

is impractical because another motor would be required to perform this work. Also, such an arrangement is unnecessary because a rotating magnetic field can be produced electrically by using phased AC voltages. In this respect, the synchronous motor is similar to the induction motor.

The synchronous motor consists of a stator field winding similar to that of an induction motor. The stator winding produces a rotating magnetic field. The rotor may be a permanent magnet, as in small, single-phase synchronous motors used for clocks and other small precision equipment, or it may be an electromagnet, energized from a DC source of power and fed through slip rings into the rotor field coils, as in an alternator. In fact, an alternator may be operated either as an alternator or a synchronous motor.

Since a synchronous motor has little starting torque, some means must be provided to bring it up to synchronous speed. The most common method is to start the motor at no load, allow it to reach full speed, and then energize the magnetic field. The magnetic field of the rotor locks with the magnetic field of the stator and the motor operates at synchronous speed.

The magnitude of the induced poles in the rotor shown in *Figure 12-313* is so small that sufficient torque cannot be developed for most practical loads. To avoid such a limitation on motor operation, a winding is placed on the rotor and energized with DC. A rheostat placed in series with the DC source provides the operator of the machine with a means of varying the strength of the rotor poles, thus placing the motor under control for varying loads.

The synchronous motor is not a self-starting motor. The rotor is heavy and, from a dead stop, it is impossible to bring the rotor into magnetic lock with the rotating magnetic field. For this reason, all synchronous motors have some kind of starting device. One type of simple starter is another motor, either AC or DC, which brings the rotor up to approximately 90 percent of its synchronous speed. The starting motor is then disconnected, and the rotor locks in step with the rotating field. Another starting method is a second winding of the squirrel cage type on the rotor. This induction winding brings the rotor almost to synchronous speed, and when the DC is connected to the rotor windings, the rotor pulls into step with the field. The latter method is the more commonly used.

AC Series Motor

An AC-series motor is a single-phase motor, but is not an induction or synchronous motor. It resembles a DC motor in that it has brushes and a commutator. The AC-series motor operates on either AC or DC circuits. Remember that the direction of rotation of a DC-series motor is independent of the polarity of the applied voltage, provided the field and armature connections remain unchanged. Hence, if a DC-series motor is connected to an AC source, a torque is developed that tends to rotate the armature in one direction. However, a DC-series motor does not operate satisfactorily from an AC supply for the following reasons:

1. The alternating flux sets up large eddy current and hysteresis losses in the unlaminated portions of the magnetic circuit and causes excessive heating and reduced efficiency.
2. The self-induction of the field and armature windings causes a low power factor.
3. The alternating field flux establishes large currents in the coils, which are short circuited by the brushes; this action causes excessive sparking at the commutator.

To design a series motor for satisfactory operation on AC, the following changes are made:

1. The eddy current losses are reduced by laminating the field poles, frame, and armature.
2. Hysteresis losses are minimized by using high permeability, transformer-type, silicon steel laminations.
3. The reactance of the field windings is kept satisfactorily low by using shallow pole pieces, few turns of wire, low frequency (usually 25 cycles for large motors), low flux density, and low reluctance (a short air gap).
4. The reactance of the armature is reduced by using a compensating winding embedded in the pole pieces. If the compensating winding is connected in series with the armature, as shown in *Figure 12-314*, the armature is conductively compensated.

   If the compensating winding is designed as shown in *Figure 12-315*, the armature is inductively compensated. If the motor is designed for operation on both DC and AC circuits, the compensating winding is connected in series with the armature. The axis of the compensating winding is displaced from the main field axis by an angle of 90°. This arrangement is similar to the compensating winding used in some DC motors and generators to overcome armature reaction. The compensating winding establishes a counter magnetomotive force, neutralizing the effect of the armature magnetomotive force, preventing distortion of the main field flux, and reducing the armature reactance. The inductively compensated armature acts like the primary of a transformer, the secondary

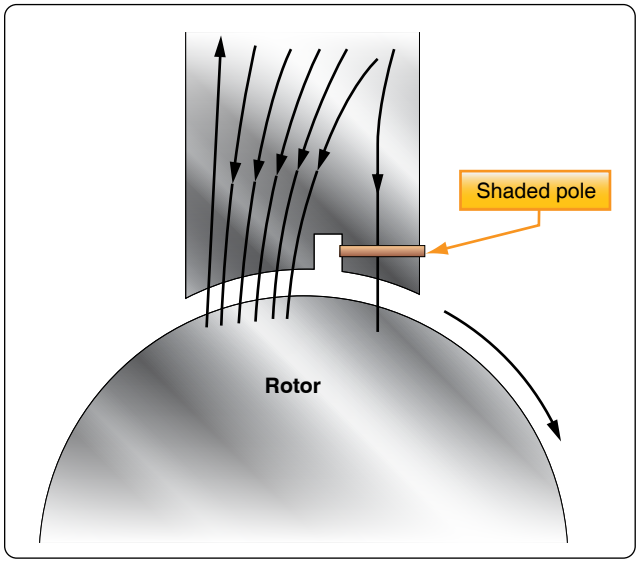


Figure 12-310. *Diagram of a shaded pole motor.*

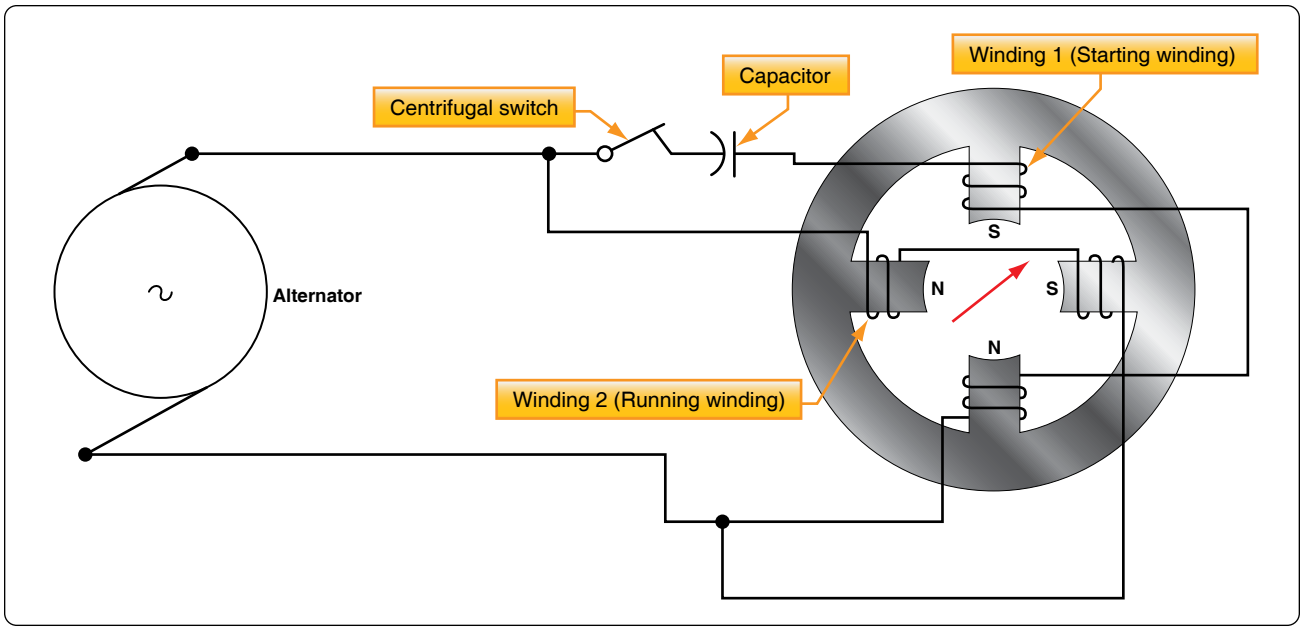


Figure 12-311. *Single-phase motor with capacitor starting winding.*

of which is the shorted compensating winding. The shorted secondary receives an induced voltage by the action of the alternating armature flux, and the resulting current flowing through the turns of the compensating winding establishes the opposing magnetomotive force, neutralizing the armature reactance.

5. Sparking at the commutator is reduced by the use of preventive leads P_1, P_2, P_3, and so forth, as shown in *Figure 12-316*, where a ring armature is shown for simplicity. When coils at A and B are shorted by the brushes, the induced current is limited by the relatively high resistance of the leads. Sparking at the brushes is also reduced by using armature coils having only a single turn and multipolar fields. High torque is obtained by having a large number of armature conductors and a large diameter armature. Thus, the commutator has a large number of very thin commutator bars and the armature voltage is limited to about 250 volts.

Fractional horsepower AC series motors are called universal motors. They do not have compensating windings or preventive leads. They are used extensively to operate fans and portable tools, such as drills, grinders, and saws.

Maintenance of AC Motors

The inspection and maintenance of AC motors is very simple. The bearings may or may not need frequent lubrication. If they are the sealed type, lubricated at the factory, they require no further attention. Be sure the coils are kept dry and free from oil or other abuse. The temperature of a motor is usually its only limiting operating factor. A good rule of thumb is that a temperature too hot for the hand is too high for safety. Next to the temperature, the sound of a motor or generator is the best trouble indicator. When operating properly, it should hum evenly. If it is overloaded it "grunts." A three-phase motor

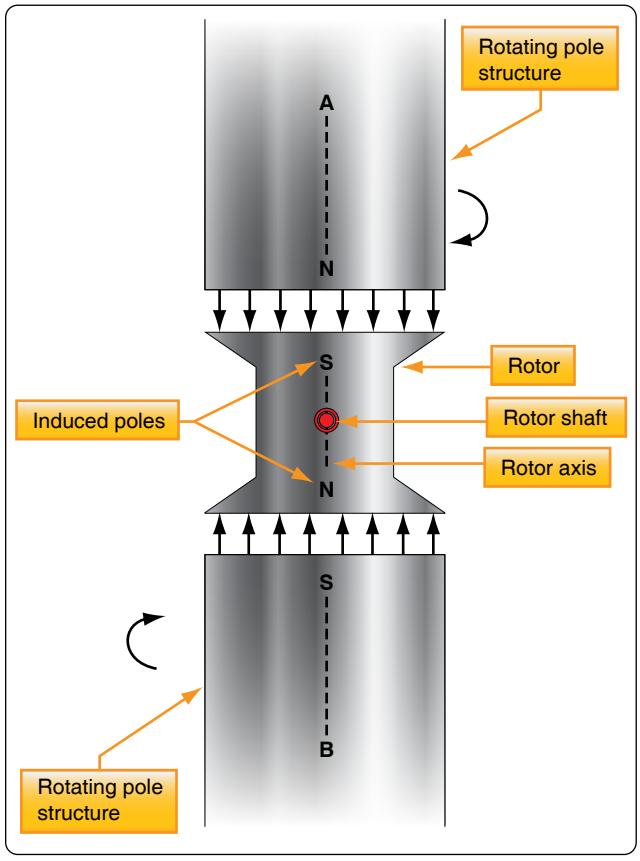


Figure 12-312. *Illustrating the operation of a synchronous motor.*

with one lead disconnected refuses to turn and "growls." A knocking sound generally indicates a loose armature coil, a shaft out of alignment, or armature dragging because of worn bearings. In all cases, the inspection and maintenance of all AC motors should be performed in accordance with the applicable manufacturer's instructions.

Alternators

Basic Alternators & Classifications

An electrical generator is a machine that converts mechanical energy into electrical energy by electromagnetic induction. A generator that produces alternating current is referred to as an AC generator and, through combination of the words "alternating" and "generator," the word "alternator" has come into widespread use. In some areas, the word "alternator" is applied only to small AC generators. This handbook treats the two terms synonymously and uses the term "alternator" to distinguish between AC and DC generators.

The major difference between an alternator *[Figure 12-317]* and a DC generator is the method of connection to the external circuit. The alternator is connected to the external circuit by slip rings, but the DC generator is connected by a commutator.

Method of Excitation

One means of classification is by the type of excitation system used. In alternators used on aircraft, excitation can be affected by one of the following methods:

1. *A direct connected, DC generator.* This system consists of a DC generator fixed on the same shaft with the AC generator. A variation of this system is a type of alternator that uses DC from the battery for excitation, after which the alternator is self-excited.
2. *By transformation and rectification from the AC system.* This method depends on residual magnetism for initial AC voltage buildup, after which the field is supplied with rectified voltage from the AC generator.
3. *Integrated brushless type.* This arrangement has a DC generator on the same shaft with an AC generator. The excitation circuit is completed through silicon rectifiers rather than a commutator and brushes. The rectifiers are mounted on the generator shaft and their output is fed directly to the AC generator's main rotating field.

Number of Phases

Another method of classification is by the number of phases of output voltage. AC generators may be single-phase, two-phase, three-phase, or even six-phase and more. In the

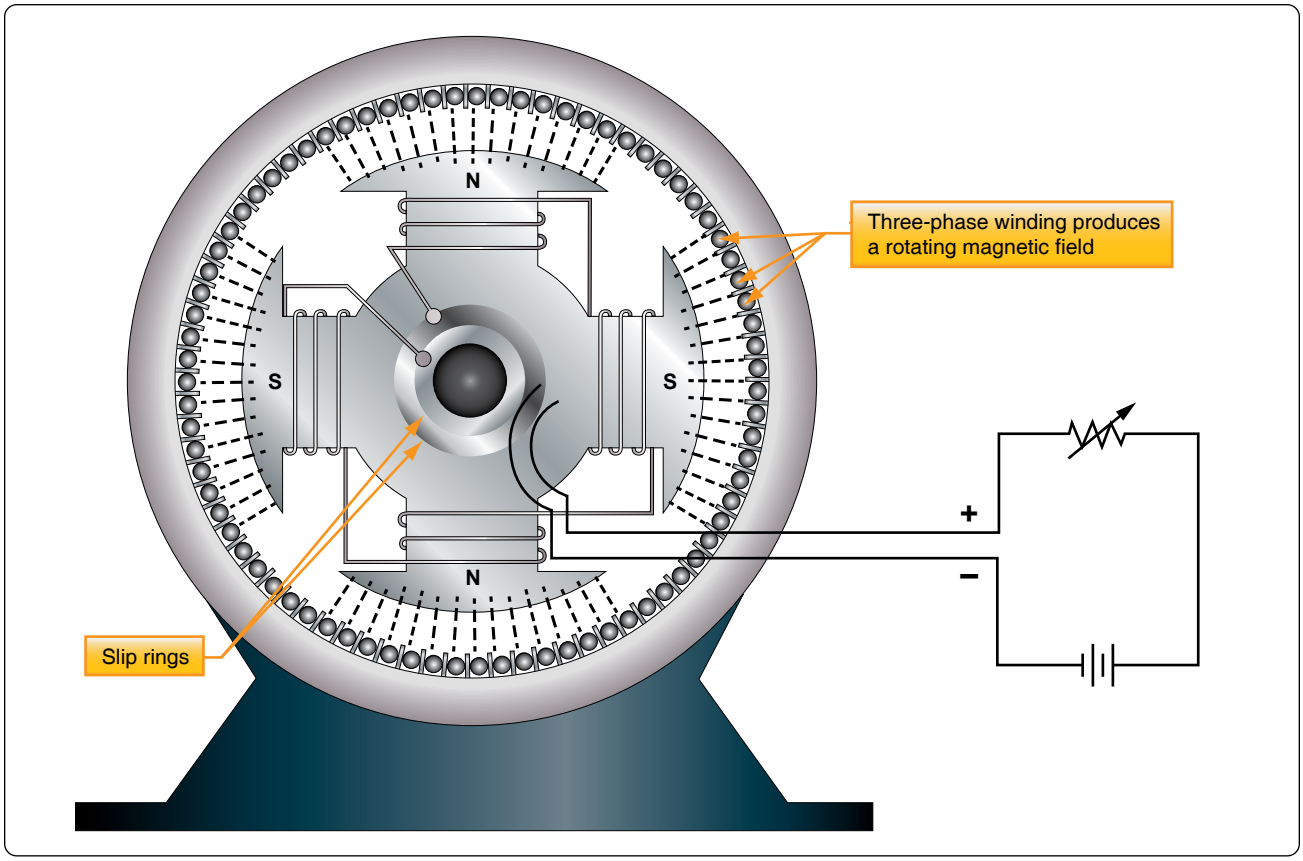


Figure 12-313. *Synchronous motor.*

electrical systems of aircraft, the three-phase alternator is by far the most common.

Armature or Field Rotation

Still another means of classification is by the type of stator and rotor used. From this standpoint, there are two types of alternators: the revolving armature-type and the revolving field-type. The revolving armature alternator is similar in construction to the DC generator in that the armature rotates through a stationary magnetic field. The revolving armature alternator is found only in alternators of low-power rating and generally is not used. In the DC generator, the EMF generated in the armature windings is converted into a unidirectional voltage (DC) by means of the commutator. In the revolving armature-type of alternator, the generated AC voltage is applied unchanged to the load by means of slip rings and brushes.

The revolving field type of alternator has a stationary armature winding (stator) and a rotating field winding (rotor). *[Figure 12-318]* The advantage of having a stationary armature winding is that the armature can be connected directly to the load without having sliding contacts in the load circuit. A rotating armature would require slip rings and brushes to conduct the load current from the armature to the external circuit. Slip rings have a relatively short service life and arc over is a continual hazard; therefore, high voltage alternators are usually of the stationary armature, rotating field-type. The voltage and current supplied to the rotating field are relatively small, and slip rings and brushes for this circuit are adequate. The direct connection to the armature circuit makes possible the use of large cross-section conductors, adequately insulated for high voltage. Since the rotating field alternator is used almost universally in aircraft systems, this type is explained in detail, as a single-phase, two-phase, and three-phase alternator.

Single-Phase Alternator

Since the EMF induced in the armature of a generator is alternating, the same sort of winding can be used on an alternator as on a DC generator. This type of alternator is known as a single-phase alternator, but since the power delivered by a single-phase circuit is pulsating, this type of circuit is objectionable in many applications.

A single-phase alternator has a stator made up of a number of windings in series, forming a single circuit in which an output voltage is generated. *[Figure 12-319]* The stator has four polar groups evenly spaced around the stator frame. The rotor has four poles with adjacent poles of opposite polarity. As the rotor revolves, AC voltages are induced in the stator windings. Since one rotor pole is in the same position relative to a stator winding as any other rotor pole, all stator polar groups are cut by equal numbers of magnetic lines of force at any time.

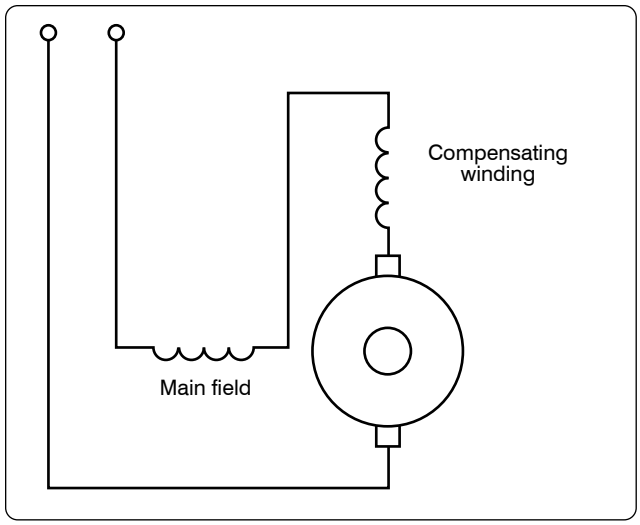


Figure 12-314. *Conductively compensated armature of AC series motor.*

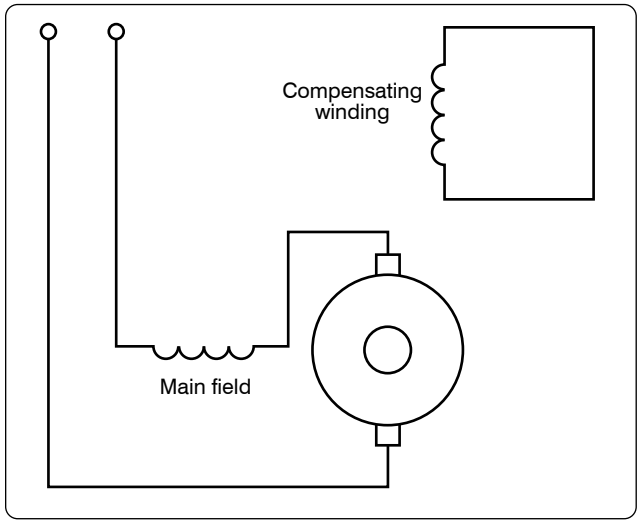


Figure 12-315. *Inductively compensated armature of AC series motor.*

As a result, the voltages induced in all the windings have the same amplitude, or value, at any given instant. The four stator windings are connected to each other so that the AC voltages are in phase or "series adding." Assume that rotor pole 1, a South pole, induces a voltage in the direction indicated by the arrow in stator winding 1. Since rotor pole 2 is a North pole, it induces a voltage in the opposite direction in stator coil 2 with respect to that in coil 1. For the two induced voltages to be in series addition, the two coils are connected as shown in *Figure 12-319*. Applying the same reasoning, the voltage induced in stator coil 3 (clockwise rotation of the field) is the same direction (counterclockwise) as the voltage induced in coil 1.

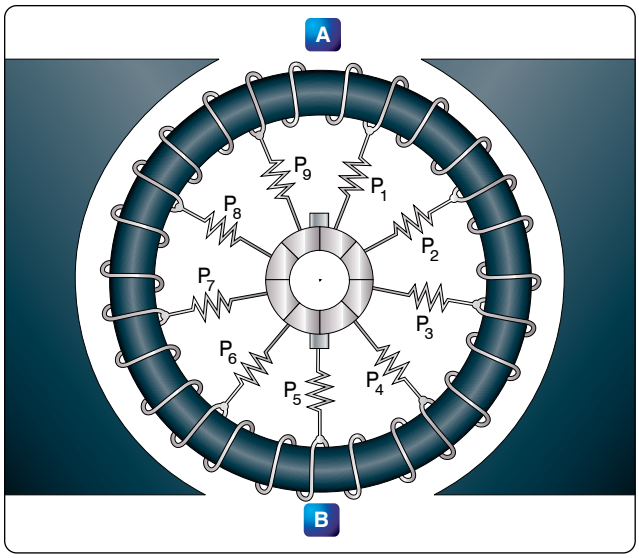


Figure 12-316. *Preventive coils in AC series motor.*

Similarly, the direction of the voltage induced in winding 4 is opposite to the direction of the voltage induced in coil 1. All four stator coil groups are connected in series so that the voltages induced in each winding add to give a total voltage that is four times the voltage in any one winding.

Two-Phase Alternator

Two-phase alternators have two or more single-phase windings spaced symmetrically around the stator. In a two-phase alternator, there are two single-phase windings spaced physically so that the AC voltage induced in one is 90° out of phase with the voltage induced in the other. The windings are electrically separate from each other. When one winding is being cut by maximum flux, the other is being cut by no flux. This condition establishes a 90° relation between the two phases.

Three-Phase Alternator

A three-phase, or polyphase circuit, is used in most aircraft alternators, instead of a single or two-phase alternator. The three-phase alternator has three single-phase windings spaced so that the voltage induced in each winding is 120° out of phase with the voltages in the other two windings. A schematic diagram of a three-phase stator showing all the coils becomes complex and difficult to see what is actually happening.

A simplified schematic diagram showing each of three phases is illustrated in *Figure 12-320*. The rotor is omitted for simplicity. The waveforms of voltage are shown to the right of the schematic. The three voltages are 120° apart and are similar to the voltages that would be generated by three single-phase alternators whose voltages are out of phase by angles of 120°. The three phases are independent of each other.

Wye Connection (Three-Phase)

Rather than have six leads from the three-phase alternator, one of the leads from each phase may be connected to form a common junction. The stator is then called wye or star connected. The common lead may or may not be brought out of the alternator. If it is brought out, it is called the neutral lead. The simplified schematic shows a wye connected stator with the common lead not brought out. *[Figure 12-321A]* Each load is connected across two phases in series. Thus, RAB is connected across phases A and B in series; RAC is connected across phases A and C in series; and RBC is connected across phases B and C in series. Therefore, the voltage across each load is larger than the voltage across a single phase. The total voltage, or line voltage, across any two phases is the vector sum of the individual phase voltages. For balanced conditions, the line voltage is 1.73 times the phase voltage. Since there is only one path for current in a line wire and the phase to which it is connected, the line current is equal to the phase current.

Delta Connection (Three-Phase)

A three-phase stator can also be connected so that the phases are connected end to end. *[Figure 12-321B]* This arrangement is called a delta connection. In a delta connection, the voltages are equal to the phase voltages; the line currents are equal to the vector sum of the phase currents; and the line current is equal to 1.73 times the phase current when the loads are balanced. For equal loads (equal output), the delta connection supplies increased line current at a value of line voltage equal to phase voltage, and the wye connection supplies increased line voltage at a value of line current equal to phase current.

Alternator Rectifier Unit

A type of alternator used in the electrical system of many aircraft weighing less than 12,500 pounds is shown in *Figure 12-322*. This type of power source is sometimes called a DC generator, since it is used in DC systems. Although its output is a DC voltage, it is an alternator rectifier unit. This type of alternator rectifier is a self-excited unit but does not contain a permanent magnet. The excitation for starting is

Figure 12-317. *Belt driven alternator for small single-engine aircraft.*

obtained from the battery; immediately after starting, the unit is self-exciting. Cooling air for the alternator is conducted into the unit by a blast air tube on the air inlet cover.

The alternator is directly coupled to the aircraft engine by means of a flexible drive coupling. The output of the alternator portion of the unit is three-phase alternating current, derived from a three-phase, delta connected system incorporating a three-phase, full-wave bridge rectifier. *[Figure 12-323]* This unit operates in a speed range from 2,100 to 9,000 rpm, with a DC output voltage of 26–29 volts and 125 amperes.

Brushless Alternator

This design is more efficient because there are no brushes to wear down or to arc at high altitudes. This generator consists of a pilot exciter, an exciter, and the main generator system. The need for brushes is eliminated by using an integral exciter with a rotating armature that has its AC output rectified for the main AC field, which is also of the rotating type. *[Figure 12-324]*

The pilot exciter is an 8-pole, 8,000 rpm, 533 cps, AC generator. The pilot exciter field is mounted on the main generator rotor shaft and is connected in series with the main generator field. The pilot exciter armature is mounted on the main generator stator. The AC output of the pilot exciter is supplied to the voltage regulator, where it is rectified and controlled, and is then impressed on the exciter field winding to furnish excitation for the generator.

The exciter is a small AC generator with its field mounted on the main generator stator and its three-phase armature mounted on the generator rotor shaft. Included in the exciter field are permanent magnets mounted on the main generator stator between the exciter poles.

The exciter field resistance is temperature compensated by a thermistor. This aids regulation by keeping a nearly constant resistance at the regulator output terminals. The exciter output is rectified and impressed on the main generator field and the pilot exciter field. The exciter stator has a stabilizing field, which is used to improve stability and to prevent voltage regulator over-corrections for changes in generator output voltage.

The AC generator shown in *Figure 12-324* is a 6-pole, 8,000 rpm unit having a rating of 31.5 kilovoltamperes (kVA), 115/200 volts, 400 cps. This generator is three-phase, 4 wire, wye connected with grounded neutrals. By using an integral AC exciter, the necessity for brushes within the generator has been eliminated. The AC output of the rotating exciter armature is fed directly into the three-phase, full-wave, rectifier bridge located inside the rotor shaft, which uses high-temperature silicon rectifiers. The DC output from the rectifier bridge is fed to the main AC generator rotating field.

Voltage regulation is accomplished by varying the strength of the AC exciter stationary fields. Polarity reversals of the AC generator are eliminated and radio noise is minimized by the absence of the brushes. A noise filter mounted on the alternator further reduces any existing radio noise. The rotating pole structure of the generator is laminated from steel punchings, containing all six poles and a connecting hub section. This provides optimum magnetic and mechanical properties.

Some alternators are cooled by circulating oil through steel tubes. The oil used for cooling is supplied from the constant speed drive assembly. Ports located in the flange connecting the generator and drive assemblies make oil flow between the constant speed drive and the generator possible.

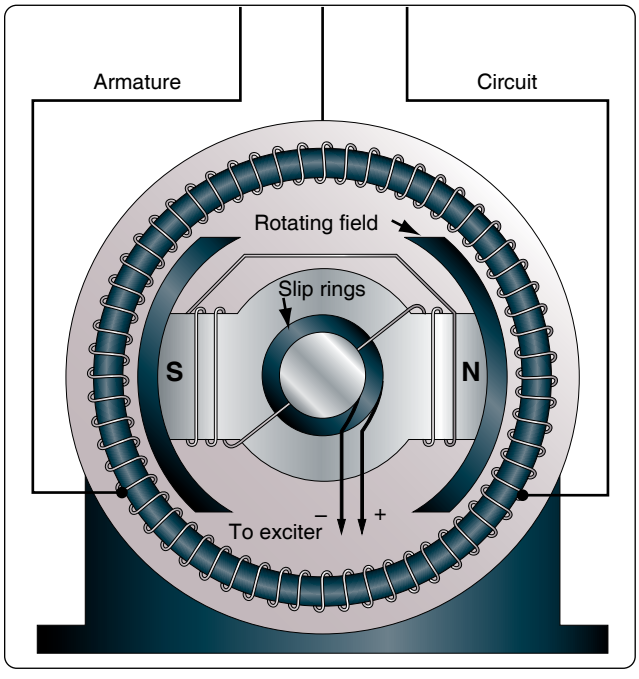


Figure 12-318. *Alternator with stationary armature and rotating field.*

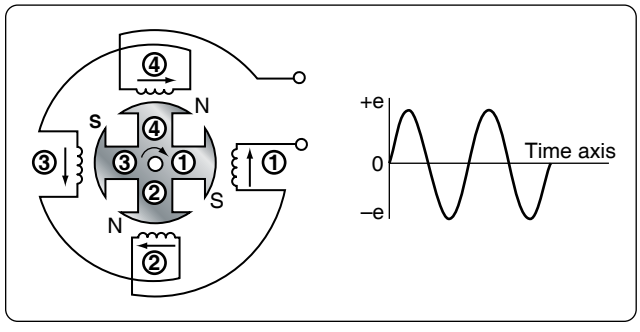


Figure 12-319. *Single-phase alternator.*

Voltage is built up by using permanent magnet interpoles in the exciter stator. The permanent magnets assure a voltage buildup, precluding the necessity of field flashing. The rotor of the alternator may be removed without causing loss of the alternator's residual magnetism.

Alternator Frequency

The frequency of the alternator voltage depends upon the speed of rotation of the rotor and the number of poles. The faster the speed, the higher the frequency; the lower the speed, the lower the frequency. The more poles on the rotor, the higher the frequency for a given speed. When a rotor has rotated through an angle so that two adjacent rotor poles (a North and a South pole) have passed one winding, the voltage induced in that winding has varied through one complete cycle. For a given frequency, the greater the number of pairs of poles, the lower the speed of rotation. A two-pole alternator rotates at twice the speed of a four-pole alternator for the same frequency of generated voltage. The frequency of the alternator in cycles per minute (cpm) is related to the number of poles and the speed, as expressed by the equation:

$$F = \frac{P}{N} \times \frac{N}{60} = \frac{PN}{120}$$

Where: P is the number of poles per phase
f is the frequency in cps
N is the rated speed in rpm

For example, a 2-pole, 3,600 rpm alternator has a frequency of:

$$\frac{2 \times 3{,}600}{120} = 60 \text{ cps}$$

A 4-pole, 1,800 rpm alternator has the same frequency; a 6-pole, 500 rpm alternator has a frequency of:

$$\frac{6 \times 500}{120} = 25 \text{ cps}$$

A 12-pole, 4,000 rpm alternator has a frequency of:

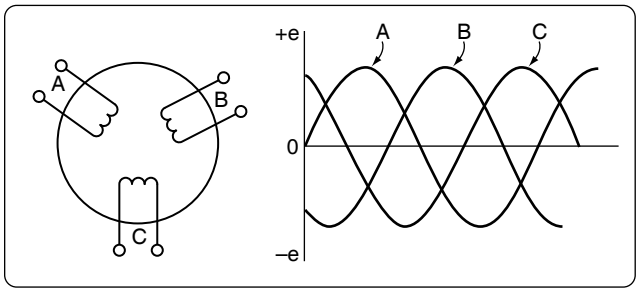


Figure 12-320. *Simplified schematic of three-phase alternator with output waveforms.*

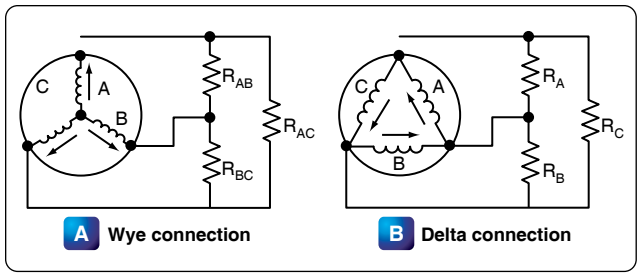


Figure 12-321. *Wye and delta connected alternators.*

$$\frac{12 \times 4{,}000}{120} = 400 \text{ cps}$$

Starter Generator

Many turbine-powered aircraft use a starter generator that acts like a starter during the start of the engine and when the engine is online it acts like a Generator. *[Figure 12-325]* The main advantage of the starter generator is saving weight by eliminating a separate starter that is only used during the start. Initially used on small turboprops and light jets but large units are now installed on the B787 aircraft engines to power the main engines and power the electrical system.

Alternator Rating

The maximum current that can be supplied by an alternator depends upon the maximum heating loss (I^2R power loss) that can be sustained in the armature and the maximum heating loss that can be sustained in the field. The armature current of an alternator varies with the load. This action is similar to that of A 12 pole, 4,000 rpm alternator has a frequency of DC generators. In AC generators, however, lagging power factor loads tend to demagnetize the field of an alternator, and terminal voltage is maintained only by increasing DC field current. For this reason, AC generators are usually rated according to kVA, power factor, phases, voltage, and frequency. One generator, for example, may be rated at 40 kVA, 208 volts, 400 cycles, three phase, at 75 percent power factor. The kVA indicates the apparent power. This is the kVA output, or the relationship between the current and voltage at which the generator is intended to operate. The power factor is the expression of the

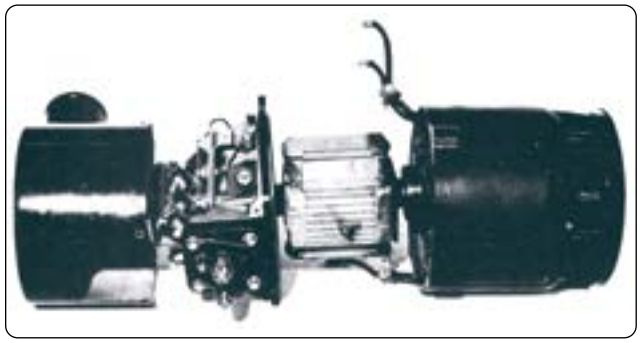


Figure 12-322. *Exploded view of alternator rectifier.*

ratio between the apparent power (volt-amperes) and the true or effective power (watts). The number of phases is the number of independent voltages generated. Three-phase generators generate three voltages 120 electrical degrees apart.

Alternator Maintenance

Maintenance and inspection of alternator systems is similar to that of DC systems. Check the exciter brushes for wear and surfacing. On most large aircraft with two or four alternator systems, each power panel has three signal lights, one connected to each phase of the power bus, so the lamp lights when the panel power is on. The individual buses throughout the airplane can be checked by operating equipment from that particular bus. Consult the manufacturer's instructions on operation of equipment for the method of testing each bus.

Alternator test stands are used for testing alternators and constant speed drives in a repair facility. They are capable of supplying power to constant speed drive units at input speeds varying from 2,400 rpm to 9,000 rpm.

A typical test stand motor uses 220/440 volt, 60 cycle, three-phase power. Blowers for ventilation, oil coolers, and necessary meters and switches are integral parts of the test stand. A load bank supplies test circuits. An AC motor generator set for ground testing is shown in *Figure 12-326*.

A typical, portable, AC electrical system test set is an analyzer, consisting of a multirange ohmmeter, a multirange combination AC DC voltmeter, an ammeter with a clip-on current transformer, a vibrating reed type frequency meter, and an unmounted continuity light.

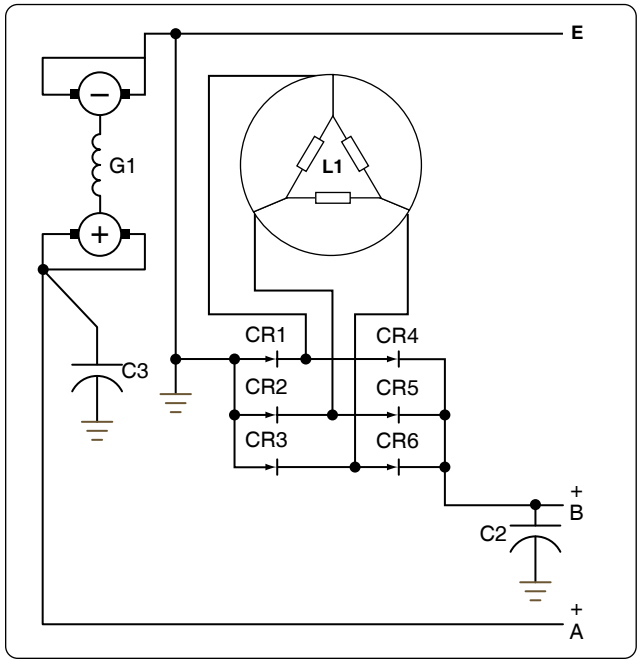


Figure 12-323. *Wiring diagram of alternator-rectifier unit.*

A portable load bank unit furnishes a load similar to that on the airplane for testing alternators, either while mounted in the airplane or on the shop test stand. A complete unit consists of resistive and reactive loads controlled by selector switches and test meters mounted on a control panel. This load unit is compact and convenient, eliminating the difficulty of operating large loads on the airplane while testing and adjusting the alternators and control equipment.

Proper maintenance of an alternator requires that the unit be kept clean and that all electrical connections are tight and in good repair. If the alternator fails to build up voltage as designated by applicable manufacturer's technical instructions, test the voltmeter first by checking the voltages of other alternators, or by checking the voltage in the suspected alternator with another voltmeter and comparing the results. If the voltmeter is satisfactory, check the wiring, the brushes, and the drive unit for faults. If this inspection fails to reveal the trouble, the exciter may have lost its residual magnetism. Residual magnetism is restored to the exciter by flashing the field. Follow the applicable manufacturer's instructions when flashing the exciter field. If, after flashing the field, no voltage is indicated, replace the alternator, since it is probably faulty.

Clean the alternator exterior with an approved fluid; smooth a rough or pitted exciter commutator or slip ring with 000 sandpaper; then clean and polish with a clean, dry cloth. Check the brushes periodically for length and general condition. Consult the applicable manufacturer's instructions on the specific alternator to obtain information on the correct brushes.

Regulation of Generator Voltage

Efficient operation of electrical equipment in an airplane depends on a constant voltage supply from the generator. Among the factors, which determine the voltage output of a generator, only one, the strength of the field current, can be conveniently controlled. To illustrate this control, refer to the diagram in *Figure 12-327*, showing a simple generator with a rheostat in the field circuit. If the rheostat is set to increase the resistance in the field circuit, less current flows through the field winding and the strength of the magnetic field in which the armature rotates decreases. Consequently, the voltage output of the generator decreases. If the resistance in the field circuit is decreased with the rheostat, more current flows through the field windings, the magnetic field becomes stronger, and the generator produces a greater voltage.

Voltage Regulation with a Vibrating-Type Regulator

Refer to *Figure 12-328*. With the generator running at normal speed and switch K open, the field rheostat is adjusted so that the terminal voltage is about 60 percent of

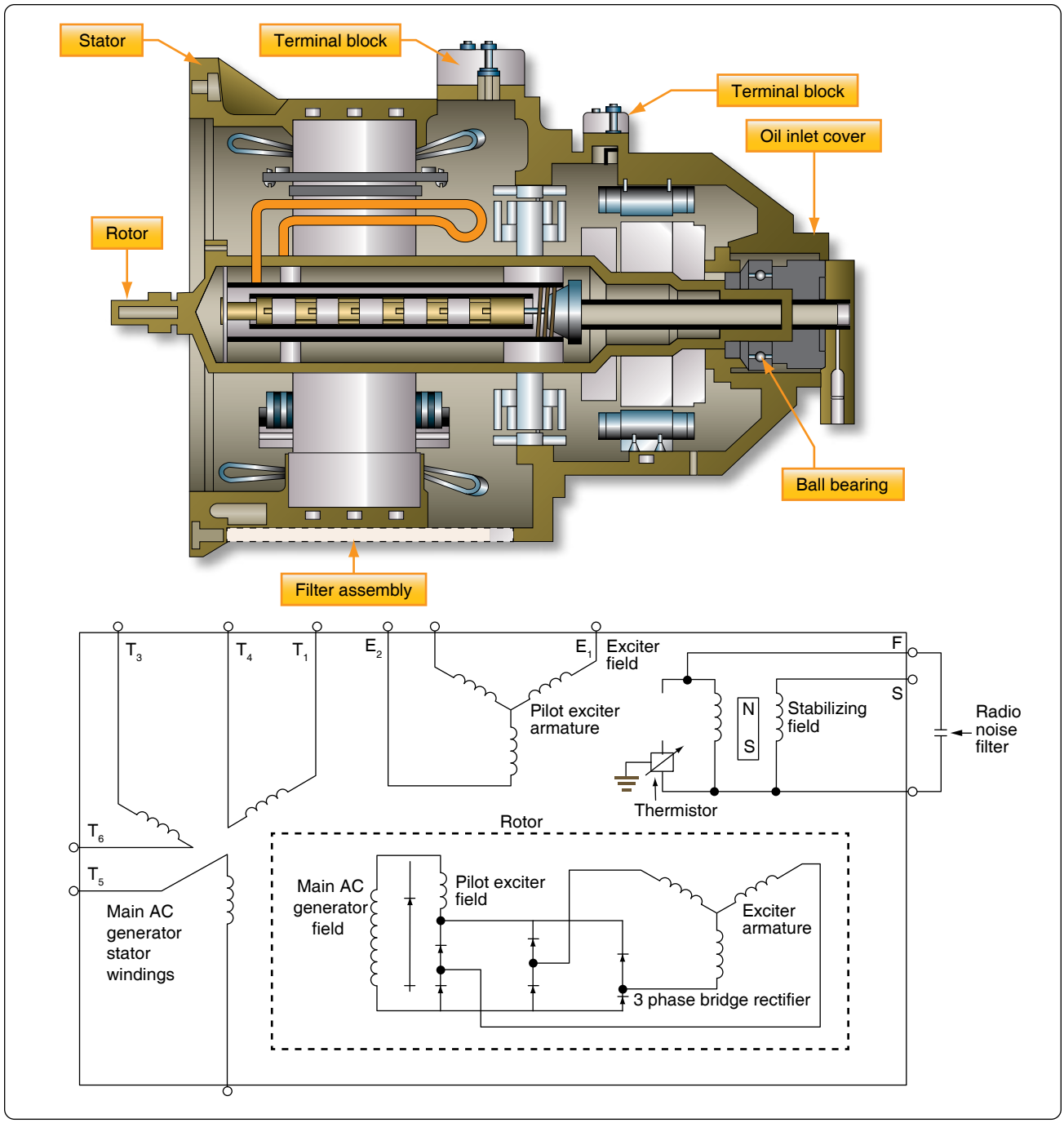


Figure 12-324. *A typical brushless alternator.*

normal. Solenoid S is weak and contact B is held closed by the spring. When K is closed, a short circuit is placed across the field rheostat. This action causes the field current to increase and the terminal voltage to rise.

When the terminal voltage rises above a certain critical value, the solenoid downward pull exceeds the spring tension and contact B opens, thus reinserting the field rheostat in the field circuit and reducing the field current and terminal voltage.

When the terminal voltage falls below a certain critical voltage, the solenoid armature contact B is closed again by the spring, the field rheostat is now shorted, and the terminal voltage starts to rise. The cycle repeats with a rapid, continuous action. Thus, an average voltage is maintained with or without load change.

The dashpot P provides smoother operation by acting as a damper to prevent hunting. The capacitor C across contact B

eliminates sparking. Added load causes the field rheostat to be shorted for a longer period of time and, thus, the solenoid armature vibrates more slowly. If the load is reduced and the terminal voltage rises, the armature vibrates more rapidly and the regulator holds the terminal voltage to a steady value for any change in load, from no load to full load, on the generator.

Vibrating-type regulators cannot be used with generators, which require a high-field current, since the contacts pit or burn. Heavy-duty generator systems require a different type of regulator, such as the carbon pile voltage regulator.

Three Unit Regulators

Many light aircraft employ a three unit regulator for their generator systems. *[Figure 12-329]* This type of regulator includes a current limiter and a reverse current cut-out in addition to a voltage regulator.

The action of the voltage regulator unit is similar to the vibrating-type regulator described earlier. The second of the three units is a current regulator to limit the output current of the generator. The third unit is a reverse current cut-out that disconnects the battery from the generator. If the battery is not disconnected, it discharges through the generator armature when the generator voltage falls below that of the battery, thus driving the generator as a motor. This action is called "motoring" the generator and, unless it is prevented, it discharges the battery in a short time.

The operation of a three unit regulator is described in the following paragraphs. *[Figure 12-330]*

The action of vibrating contact C1 in the voltage regulator unit causes an intermittent short circuit between points R1 and L2. When the generator is not operating, spring S1 holds C1 closed; C2 is also closed by S2. The shunt field is connected directly across the armature.

When the generator is started, its terminal voltage rises as the generator comes up to speed, and the armature supplies the field with current through closed contacts C2 and C1.

Figure 12-325. *Starter generator for small business jet.*

As the terminal voltage rises, the current flow through L1 increases and the iron core becomes more strongly magnetized. At a certain speed and voltage, when the magnetic attraction on the movable arm becomes strong enough to overcome the tension of spring S1, contact points C1 are separated. The field current now flows through R1 and L2. Because resistance is added to the field circuit, the field is momentarily weakened and the rise in terminal voltage is checked. Also, since the L2 winding is opposed to the L1 winding, the magnetic pull of L1 against S1 is partially neutralized, and spring S1 closes contact C1. Therefore, R1 and L2 are again shorted out of the circuit, and the field current again increases; the output voltage increases, and C1 is opened because of the action of L1. The cycle is rapid and occurs many times per second. The terminal voltage of the generator varies slightly, but rapidly, above and below an average value determined by the tension of spring S1, which may be adjusted.

The purpose of the vibrator-type current limiter is to limit the output current of the generator automatically to its maximum rated value in order to protect the generator. As shown in *Figure 12-330*, L3 is in series with the main line and load. Thus, the amount of current flowing in the line determines when C2 is opened and R2 placed in series with the generator field. By contrast, the voltage regulator is actuated by line voltage, whereas the current limiter is actuated by line current. Spring S2 holds contact C2 closed until the current through the main line and L3 exceeds a certain value, as determined by the tension of spring S2, and causes C2 to be opened. The increase in current is due to an increase in load. This action inserts R2 into the field circuit of the generator and decreases the field current and the generated voltage. When the generated voltage is decreased, the generator current is reduced. The core of L3 is partly demagnetized and the spring closes the contact points. This causes the generator voltage and current to rise until the current reaches a value sufficient to start the cycle again. A certain minimum value of load current is necessary to cause the current limiter to vibrate.

The purpose of the reverse current cut-out relay is to automatically disconnect the battery from the generator when the generator voltage is less than the battery voltage. If this device were not used in the generator circuit, the battery would discharge through the generator. This would tend to make the generator operate as a motor, but because the generator is coupled to the engine, it could not rotate such a heavy load. Under this condition, the generator windings may be severely damaged by excessive current.

There are two windings, L4 and L5, on the soft iron core. The current winding, L4, consisting of a few turns of heavy wire, is in series with the line and carries the entire line current.

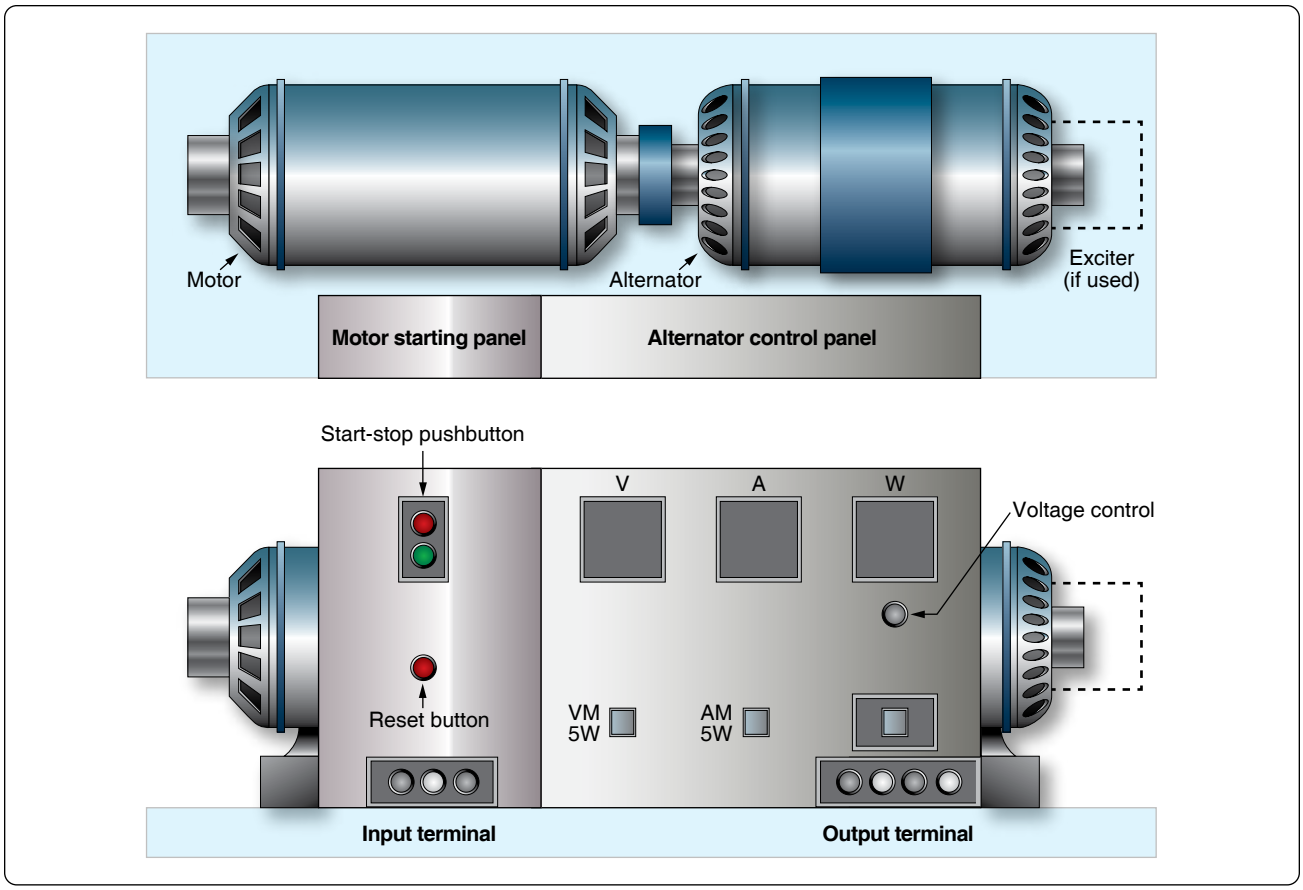


Figure 12-326. *AC motor generator set for ground testing.*

The voltage winding, L5, consisting of a large number of turns of fine wire, is shunted across the generator terminals.

When the generator is not operating, the contacts, C3 are held open by the spring S3. As the generator voltage builds up, L5 magnetizes the iron core. When the current (as a result of the generated voltage) produces sufficient magnetism in the iron core, contact C3 is closed, as shown. The battery then receives a charging current. The coil spring, S3, is so adjusted that the voltage winding does not close the contact points until the voltage of the generator is in excess of the normal voltage of the battery. The charging current passing through L4 aids the current in L5 to hold the contacts tightly closed. Unlike C1 and C2, contact C3 does not vibrate. When the generator slows down or, for any other cause, the generator voltage decreases to a certain value below that of the battery, the current reverses through L4 and the ampere turns of L4 oppose those of L5. Thus, a momentary discharge current from the battery reduces the magnetism of the core and C3 is opened, preventing the battery from discharging into the generator and motoring it. C3 does not close again until the generator terminal voltage exceeds that of the battery by a predetermined value.

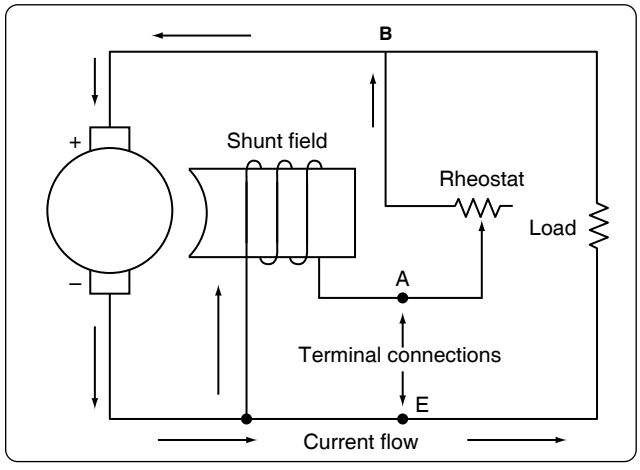


Figure 12-327. *Regulation of generator voltage by field rheostat.*

Differential Relay Switch

Aircraft electrical systems normally use some type of reverse current relay switch, which acts not only as a reverse current relay cut-out but also serves as a remote control switch by which the generator can be disconnected from the electrical system at any time. One type of reverse current relay switch operates on the voltage level of the generator, but the type most commonly used on large aircraft is the differential relay switch, which is controlled by the difference in voltage

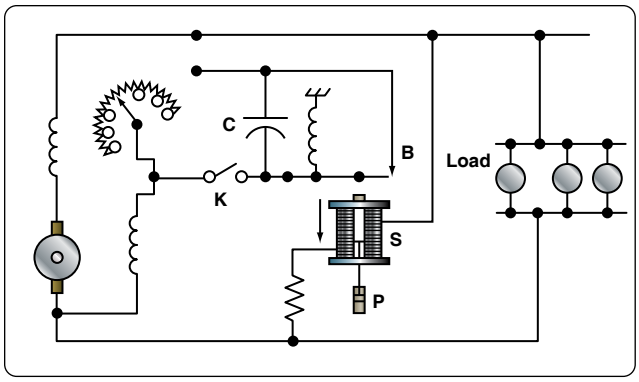


Figure 12-328. *Vibrating-type voltage regulator.*

between the battery bus and the generator.

The differential type relay switch connects the generator to the main bus bar in the electrical system when the generator voltage output exceeds the bus voltage by 0.35 to 0.65 volt. It disconnects the generator when a nominal reverse current flows from the bus to the generator. The differential relays on all the generators of a multiengine aircraft do not close when the electrical load is light. For example, in an aircraft having a load of 50 amperes, only two or three relays may close. If a heavy load is applied, the equalizing circuit lowers the voltage of the generators already on the bus and, at the same time, raise the voltage of the remaining generators, allowing their relays to close. If the generators have been paralleled properly, all the relays stay closed until the generator control switch is turned off or until the engine speed falls below the minimum needed to maintain generator output voltage.

The differential generator control relay shown in *Figure 12-331* is made up of two relays and a coil-operated contactor. One relay is the voltage relay and the other is the differential relay. Both relays include permanent magnets that pivot between the pole pieces of temporary magnets wound with relay coils. Voltages of one polarity set up fields about the temporary magnets with polarities that cause the permanent magnet to move in the direction necessary to close the relay contacts; voltages of the opposite polarity establish fields that cause the relay contacts to open. The differential relay has two coils wound on the same core. The coil-operated contactor, called the main contactor, consists of movable contacts that are operated by a coil with a movable iron core.

Closing the generator switch on the control panel connects the generator output to the voltage relay coil. When generator voltage reaches 22 volts, current flows through the coil and closes the contacts of the voltage relay. This action completes a circuit from the generator to the battery through the differential coil.

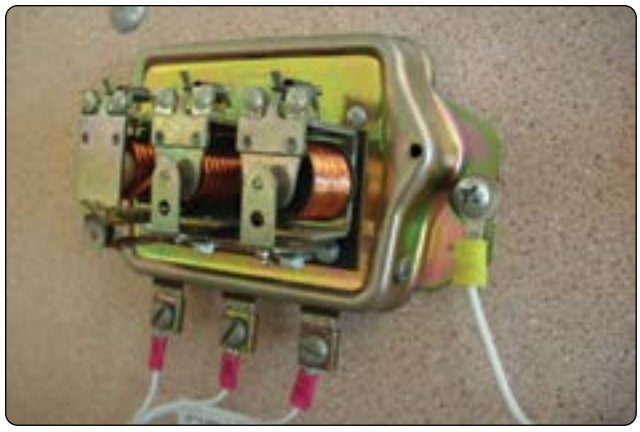


Figure 12-329. *Three unit regulator.*

When the generator voltage exceeds the bus voltage by 0.35 volt, current flows through the differential coil, the differential relay contact closes and, thus, completes the main contractor coil circuit. The contacts of the main contactor close and connect the generator to the bus.

When the generator voltage drops below the bus (or battery) voltage, a reverse current weakens the magnetic field about the temporary magnet of the differential relay. The weakened field permits a spring to open the differential relay contacts, breaking the circuit to the coil of the main contactor relay, opening its contacts, and disconnecting the generator from the bus. The generator battery circuit may also be broken by opening the flight deck control switch, which opens the contacts of the voltage relay, causing the differential relay coil to be de-energized.

Overvoltage & Field Control Relays

Two other items used with generator control circuits are the overvoltage control and the field control relay. As its name implies, the overvoltage control protects the system when

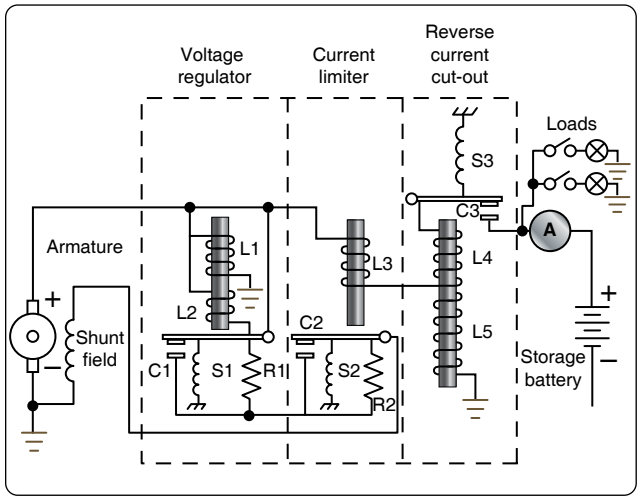


Figure 12-330. *Three unit regulator for variable speed generators.*

excessive voltage exists. The overvoltage relay is closed when the generator output reaches 32 volts and completes a circuit to the trip coil of the field control relay. The closing of the field control relay trip circuit opens the shunt field circuit and completes it through a resistor, causing generator voltage to drop; also, the generator switch circuit and the equalizer circuit (multiengine aircraft) are opened. An indicator light circuit is completed, warning that an overvoltage condition exists. A "reset" position of the flight deck switch is used to complete a reset coil circuit in the field control relay, returning the relay to its normal position.

Generator Control Units (GCU)

Basic Functions of a Generator Control Unit (GCU)

The generator control unit (GCU) is more commonly found on turbine power aircraft. The most basic GCU perform a number of functions related to the regulation, sensing, and protection of the DC generation system. *[Figure 12-332]*

Voltage Regulation

The most basic of the GCU functions is that of voltage regulation. Regulation of any kind requires the regulation unit to take a sample of an output and to compare that sample with a controlled reference. If the sample taken falls outside of the limits set by the reference, then the regulation unit must provide an adjustment to the unit generating the output so as to diminish or increase the output levels. In the case of the GCU, the output voltage from a generator is sensed by the GCU and compared to a reference voltage. If there is any difference between the two, the error is usually amplified and then sent back to the field excitation control portion of the circuit. The field excitation control then makes voltage/excitation adjustments in the field winding of the generator in order to bring the output voltage back into required bus tolerances.

Overvoltage Protection

Like the voltage regulation feature of the GCU, the overvoltage protection system compares the sampled voltage to reference voltage. The output of the overvoltage protection circuit is used to open the relay that controls the output for the field excitation. These types of faults can occur for a number of reasons. The most common, however, is the failure of the voltage regulation circuit in the GCU.

Parallel Generator Operations

The paralleling feature of the GCU allows for two or more GCU/generator systems to work in a shared effort to provide current to the aircraft electrical system. Comparing voltages between the equalizer bus and the interpole/compensator voltage, and amplifying the differences accomplishes the control of this system. The difference is then sent to the voltage regulation circuit, where adjustments are then made in the regulation output. These adjustments continue until all of the busses are equalized in their load sharing.

Over-Excitation Protection

When a GCU in a paralleled system fails, a situation can occur where one of the generators becomes overexcited and tries to carry more than its share of the load, if not all of the loads. When this condition is sensed on the equalizing bus, the faulted generation control system shuts down by receiving a de-excitation signal. This signal is then transmitted to the overvoltage circuit, and then opens the field excitation output circuit.

Differential Voltage

When the GCU allows the logic output to close the generator line contactor, the generator voltage must be within a close tolerance of the load bus. If the output is not within the specified tolerance, then the contactor is not allowed to connect the generator to the bus.

Reverse Current Sensing

If the generator is unable to maintain the required voltage level, it eventually begins to draw current instead of providing it. In this case, the faulty generator is seen as a load to the other generators and will need to be removed from the bus. Once the generator is off-line, it is not permitted to be reconnected to the bus until such time that the generator faults are cleared and the generator is capable of providing a current to the bus. In most cases, the differential voltage circuit and the reverse current sensing circuit are one in the same.

Alternator Constant Speed Drive System

Alternators are not always connected directly to the airplane engine like DC generators. Since the various electrical devices operating on AC supplied by alternators are designed to operate at a certain voltage and at a specified frequency, the speed of the alternators must be constant; however, the speed of an airplane engine varies. Therefore, the engine, through a constant speed drive installed between the engine and the alternator, drives some alternators.

A typical hydraulic-type drive is shown in *Figure 12-333*. The following discussion of a constant speed drive system is based on such a drive found on large multiengine aircraft. The constant speed drive is a hydraulic transmission that may be controlled either electrically or mechanically.

The constant speed drive assembly is designed to deliver an output of 6,000 rpm, provided the input remains between 2,800 and 9,000 rpm. If the input, which is determined by engine speed, is below 6,000 rpm, the drive increases the speed in order to furnish the desired output. This stepping up of speed is known as overdrive.

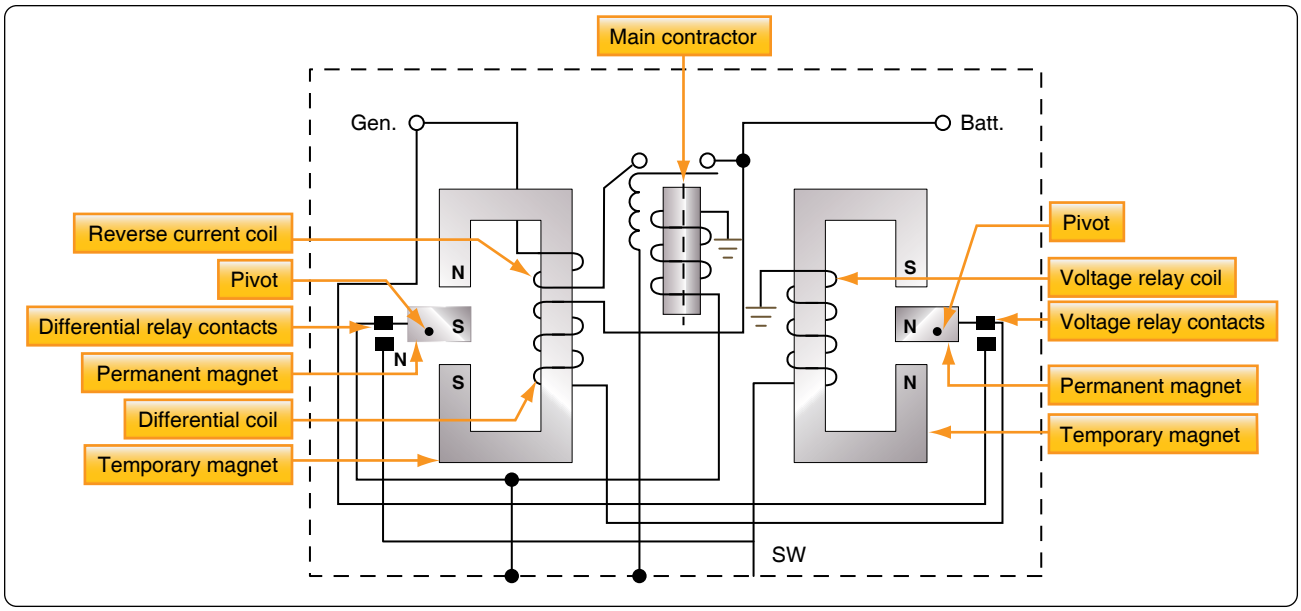


Figure 12-331. *Differential generator control relay.*

In overdrive, an automobile engine operates at about the same rpm at 60 mph as it does in conventional drive at 49 mph. In aircraft, this principle is applied in the same manner. The constant speed drive enables the alternator to produce the same frequency at slightly above engine idle rpm as it would at takeoff or cruising rpm.

With the input speed to the drive set at 6,000 rpm, the output speed is the same. This is known as straight drive and might be compared to an automobile in high gear. However, when the input speed is greater than 6,000 rpm, it must be reduced to provide an output of 6,000 rpm. This is called underdrive, which is comparable to an automobile in low gear. Thus, the large input, caused by high engine rpm, is reduced to give the desired alternator speed.

As a result of this control by the constant speed drive, the frequency output of the generator varies from 420 cps at no load to 400 cps under full load. This, in brief, is the function of the constant speed drive assembly. Before discussing the various units and circuits, the overall operation of the transmission should be discussed as follows.

Hydraulic Transmission

The transmission is mounted between the generator and the aircraft engine. Its name denotes that hydraulic oil is used, although some transmissions may use engine oil. Refer to the cutaway view of such a transmission in *Figure 12-334*. The input shaft D is driven from the drive shaft on the accessory section of the engine. The output drive F, on the opposite end of the transmission, engages the drive shaft of the generator.

The input shaft is geared to the rotating cylinder block gear, which it drives, as well as to the makeup and scavenger gear pumps E.

The makeup (charge) pump delivers oil (300 psi) to the pump and motor cylinder block, to the governor system, and to the pressurized case, whereas the scavenger pump returns the oil to the external reservoir.

The rotating cylinder assembly B consists of the pump and motor cylinder blocks, which are bolted to opposite sides of a port plate. The two other major parts are the motor wobbler A and the pump wobbler C. The governor system is the unit at the top of the left side in *Figure 12-334*.

The cylinder assembly has two primary units. The block assembly of one of the units, the pump, contains 14 cylinders, each of which has a piston and pushrod. Charge pressure from the makeup pump is applied to each piston in order to force it outward against the pushrod. It, in turn, is pushed against the pump wobble plate.

If the plate remained as shown in *Figure 12-335A*, each of the 14 cylinders would have equal pressure, and all pistons would be in the same relative position in their respective cylinders. But with the plate tilted, the top portion moves outward and the lower portion inward. *[Figure 12-335B]* As a result, more oil enters the interior of the upper cylinder, but oil is forced from the cylinder of the bottom piston.

If the pump block were rotated while the plate remained stationary, the top piston would be forced inward because of the angle of the plate. This action would cause the oil confined within the cylinder to be subjected to increased pressure great

Figure 12-332. *Generator Control Unit (GCU).*

enough to force it into the motor cylinder block assembly.

Before explaining what the high-pressure oil in the motor unit does, it is necessary to know something about this part of the rotating cylinder block assembly. The motor block assembly has 16 cylinders, each with its piston and pushrod. These are constantly receiving charge pressure of 300 psi. The position of the piston depends upon the point at which each pushrod touches the motor wobble plate. These rods cause the wobble plate to rotate by the pressure they exert against its sloping surface.

The piston and pushrod of the motor are pushed outward as oil is forced through the motor valve plate from the pump cylinder. The pushrods are forced against the motor wobble plate, which is free to rotate but cannot change the angle at which it is set. Since the pushrods cannot move sideways, the pressure exerted against the motor wobble plate's sloping face causes it to rotate.

In the actual transmission, there is an adjustable wobble plate. The control cylinder assembly determines the tilt of the pump wobble plate. For example, it is set at an angle, which causes the motor cylinders to turn the motor wobble plate faster than the motor assembly if the transmission is in overdrive. The greater pressure in the pump and motor cylinders produces the result described.

With the transmission in underdrive, the angle is arranged so there is a reduction in pumping action. The subsequent slippage between the pushrods and motor wobble plate reduces the output speed of the transmission. When the pump wobble plate is not at an angle, the pumping action is at a minimum and the transmission has what is known as hydraulic lock. For this condition, the input and output speed is about the same, and the transmission is considered to be in straight drive.

To prevent the oil temperature from becoming excessively high within the cylinder block, the makeup pressure pump forces oil through the center of this block and the pressure relief valve. From this valve, the oil flows into the bottom of the transmission case. A scavenger pump removes the oil from the transmission case and circulates it through the oil cooler and filters before returning it to the reservoir. At the start of the cycle, oil is drawn from the reservoir, passed through a filter, and forced into the cylinder block by the makeup pressure pump.

The clutch, located in the output gear and clutch assembly, is an overrunning one way, sprag-type device. Its purpose is to ratchet if the alternator becomes motorized; otherwise, the alternator might turn the engine. Furthermore, the clutch provides a positive connection when the transmission is driving the alternator.

There is another unit of the drive that must be discussed—the governor system. The governor system, which consists of a hydraulic cylinder with a piston, is electrically controlled. Its duty is to regulate oil pressure flowing to the control cylinder assembly. *[Figure 12-336]*

The center of the system's hydraulic cylinder is slotted so the arm of the pump wobble plate can be connected to the piston. As oil pressure moves the piston, the pump wobble plate is placed in either overspeed, underspeed, or straight drive.

Figure 12-337 shows the electrical circuit used to govern the speed of the transmission. First, the main points of the complete electrical control circuit are discussed. *[Figures 12-337* and *12-338]* For simplification, two portions, the overspeed circuit and the load division circuit, are considered as individual circuits.

Note, in *Figure 12-337*, that the circuit has a valve and solenoid assembly (O) and a control cylinder (E), and that it contains such units as the tachometer generator (D), the rectifier (C), and adjustable resistor (B), rheostat (A), and the control coil (Q).

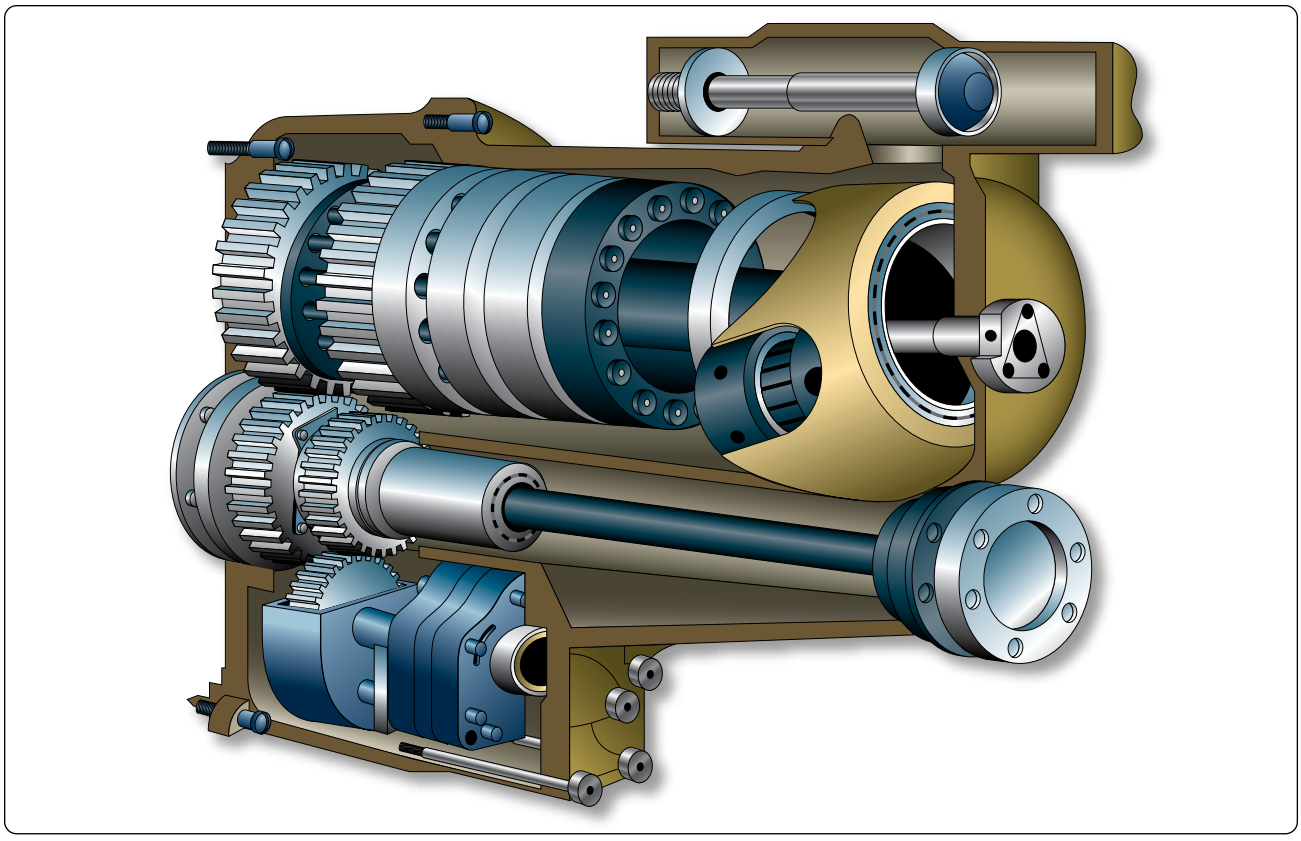


Figure 12-333. *Constant speed drive.*

Since it is driven by a drive gear in the transmission, the tachometer (often called tach) generator, a three-phase unit, has a voltage proportional to the speed of the output drive. The rectifier changes its voltage from AC to DC. After rectification, the current flows through the resistor, rheostat, and valve and solenoid. Each of these units is connected in series. *[Figure 12-338]*

Under normal operating conditions, the output of the tach generator causes just enough current to enter the valve and solenoid coil to set up a magnetic field of sufficient strength to balance the spring force in the valve. When the alternator speed increases as the result of a decrease in load, the tach generator output also increases. Because of the greater output, the coil in the solenoid is sufficiently strengthened to overcome the spring force. Thus, the valve moves and, as a result, oil pressure enters the reduced speed side of the control cylinder.

In turn, the pressure moves the piston, causing the angle of the pump wobble plate to be reduced. The oil on the other side of the piston is forced back through the valve into the system return. Since the angle of the pump wobble plate is smaller, there is less pumping action in the transmission. The result is decreased output speed. To complete the cycle, the procedure is reversed.

With the output speed reduction, tach generator output decreases; consequently, the flow of current to the solenoid diminishes. Therefore, the magnetic field of the solenoid becomes so weak that the spring is able to overcome it and reposition the valve.

If a heavy load is put on the AC generator, its speed decreases. The generator is not driven directly by the engine; the hydraulic drive allows slippage. This decrease causes the output of the tach generator to taper off and, as a result, weakens the magnetic field of the solenoid coil. The spring in the solenoid moves the valve and allows oil pressure to enter the increase side of the control cylinder and the output speed of the transmission is raised.

There are still two important circuits that must be discussed: the overspeed circuit and the load division circuit. The generator is prevented from overspeeding by a centrifugal switch (S) in *Figure 12-339* and the overspeed solenoid coil (R), which is located in the solenoid and valve assembly. The centrifugal switch is on the transmission and is driven through the same gear arrangement as the tach generator.

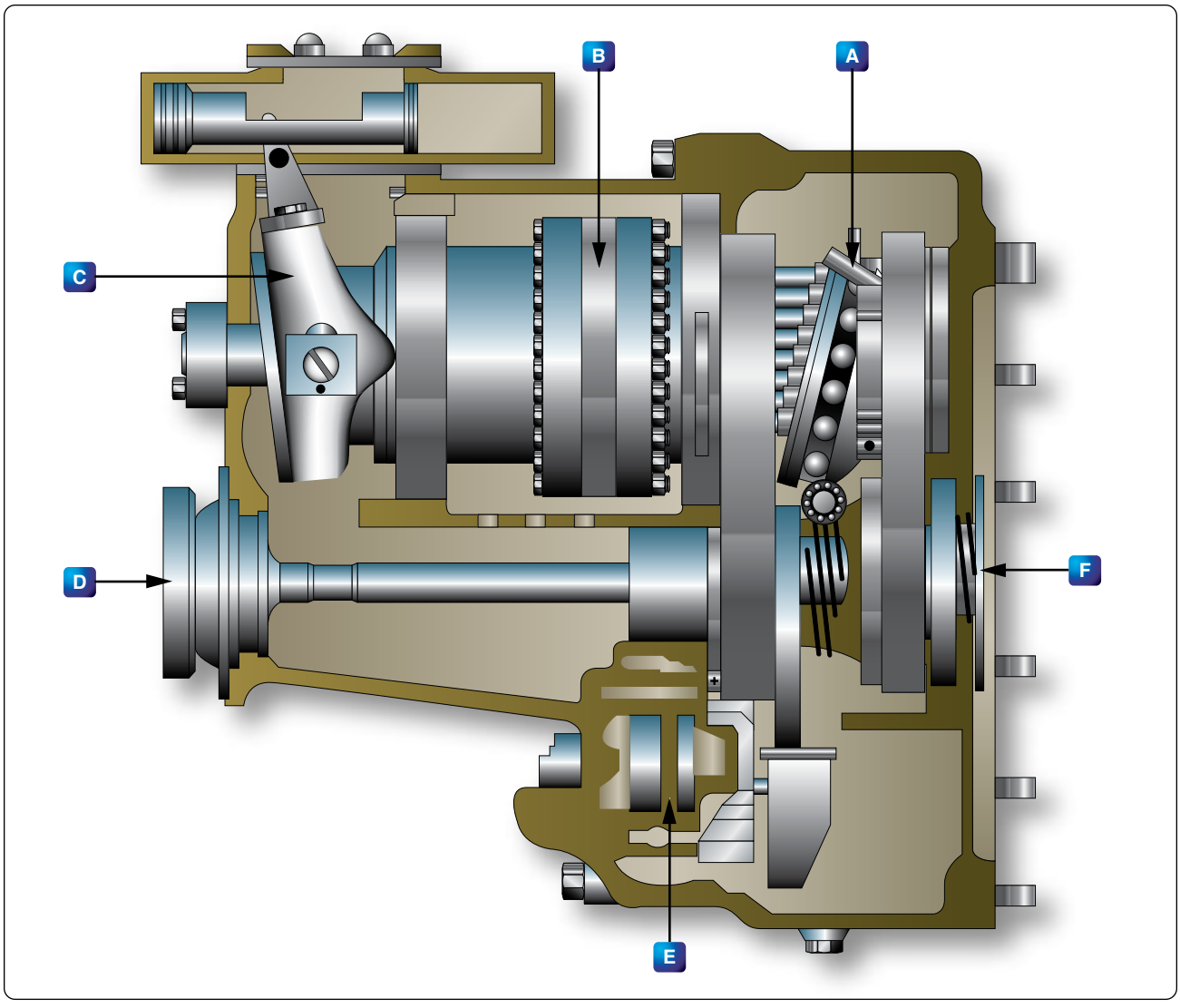


Figure 12-334. *Cutaway of a hydraulic transmission.*

The aircraft DC system furnishes the power to operate the overspeed coil in the solenoid and coil assembly. If the output speed of the transmission reaches a speed of 7,000 to 7,500 rpm, the centrifugal switch closes the DC circuit and energizes the overspeed solenoid. This component then moves the valve and engages the latch that holds the valve in the underdrive position. To release the latch, energize the underdrive release solenoid.

The load division circuit's function is to equalize the loads placed on each of the alternators, which is necessary to assure that each alternator assumes its share; otherwise, one alternator might be overloaded while another would be carrying only a small load.

In *Figure 12-340*, one phase of the alternator provides power for the primary in transformer (G), whose secondary supplies power to the primaries of two other transformers (J_1 and J_2). Rectifiers (K) then change the output of the transformer secondaries from AC to DC.

The function of the two capacitors (L) is to smooth out the DC pulsations.

The output of the current transformer (F) depends upon the amount of current flowing in the line of one phase. In this way, it measures the real load of the generator. The output voltage of the current transformer is applied across resistor (H). This voltage is added vectorially to the voltage applied to the upper winding of transformer (J) by the output of transformer (F). At the same time as it adds vectorially to the upper winding of transformer (J), it subtracts vectorially from the voltage applied to the lower winding of (J).

This voltage addition and subtraction depends on the real load of the generator. The amount of real load determines the phase

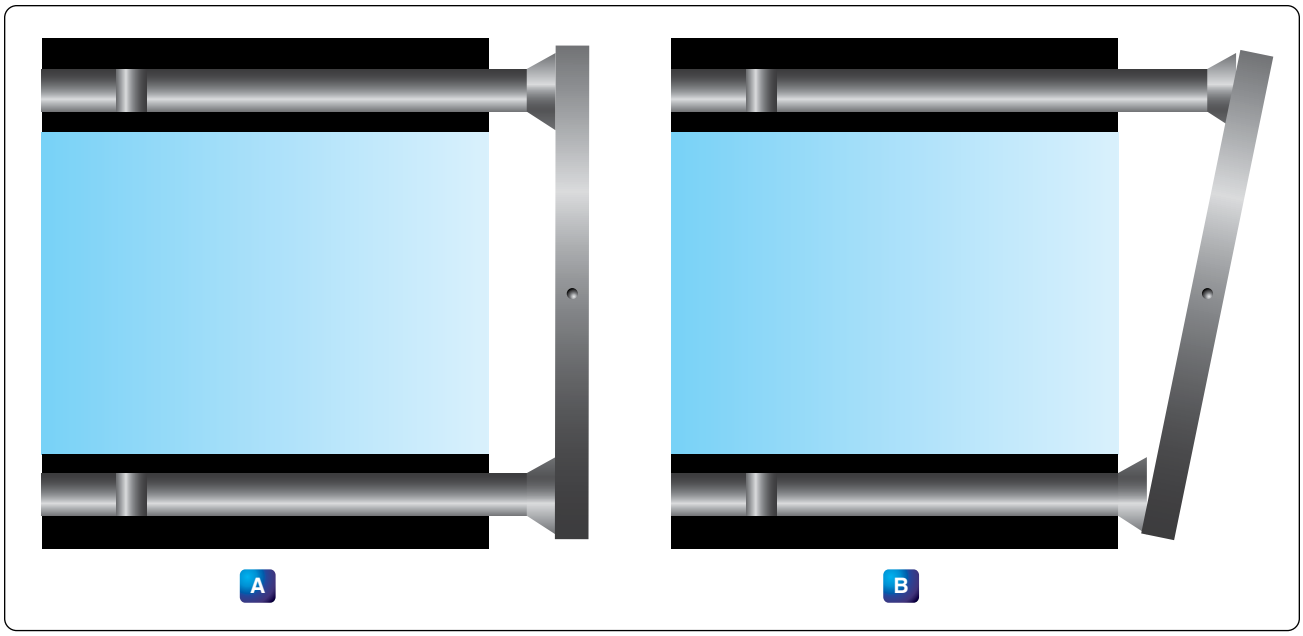


Figure 12-335. *Wobble plate position.*

angle and the amount of voltage impressed across resistor (H). The greater the real load, the greater the voltage across (H), and hence, the greater the difference between the voltages applied to the two primaries of transformer (J). The unequal voltages applied to resistor (M) by the secondaries of transformer (J) cause a current flow through the control coil (P).

The control coil is wound so that its voltage supplements the voltage for the control coil in the valve and solenoid assembly. The resulting increased voltage moves the valve and slows down the generator's speed. Why should the speed be decreased if the load has been increased? Actually, systems using only one generator would not have decreased speed, but for those having two or more generators, a decrease is necessary to equalize the loads.

The load division circuit is employed only when two or more generators supply power. In such systems, the control coils are connected in parallel. If the source voltage for one of these becomes higher than the others, it determines the direction of current flow throughout the entire load division circuit. As explained before, the real load on the generator determines the amount of voltage on the control coil; therefore, the generator with the highest real load has the highest voltage.

As shown in *Figure 12-341*, current through No. 1 control coil, where the largest load exists, aids the control coil of the valve and solenoid, thereby slowing down the generator. (The source voltage of the control coils is represented by battery symbols in *Figure 12-341*.) The current in the remaining control coils opposes the control coil of the valve and solenoid, in order to increase the speed of the other generators so the load is more evenly distributed.

On some drives, instead of an electrically-controlled governor, a flyweight-type governor is employed, which consists of a recess-type revolving valve driven by the output shaft of the drive, flyweights, two coil springs, and a nonrotating valve stem. Centrifugal force, acting on the governor flyweights, causes them to move outward, lifting the valve stem against the opposition of a coil spring.

The valve stem position controls the directing of oil to the two oil outlines. If the output speed tends to exceed 6,000 rpm, the flyweights lift the valve stem to direct more oil to the side of the control piston, causing the piston to move in a direction to reduce the pump wobble plate angle. If the speed drops below 6,000 rpm, oil is directed to the control piston so that it moves to increase the wobble plate angle.

Overspeed protection is installed in the governor. The drive starts in the underdrive position. The governor coil springs are fully extended and the valve stem is held at the limit of its downward travel. In this condition, pressure is directed to the side of the control piston giving minimum wobble plate angle. The maximum angle side of the control piston is open to the hollow stem. As the input speed increases, the flyweights start to move outward to overcome the spring bias. This action lifts the valve stem and starts directing oil to the maximum side of the control piston, while the minimum side is opened to the hollow stem.

At about 6,000 rpm, the stem is positioned to stop drainage of either side, and the two pressures seek a balance point as

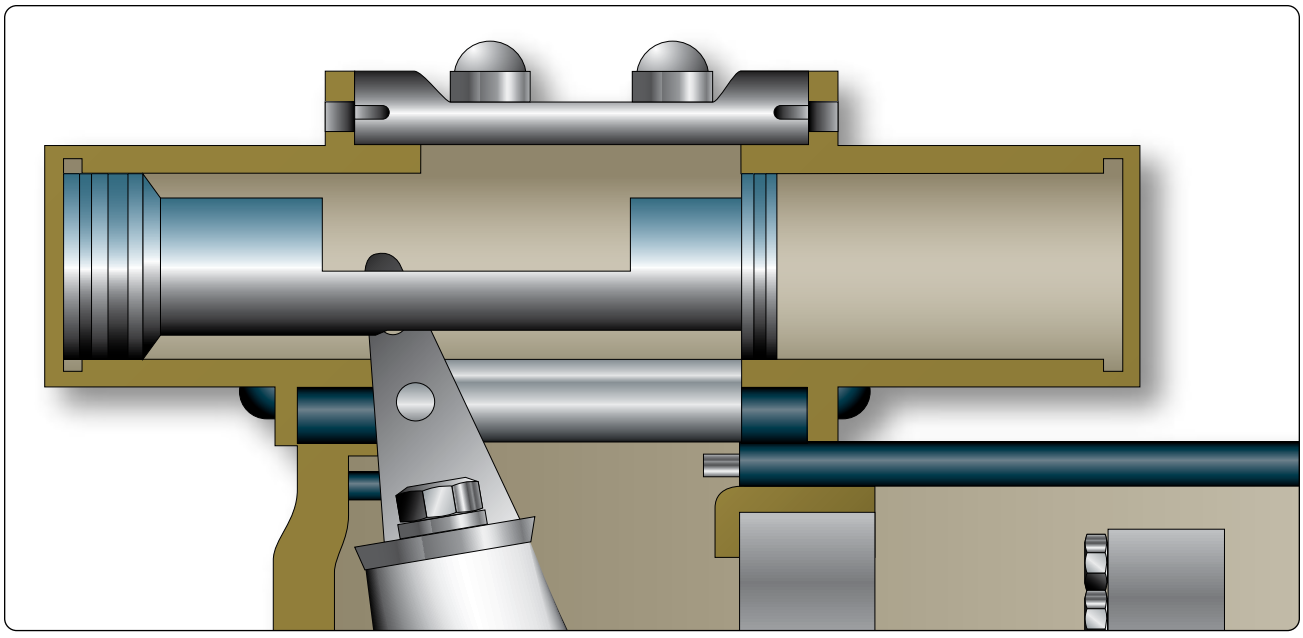


Figure 12-336. *Control cylinder.*

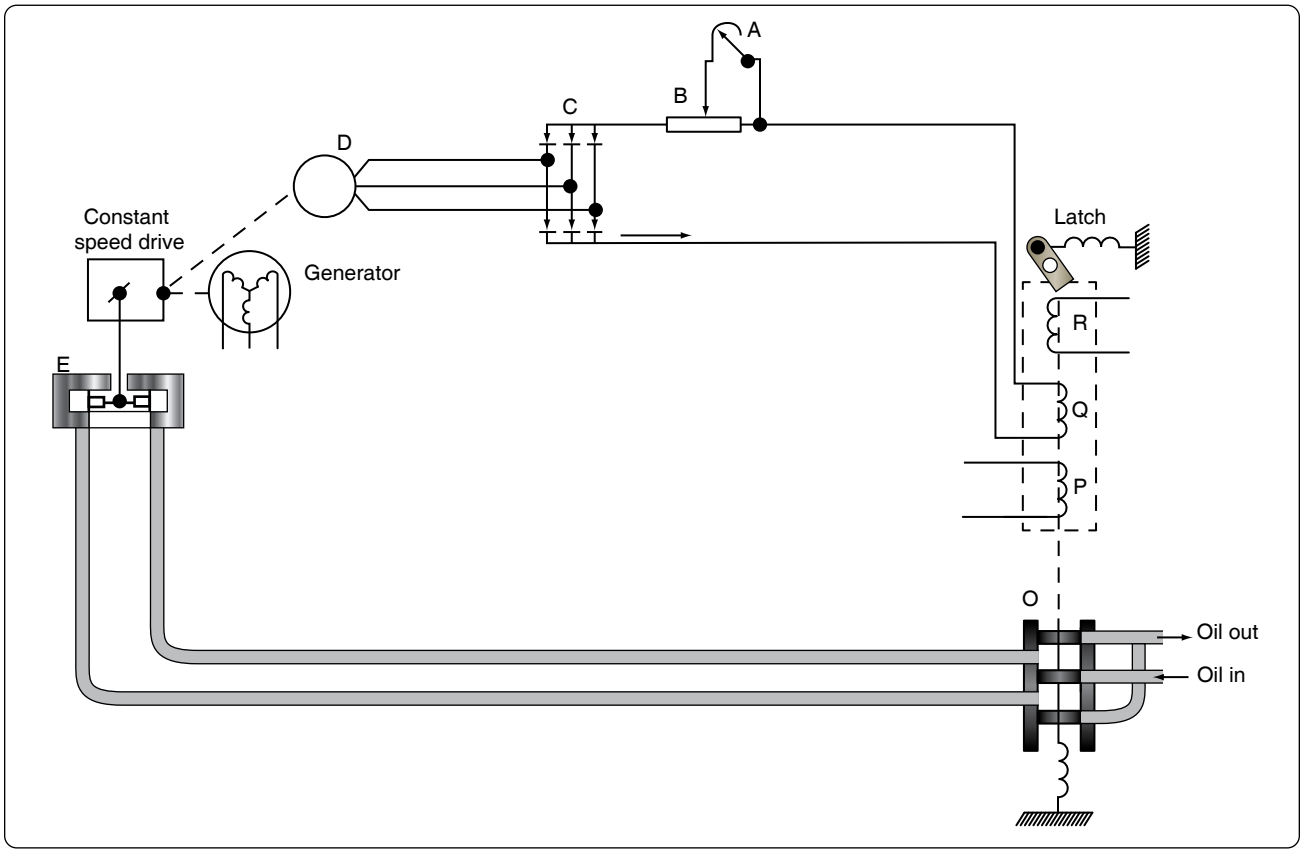


Figure 12-337. *Electrical hydraulic control circuit.*

the flyweight force is balanced against the spring bias. Thus, a mechanical failure in the governor causes an underdrive condition. The flyweight's force is always tending to move the valve stem to the decrease speed position so that, if the coil spring breaks and the stem moves to the extreme position in that direction, output speed is reduced. If the input to the governor fails, the spring forces the stem all the way to the start position to obtain minimum output speed.

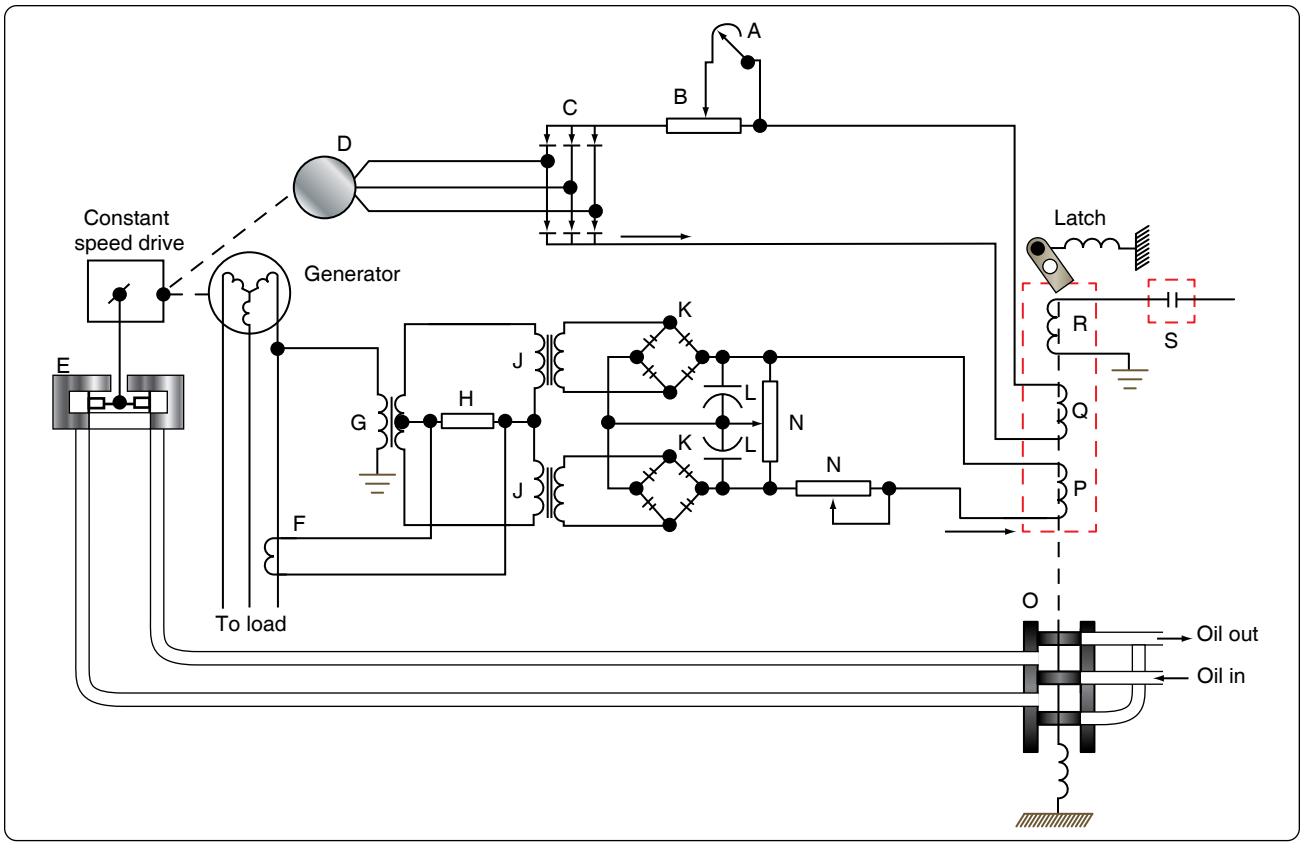


Figure 12-338. *Speed control circuit.*

An adjustment screw on the end of the governor regulates the output speed of the constant speed drive. This adjustment increases or decreases the compression of a coil spring, opposing the action of the flyweights. The adjustment screws turn in an indented collar, which provides a means of making speed adjustments in known increments. Each "click" provides a small change in generator frequency.

The constant speed drive (CSD) can be an independent unit or mounted within the alternator housing. When the CSD and the alternator are contained within one unit, the assembly is known as an integrated drive generator (IDG).

Voltage Regulation of Alternators

The problem of voltage regulation in an AC system does not differ basically from that in a DC system. In each case, the function of the regulator system is to control voltage, maintain a balance of circulating current throughout the system, and eliminate sudden changes in voltage (anti-hunting) when a load is applied to the system. However, there is one important difference between the regulator system of DC generators and alternators operated in a parallel configuration. The load carried by any particular DC generator in either a two or four generator system depends on its voltage as compared with the bus voltage, while the division of load between alternators depends upon the adjustments of their speed governors, which are controlled by the frequency and droop circuits discussed in the previous section on alternator constant-speed drive systems.

When AC generators are operated in parallel, frequency and voltage must both be equal. Where a synchronizing force is required to equalize only the voltage between DC generators, synchronizing forces are required to equalize both voltage and speed (frequency) between AC generators. On a comparative basis, the synchronizing forces for AC generators are much greater than for DC generators. When AC generators are of sufficient size and are operating at unequal frequencies and terminal voltages, serious damage may result if they are suddenly connected to each other through a common bus. To avoid this, the generators must be synchronized as closely as possible before connecting them together.

Regulating the voltage output of a DC exciter, which supplies current to the alternator rotor field, best controls the output voltage of an alternator. This is accomplished by the regulation of a 28-volt system connected in the field circuit of the exciter. A regulator controls the exciter field current and thus regulates the exciter output voltage applied

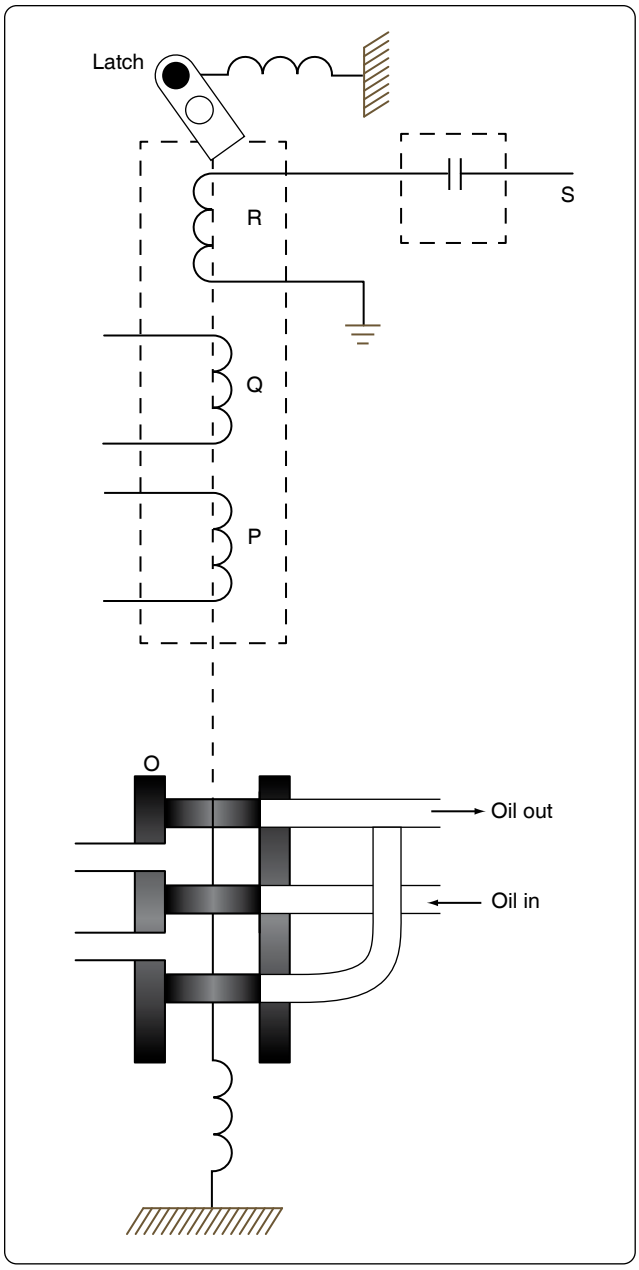


Figure 12-339. *Overspeed circuit.*

to the alternator field.

Alternator Transistorized Regulators

Many aircraft alternator systems use a transistorized voltage regulator to control the alternator output. Before studying this section, a review of transistor principles may be helpful.

A transistorized voltage regulator consists mainly of transistors, diodes, resistors, capacitors, and, usually, a thermistor. In operation, current flows through a diode and transistor path to the generator field. When the proper voltage level is reached, the regulating components cause the transistor to cut off conduction to control the alternator field strength. The regulator operating range is usually adjustable through a narrow range. The thermistor provides temperature compensation for the circuitry. The transistorized voltage regulator shown in *Figure 12-342* will be referred to in explaining the operation of this type of regulator.

The AC output of the generator is fed to the voltage regulator, where it is compared to a reference voltage, and the difference is applied to the control amplifier section of the regulator. If the output is too low, field strength of the AC exciter generator is increased by the circuitry in the regulator. If the output is too high, the field strength is reduced.

The power supply for the bridge circuit is CR1, which provides full-wave rectification of the three phase output from transformer T1. The DC output voltages of CR1 are proportional to the average phase voltages. Power is supplied from the negative end of the power supply through point B, R2, point C, zener diode (CR5), point D, and to the parallel hookup of V1 and R1. Takeoff point C of the bridge is located between resistor R2 and the zener diode. In the other leg of the reference bridge, resistors R9, R7, and the temperature compensating resistor RT1 are connected in series with V1 and R1 through points B, A, and D. The output of this leg of the bridge is at the wiper arm of R7.

As generator voltage changes occur, for example, if the voltage lowers, the voltage across R1 and V1 (once V2 starts conducting) remains constant. The total voltage change occurs across the bridge circuit. Since the voltage across the zener diode remains constant (once it starts conducting), the total voltage change occurring in that leg of the bridge is across resistor R2. In the other leg of the bridge, the voltage change across the resistors is proportional to their resistance values. Therefore, the voltage change across R2 is greater than the voltage change across R9 to wiper arm of R7. If the generator output voltage drops, point C is negative with respect to the wiper arm of R7. Conversely, if the generator voltage output increases, the polarity of the voltage between the two points is reversed.

The bridge output, taken between points C and A, is connected between the emitter and the base of transistor Q1. With the generator output voltage low, the voltage from the bridge is negative to the emitter and positive to the base. This is a forward bias signal to the transistor, and the emitter to collector current therefore increases. With the increase of current, the voltage across emitter resistor R11 increases.

This, in turn, applies a positive signal to the base of transistor

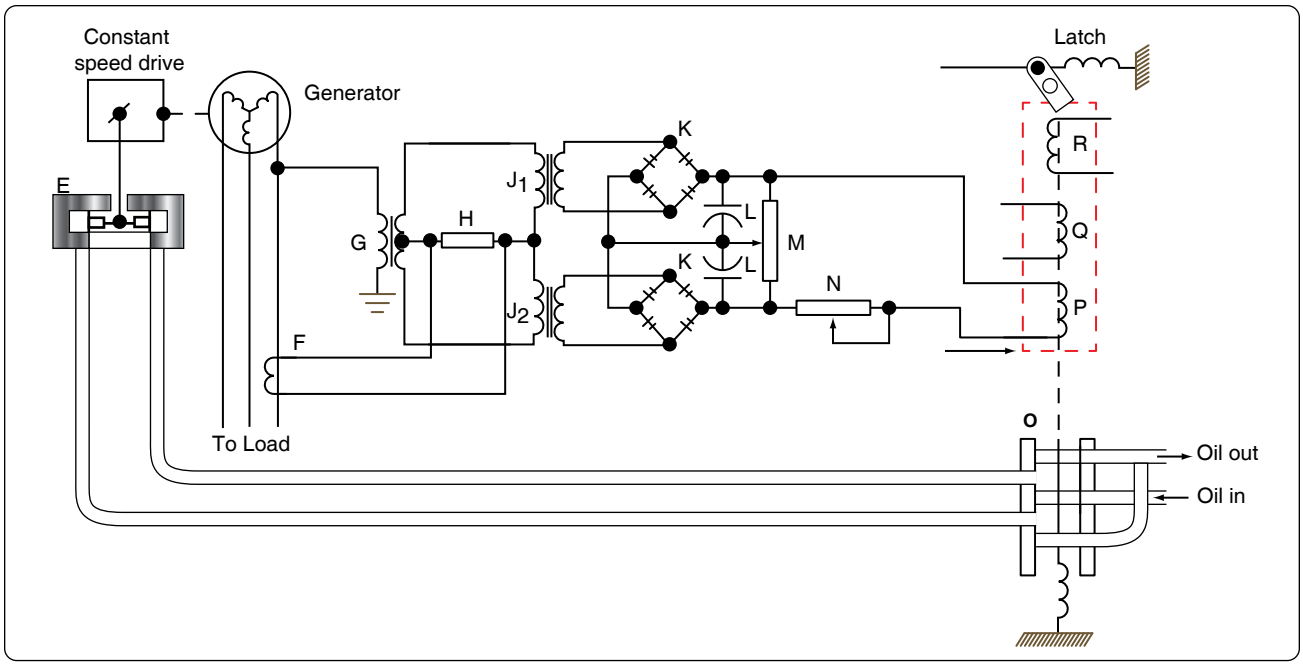


Figure 12-340. *Droop circuit.*

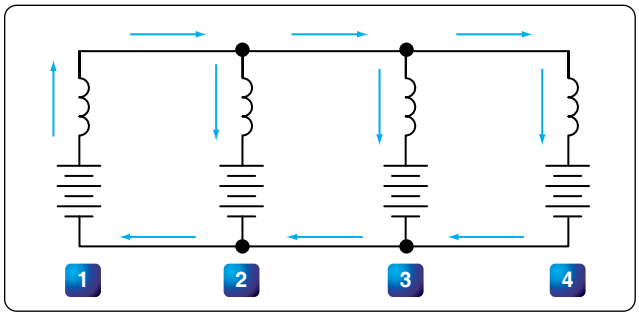


Figure 12-341. *Relative direction of current in droop coil circuit with unequal loads.*

Q4, increasing its emitter to collector current and increasing the voltage drop across the emitter resistor R10.

This gives a positive bias to the base of Q2, which increases its emitter to collector current and increase the voltage drop across its emitter resistor R4. This positive signal controls output transistor Q3. The positive signal on the base of Q3 increases the emitter to collector current.

The control field of the exciter generator is in the collector circuit. Increasing the output of the exciter generator increases the field strength of the AC generator, which increases the generator output.

To prevent exciting the generator when the frequency is at a low value, there is an underspeed switch located near the F+ terminal. When the generator reaches a suitable operating frequency, the switch closes and allows the generator to be excited.

Another item of interest is the line containing resistors R27, R28, and R29 in series with the normally closed contacts of the K1 relay. The operating coil of this relay is found in the lower left part of the schematic. Relay K1 is connected across the power supply (CR4) for the transistor amplifier. When the generator is started, electrical energy is supplied from the 28-volt DC bus to the exciter generator field to "flash the field" for initial excitation. When the field of the exciter generator has been energized, the AC generator starts to produce, and as it builds up, relay K1 is energized, opening the "field flash" circuit.

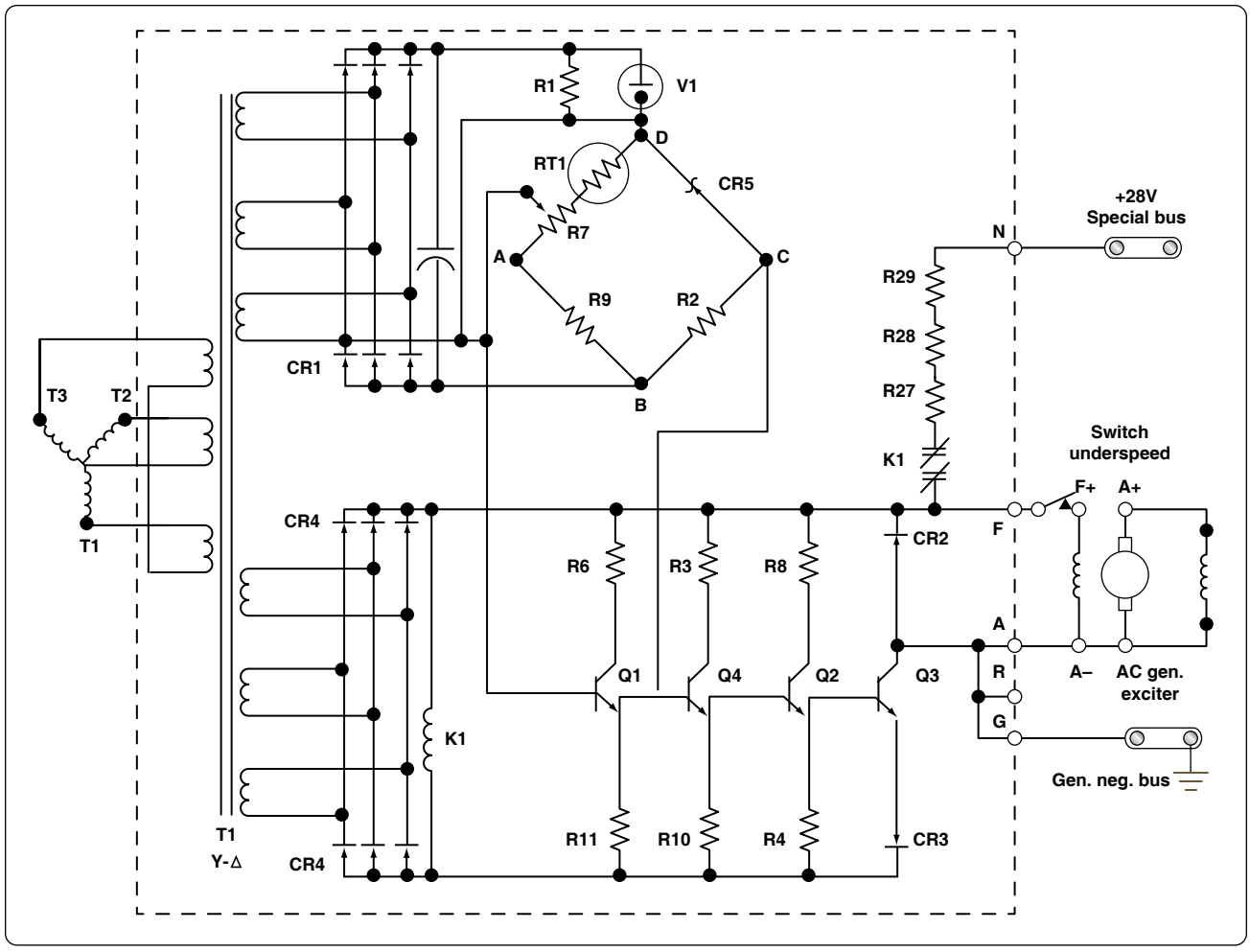


Figure 12-342. *Transistorized voltage regulator.*